Analysis Note:
Unbinned OmniFold analysis of MINERvA charged-current inclusive data

Joseph Bailey*[1] and Benjamin Nachman†[2]

[1]Department of Physics, Stanford University
[2]Department of Particle Physics and Astrophysics, Stanford University and SLAC National Laboratory

Version 1.0 (review), 20 October 2026

Code and file references: github.com/josephbaileyy/MINERvA-OmniFold

Abstract

We present a simultaneous measurement of the inclusive muon-neutrino charged-current cross section in muon transverse and longitudinal momenta, available hadronic energy, three-momentum transfer and hadronic invariant mass. The measurement uses OmniFold to unfold MINERvA medium-energy open data without binning and reports cross sections and their covariance in sixteen cells. The median relative uncertainty per cell is 7.86%, including a run-period contribution estimated after differences between periods were observed. Three external generator predictions give χ^2 values of 107–229 for 16 degrees of freedom; GiBUU gives 48 on its eight supported cells. Bias tests at two held-out GENIE v3 truths find residuals below the reported uncertainty in every cell. At a GiBUU-like truth and a synthetic momentum-transfer tilt, residual bias exceeds predeclared uncertainty-based thresholds. The coverage of the reported uncertainty intervals has not been validated.

Executive summary

We apply *unbinned machine-learning unfolding* (OmniFold [1, 2]) to MINERvA's medium-energy neutrino data. The principal result is the sixteen-cell cross section of §10 in the five observables p_T, $p_\parallel$, E_{avail}, q_3 and W. Its covariance gives a median per-cell uncertainty of 7.86%. Three external generator predictions, compared on all sixteen cells, give χ^2 between 107 and 229 for 16 degrees of freedom. GiBUU, compared only on the 8 cells its sample covers, gives $\chi^2 = 48$ on those cells (§10.6).

The unbinned result for $d^2\sigma/(dp_T\, dp_\parallel)$ reproduces the published integrated cross section [3] within about one percent, with a correlated shape difference (§6). Whether it is ready for publication as a measurement has not been established. LightGBM was chosen over the exact-split estimator, which is kept as a cross-check, after both comparisons with the publication were known (§3.2).

*josephrb@stanford.edu
†nachman@stanford.edu

Adding E_{avail}, q_3 and W gives simultaneous unfolds in three, four and five dimensions; the 3D, 4D and configuration-(a) five-dimensional unfolds are reported as central values. In configuration (a), the purity-subtracted estimator of the five-dimensional central value, the 4D and 5D marginal-normalization checks against the lower-dimensional result pass to ≈1–2 %. The 3D marginal of configuration (e), a deterministic estimator with refined negative-weight subtraction, agrees to 0.141 % in total with the 2D result of configuration (f), the nominal estimator with that subtraction, a cross-configuration comparison (§7). Reconstructed W is fixed by $(p_T, p_\parallel, q_3)$, so the five are simultaneous, not independent, observables (§8.2).

The generator comparisons of the central values are descriptive. All four truth-level generators, GENIE [4], MINERvA Tune [5], NuWro [6] and GiBUU [7], under-predict the unfolded central value in the quasielastic–Δ dip ($0.1 \leq E_{\text{avail}} < 0.4\,\text{GeV}$), the region of the established MINERvA low-recoil discrepancy [5, 8] (§7.6, §7.7). In cell-integrated terms, most of the positive difference from the Tune v1 prior lies at $E_{\text{avail}} \geq 0.8\,\text{GeV}$, much of it at $W \geq 1.8\,\text{GeV}$ (§8.4). The high-E_{avail}, high-W region carries the largest regularization bias measured in §9. The four external generators also fall below the data in this region, each by about its overall shortfall once GiBUU's sample limit is allowed for (§8.5).

The generator tests are an application that illustrates calibrated inference with a biased estimator. Calibrated joint five-dimensional tests of hybrid null hypotheses for five predictions, in total (rate and shape) and in shape, reject all ten under six stated conditions (§11.4). The rejections are not physical exclusions, and matched coarse projections reject as well (§11.7). The pseudo-experiments lost to time limits have been rerun, for reporting only, so every submitted draw has been observed. No recovered draw changes a decision (§11.6). Correcting the carbon-only NuWro and GiBUU predictions to the CH target moves both toward the data. The ten decisions, made with the carbon-only predictions, are unchanged. In the descriptive re-evaluation with corrected predictions, NuWro's total-test rejection is not robust to the correction (§11.9).

A methodological finding concerns the standard covariance, which sums single-source systematic matrices, each with all other sources nominal, and so omits nonlinear cross-source response terms. A corrected unified-throw test, which re-unfolds with all reweightable sources perturbed together, stores the joint mean shift separately (§4.5).

A preliminary extended-fiducial study removes the truth-muon cuts and keeps the fiducial volume and MINOS-based selection. It puts about 6 % of the scalar 2D rate in newly acceptance-supported cells (part of tier 1, the interpreted measurement) and about 28 % in low- or zero-efficiency cells (tier 2, extrapolated from the prior; §12).

The body states each limitation with the result it qualifies; the principal ones are these. C_k denotes coverage within $\pm k$ standard deviations. The quoted 2D statistical block has had no coverage test; the fixed-truth coverage studies scored two earlier statistical bands, of which the first-built undercovers in its tails ($C_2 = 0.912$, vs. 0.954; §4.2, §5.2). No result's intervals have validated coverage (two dimensions: §4.6; five dimensions: §9.5; the sixteen cells: §10.5). The others are stated with their results: the generator comparisons (§§7–8); the absence of a 3D or 4D covariance (§7); the five-dimensional estimator's regularization bias (§9.3) and the purity-subtraction bias of configurations (a) and (c) (§9.2); five-dimensional intervals and their coverage (§9.4, §9.5); the five-axis covariance of the configuration-(a) central value (§9.6); the calibration and nuisance model of the generator tests (§11.3, §11.8); tier 2 of the extended-fiducial study (§12); the fake-data limitation of the sixteen-cell result (§10.5); and point-cloud work (App. F).

Contents

Executive summary **1**

1 Introduction and motivation **5**

2 The MINERvA experiment, data and simulation samples **7**
2.1 Detector and beam . . . 7
2.2 Data and simulation samples . . . 7
2.3 Signal definition and event selection . . . 8
2.4 Per-event analysis trees . . . 9
2.5 Reco-level control distributions . . . 9

3 The unbinned OmniFold procedure **11**
3.1 Algorithm . . . 11
3.2 Estimators and configuration . . . 13
3.3 MINERvA driver and pipeline conventions . . . 16
3.4 Cross-section extraction . . . 16
3.5 Binning . . . 17
3.6 Related developments and estimator scope . . . 18

4 Uncertainty quantification: constructing the covariance blocks **18**
4.1 Systematic universes . . . 19
4.2 Statistical uncertainty: Poisson bootstrap . . . 20
4.3 ML/optimization uncertainty: training-seed variation . . . 20
4.4 Further checks of the 2D covariance . . . 21
4.5 Combination and the unified-throw test . . . 21
4.6 Scope, conventions and limitations of the 2D covariance . . . 22
4.7 What changes for the higher-dimensional products . . . 25

5 Validation of the two-dimensional measurement **25**
5.1 Closure tests across dimensions . . . 27
5.2 Fixed-truth coverage of the 2D statistical band . . . 29
5.3 Other validation diagnostics . . . 30

6 Results **31**
6.1 Double-differential cross section . . . 32
6.2 Comparison to the published measurement . . . 34
6.3 Uncertainty budget . . . 36
6.4 Eigenmode analysis of the published-covariance tension, $\chi^2 = 2.27$ . . . 37
6.5 Comparison to GENIE MINERvA Tune v1 . . . 38

7 Higher-dimensional extensions: available energy and momentum transfer **40**
7.1 Operational definition and provenance . . . 41
7.2 The available-energy spectrum . . . 42
7.3 Validation against the 2D measurement . . . 43
7.4 Closure: the one injected E_{avail} deformation is recovered . . . 44
7.5 Classifier-family cross-check (GBDT vs. NN) . . . 44
7.6 Generator comparison in 3D . . . 45

7.7 The low-available-energy excess and the known low-recoil deficit . . . 46
7.8 Fourth axis: three-momentum transfer q_3 . . . 48

8 Fifth axis: W and the (E_{avail}, W) plane 50
8.1 W as a fifth axis . . . 51
8.2 Implemented definitions of the five observables . . . 51
8.3 Generator predictions and the target difference . . . 52
8.4 Distribution of the excess in the (E_{avail}, W) plane . . . 53
8.5 Comparison with four external generators . . . 54

9 Five-dimensional estimator: validation and uncertainty 56
9.1 Which source of error is being tested . . . 56
9.2 Background subtraction and generator-sized truth changes . . . 56
9.3 Why the estimator does not follow a generator-sized change . . . 57
9.4 Model dependence, variability and reporting resolution . . . 59
9.5 Five-dimensional coverage test . . . 64
9.6 The five-axis covariance of the configuration-(a) central value and its (E_{avail}, W) projection . . . 65

10 The sixteen-cell five-observable measurement 67
10.1 Cells and result . . . 67
10.2 Quadrant sums against the lower-dimensional results . . . 67
10.3 The reported covariance . . . 69
10.4 The run-period block . . . 72
10.5 Fake-data tests of the estimator's bias . . . 73
10.6 Comparison with generator predictions . . . 76
10.7 The predictions' target and definitions . . . 77

11 Calibrated joint tests of generator predictions 83
11.1 Hypotheses, grid and statistics . . . 83
11.2 Calibration within the implemented simulation model . . . 84
11.3 Multiplicity, stopping rule and design . . . 85
11.4 Conditions . . . 86
11.5 Decisions . . . 88
11.6 Lost pseudo-experiments and their recovery . . . 90
11.7 Matched coarse projections . . . 90
11.8 Hadronic-response condition and recoil-energy sensitivity . . . 91
11.9 Hydrocarbon target correction for NuWro and GiBUU . . . 92
11.10 What the tests establish . . . 93

12 Extended fiducial phase-space study 93
12.1 Acceptance and the extended domain . . . 93
12.2 Validation tests . . . 94
12.3 Central value and two-tier reporting . . . 95

13 Summary and outlook 96

Guide to the appendices 97

A Statistical methods and the 2D comparison **98**
A.1 Numerical conventions: inverting a covariance (Cholesky, ridge jitter, pseudo-inverse) 98
A.2 Variance-source procedures . . . 99
A.3 Comparing to the paper . . . 104
A.4 Ill-conditioning and rank deficiency . . . 107
A.5 Anatomy of the paper-covariance tension . . . 108
A.6 Closure tests: detecting bias . . . 110
A.7 Summary of the 2D comparison numbers . . . 111

B Statistical methods for the multidimensional results and inference **112**
B.1 Rank, ndf and finite-ensemble declarations for N-D covariances . . . 112
B.2 Scalar-5D diagnostic reductions and interval definitions . . . 113
B.3 Extended-fiducial statistical-band and pull fractions . . . 114
B.4 Inference from unfolded events and selected projections . . . 114
B.5 Familywise error of the sequential joint-test decisions . . . 115

C Unbinned background subtraction by negative-weight injection **116**
C.1 Estimator . . . 116
C.2 Binned limit: exact reduction to the purity correction . . . 117
C.3 Closed-form toy . . . 118
C.4 Comparison on the 2D measurement . . . 118
C.5 Propagation through systematics and statistics . . . 119
C.6 Which background treatment each production chain uses . . . 120
C.7 Displacement argument and scope of the agreement . . . 121
C.8 Five-dimensional pseudo-experiments with both background treatments . . . 121

D Detector-response-mismatch closure test **122**

E The MINERvA measurement-dimensionality survey **124**

F Future work: point-cloud representations **125**

G Result package and reproduction limits **125**
G.1 Scope of reproduction and required inputs . . . 126
G.2 Release schema . . . 126
G.3 Bin edges . . . 127
G.4 One minimal projection example . . . 127

H Codebase layout and terminology **128**

I Supplementary figure: coverage of a purity-subtracted variant **129**

J Supplementary figure: closure with backgrounds **129**

1 Introduction and motivation

MINERvA (Main Injector Experiment for ν-A) is a dedicated neutrino–nucleus scattering experiment in Fermilab's NuMI beamline, immediately upstream of the MINOS near detector [9]. During roughly a decade of operation ending in 2019, it recorded high-statistics samples in low-energy and

medium-energy (ME) beam configurations [10] and in neutrino-dominated (forward horn current, FHC) and antineutrino-dominated modes. MINERvA measures neutrino–nucleus cross sections on a range of nuclear targets in the few-GeV region most relevant to accelerator-based oscillation experiments. These cross sections and their nuclear effects are a leading systematic uncertainty for T2K [11], NOvA [12], Hyper-Kamiokande [13], and DUNE [14]. The few-GeV regime combines quasielastic, resonant, and deep-inelastic processes that no single model describes well. This analysis uses the public ME FHC CC-inclusive data product (detector details in §2).

The MINERvA ME FHC charged-current (CC) inclusive double-differential cross section in muon transverse and longitudinal momentum, $d^2\sigma/(dp_T\, dp_\parallel)$, was published by Ruterbories *et al.* [3] using D'Agostini iterative Bayesian unfolding (IBU) [15] with a binned response matrix. We reproduce it with *unbinned* OmniFold [1, 2], which iteratively reweights events using machine-learned likelihood ratios on per-event observables rather than on a fixed histogram binning.

Machine-learning advances have enabled unbinned unfolding (overviews in Refs. [16, 17]; cross-experiment guidance in Ref. [18]). One broad family iteratively reweights a starting simulation; OmniFold [1, 2, 19] is its most widely adopted method. The other generates new samples directly with a neural network [20–23]. Both lift the binned-response limitations of widely-used classical methods [24] such as IBU, the SVD method [25], and TUnfold [26].

Prior OmniFold applications. In the last few years OmniFold has progressed from a method proposal to published measurements. H1 published the first real-data applications in deep-inelastic scattering: lepton–jet correlations unfolded simultaneously in eight dimensions [27], followed by jet substructure at high Q^2 [28]. LHCb unfolded identified-hadron distributions in Z-tagged jets [29]. ATLAS performed a simultaneous unbinned measurement of twenty-four Z+jets observables [30]. Ref. [18] collects these and other applications and distills cross-experiment practice. Three of the practices it audits appear in our uncertainty budget: the ensemble-mean cross-check of the central value, the bootstrap statistical covariance, and the joint-perturbation (unified-throw) covariance check of §4.5.

H1 cut backgrounds to a negligible level; LHCb corrected per track before unfolding. ATLAS left its less-than-0.25 % background in place and assigned it as an uncertainty, while identifying negatively-weighted simulated events as the natural in-method subtraction [2, 30]. Ref. [18] recommends this in-method subtraction for nontrivial backgrounds. Recent methodological work explicitly treats backgrounds, fakes, acceptance, and inefficiency in OmniFold-HI [31] and generative unfolding [32]. These constructions address related problems; their subtraction prescriptions can differ.

Closest to this analysis is the study of Huang *et al.* [33], the first application of OmniFold to a neutrino cross section: a mock-data unfolding of muon kinematics on the public T2K near-detector simulated dataset, benchmarked against traditional binned approaches. Two of its findings bear on neutrino applications. The first is its iteration count, which we do not adopt (§3.2). The second is an ensemble of reweighting trials with different train/test splits that controls the ML stochastic component; we include that component explicitly, as the training-seed variation band (§4). The present work differs from that study in three ways. It unfolds *real* MINERvA data to an absolutely normalized cross section, with a 2D uncertainty budget whose covariance components are all built with the central value's own learner (§3.2). Its gradient-boosted decision-tree (GBDT) learners suit a low-dimensional tabular feature set. It validates the method directly against a published binned measurement of the same data (§6).

Motivation for unbinned unfolding. First, OmniFold learns event weights in the full observable space and histograms only the final result, so reported bins can be chosen or refined after unfolding

without rerunning it. Background subtraction is unbinned too in the 2D result and its covariance, in the 3D and 4D central values and in the sixteen-cell five-dimensional result (§10), which use refined negative-weight subtraction (§3.1). The configuration-(a) five-dimensional central value and its five-axis covariance, and the extended-fiducial value, use the binned purity weight, a per-reco-bin $(D - B)/D$ weight fixed *before* training. Second, additional observables enter as input features, without multiplying response-matrix bins. The available-energy extension (§7) is built this way; a binned method such as IBU would need a correspondingly larger multidimensional response matrix to include it. Third, OmniFold regularizes through model capacity and through an iteration count, which it shares with IBU. Comparing the two methods on the same data probes the result's method dependence (§6).

Analysis strategy. The published 2D cross section provides a quantitative validation of unbinned OmniFold on MINERvA data and a size comparison for its uncertainty budget (§§2–6). The same pipeline then adds E_{avail}, q_3 (§7) and W (§8) as classifier inputs, tested by marginalization to each lower-dimensional result, closure, coverage and classifier-family comparisons. The validation and uncertainty of the five-dimensional estimator (§9), the sixteen-cell result (§10) and the calibrated generator tests built on the estimator (§11) follow these central values. A scalar study with relaxed truth-muon cuts (§12) probes an alternative signal definition. The companion journal article is a shortened account, and this note gives the detail it omits.

Scope. The results hold for one implementation of unbinned OmniFold, applied to one data set, and for the configurations, observables and deformations tested. The principal limitations are listed in the executive summary (p. 2). The generator tests of §11 evaluate hybrid null hypotheses and add no reportable uncertainty. Full-event unfolding with point-cloud transformer (PET) representations is in progress and documented separately (App. F), and no result in this note depends on it.

2 The MINERvA experiment, data and simulation samples

2.1 Detector and beam

MINERvA [9] was a fine-grained scintillator-tracker neutrino experiment on Fermilab's NuMI beamline. Its central tracker comprises hexagonal planes of triangular polystyrene scintillator strips in three stereo orientations, surrounded by electromagnetic and hadronic calorimetry. The magnetized MINOS near detector, 2 m downstream, serves as the muon spectrometer, measuring momentum and charge for MINOS-matched tracks. We use the *medium-energy* (ME) neutrino-mode (FHC) exposure with $\langle E_\nu \rangle \sim 6\,\text{GeV}$. The flux is the *a posteriori* constrained NuMI ME prediction of Ref. [10], combining PPFX hadron-production weights and the neutrino–electron-scattering constraint, identical to the published measurement's flux [3].

2.2 Data and simulation samples

The twelve ME FHC playlists (ME1A–ME1G, ME1L–ME1P) contain 1.057×10^{21} protons on target, the same exposure as Ref. [3]. The selection below leaves 4 119 797 reconstructed data events.

The reference simulation uses GENIE 2.12.6 [4] and MINERvA Tune v1, applied as per-event weights by the reweighting tools of the MINERvA Analysis Toolkit (MAT). These are the flux-and-central-value (flux-and-CV) weight, the GENIE model-parameter ("dial") weights, the low-recoil 2p2h enhancement fit, and the non-resonant-pion reduction. The full GEANT4-based [34] MINERvA detector simulation underlies the simulated pre-selected analysis ntuples (AnaTuples). The simulated

sample contains 32 849 103 true signal events, POT-normalized event-by-event to the data exposure and processed with identical reconstruction and selection.

2.3 Signal definition and event selection

The signal definition and phase space follow Ref. [3] exactly. Truth-level signal events are charged-current ν_μ interactions in the central-tracker fiducial volume with muon phase space

$$\theta_\mu < 20^\circ, \qquad 1.5 < p_\parallel < 60\,\mathrm{GeV/c}, \qquad p_T < 4.5\,\mathrm{GeV/c}.$$

The fiducial volume spans $5980 < z < 8422$ mm within an 850 mm apothem and contains 3.2353×10^{30} nucleons, the fixed tracker-geometry constant.

Reconstructed events require a vertex in the same fiducial volume and apothem, a muon track with $\theta_\mu < 20^\circ$ matched to a track in the MINOS near detector whose MINOS track fit succeeded, no dead-time overlap, and a neutrino (rather than antineutrino) candidate.

Genuine backgrounds are reconstructed events whose true interaction fails the signal definition: charged-current events of other flavors or wrong sign, neutral-current interactions, and out-of-fiducial-vertex events. Their predicted rate is 0.35 % of selected events, evaluated on playlist 1A. This is the same sub-percent scale as the 8655 (0.2 %) predicted background events of Ref. [3], which counts only the wrong-flavor, wrong-sign and neutral-current components; exact agreement is not expected given the differences in definition and selection, and neither prediction carries an uncertainty. Out-of-phase-space signal, reconstructed inside the phase space with its true muon outside it, is treated separately. Both are POT-scaled and subtracted from the reconstructed-data target before unfolding (§3).

Event selection determines which observations enter the analysis and which truth population is being estimated. Classifier features determine what the learner can use within that population. A muon can therefore be required for selection even when its momentum is absent from the classifier.

Analysis	Selected observations	Truth domain	Classifier information
Scalar 2D–5D	Tracker vertex and MINOS-matched muon; common reconstructed selection	Published muon cuts	Muon $(p_T, p_\parallel)$; successively E_{avail}, q_3, W
Scalar extended fiducial	Same reconstructed selection	Tracker CC ν_μ; truth-muon cuts removed	Muon kinematics; extended-domain study
Historical recoil-only point-cloud (PET)	MINOS-based selection, including the muon	Standard or extended domain, as declared for each cross-check	Recoil clusters at reco; hadron four-momenta at truth; no muon features
Full-event PET candidate	Same MINOS-based reconstructed selection	Extended tracker CC ν_μ domain	Muon event features plus recoil/hadron clouds
PET design studies	Signal-only simulation samples with fixed selection and sampling	Declared sample population	Muon and cloud features; finalists add detector energy summaries

Table 1: Selection and classifier information are separate choices. Reporting bins are not classifier inputs. Extended fiducial domain: §12. PET remains diagnostic method development; its design-study recovery does not validate real-data background subtraction.

2.4 Per-event analysis trees

Unbinned unfolding needs per-event records rather than histograms. A dedicated event loop (App. H), built on MAT's systematic-universe tools, writes four per-event samples for each playlist: all true signal events (the efficiency denominator), reconstructed truth-matched simulated events, simulated background, and data. Entries carry truth and reconstructed observables (p_T, $p_\parallel$, and in extended runs E_{avail}, q_3, W) and the MINERvA Tune v1 central-value weight. The systematic-universe samples also carry every universe's weight and the shifted observables of each lateral band, a variation that shifts reconstructed kinematics rather than reweighting events (a vertical band; §4). Two conditions make the truth sample the complete reference for the simulated signal:

- A reconstructed signal event is filled only if its (run, subrun, event) key is in the truth denominator. Truth events without a reconstructed partner enter as native OmniFold "miss" entries.
- The simulated signal sample is complete by construction at ME FHC (completeness $c = 1.000000$; §3.4).

The loop reads one playlist-wide flux and calibration from the first event, so it is run separately for each playlist, and the twelve per-playlist outputs are combined. The corresponding baseline histogram event loop supplies the flux histogram, POT-weighted across playlists; playlist-to-playlist flux differences are a $\sim 0.2\,\%$ effect.

2.5 Reco-level control distributions

Selected data exceed the POT-scaled MINERvA Tune v1 prediction (signal + background) by $\sim 12\,\%$ in overall normalization. Figure 1 compares all five reconstructed observables. The data/MC ratio rises with p_T, from 0.88 in the lowest bin to ~ 1.2 above 1 GeV/c. Excess and trend together reflect the truth-level normalization deficit of §6.5 and §7.6. The simulation describes the *shapes* well, and the unfolding relies on the shapes; its reweighting toward the data corrects the rate offset.

The beam and acceptance set the shapes. The $p_\parallel$ peak at 4–6 GeV/c reflects the $\langle E_\nu \rangle \approx 6$ GeV NuMI medium-energy peak; spectrometer matching removes the lowest-$p_\parallel$ muons. The p_T spectrum peaks near 0.7 GeV/c and falls by over an order of magnitude across the measured range. Low-recoil (quasi-elastic and $2p2h$) events dominate the sample. Available energy E_{avail} concentrates below $\sim$0.5 GeV, with a long inelastic tail in the wide upper bins; q_3 tracks this population's momentum transfer. Reconstructed W rises through the smeared quasi-elastic and $\Delta(1232)$ region before the deep-inelastic continuum takes over above $\sim$2 GeV. Absolute background is at the few-percent level under the muon projections and concentrates at high recoil (highest E_{avail}, q_3 and W bins). Beyond the p_T trend noted above, the ratio panels also show a mild enhancement in the lowest E_{avail} bins, reflecting the low-recoil excess examined at truth level in §7.7.

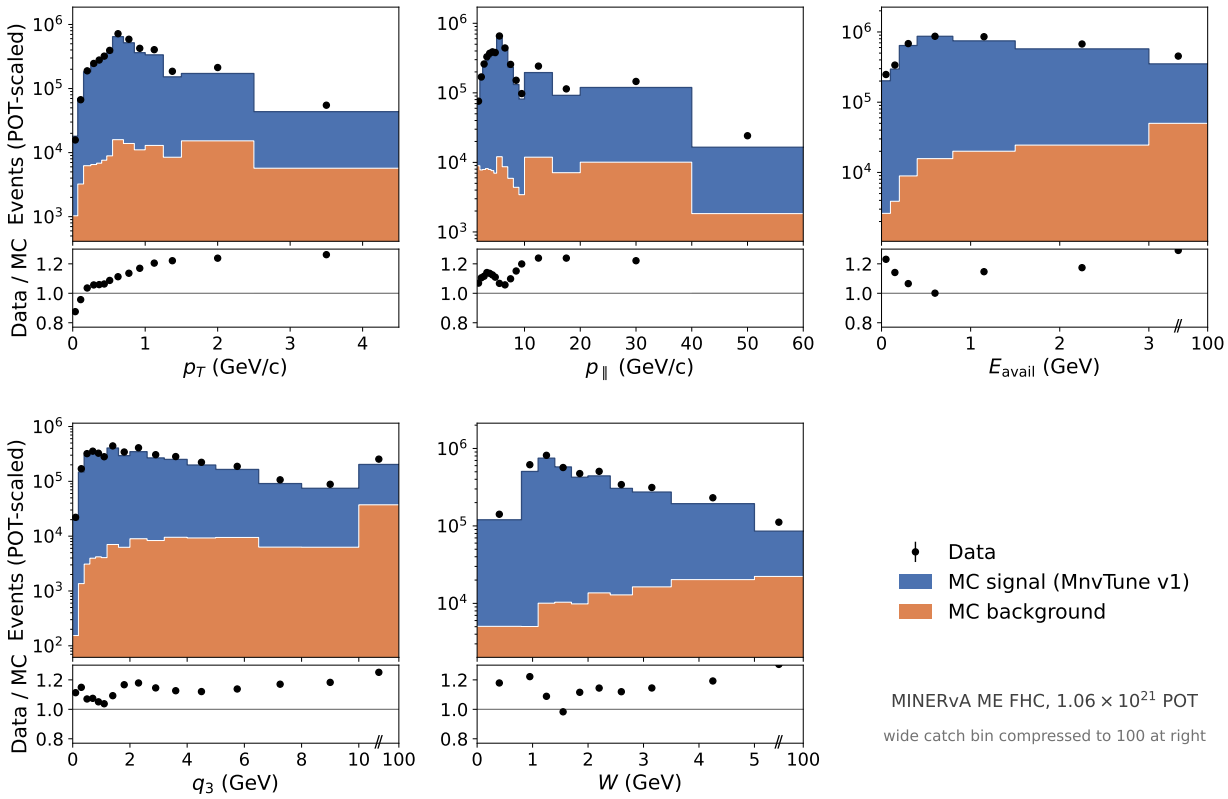


Figure 1: Reconstructed p_T, $p_\parallel$, E_{avail}, q_3 and W for the analysis selection. Data points overlay POT-scaled simulated signal (blue) stacked on background (orange); data/MC ratios appear below. Wide hadronic catch bins extend to 100 GeV and are compressed at each panel's right edge. The gray MC-statistical band on the ratio is mostly narrower than the points. The $\sim 12\,\%$ data normalization excess over MINERvA Tune v1 reflects the truth-level generator deficits in §7.6.

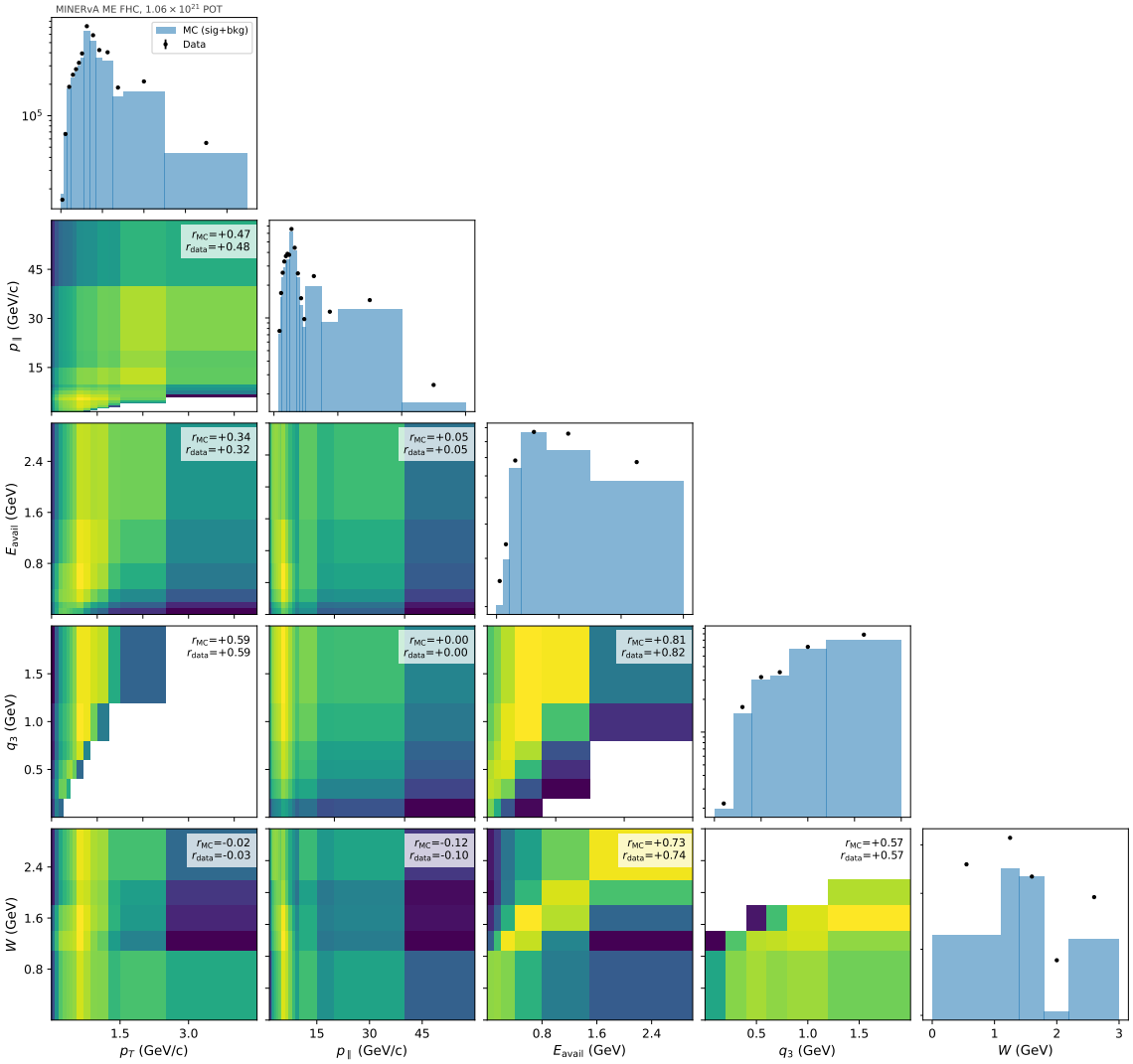


Figure 2: Reco-level corner plot of five observables. The lower triangle shows pairwise POT-scaled MC (signal+background) densities on a logarithmic color scale, with wide catch bins dropped. Each panel annotates the simulation's weighted Pearson correlation and the corresponding data value. The diagonal shows 1D marginals (data points over MC). The strongest correlations follow from kinematics: q_3 with E_{avail} ($r \approx 0.81$) and W with E_{avail} ($r \approx 0.73$) share recoil energy; q_3 with p_T ($r \approx 0.59$) shares the muon transverse kick. Both E_{avail} and W are nearly uncorrelated with the muon variables. Data reproduce every simulated coefficient within 0.02.

3 The unbinned OmniFold procedure

3.1 Algorithm

OmniFold [1, 2] unfolds by iterated reweighting with machine-learned likelihood ratios, operating on per-event observables rather than on a fixed histogram binning. Each iteration has two steps:

1. **Step 1 (detector space).** A classifier separates measured data from reconstructed simulation weighted by the current push-weights. Its calibrated output p gives the likelihood-ratio reweight

$w = p/(1-p)$ for each reconstructed simulated event. Each simulated event is a (truth, reco) pair, allowing the weight to be *pulled back* directly to its truth record. Truth events without a reconstructed partner ("misses") receive step-1 weights from a regressor trained on the reconstructed subsample, w(truth features). This native OmniFold acceptance treatment requires no separate efficiency correction.

2. **Step 2 (truth space).** A second classifier separates the original truth-level simulation from the same sample carrying the pulled-back step-1 weights. Its smooth truth-space reweight becomes the next iteration's push-weight.

After n iterations, the push-weights define an efficiency-corrected truth-level reweighted simulation ensemble. Histogramming this ensemble in any binning chosen afterwards gives the unfolded spectrum. The iteration count and learner capacity regularize the unfolding.

Training details. Each classifier is trained on the full sample whose weights it then sets, with no held-out or cross-fitted split and no early stopping. The two classes keep their physical weights, with no class balancing and no renormalization between iterations, so the learned ratio $p/(1-p)$, with p restricted to $[10^{-6}, 1-10^{-6}]$, carries the normalization. Truth events without a reconstructed partner receive the prediction of an unweighted regressor of the step-1 weight on the truth observables. In the refined configurations, a classifier separating positive from negative weights, trained with $|w|$, gives each event the weight $|w| \max(0, 2g-1)$. No effective sample size of the unfolded ensemble is computed.

Background subtraction. In the binned purity subtraction, all selected data events enter step 1; none is discarded and no negative-weight events are injected. Each receives a reco-bin *purity* weight. The POT-scaled background prediction includes genuine backgrounds and out-of-phase-space signal (§2.3) and is binned on the reconstructed analysis grid. A data event in reco bin b receives

$$w_{\mathrm{pur}}(b) \;=\; \max\!\big(0,\; N_{\mathrm{data}}(b) - N_{\mathrm{bkg}}(b)\big)\big/N_{\mathrm{data}}(b),$$

the predicted signal purity, floored at zero in the rare bins where the background prediction exceeds the data. The step-1 classifier retains the full unbinned kinematics. Before the final histogramming, binning enters, by design, only through this multiplicative correction, an $\sim$3 % effect overall, largest where backgrounds concentrate (§2.3).

Other experiments' treatments are summarized in §1. At the sub-percent genuine-background scale, a per-reco-bin weight is a conservative middle ground. A fully unbinned alternative injects negatively weighted background simulation on the measured side, targeting the identical background-subtracted density $(D-B)/S$. It reproduces the binned correction to -0.13 % on the total 2D cross section and its systematic and statistical covariances to within 2 % (App. C). It removes residual binning dependence and suits the extended-fiducial analysis, where backgrounds are larger. The quoted 2D result and its covariance (App. C, §6), the 3D and 4D central values (§7) and the sixteen-cell 5D result and its covariance (§10) use its refined form; so does the refined arm of the background-bias and generator-sized-change studies (§9.2), the held-out-shape and precision studies (§§9.3–9.4) and the generator tests. Every other quoted cross section and uncertainty, including the configuration-(a) five-dimensional central value and the extended-fiducial value, uses the binned correction. Table 2 (§3.2) gives the configuration behind each result.

3.2 Estimators and configuration

Gradient-boosted decision trees (GBDTs) replace the original OmniFold neural networks for the unfolding of scalar observables. With at most five high-level features per event, tree ensembles typically match or outperform neural networks [35] and train deterministically and far faster on CPU. Ref. [36] demonstrates their effectiveness for unbinned unfolding with few high-level features, complementing deep models for low-level event representations. We use GBDTs for all unfolding results in this note (point-cloud representations: App. F). Calibration tests and MLP/transformer cross-checks test the implementation in §7.5.

The unfolding code supports interchangeable learners (component terminology and code paths in App. H). The quoted 2D central value and all of its uncertainty ensembles use LightGBM with refined negative-weight background subtraction (App. C), configuration (f) of Table 2. These ensembles are the systematic-universe sweep, the Poisson bootstrap, the training-seed variations and the run-period unfoldings. The step-1 classifier, the step-2 classifier and the regressor for unreconstructed truth events share one setting: 100 trees, 8 leaves and learning rate 0.1. The central value is the universe sweep's own central value at seed 42, so the systematic covariance shares its estimator and seed (§4.1). The quoted statistical covariance, from a bootstrap drawn in three independent streams (data, signal and background simulation), also uses seed 42, and the training-seed covariance uses five seed settings that include seed 42 (§4.2, §4.3).

A central unfolding at seed 45 differs from the seed-42 one by at most 1.4 % in a cell. With the purity-based subtraction (configuration (a)), the seed-42 central value differs from the seed-1 one by a per-bin median of 0.18 % (maximum 1.79 %), or 0.30 (maximum 1.28) of the per-bin standard deviation of the earlier statistical band, built at seed 1. Two regenerations, with one driver and one input and differing only in the seed, reproduce the two central values to $\lesssim 2 \times 10^{-11}$, so the difference is attributable to the seed. The quoted central value records its run configuration; it repeats to 9×10^{-12} and equals the refined 2D data unfold of App. C.6 to 7×10^{-12}. The purity-subtracted seed-42 central value has no recorded run configuration. An independent rerun reproduces it to 1.4×10^{-11}, and so does a regeneration whose run configuration is recorded (Ref. [37], R1.12). Against the ten purity-subtracted LightGBM central values at seeds 1–10 it is mildly over-dispersed (RMS standardized difference 1.28, maximum 4.25). Seed 42 was chosen on computing cost: building the statistical band at seed 42 was the cheaper of the two costed ways to give the central value and the statistical band a common seed.

The exact-split cross-check. The exact-split configuration (`exact`) uses scikit-learn `GradientBoosting` with single-threaded exact-split CART. Its seed was not fixed, and its run log was not found; two driver revisions regenerate it to 7×10^{-15}. It is not invariant to the estimator seed, however: one reseeded run (`random_state` 1/2/3) moves every bin, by a median of 0.18 % (maximum 1.6 %), and gives $\chi^2/\mathrm{ndf} = 3.57$. The spread over seeds is unmeasured. The reseeded run's median agrees with the 0.18 % LightGBM seed-1 to seed-42 median above, but the agreement is coincidental. The exact-split central value differs from the purity-subtracted LightGBM seed-42 central value by a per-bin median of 0.98 % (maximum 12.2 %), 1.32 statistical standard deviations. It differs from the quoted central value, which also differs from it in background treatment, by a per-bin median of 1.03 % (maximum 17.6 %). It gives $\chi^2/\mathrm{ndf} = 3.66$ (unseeded, reproducible) against the published covariance, against 2.65 for the purity-subtracted LightGBM central value and 2.27 for the quoted one (§6.4). It has no matched covariance: no exact-split bootstrap, universe sweep or seed scan exists. LightGBM was chosen for the central value after both comparisons with the publication were known, so that the central value and its covariance share an estimator. A third learner, the scikit-learn histogram backend (`hist`, `HistGradientBoosting`, matched 8-leaf configuration), is retained for

method-dependence studies (§6.4).

The estimator configurations. Six estimator configurations, labeled (a)–(f), are used; Table 2 gives each one's learner, background treatment and results, and each result names the configuration it uses. All use five OmniFold iterations and gradient-boosted trees of nominal capacity (100 trees, 8 leaves) in both OmniFold steps; "deterministic" means fixed seeds and reproducible training. Results from different configurations are not combined. Four are described in this section:

- (f), the nominal LightGBM configuration above with refined negative-weight subtraction and the refinement classifier of (e), seeded at the estimator seed plus three: the 2D central value (seed 42) and its uncertainty ensembles (§6);
- (a), the same learner with purity-based background subtraction (§3.1): the five-dimensional central value (§8), the 3D and 4D injected-bump closures and Figs. 17 and 23 (§7), and the purity-subtracted 2D comparisons of this section and §6.4;
- (e), the joint-test estimator: a deterministic LightGBM variant with the unbinned negative-weight background treatment (App. C), whose refinement classifier has 400 trees and 31 leaves. The negative-weight treatment reduces the purity method's nominal-truth bias in the highest-W column at least fifteen-fold (from 3.5 % to 4.1 % to 0.11 % to 0.24 %, the level of the no-background closure), and the larger refinement classifier removes most of the background-related bias that remains (§§9.2–9.3). It gives the 3D and 4D central values (§7) and the sixteen-cell five-dimensional result (§10). It is used for the generator tests (§11), the study of generator shapes held out of its development (§9.3) and the precision study (§9.4). It was selected before the joint-test design was fixed (§11);
- (b), the exact-split cross-check above.

Configuration (c) is the deterministic variant with purity-based subtraction, and (d) is the same variant with negative weights and a nominal-capacity refinement; they serve the background-bias, generator-sized-change and coverage studies listed in the table. The five-dimensional central value of §8 and its five-axis covariance use configuration (a): they were produced before the purity bias was found and the refinement was developed. A refined five-dimensional data unfolding (configuration (e)) gives the sixteen-cell result; its reported covariance on the sixteen cells is described in §10.3. The 2D central value and its covariance were rebuilt with configuration (f) after a two-dimensional closure test with backgrounds found the purity bias (App. C.6). The change was decided on that closure test, before the rebuilt result's distance from the published result was computed, and the rebuilt result was adopted after that distance was known. The exact-split central value was the quoted 2D central value until 2026-10-11; studies labeled *exact-split* use it and have not been recomputed. The 3D and 4D central values have been regenerated with configuration (e); no 3D or 4D covariance is quoted.

Hyperparameter choice. The learner settings were fixed *a priori*; no hyperparameter search was performed on data, which avoids tuning on the result. They are the scikit-learn `GradientBoosting` defaults: 100 trees, learning rate 0.1, and depth-3 capacity. The LightGBM learner of the central value and its ensembles uses matched capacity (`num_leaves` $= 8 = 2^3$), giving every learner the same deliberately modest capacity, one of OmniFold's regularizers (§3.1). The configuration is instead validated after the fact by closure and classifier-calibration tests (§5, §7.5), insensitivity to doubling the iteration count, measured for the exact-split cross-check (below), backend agreement in method-dependence studies (§6.4), and an explicit training-seed variation band for residual ML stochasticity (§4).

Table 2: Estimator configurations.

Configuration	Learner	Background treatment	Results
(a) central-value estimator	LightGBM; seed 42 in two dimensions	purity-based subtraction	five-observable central value of §8 and its five-axis covariance (§9.6); three- and four-observable closures and Figs. 17, 23 (§7); two-observable purity-subtracted comparisons (§6.4)
(b) exact-split cross-check	scikit-learn exact splits; seed not fixed	purity-based subtraction	two-observable cross-check (above); studies labeled *exact-split*
(c) deterministic variant	LightGBM, deterministic	purity-based subtraction	background bias (§9.2); coverage test (§9.5)
(d) deterministic variant with refinement	as (c)	negative weights; refinement classifier 100 trees, 8 leaves	background bias; generator-sized E_{avail} change (§9.2)
(e) joint-test estimator	as (c)	negative weights; refinement classifier 400 trees, 31 leaves	three- and four-observable central values (§7); sixteen-cell five-observable result (§10); generator shapes held out of development; precision study (§9.4); generator tests (§11)
(f) refined central-value estimator	as (a); refinement seeded at the estimator seed plus three	negative weights; refinement classifier 400 trees, 31 leaves	two-observable central value and covariance (§6)

Production uses $n = 5$ iterations. Huang *et al.* [33] select 45 iterations for OmniFold-family methods (20 for their binned benchmark) using a binned χ^2 against known mock-data truth, a criterion they note is infeasible on real data. We do not adopt that recommendation. Their mock data injects a large prior–data *shape* discrepancy through an RPA-modified quasielastic model, requiring many iterations. The simulation prior already describes the measured shapes well (§2.5); the correction the data impose on it is dominantly one of normalization. Cross-experiment production also typically uses ∼5 iterations [18]. Our $n = 5$ is instead justified empirically: for the exact-split cross-check, the total cross section agrees to within 0.026 % between five and ten iterations (§5), and the unfolded result reproduces an independently published binned measurement of the same data (§6).

3.3 MINERvA driver and pipeline conventions

The higher-dimensional MINERvA driver (App. H) extends the 2D driver using an axis list and the same 2D helpers. The first two axes are always $(p_T, p_\parallel)$, carrying the fiducial phase-space cut and the per-p_T flux. Higher-dimensional runs extend the muon observables $(p_T, p_\parallel)$ with E_{avail}, q_3, W in the phase space of §2.3. Every production, universe-sweep, and closure run follows these conventions:

- **Reco-space masking.** The measured events entering step-1 training are exactly those passing the reconstructed-level selection; nothing outside the reconstructed phase space reaches the classifier.
- **Step-1 training within the phase space.** Step 1 targets the background-subtracted data of §3.1, purity-weighted or with background simulation injected at negative weight (refined to non-negative weights in the refined configurations). Genuine backgrounds *and* out-of-phase-space signal are removed from the data side. The simulated reconstructed sample contains no out-of-phase-space signal by construction, so neither side of the step-1 training contains events whose truth lies outside the measurement phase space.
- At histogramming, each simulated truth event carries its final OmniFold push-weight times its central-value simulation weight, including MINERvA Tune v1 reweighting. No efficiency-histogram division is applied: the step-1 regression for unreconstructed truth events already gives an efficiency-corrected truth spectrum, and dividing again would double-correct.
- **Closure mirrors production.** In closure tests the pseudo-data are simulated events passing both the reconstructed and truth selections, weighted identically to the production data path, so closure exercises the same code path as the measurement.

3.4 Cross-section extraction

The differential cross section follows from unfolded truth-level counts:

$$\frac{d^n\sigma}{dx_1 \cdots dx_n} = \frac{U}{c\,\Phi\,N\,\mathrm{POT}\,\Delta x_1 \cdots \Delta x_n} \times 10^4,$$

where U is the unfolded yield and c the per-bin *completeness*, the fraction of truth events visible to the unfolding input. The event loop appends the truth events without a reconstructed partner, so every truth event enters and $c \equiv 1$ bin-by-bin. The factor remains a monitored self-check; dropping those events would give $c < 1$. Φ is the per-p_T flux integral, broadcast over other axes, N the fiducial nucleon count (3.2353×10^{30}), and POT the data exposure (1.057×10^{21}). The factor 10^4 converts flux tabulated per square meter to the reported cm^2 ($1\,\mathrm{m}^2 = 10^4\,\mathrm{cm}^2$), giving units of cm^2 per axis

unit per nucleon. The implementation (App. H) is self-tested to reproduce stored lower-dimensional reference extractions to $< 10^{-12}$. Since $\sigma \propto 1/\Phi$, Φ varies per flux universe when building the systematic covariance. This variation dominates the Flux band (§4).

For the higher-dimensional extractions the integrated flux is $\Phi = 6.243 \times 10^{-8}\,\mathrm{cm}^{-2}$ per proton on target, from 0 to 100 GeV after the neutrino–electron-scattering constraint, and the target contains 3.235×10^{30} nucleons.

3.5 Binning

The 2D grid has 14 p_T × 16 $p_\parallel$ bins; 205 of 224 are reported. The 19 unreported diagonal cells have $p_T^{\mathrm{hi}}/p_\parallel^{\mathrm{lo}} > \tan 20°$. The extension axes use E_{avail}: $[0, 0.1, 0.2, 0.4, 0.8, 1.5, 3.0, 100]$ GeV; q_3: $[0, 0.2, 0.4, 0.6, 0.8, 1.2, 2.0, 100]$ GeV; W: $[0, 1.1, 1.4, 1.8, 2.2, 3.0, 100]$ GeV. Each ends in a wide catch bin to capture the full CC-inclusive tails in marginals; truncating the top edge would break the marginalization checks of §7.3. Binning after unfolding allows these choices to be revisited without re-running the unfolding. App. G (§G.3) gives all five axes' edges. §A.3.1 defines the 205 reported cells; the authoritative mapping is the bin-mapping file of the published data release [3].

Neither underflow nor overflow appears in the reported product, by construction rather than by truncation: the unbinned unfolding has no response-matrix under- or overflow, and the reporting bins are applied once, to events that all lie inside the grids. Truth-level muon edges coincide with the measurement phase space (§2.3); no signal is defined beyond them. Hadronic axes span $[0, 100]$ GeV, with catch-bin edges far beyond the kinematic reach at $\langle E_\nu \rangle \sim 6$ GeV, so the truth grids are exhaustive. A selected reco event outside the grid is treated as unaccepted and excluded from the measured sample by §3.3's masking convention. Its in-phase-space truth content is recovered, as for any unreconstructed event, by the treatment of truth events without a reconstructed partner.

Truth→reco migration on the analysis binning (Fig. 3) summarizes the smearing OmniFold must undo, although it uses no response matrix. Median diagonal purity is ~ 0.6 on every axis. At migrations of this size a bin-by-bin correction would be badly biased, so unfolding is required.

The structure differs qualitatively between the muon and hadron sides. The spectrometer muon measurement makes the p_T and $p_\parallel$ maps nearly tridiagonal, with migrations rarely exceeding one bin. Purity dips to ∼0.45–0.5 for p_T and ∼0.35 for $p_\parallel$ where bins are finest, recovering in wide edge bins. The calorimetric axes have coarser bins but relatively broader, visibly asymmetric migrations. E_{avail} preferentially migrates *downward* because energy carried by neutrons and other unresponsive particles is not collected. Diagonal purity falls from 0.61 in the lowest bins to 0.43 in the highest. q_3, built from E_{avail} and the muon, inherits a milder version of the same pattern. W compounds E_{avail} and muon smearing and has the broadest migrations; two-bin shifts are common in the resonance region. The lowest-W (quasi-elastic) bin remains purest at 0.71.

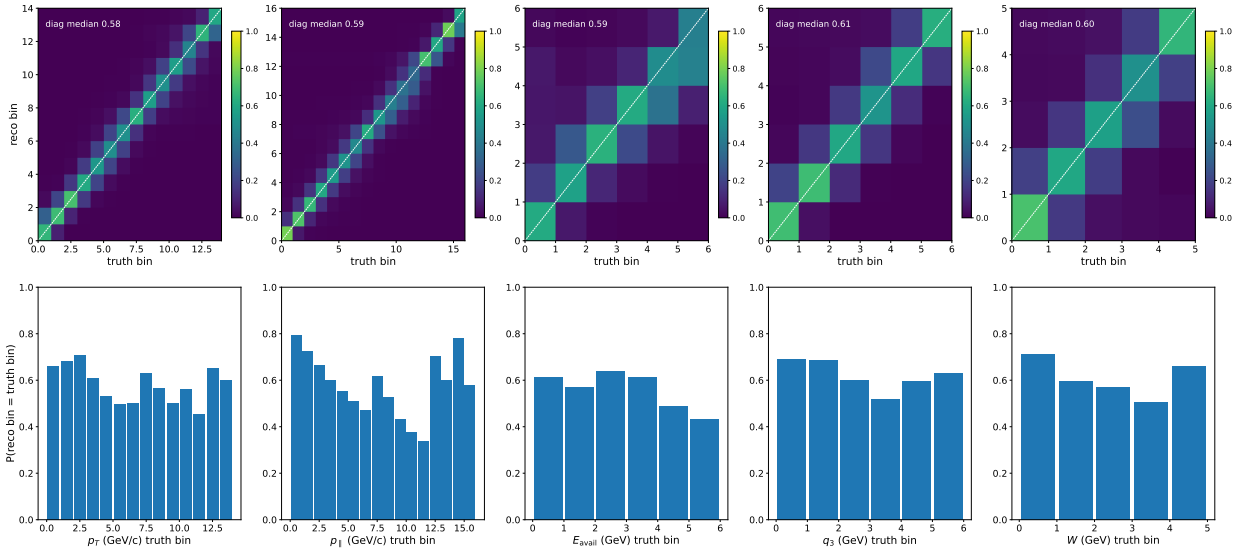


Figure 3: Simulated signal truth→reco migration on the analysis binning for five observables. Top: column-normalized maps; bottom: diagonal purity per truth bin. Median diagonal purity is 0.58–0.61 per axis. The figure is illustrative only; the unfolding is unbinned.

3.6 Related developments and estimator scope

Profile OmniFold [38] extends classifier-based EM to forward-model nuisance parameters. Its distinct algorithmic construction complements earlier unbinned profiled unfolding [19]. Profiling nuisances with observed data would change our inference model, which propagates an external nuisance ensemble through a fixed procedure. Neither a profiled estimator nor its uncertainty is used here.

SPINUP learns a forward mapping [39]; AUSSIE replaces OmniFold's second step with a different loss and avoids the outer iteration [40]. Both offer possible future cross-checks with different algorithmic assumptions. Reduced simulation-prior dependence does not establish detector-model independence, adequate phase-space support, or coverage for this measurement. Comparisons would require matched signal, selection, normalization, and uncertainty definitions.

4 Uncertainty quantification: constructing the covariance blocks

The uncertainty budget sums four blocks: systematic universes, statistical bootstraps, ML training-seed variation, and a run-period block (§4.5). Their independence is an assumption: distinct random streams and physics sources do not by themselves establish it after unfolding. The universe, statistical and training-seed blocks are computed with the LightGBM estimator of the quoted 2D central value (§3.2); the universe and statistical blocks use its training seed (seed 42), and the training-seed block uses five seed settings that include it. A 3D budget built with the analogous 3D grid construction is not quoted, and no 4D uncertainty magnitude is reported (§7.8). App. A gives the full mathematics and worked examples using the reported numbers (N-D declarations in App. B). The resulting budget and its agreement with the published total are in §6.3; what the covariance includes and omits, and the conventions it follows, are in §4.6.

4.1 Systematic universes

Detector and model systematics use the July universe sweep: the 44 error bands and 187 universes of the MINERvA Analysis Toolkit (MAT), identical to the standard systematics set of the published analysis. Each universe is re-unfolded with LightGBM at seed 42 and compared with the quoted 2D central value, unfolded with the same estimator and seed. The five lateral bands, which shift reconstructed kinematics rather than event weights, are taken from re-unfolds of re-selected universes (§4.7) at the same seed. The band sum omits the nonresonant-pion band Rvp1pi, which encodes the same shift as Rvn1pi (§4.6); it holds 43 bands plus the flat normalization. Per-band covariances follow MAT's universe-mean centering and biased $1/N$ normalization. For a $\pm 1\sigma$ pair the covariance is the outer product of the half-difference of the two shifts, excluding any common endpoint displacement from the central value (CV). The 100 PPFX hadron-production flux universes use the same mean-centered $1/N$ formula.

Ranked by per-bin median fractional standard deviation, the largest bands of the 2D universe block are the flux (3.71 %, with MAT's ν–e and inverse-muon-decay constraint weights), the MINOS muon energy scale (Muon_Energy_MINOS, 2.36 %), the MINOS muon-match efficiency (MinosEfficiency, 1.50 %) and the MINERvA muon energy scale (Muon_Energy_MINERvA, 1.44 %). Figure 4 shows the fractional uncertainty grouped as in Figs. 6–7 of the published measurement; the full 205×205 covariance is projected exactly, $C_{1D} = PC_{2D}P^{\top}$, before division by the 1D central value.

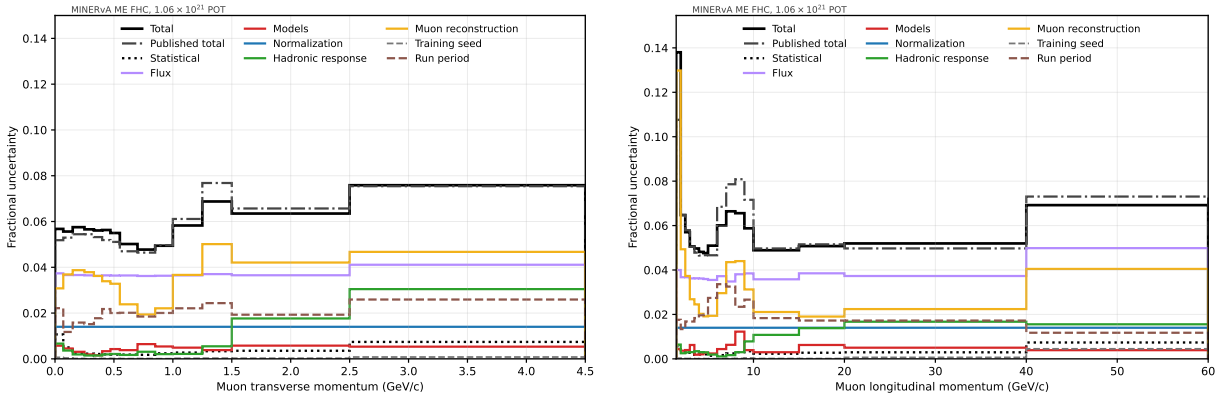


Figure 4: 2D cross-section fractional uncertainty projected onto p_T (left) and $p_{\parallel}$ (right), grouped as in Figs. 6–7 of Ref. [3] by flux, interaction models, normalization, hadronic response, muon reconstruction, the statistical band, the training-seed block and the run-period block, with the total and the published total.

Flux $1/\Phi$ normalization. Varying Φ per flux universe in the dominant $\sigma \propto 1/\Phi$ term gives a Flux band of 3.71 % with the constraint weights, and a combined budget, from our covariance alone, of 6.41 %. Target-nucleon normalization contributes a fully-correlated rank-1 band of 1.4 %. The two MINOS muon-energy-scale universes carry the correlated-flux weight, so each is divided by its own flux, not by the CV flux; that band's integral σ is 0.30 %. The unweighted 100-universe flux covariance has participation-ratio effective rank ≈ 1. Its leading mode correlates at $\rho = 0.900$ with the rank-1 muon-energy-scale band of the joint flux and muon-energy-scale block of Ref. [41], as rederived here. The cross term is rank-2; two independent estimators give $\rho = 0.87$–0.90. Adding this block leaves the systematic rank unchanged because its directions already lie in the universe-covariance span (App. A.4). The quoted budget does not include this cross block: with the MINOS universes divided by their own flux, the correlated-flux effect is already in the MINOS band, and

adding the block would count it twice.

4.2 Statistical uncertainty: Poisson bootstrap

The statistical block jointly bootstraps data and simulation with 300 per-event Poisson(1) weight replicas, each fully re-unfolded with the LightGBM estimator at the central value's training seed (seed 42), and is their sample covariance about the replica mean (Fig. 5). Each replica draws its weights in three independent streams, for data, signal simulation and background simulation; possible shared events between the two simulations were not assessed. Because the background simulation is resampled in every replica, the separable background-only term is contained in the statistical block. The reconstructed-level and truth-level weights of a simulated event share one draw, which preserves within-event migration correlations. Fixing the GBDT seed per replica separates bootstrap and training-seed variances. With 300 replicas the Gaussian formula $1/\sqrt{2(N-1)}$ gives a relative standard error of about 4.1 % on each per-bin σ, and the per-bin σ from a nested half (150) of the replicas is a median 1.000 times that from all 300. The replica mean differs from the central value by a per-bin median of 4.37 standard errors of the mean ($\sigma/\sqrt{300}$), or 0.25 of the statistical standard deviation. Because the block is the covariance about the replica mean, it does not include this offset. The bootstrap block is smaller than the published statistical block (ratio of square-root traces 0.407; App. A.4).

The statistical block quoted here has had no coverage test: the fixed-truth coverage studies of §5.2 scored two earlier statistical bands, not this block. No coverage test on held-out simulation has been run, and no design for one has been registered in advance. Because the fixed-truth test's truth equals the prior, it also cannot decide whether the smaller statistical block reflects an efficiency gain, regularization bias or undercoverage.

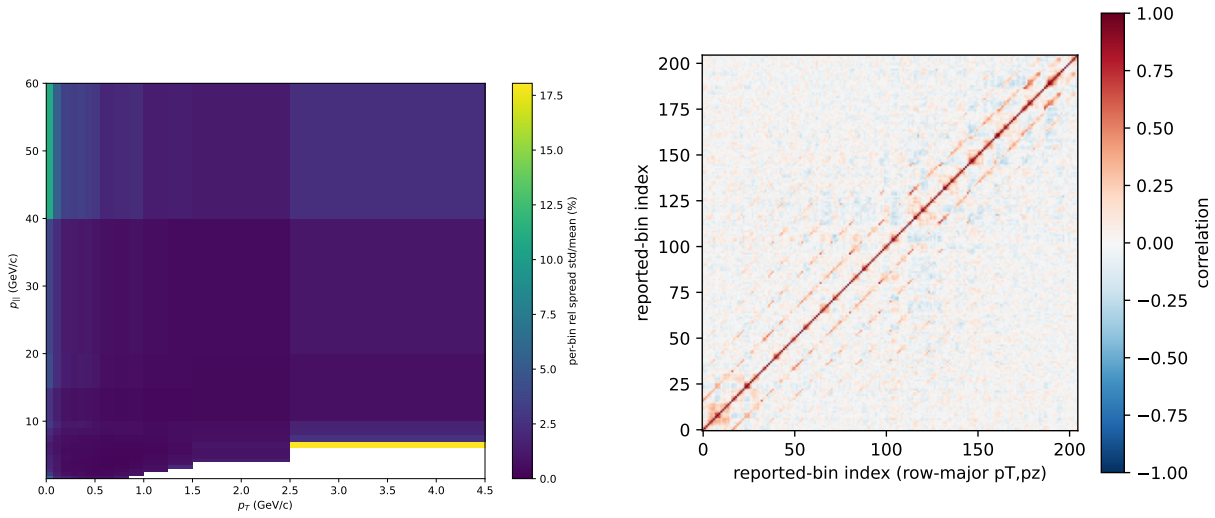


Figure 5: 300-replica Poisson bootstrap of the statistical block: per-bin spread σ relative to the central value of the unfolded 2D cross section, labeled std/mean on the color bar (left), and correlation matrix for the 205 reported bins (right).

4.3 ML/optimization uncertainty: training-seed variation

The training-seed block is the sample covariance, over five training-seed settings, of the mean of 20 bootstrap replicas at each setting. Each setting is a triplet $(s, s{+}1, s{+}2)$ of seeds for the step-1

classifier, step-2 classifier and miss regressor, with $s \in \{42, 45, 48, 51, 54\}$; the block has rank 4 (§A.2.1). It is the smallest block, with a per-bin median fractional standard deviation of 0.048 %. With four degrees of freedom, its variance has a Gaussian relative standard error of about 71 %. Its per-bin σ is a median 0.27 of that of the earlier ten-seed block, because the bootstrap × seed interaction, which dominated the earlier scan, is already inside the fixed-seed statistical block; the training-seed block carries it only at its 1/20 share, a median 0.763 of its diagonal. The interaction itself and the universe × retraining term are reported and not added (§4.6).

No block contains the dependence on the learner. The central value and the exact-split cross-check differ by a per-bin median of 1.03 % (maximum 17.6 %; §3.2); they differ in background treatment as well as in learner. One reseeded exact-split run differs from the unseeded one by a per-bin median of 0.18 % (maximum 1.6 %). The spread of the exact-split estimator over seeds is unmeasured (§3.2).

4.4 Further checks of the 2D covariance

Seven universes of the July sweep (§4.1) are reproduced to $\leq 1.4 \times 10^{-11}$ by the committed driver; the others are accepted by inference, because the same code produced them (Ref. [37], R1.16).

For two bands whose universes barely change the unfolding inputs, EtaNCEL and NormNCRES, the responses are consistent with instability of the GBDT training contributing at the scale of the training-seed variation (§4.3). They neither establish the mechanism for every small band nor justify subtracting an overlap from the budget.

Each lateral band of the quoted universe block uses a single unfold at the central training seed, so it includes that seed's training noise. The flux-universe index identity behind the $1/\Phi_u$ rescale is established by the code chain and by an exact test: 900 events × 100 universes match to 5.5×10^{-14}, and a control shifted by one universe fails at ≥ 0.245 (Ref. [37], R1.13). In all 12 playlists the production flux weights equal those of the constraint file and of the earlier 2D budget, with which §4.6 compares the quoted construction.

4.5 Combination and the unified-throw test

The nominal covariance is the block sum $C = C_{\text{syst}} + C_{\text{stat}} + C_{\text{ML}} + C_{\text{PL}}$, which assumes the blocks are independent and the unfolding responds linearly to simultaneous perturbations, so that per-band covariances add. The *unified throw* tests linearity directly (App. A.2.4; §9.6) by perturbing all reweightable bands and the flux universes simultaneously in each event and re-unfolding jointly. The procedure interpolates asymmetrically between actual -1σ and $+1\sigma$ event-weight endpoints, fixes the estimator seed for every throw, and centers the joint covariance on the throw mean. That mean's displacement from CV is stored separately as a per-bin vector and its norm; per-band endpoint displacements are not stored. The matched block-sum comparator re-unfolds both endpoints of each band using MAT's universe-mean centering and biased $1/N$ normalization. A single-throw superposition probe is insufficient to test nonlinear response. No scalar run-to-run jitter floor is subtracted. The configuration-(a) 5D result uses the strict joint ensemble and matched-endpoint comparator.

In the five-axis covariance of the configuration-(a) central value, C_Z, the per-source decomposition is retained for attribution and the joint throws enter through the inflation of Eq. (5) (§9.6).

The run-period block C_{PL} is an uncertainty assigned for an observed inconsistency between run periods whose cause is unknown; it is not a measured systematic uncertainty from an identified source. Data taken in different run periods are not mutually consistent within the other uncertainties. After unfolding, the double ratio (data over simulated truth) of the data-taking periods (playlists) 1G–1M to the earliest periods, 1A–1F, has a $p_{\parallel}$ step of $(+6.89 \pm 0.46)\%$ at 8 GeV/c. Diluted into

the combined result, the raw shift of those playlists is small in each bin: every marginal z is below 0.7. Its $p_{\parallel}$ χ^2 against the earlier 2D budget is nevertheless 34.4, where noise would give 2.0. These data cannot establish which period, if any, is correct.

The block is built from separate seed-42 unfolds of three run periods, the playlist groups 1A–1F, 1G–1M and 1N–1P, each unfolded through its own simulated detector response. The block uses each period's unfolded result over that period's own simulated truth. Each period is compared with the exposure-weighted mean of the three, so no reference period is needed; the spread, applied to the central value, gives the block. It is added after the other blocks. It has rank two, and with three period groups its own size is poorly determined. Its per-bin median fractional standard deviation is 2.09 % (84th percentile 3.34 %, maximum 6.48 %); it is 1.17 % to 3.36 % on the $p_{\parallel}$ projections and 1.97 % on the integrated cross section. Added to the other blocks, it raises the per-bin σ by a median of 5.8 % (Table 3). Taking 1A–1F as the reference would give a smaller block (per-bin median 1.38 %, integral 1.09 %).

	Increase in total σ	Block's share of total variance
Per bin, median	5.8 %	10.6 %
Per bin, maximum	39.1 %	—
$p_{\parallel}$ projections, maximum	21 %	—
Integrated cross section	9.9 %	17.1 %

Table 3: The run-period block added to the other blocks.

The block's interpretation is limited by the small number of period groups, by possible shrinkage from the regularization, and by the statistical and training noise it retains. Because the block is built from the observed differences among the periods, it contains them by construction; this carries no information about its coverage, that is, about whether its size is right. For the same reason, the double-ratio χ^2 against any total that contains the block is bounded whatever the data, so it does not test the block. The raw 1G–1M $p_{\parallel}$ χ^2 is 34.8 against the other blocks alone and 2.59 with the block added; these values indicate sensitivity only.

4.6 Scope, conventions and limitations of the 2D covariance

Status of the 2D result. We report a reproducible two-dimensional central value, with a covariance computed with the central value's own estimator, under the conventions stated below. The result's intervals, bias and covariance have not been independently validated. In particular, the coverage of its intervals has not been validated on independent simulated samples, nor simultaneously over all reported bins and projections. The covariance also gives no calibrated significance for comparisons with generator predictions. Such validation would need an independent simulation sample of production size, a validated generator of pseudo-data with continuously varied nuisance parameters, and samples from independent event generators beyond the truth-level ones already examined; none is available. We therefore make no claim about coverage, and goodness-of-fit values computed with this covariance are indicative only. The following are not included in the covariance:

- the approximation in how the systematic-universe unfolds weight events that fail reconstruction, whose effect has not been measured beyond the low-sensitivity test below;
- the dependence on the machine-learning estimator (§4.3);
- the dependence on the regularization, and departures of the detector response from its parametrized family, which we assume absent; we assign no uncertainty for either;

- the dependence on the generator shape of the prior, which is reported beside the covariance as C_{model} (below).

Generator-shape model dependence. The dependence of the result on the generator shape of the prior is reported beside the covariance, as a separate covariance $C_{\text{model}} = \sum_k s_k s_k^{\mathsf{T}}$. For each of three generators k (GENIE with meson-exchange currents, NuWro and GiBUU), the prior's truth distribution is reweighted by the ratio of that generator's $(p_T, p_\parallel)$ shape to GENIE's, with its normalization kept, and the data are unfolded again with the central value's estimator; s_k is the per-cell relative change of the result. Each generator enters as one fully correlated component, unscaled, so C_{model} has rank three. Its median fractional standard deviation over the 205 reported cells is 3.71 % (84th percentile 7.10 %, maximum 21.10 %). For the sum of the combined covariance and C_{model}, the total for users who include this dependence, the per-cell median is 7.90 % (84th percentile 12.32 %, maximum 24.49 %). On the integrated cross section C_{model} gives 0.03 %, and the total stays at 4.76 %: the reweightings change only the prior's shape, so C_{model} does not test the model dependence of the normalization. The comparison with the published uncertainty (§6.3) uses the combined covariance alone, because the published uncertainty has no component for this dependence either.

C_{model} measures generator-shape model dependence; it is not a calibrated bound on the bias in each cell. A closure test with NuWro-shaped pseudo-data and the configuration-(a) estimator has a bias above 1 % in 127 cells. In 67 of them the NuWro shift of the quoted result reaches the bias, and in 72 the quadrature sum of the other two generators' shifts reaches it; these counts are descriptive. Where a pooled cell, a noise-gated cell, or an unreported cell meets another cell, the construction can draw a step of more than 0.1 in a generator's reweighting ratio; the cells beside such steps enter with their measured shifts. Identified after the measurement, these 52 cells hold 59.4 % of the trace of C_{model} and carry most of its large shifts.

Conventions. Each systematic band's covariance is computed about the mean of its universes (constraint-weighted for the flux), with $1/N$ normalization, not about the central value. This follows the MAT treatment of universes (§4.1), which we use as a stated convention. Across bands, the per-bin median offset of a band's center from the central value is 0.10 % to 0.55 % of the central value, or 0.09–2.85 times the band's σ. The statistical block is the covariance about the replica mean (§4.2).

Approximations and omitted terms. The systematic-universe unfolds approximate the weighting of events that fail reconstruction, which enters through each universe's completeness; the approximation is not corrected. It was tested in two of the 187 universes, `MaRES_0` and the flux universe with the largest median completeness change (`Flux:23`), each at seeds 42 and 43. In neither was an effect resolved above the training-seed noise. The tests compared the difference with the σ of the earlier ten-seed block, which is about 3.7 times that of the training-seed block used here. They exclude only a median effect above twice that σ, so an effect below the training noise is not excluded. Separately, retraining moves these two universes' shifts by a median 1.13–1.23 times that σ.

One systematic category provided in the input products, `NormDISCC`, is identically zero; we carry it as zero, which we have not been able to confirm is intended. No separate uncertainty is assigned to the counting of protons on target: the NuMI flux prediction carries a 2 % counting term by convention, so a separate band would risk counting it twice. Two links in that argument, that the flux universes carry the term and that the neutrino–electron constraint absorbs it, are inferred, not verified. The twelve playlists are combined with one global exposure ratio rather than per-period

ratios, and the resulting bias is not corrected. At reconstruction level the pooled result is high, by 0.18 % in total and by at most 0.32 % in a marginal bin. After unfolding, the pooled result is 0.076 % above the exposure-weighted combination of the period results on the integral, the same sign as at reconstruction level. It differs from that combination by at most 0.39 % in a marginal bin (the highest $p_{\parallel}$ bin) and by at most 0.33 % in p_T. In single cells the combination differs from the pooled result by -1.83 % to 3.78 %, each with $|z|$ at most 0.31.

Other limitations. Both programs that use the covariance for the reported results, the χ^2 comparison (§A.3.2) and the program that draws Fig. 4, check that its stored set of cells equals the published set, and the check passes in every run of both on the quoted covariance. Limitations specific to one block are stated with that block (§4.2, §4.3, §4.4, §6.3) and, for the duplicated pion band, below.

Seed-related terms reported, not added. The variance of the bootstrap × seed interaction is a per-bin median 0.072 of the statistical block's variance. For the universe × retraining term, in the six tested universes (of 187) with a non-null shift, the standard deviation of each universe's shift over training seeds, with four degrees of freedom, is a median 1.08–1.36 times the σ of the earlier ten-seed block. In those universes the universe block, built at one training seed, therefore carries single-seed training noise of about the earlier seed block's size; it is part of the block as built and is not separated from the physical variation. In two of them the noise is comparable to or larger than the shift: in `2p2h:0` the shift is 0.097 % against a seed standard deviation of 0.192 %, and in `MaCCQE:0` 0.239 % against 0.210 % (per-bin medians). We infer that in expectation such noise adds variance to a band rather than removing it, and, from the medians, that these two bands are largely training noise. The universe × retraining values and the band-centering numbers cannot be recomputed from the saved covariance products; they were therefore not recomputed independently, and their verification is limited to the checksums of their inputs.

Comparison with the earlier 2D budget. The saved products and code of the earlier 2D budget [42] show the following defects in its construction; the quoted budget corrects each. First, the systematic sweep of the earlier budget (the May sweep) has a simulated truth input that differs from that of its seed-42 reference: even an EtaNCEL universe that leaves the event weights essentially unchanged differs in all 205 truth-input cells, while its saved reconstructed inputs agree. A common displacement cancels to first order in the MAT mean-centered covariance. The mismatch does not invalidate the entire covariance, and it supplies no correction to subtract. Because the May input and code revision are not available, the mismatch cannot be attributed more specifically. The July sweep, which the quoted universe block uses, does not have this mismatch. Second, the two bands Rvn1pi and Rvp1pi encode the same shift under the configured nonresonant-pion tune, and the earlier budget counted it twice; the quoted universe block drops Rvp1pi and counts it once (§4.1). The band is small, 0.088 %, but the double count is a defect in the construction of the covariance; no conclusion about the published analysis's implementation follows. Third, the earlier budget did not propagate background-template statistics. In the saved reconstructed histograms their error exceeds one tenth of the data's Poisson error in 60 of 205 bins, so neglecting them everywhere is unsupported; this is a reconstructed-space diagnostic, not a measured unfolded uncertainty. The quoted statistical block draws the background simulation as its own bootstrap stream (§4.2), and the quoted universe block takes its lateral bands from re-unfolds of re-selected universes (§4.1). The values printed for the earlier budget were not changed by these findings.

4.7 What changes for the higher-dimensional products

Two event-loop modes supply the lateral bands. The single-pass event loop, which writes every universe in one pass, stores each lateral universe's shifted observables only for events that pass the selection at the central value, so it omits events that the shift moves into the selection. Re-selected event loops re-run the selection with one lateral universe active at a time, so they include these migrations. For the five kinematic bands (beam angle BeamAngleX/Y, muon resolution MuonResolution, and the muon energy scales Muon_Energy_MINERvA/MINOS), the re-selected event loops supply the lateral-band block of C_Z; higher-dimensional products built from the single-pass shifted observables are not used for any quoted uncertainty. Replacing the single-pass lateral-band block by the re-selected one moves that block's square-root trace by $-0.0288\,\%$. This aggregate change does not bound the change in any individual bin. Across the 10 694 reported bins the per-bin ratio of re-selected to single-pass standard deviation runs 0.177 to 3.062; within the 1 % of bins with the largest single-pass variance, which carry 74.11 % of the total variance, it runs 0.687 to 1.153. A small share of the aggregate variance does not make a bin irrelevant to a quoted result.

Scope of the re-selected replacement. The replacement covers the five detector bands using shifted reconstructed kinematics. MINOS efficiency and the three GEANT hadron-interaction bands (GEANT_Neutron, GEANT_Pion, GEANT_Proton) keep their weight-based treatment and are therefore outside the kinematic replacement. This classification does not independently validate hadronic-response model completeness for the E_{avail} and W measurement.

The ML term C_{ML} of C_Z (Eq. (4)) is the split-varied band: its ensemble fixes the estimator seed at 42 and varies the train/test split seed. Each of its replicas refits every classifier on a random training subset of the events and still reweights every event, the only use of a training subset in this note (training details: §3.1). The band therefore measures split response and not the full estimator stochasticity. A separate scan of twelve estimator seeds at fixed data and split covers the remaining stochasticity. Its covariance has $\sqrt{\operatorname{Tr} C} = 1.306 \times 10^{-39}\,\text{cm}^2/\text{nucleon}$, which is 87.5 % of that of the split-varied band ($1.493 \times 10^{-39}\,\text{cm}^2/\text{nucleon}$). Estimator stochasticity, not the particular train/test partition, therefore dominates the split-varied band. The scan is an auxiliary robustness check, not an independent covariance term, since split and estimator responses are not assumed uncorrelated. Both bands are minor components, about three percent of the combined budget in $\sqrt{\operatorname{Tr} C}$.

C_Z is a separate construction in which the unified-throw covariance is centered on the central value (the CV-centered variant, `cv`; §9.6). A 5D covariance construction that preceded it used both mean-centered and CV-centered unified throws (§4.5); neither variant is used, and no magnitude from either is quoted.

Subtracting the background separately in each of the 187 universe unfolds and in the CV changes the square-root trace of the combined block sum by 0.28 % and C_Z by 0.18 %, a numerically negligible bound on the background-CV choice (Ref. [37], superseded values). The 4D and extended-fiducial results have no covariance from this procedure, and C_Z is not transferred to them.

5 Validation of the two-dimensional measurement

In a simulated experiment, *truth* is the known target distribution, the *prior* is the starting simulated truth distribution, and the *response* describes reconstruction and selection conditional on truth. A closure test compares the unfolded estimate with its known target under a specified deformation. Bias is the mean estimate minus that target over repeated experiments; a single closure residual also contains sampling fluctuations. Coverage is the fraction of repeated intervals containing the specified truth, and tests the estimator and interval procedure together. Table 4 separates these questions

before the results. Repeating a numerical calculation checks neither the coverage and width of the intervals nor the adequacy of the physical model.

Check	Target and estimator	Sample relation / measured quantity	Rule and conclusion
Injected closure	Reweighted simulated truth; LightGBM, five iterations, ME1A simulation	Same simulated response; residual under named truth or hidden-variable deformation	Thresholds below; passes depend on deformation and amplitude
Miss completeness	Full truth-event inventory; production truth–reco event matching	Same event keys; truth-only entries retained	Membership and completeness conditions; no recovery or coverage claim
Iteration stability	Same data, exact-split cross-check; five versus ten iterations	Shared sample; movement of total and shape	Small observed movement; not a known-truth accuracy test
Fixed-truth coverage	Fixed unfluctuated MC truth; LightGBM statistical estimator	Poisson pseudo-data; MC resampling shares the band's mechanism; pooled toy–bin containment	Predeclared coverage windows; earlier (completeness-divided) band undercovers in its tails (§5.2)

Table 4: Validation questions, targets and limits. Coverage pooled over bins is not a simultaneous probability that every bin is covered.

Closure. Three families of closure tests use the simulation of playlist ME1A (§2), unfolded with LightGBM and five iterations; pass thresholds are in parentheses.

- **Truth reweighting** (the same factor on truth and reco weights): a Gaussian in p_T and a tilt in $p_\parallel$ leave residuals of 0.046 % and 0.013 %, within the 1.5 % threshold.
- **Hidden variable** (a bias in $\Delta p_T = p_T^{\text{reco}} - p_T^{\text{truth}}$, a resolution variable outside the feature set): the leakage is linear in amplitude, $A = 0.05/0.10/0.30 \to 0.57/1.14/3.43\,\%$ (threshold 3.0 %); the first two amplitudes pass, and the deliberately large $A = 0.30$ arm exceeds the threshold.
- **Alternative model** (training on the central-value simulation, pseudo-data from an alternative systematic universe): MaCCQE 0.068 %, Flux 0.376 % (threshold 2.0 %).

Injected-feature recovery. The 2D truth-reweighting closure with a Gaussian bump passes the recovery test of the unbinned-unfolding guide [18]. In the feature bins the residual from the injected truth is 1.69 % for an injected 17.2 % feature (ratio 0.098), about ten times smaller. The test requires the residual left after recovering a deviation from the prior to be far smaller than the deviation itself; unfolding cannot create discriminating power (3D: §7.4).

A data-versus-prior χ^2 with a diagonal covariance of statistical variances only is larger at truth level than at reco level. This is expected at high resolution: detector smearing lowers the reco-level value, and the diagonal covariance omits the unfolding covariance. The comparisons that use a full covariance are the χ^2 against the published result (§6.4) and the generator comparisons.

These closure tests validate the estimator; no non-closure uncertainty is assigned from them. Assigning one would require a predeclared suite of injection shapes and amplitudes, not calibration to a single Gaussian bump.

None of these tests uses pseudo-data whose detector response differs from the unfolding response. Appendix D defines such a test, a closure that rescales only the reconstructed E_{avail} of the pseudo-

data, together with a more complete leading-nucleon follow-up. No result from it is reported here. The five-dimensional estimator, including its background subtraction, is validated in §9.1.

Requirements for validated total uncertainties. The follow-up assessment [42] distinguishes a proposed split-sample width check from repeated-sampling coverage. With the MC sample, binning maps and background subtraction held fixed, such a check could establish only the precision conditional on them, at its tested exposure. Nonlinear unfolding does not guarantee a simple scaling from half to full exposure. Nor can the check supply the variation across independent MC populations. With the current inputs, validated total uncertainties for all 205 bins, the integral and the projections therefore cannot be obtained. They would require independent populations or a validated generative law, a continuous nuisance generator for tests of total intervals, and an untouched generator-truth domain for bias checks (§4.6). This is a limit of the available inputs, not a general impossibility result for 2D unfolding.

5.1 Closure tests across dimensions

Only the MINERvA simulation supplies a detector response, the conditional distribution of reconstructed events given truth. Other generators (GiBUU, NuWro, GENIE variants) supply truth only. Every closure test here therefore keeps the simulated MINERvA response and changes the truth distribution, by reweighting the simulated events, or changes the analysis chain around it. Reweighting to another generator's shape is a valid known-truth test of the unfolding; it is not a test of the detector model. An ordinary closure, whose pseudo-data are the simulation itself, is an identity check that a constant estimator would also pass. Table 5 lists, for each test, what it changes relative to the unfolding's simulation and what its result can detect.

Table 5: Closure tests by what they change relative to the unfolding's own simulation. "Marginal" means the metric is a one-dimensional projection; "joint" means multidimensional cells.

Test	What changes	Dim.	Result and what it can detect
Ordinary closure	Nothing: pseudo-data are the simulated reco	any	Bookkeeping only; cannot detect estimator bias
Truth reweight	Gaussian in p_T, tilt in $p_\parallel$, applied to truth and reco weights	2D	Residuals 0.046 % and 0.013 % (thr. 1.5 %); pass
Injected feature	Gaussian truth feature of 17.2 %	2D	Residual 1.69 % (ratio 0.098); passes the recovery test
Alternative universe	Pseudo-data from MaCCQE or flux universe	2D	0.068 %, 0.376 % (thr. 2.0 %); pass
Hidden variable	Bias in Δp_T, a resolution variable outside the features	2D	Leakage linear in amplitude; $A = 0.30$ exceeds its 3 % threshold

Test	What changes	Dim.	Result and what it can detect
Injected E_{avail} bump	+30 % Gaussian in truth E_{avail} at 0.3 GeV, width 0.15 GeV; configuration (a) (§3.2)	3D	Marginal residual 1.84 % of 18.1 % injected (ratio 0.102); no predeclared threshold. Per-3D-bin spread 3.2 % (§7.4)
Injected q_3 bump	Same Gaussian form in truth q_3 (mean weight 1.014); configuration (a)	4D	Marginal q_3 ratios within 1.1 %; no predeclared threshold
Injected W bump	+30 % Gaussian in truth W at 1.23 GeV (the Δ peak), width 0.15 GeV; mean weight 1.053	5D	Marginal residual 4.2 % of 23.1 % injected in the 1.1 GeV to 1.4 GeV bin (ratio 0.180; threshold 0.20, set in advance); pass, by a narrow margin. Neighbouring W bins are 3.1 % and 2.7 % high, and joint (E_{avail}, W) cells in the peak column 2.2 % to 9.4 % low. Per-5D-bin spread 3.7 %. Signal only
Background-inclusive	Adds background simulation and rebuilds purity weights; nominal truth	5D	Highest-W column 3.5 % to 4.1 % low; negative-weight treatment 0.11 % to 0.24 %, with joint cells 0.7 % to 1.1 % high (§9.2)
Generator-shaped	Truth reweighted by GiBUU/GENIE ratio in E_{avail} (0.70–1.33 per bin)	5D	Joint (E_{avail}, W) residuals median 9.7 %, up to 74 % at highest W; regularization bias outside every uncertainty quoted for the configuration-(a) central value (sixteen-cell result: §§10.3, 10.5)
Withheld generator shapes	NuWro/GENIE and GENIE MEC/CV ratios	5D	Median 2 % to 6 %, at most 16 % to 31 % per joint cell; within the configuration-(a) total uncertainty in 20 of 42 (E_{avail}, W) cells (§9.3)
Hidden q_3	q_3 at fixed (E_{avail}, W) truth, amplitude 0.3	5D	(E_{avail}, W) cells shift by up to 37 %
Response mismatch	Reconstructed E_{avail} of pseudo-data scaled	5D	Protocol implemented, not run (App. D)

Comparison of injected-bump and generator-shaped closure tests. The injected-bump and generator-shaped tests differ in deformation, metric and iteration dependence, so their results do not contradict each other.

- *Deformation.* The closing injections are single smooth bumps in one variable the classifiers see directly. A generator ratio changes correlated structure across E_{avail}, W and q_3, including directions the reconstructed observables constrain weakly (§9.5, response-information diagnostic).

- *Metric.* The 3D, 4D and 5D closures are judged on one-dimensional marginals, where per-cell errors average out; the 3D test's per-bin spread is already 3.2 %, and the 5D test's is 3.7 %. The generator-shaped failures are measured in joint (E_{avail}, W) cells.

- *Iterations.* At five iterations the estimator retains part of the prior wherever the reconstructed data constrain the truth weakly. Thirty iterations reduce the E_{avail} residual by 26 % while the q_3 residual grows (§9.3).

- *Five-dimensional W test.* A W bump at the Δ peak is recovered to a ratio of 0.180 against a threshold of 0.20, a narrow pass in the marginal only. The same unfolding result leaves joint (E_{avail}, W) cells in the peak column up to 9.4 % low and moves part of the bump into the neighbouring W bins. A single smooth bump in a marginal is therefore weaker evidence than the generator-shaped tests, which remain the main five-dimensional recovery evidence of §9.

5.2 Fixed-truth coverage of the 2D statistical band

This test asks whether a 2D statistical band covers a *fixed* truth at its nominal rate. The test scored the *completeness-divided band*, the first-built statistical band, quoted when the test ran, whose replicas divide by a bootstrapped completeness. The *undivided band* is the rebuilt band, without the division, since superseded. The quoted statistical block (§4.2) is a separate ensemble of 300 replicas at the central value's training seed. The design, criteria and positive control of the test were specified before any toy was generated (Ref. [37], R1.1).

Each toy draws pseudo-data event by event as Poisson counts with means equal to the simulated reconstructed weights. It resamples the simulation with the per-event Poisson(1) bootstrap of the statistical band and unfolds with that band's estimator, LightGBM with five iterations at a fixed training seed, the learner of the quoted central value at a different seed (§3.2). The reference T is the unfluctuated simulated truth, identical in every toy. The no-fluctuation configuration reproduces both the production closure output and T exactly.

In reported bin b of toy t, $z_{tb} = (U_{tb} - T_b)/(r_b T_b)$, where r_b is the production band's relative spread. Coverage C_k is the fraction of all (toy, bin) pairs with $|z| \le k$, with intervals from resampling whole toys. The band counts as nominal if its ratio to the true scatter lies in $[0.9, 1.1]$, which gives the windows $C_1 \in [0.632, 0.729]$ and $C_2 \in [0.928, 0.972]$. A failure at the interim analysis of 100 toys, with 99.5% intervals, stops the study.

At the interim analysis ($N_{\text{toy}} = 100$, 205 bins) the completeness-divided band undercovers in its tails. Its one-standard-deviation coverage $C_1 = 0.679$ $[0.669, 0.690]$ is nominal (Gaussian 0.683), but its two-standard-deviation coverage $C_2 = 0.912$ $[0.907, 0.918]$ lies entirely below its window (Gaussian 0.954). The pull RMS is 1.60, and the mean pull is consistent with zero. By the criteria above this is a failure by undercoverage, and the study stopped at the interim analysis. An independent reimplementation of the scoring reproduces the pooled coverages, their intervals, the controls and the secondary analysis below.

The positive control rescales the band on the same toys. At $\times 0.7$ and $\times 1.3$, C_2 moves to 0.809 and 0.955, and the test flags both directions.

The failure is one of width, not bias. The per-bin pattern, which is descriptive and post hoc, is uneven in both directions. The band is too wide in 82 bins (pull RMS below 0.9) and too narrow in 71 (above 1.1). The pooled C_1 is nominal partly through this cancellation; the median bin's pull RMS is 0.97. 35 bins have a per-bin C_2 below 0.85. Of these, 21 form a block at $p_\parallel \ge 15\,\text{GeV/c}$ and $p_T < 0.55\,\text{GeV/c}$. Six more lie in the highest-p_T row (2.5 GeV/c to 4.5 GeV/c) at $p_\parallel$ 6 GeV/c to 20 GeV/c. The 14 bins with pull RMS above 2 carry 72 % of the C_2 deficit.

Unlike the band's replicas, the central value has completeness exactly 1. A secondary analysis, added after the pilot, applies the band's division to the toys. Scored against the same band, the divided toys overcover: $C_1 = 0.833$, with its interval above the window. The attribution rule was fixed before the full run, but its clause for the undercoverage outcome was written after the pilot pulls had been seen. Under this rule the primary undercoverage is attributed to the completeness term of the band. The attribution is partial: in the replica form, the $p_{\|}$ 40 GeV/c to 60 GeV/c column still undercovers.

The band was then rebuilt without this term. The resulting undivided band was rescored only post hoc (below), and the quoted statistical block has not been tested.

Post hoc rescoring against the undivided band. Scoring the same 100 toys against the undivided band removes the tail undercoverage: $C_2 = 0.977$ $[0.974, 0.980]$ and $C_1 = 0.781$ $[0.771, 0.790]$, both above their nominal windows, so the undivided band is wider than the toy scatter (pull RMS 0.84, mean pull consistent with zero). The rescoring is post hoc and partly circular, because the band's MC part reuses the toys' resampling mechanism.

The excess width lies in the data-statistics component. The per-bin pull RMS falls as that component's share of the band's variance rises, which corresponds to a data-statistics part about $1.6\times$ wider in σ than the toys produce. Background purity does not explain it, and its cause is not established. A post hoc diagnostic bootstraps a single toy; its decision rule was recorded before it ran. Its result is consistent with the closure toys scattering too little, rather than with the data bootstrap scattering too much. The toys would scatter too little because their pseudo-data are built on the training MC events. The diagnostic is neither an independent-population coverage test nor a calibration on real data. A re-test on held-out simulation would separate the two explanations; it has not been run.

Scope. The test concerns the statistical band alone and changes no central value or quoted uncertainty. It tests the LightGBM statistical estimator at the band's training seed (seed 1), not at the quoted central value's seed 42, and does not address systematic, total or five-dimensional coverage. Its simulation-side scatter reuses the resampling that built the band, so only the data-side fluctuation is tested by an independent mechanism. Because T equals the prior, the test cannot detect regularization bias toward the prior.

5.3 Other validation diagnostics

Unbinned diagnostic (C2ST). In the classifier two-sample test (C2ST), a classifier is trained to separate the data from the (reweighted) simulation in the full unbinned feature space, and its held-out area under the ROC curve (AUC) or accuracy is compared with the chance value 0.5. The AUC drops from ≈ 0.535 before unfolding to ≈ 0.501 after, indicating removal of the reconstructed data $\leftrightarrow$ MC mismatch. This binning-free check is descriptive, and no calibrated p-value is quoted. As run, the C2ST trains and evaluates on the same sample without cross-fitting; a valid null would require re-running the full cross-fitted OmniFold pipeline per pseudoexperiment. Calibrated unbinned goodness-of-fit remains open [18].

Same-ensemble pull diagnostic (not a coverage test). An earlier ensemble of $N_{\text{toy}} = 200$ 2D closure unfolds (Ref. [37], R1.2) uses weighted MC reco pseudo-data and stores the truth marginals in truth histograms. Each toy has its own bootstrap seed k and per-event Poisson(1) draws, as in production, with seeds chosen not to collide with the production replicas. Toy seeds are independent; bins within a toy are correlated. The MC bootstrap multiplies the truth weight, so it fluctuates both the unfolded result and the stored truth. The stored truth varies in all 205 reported bins, and

its all-toy mean $\bar{X}^{\text{truth}}$ is therefore not an independent fixed truth for a split-sample coverage claim; it only centers a same-ensemble diagnostic.

In reported bin i with $\bar{X}_i^{\text{truth}} > 0$, define σ_i as the across-toy sample standard deviation (denominator $N_{\text{toy}} - 1$) and standardized residual $r_i^{(t)} = (U_i^{(t)} - \bar{X}_i^{\text{truth}})/\sigma_i$. Compare the per-bin fraction c_i of toys with $|r_i^{(t)}| \leq 1$ to the unit-normal $\Phi(1) - \Phi(-1) = \text{erf}(1/\sqrt{2}) \approx 0.6827$ and the mean absolute pull $\langle|r|\rangle$ to $\sqrt{2/\pi}$. The fraction with $|r| \leq 1$ is 68.71 % (Gaussian reference 68.27%; $\Delta = +0.44\%$; median 68.50%), and $\langle|r|\rangle = 0.794$ against $\sqrt{2/\pi} = 0.798$ (within 0.5%). The signed residual mean is $+0.006 \pm 0.082\,\sigma$, consistent with zero. The within-bin fraction is above 65 % in 97.6 % of bins (only 5 of 205 fall below). These show that the standardized toy residuals are approximately normal.

Because σ_i is estimated from the same toys that are evaluated, these figures diagnose standardized pulls and Gaussianity only and are not a coverage test of the bootstrap band; no coverage number is quoted from this ensemble, and the 2D coverage statement is the fixed-truth test of §5.2.

Stability and efficiency. With five and with ten iterations the total cross section agrees to within 0.026 %; the per-bin shape differs by 1.54 % RMS. The histogram-based and exact-split backends agree on the total cross section to 0.04 %. Both comparisons were measured for the exact-split cross-check. The selection efficiency (Fig. 6) reproduces Fig. 5 of the published measurement.

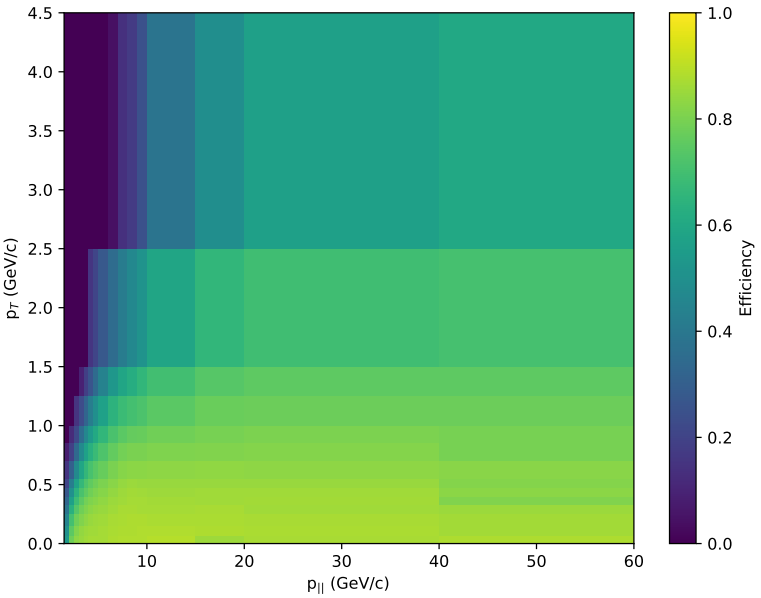


Figure 6: Selection efficiency on the $(p_T, p_{\|})$ truth grid. Efficiency is diagnostic only; OmniFold's miss treatment natively corrects acceptance (§3.1).

6 Results

The error bars on the unfolded points of Fig. 7 are the bin errors stored with the unfolded histogram and its projections, not the covariance of §4. The published points in its top panel carry the square root of the diagonal of the published statistical covariance, and Figs. 9–10 carry the same bin errors as the unfolded points. The uncertainty budget of §4 is shown in Fig. 4 and Table 7.

6.1 Double-differential cross section

Figure 7 shows the unfolded $d^2\sigma/(dp_T\,dp_\parallel)$ and its 1D projections. The total cross section is $3.069 \times 10^{-38}\,\mathrm{cm}^2$/nucleon (published: $3.039 \times 10^{-38}\,\mathrm{cm}^2$/nucleon).

The shapes are those expected for CC-inclusive scattering at $\langle E_\nu \rangle \sim 6\,\mathrm{GeV}$. The p_T projection vanishes at $p_T = 0$ and peaks broadly at $p_T \approx 0.55$–$0.70\,\mathrm{GeV/c}$, reflecting the $Q^2 \lesssim 1\,\mathrm{GeV}^2$ scale of few-GeV scattering through $p_T^2 \simeq Q^2\,E_\mu/E_\nu$, with $E_\mu/E_\nu \sim 0.6$ for these forward muons. It falls by more than two orders of magnitude toward the $4.5\,\mathrm{GeV/c}$ edge. The $p_\parallel$ projection rises from the $1.5\,\mathrm{GeV/c}$ threshold to a broad $p_\parallel \approx 3$–$4.5\,\mathrm{GeV/c}$ maximum (the NuMI medium-energy flux peak at $E_\nu \approx 6\,\mathrm{GeV}$), then falls by two orders of magnitude by $p_\parallel \approx 20\,\mathrm{GeV/c}$. The $\theta_\mu < 20^\circ$ acceptance empties the 2D map's high-p_T, low-$p_\parallel$ corner (19 unreported diagonal cells; §3.5).

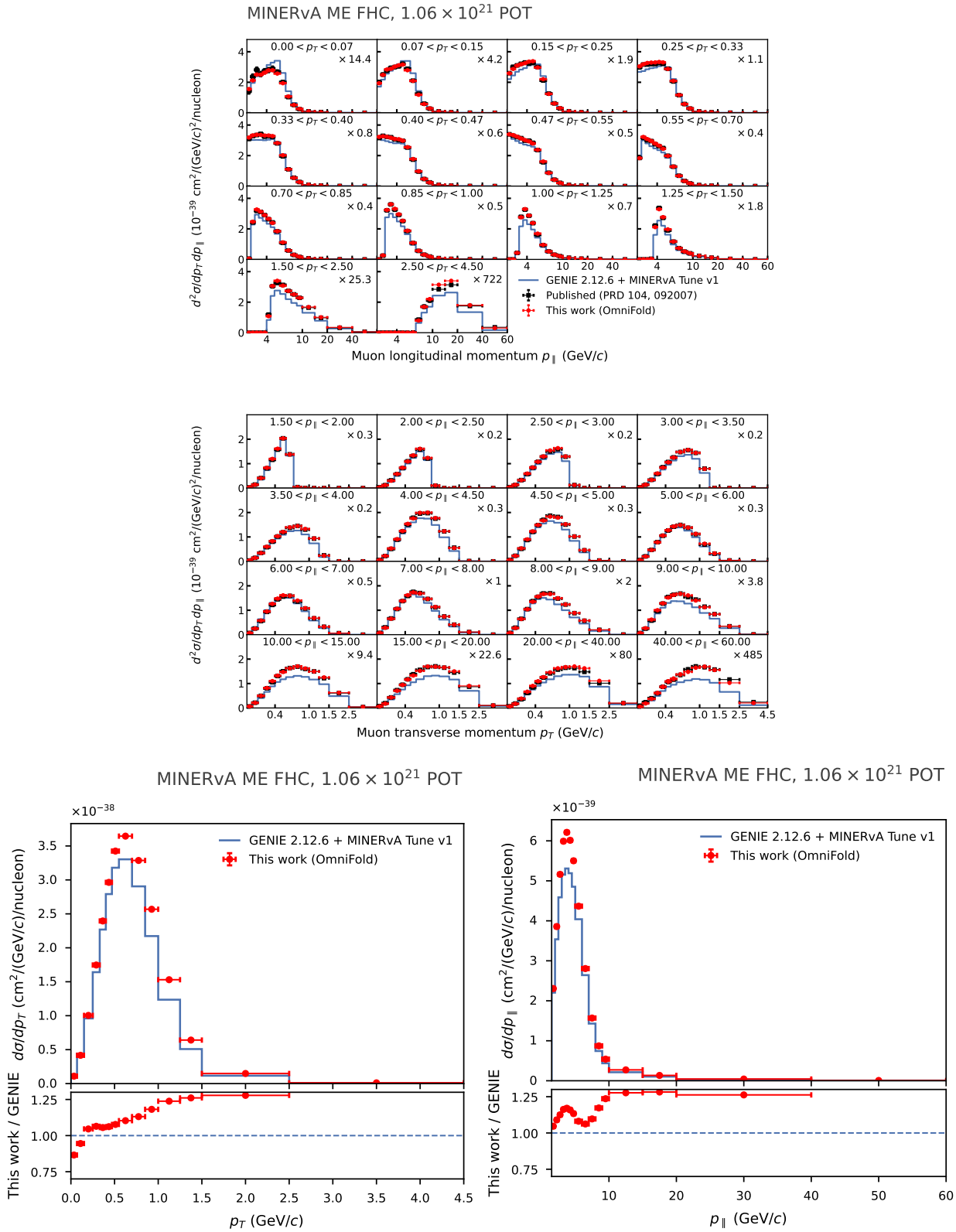


Figure 7: Unfolded 2D cross section (top, in the format of Fig. 13 of Ref. [3]) and its p_T and $p_\parallel$ projections (bottom). The p_T peak at $\sim$0.6 GeV/c reflects the $Q^2 \lesssim 1\,\mathrm{GeV}^2$ momentum-transfer scale; the $p_\parallel$ peak reflects the medium-energy flux maximum. The vertical error bars on this measurement are the weight statistics of the reweighted simulated sample (the square root of the summed squared event weights, combined in quadrature with the bin error propagated for the completeness ratio, which is one by construction); they are neither the data's statistical uncertainty nor the reported covariance. Those on the published data (top) are the square roots of the diagonal of its statistical covariance. Horizontal bars span the bins.

6.2 Comparison to the published measurement

The published and unbinned results share the data and the flux, detector and generator systematics, and their cross-covariance is not available. Every χ^2 and standardized difference here is therefore descriptive, not a calibrated goodness-of-fit test.

Over the 205 reported bins, the total cross section reproduces the published result [3] to 1.0 % (ratio 1.010; Table 6). The median per-bin ratio is 1.005; 86.8% of bins agree within 5 % and 95.1% within 10 %. Indicative values of χ^2 per reported bin are 2.27 with the published covariance alone (analysed in §6.4) and 0.929 with the sum of the published covariance and that of this measurement.

The latter, $\chi^2/\mathrm{ndf} = 0.929$, double-counts the flux, GENIE and MINOS systematics that the two results share. With the covariance of this measurement alone, χ^2 is computed only in the covariance's well-constrained subspace, with a truncated spectral pseudo-inverse. The per-bin pulls and the truncated result are therefore quoted together, and an unconstrained pseudo-inverse is not used as a goodness-of-fit test on its own. The regularization scan (App. A.4) was made for the exact-split cross-check on an earlier covariance and was not repeated for the quoted one.

Per-bin differences standardized by the published uncertainty (Fig. 8) are centered (mean 0.060), with RMS 0.54, well inside the published uncertainties. Figures 9–10 compare the unfolded result with *simulated truth*, the MC prior, in the published panel format.

Quantity (205 reported bins)	Value
σ_{tot}(this)/σ_{tot}(published)	1.010
Median bin ratio (this/published)	1.005
Bins within 5 %/10 %/20 %	86.8/95.1/99.0 %
Indicative χ^2/bin, published covariance	2.27
Indicative χ^2/bin, published + this covariance	0.929
Published-σ standardized-difference mean / RMS	0.060/0.535

Table 6: Agreement of this measurement with the published measurement of Ref. [3]. Per-bin differences standardized by published uncertainties (Fig. 8) are sub-1σ; the χ^2 values are descriptive.

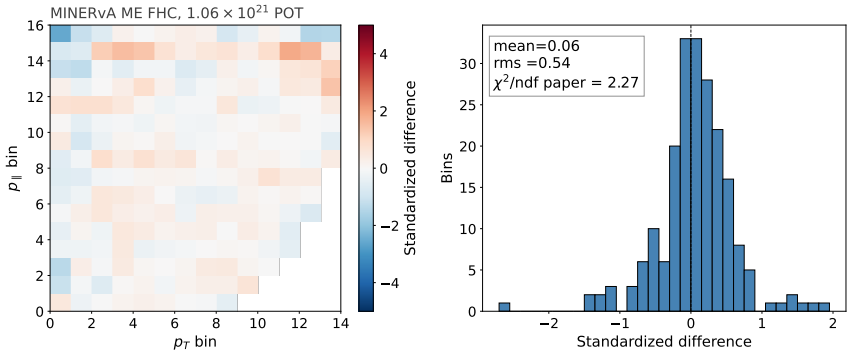


Figure 8: Per-bin standardized difference (this − published)/$\sigma_{\mathrm{published}}$ on the 14 × 16 grid, and its distribution. It is descriptive, not a pull between independent results.

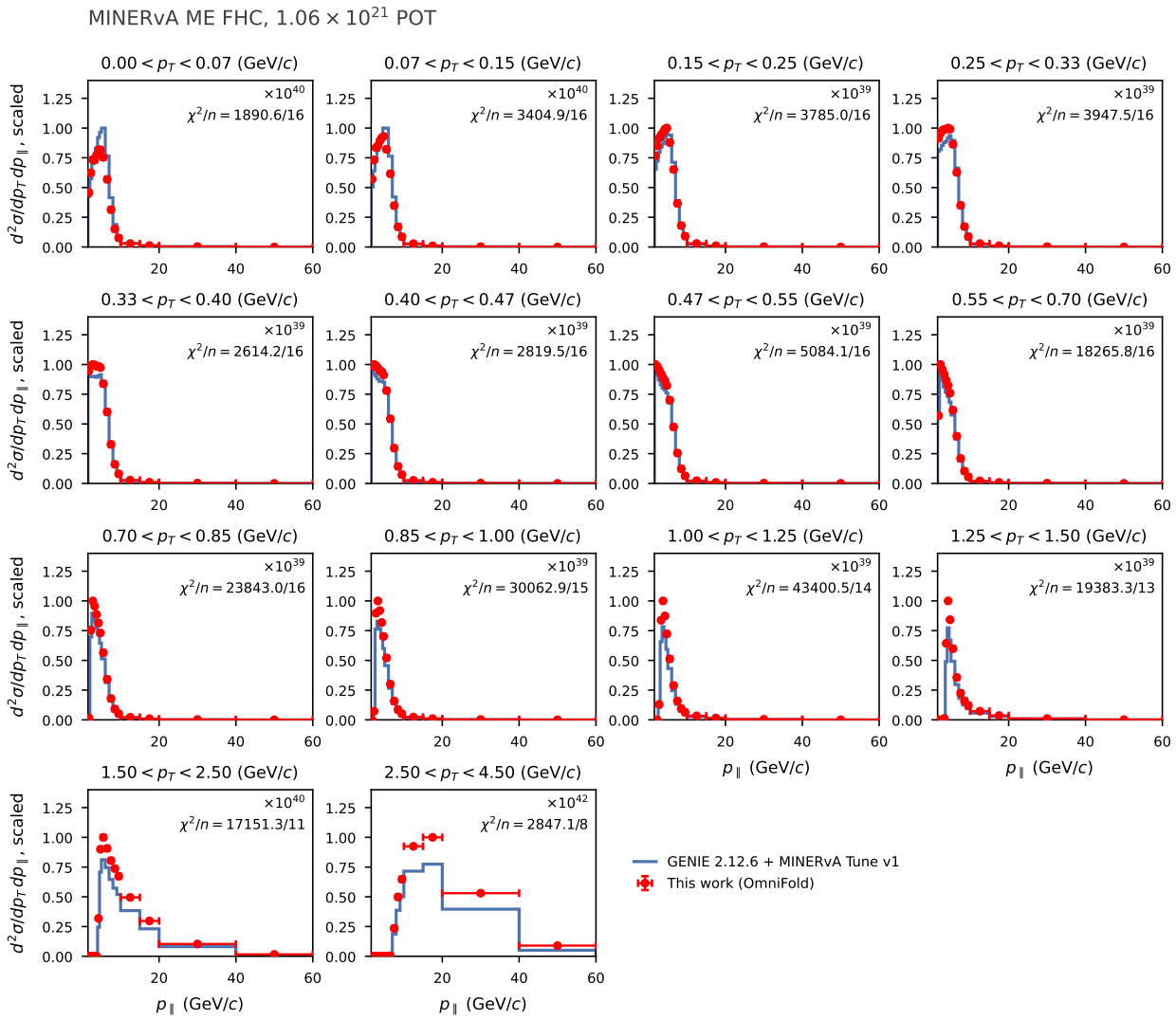


Figure 9: Unfolded cross section (points) and GENIE 2.12.6 + MINERvA Tune v1 *simulated truth* (histogram), sliced in p_T versus $p_\parallel$. The histogram is the analysis MC truth denominator with the unfolded result's flux, nucleon-count and efficiency normalization, so the two differ only as data versus MC prior; no published measurement is used, and only the *panel format* follows Fig. 13 of Ref. [3]. The vertical error bars are the weight statistics of the reweighted simulated sample, as in Fig. 7, neither the data's statistical uncertainty nor the reported covariance, and only these enter the annotated per-panel χ^2, which is a descriptive scale. Comparisons *with the published data* are in Figs. 8 and 13 and Table 6.

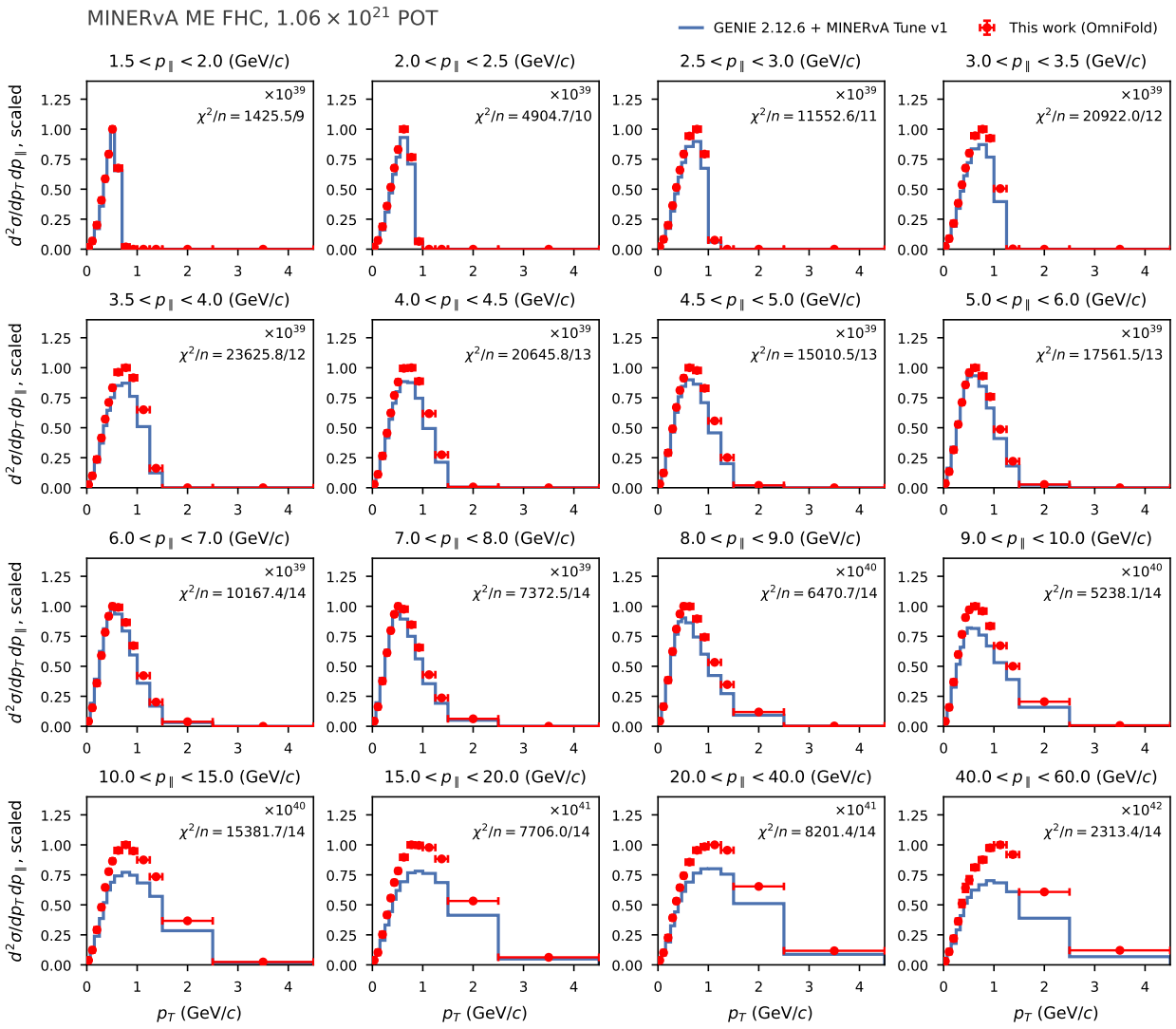


Figure 10: Unfolded cross section versus simulated truth (Fig. 9), sliced in $p_{\parallel}$ versus p_T, with the same error bars and no published measurement.

6.3 Uncertainty budget

Table 7 gives the budget over the 205 reported bins, constructed as described in §4. Our own combined budget, without the published covariance, is comparable to the published total: its per-bin median, 6.41 %, is smaller than the published 6.85 %, and its value on the integrated cross section, 4.76 %, is larger than the published 4.61 % (the width-weighted total σ of App. A.3.2). The comparison leaves out the generator-shape component C_{model} (§4.6), because the published uncertainty has no such component either; with it, the per-bin median is 7.90 %. The systematic-universe block accounts for a median 0.87 of the per-bin variance and 0.83 of the integral's variance, and the run-period block (§4.5) raises the per-bin median from 5.98 % to 6.41 %. The largest band is the flux, as in the publication (Fig. 4), and it accounts for part of the difference between the per-bin medians: its median is 3.71 %, against 4.01 % for the published flux block. The public release cannot establish the origin of this difference, for which the FHC-only constraint file remains the leading, unconfirmed hypothesis of this analysis. Our flux band carries the per-universe ν–e and inverse-muon-decay

constraint weights of the MINERvA Analysis Toolkit. The statistical and ML blocks together contribute under a tenth of the combined $\sqrt{\mathrm{Tr}\, C}$. Every block is computed with the LightGBM estimator of the quoted central value (§3.2), and relative uncertainties are taken with respect to that central value, which is the universe sweep's central value at training seed 42. Taken with respect to the exact-split cross-check instead, the same combined covariance has a per-bin median of 6.462 %.

Component	N	$\sqrt{\mathrm{Tr}\, C}$	per-bin median rel. σ
ML noise (LightGBM training-seed settings)	5	1.35×10^{-41}	0.048 %
Statistical (Poisson bootstrap)	300	1.89×10^{-40}	0.676 %
Systematic (universe sweep + 1.4 % normalization)	43 bands+1	2.69×10^{-39}	5.935 %
Run period	3	8.31×10^{-40}	2.087 %
Combined (block sum)	—	2.82×10^{-39}	6.414 %
Published total covariance (reference)	—	2.68×10^{-39}	6.85 %

Table 7: 2D uncertainty budget over 205 reported bins. Each component has a 205×205 covariance matrix C; $\sqrt{\mathrm{Tr}\, C}$ gives its absolute per-bin uncertainties in quadrature. The systematic row combines 43 of the 44 bands of the systematic-universe sweep (187 universes) with the normalization; the ML row combines five training-seed settings, and the run-period row three run periods. Our combined median relative uncertainty (6.414 %) is below the published total (6.85 %).

6.4 Eigenmode analysis of the published-covariance tension, $\chi^2 = 2.27$

With the published uncertainties as a descriptive scale, the central values have a pull RMS of 0.535 (a diagonal-only χ^2/ndf would be 0.29), whereas χ^2 per bin with the full published covariance is 2.27. The eightfold increase lies entirely in the off-diagonal structure of the published covariance, and it is a difference in *shape*, not normalization ($\chi^2_{\mathrm{norm}} = 0.04$ against $\chi^2_{\mathrm{shape}} = 465$). The eigenmode decomposition (Fig. 11) spreads it over many modes: the largest has a paper-covariance-standardized component of magnitude 4.8, and the next nine 3.4–3.8. The 10 smallest-λ modes carry 9.6 % of the total χ^2. Of the five largest contributors, two lie in the smallest-λ tenth of the spectrum. Restricted to the 73 best-measured directions, $\chi^2 = 136$ ($\chi^2/205 = 0.66$; the divisor is the number of reported bins, not the number of directions kept). Of the six leading modes, the three in the highest-$p_\parallel$ bins are all poorly determined directions ($\lambda/\lambda_{\max} \lesssim 4 \times 10^{-9}$); the other three are moderate-λ modes ($\lambda/\lambda_{\max}$ between 4×10^{-6} and 6×10^{-5}) and lie below the highest $p_\parallel$ bins (Fig. 12). No single p_T or $p_\parallel$ band carries more than 15.2 % of the total.

With the purity-based subtraction, the choice of GBDT estimator spreads the distance over about one unit, a methodological band: the exact-split cross-check gives 3.66, HistGBT 2.70 ± 0.04, and LightGBM with seeds 1–10 gives 2.64–2.75. Beyond this estimator dependence, the source of the shape difference has not been identified. At the central value's seed, the refined negative-weight subtraction of the quoted result lowers the LightGBM χ^2 per bin from 2.65 to 2.27. For the exact-split cross-check, χ^2 is larger and distributed differently: its components of magnitude 5–6 lie in moderate-λ modes on the low-p_T ridge of the cross-section peak. Its 10 smallest-λ modes carry 3 % of the χ^2, and the 73 best-measured directions give $\chi^2 = 291$ ($\chi^2/205 = 1.42$, again divided by the number of reported bins; App. A.5). Going from 5 to 10 iterations changes the LightGBM (seed 1) value by a negligible +0.01. App. A.5 gives the full treatment and Figs. 11–12 the eigenmode-localization maps.

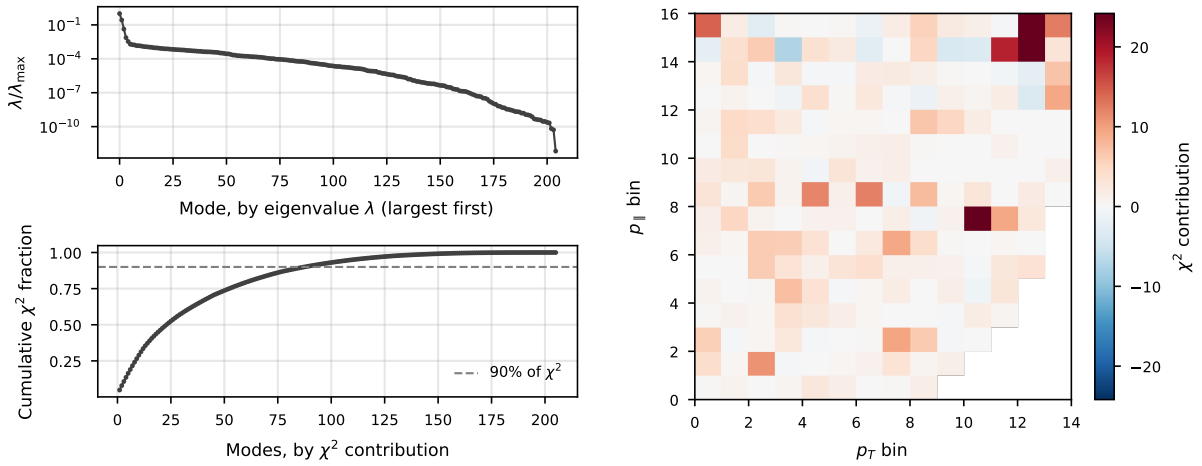


Figure 11: Eigenmode decomposition of the published-covariance χ^2: eigenvalue spectrum with cumulative χ^2 (left) and the per-bin χ^2-contribution map (right).

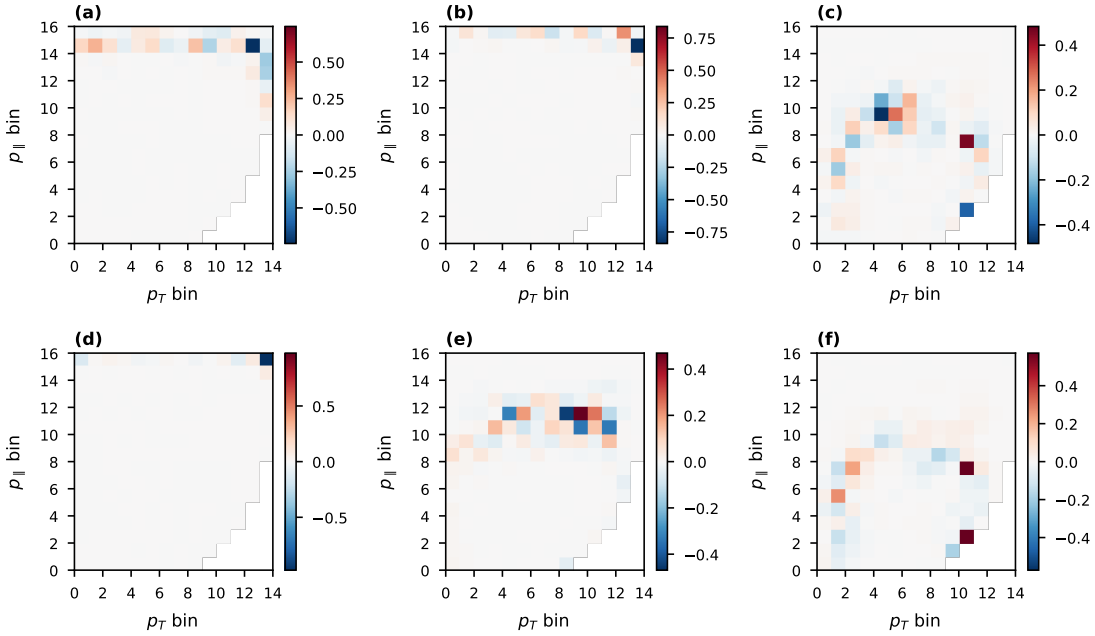


Figure 12: Leading χ^2-carrying eigenmodes of the published covariance in bin space, for the LightGBM central value. Three of the six, all smallest-λ directions, concentrate in the highest-$p_\parallel$ bins; the other three, moderate-λ modes, lie below them.

6.5 Comparison to GENIE MINERvA Tune v1

GENIE 2.12.6 [4] with MINERvA Tune v1 under-predicts the total cross section by $\sim 9\,\%$ ($\sigma_{\text{tune}}/\sigma_{\text{data}} = 0.91$), and the deficit is concentrated in the low-p_T peak. Figure 13 compares the prediction from the published ancillary files with both the published data and this measurement. The data covariance strongly disfavors the tune, with $\chi^2/\text{ndf} = 33.0$ for the published data and 31.3

for this measurement (26.5 for the exact-split cross-check), and our result tracks the data far more closely than the model does (Fig. 14). The published analysis [3] also compares with NuWro [6], GiBUU [7] and NEUT [43]. OmniFold's per-event reweighted truth ensemble can be compared with any such prediction in *any* projection of the classifier features without re-unfolding. An observable outside that space requires a new unfolding, as in the 3D extension (§7), which adds E_{avail} as a classifier feature.

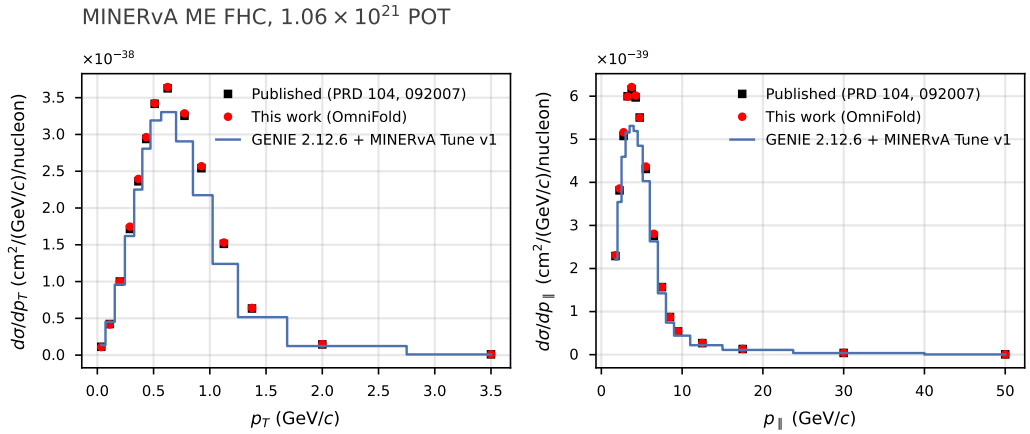


Figure 13: Projections of this measurement (red circles) and the published data (black squares), with GENIE 2.12.6 + MINERvA Tune v1 (blue line), which lies below the data in the peak. Uncertainties are not drawn.

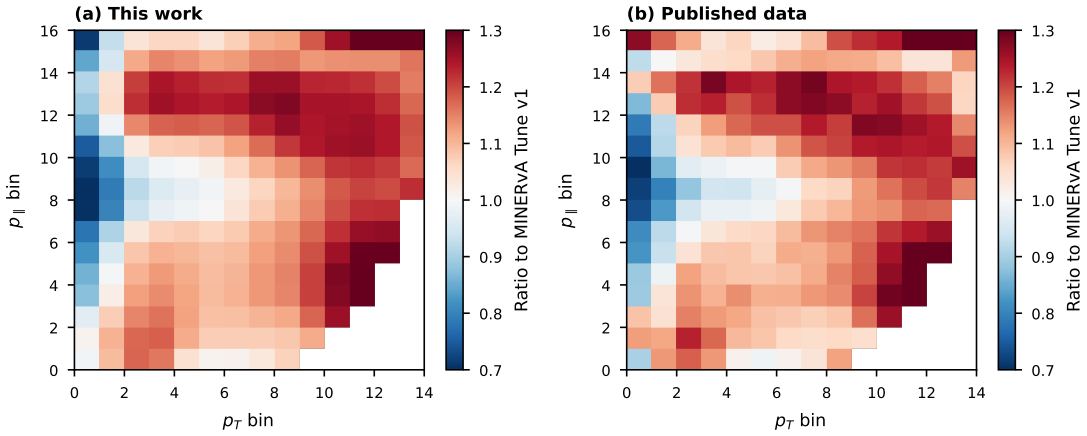


Figure 14: Bin-by-bin ratios to MINERvA Tune v1 on the $(p_T, p_\parallel)$ grid of our result (a, left) and of the published data (b, right), on a common colour scale of 0.7–1.3. Both maps show the same tune deficit, with ratios above unity over most of the plane and below it at the lowest p_T. The maps are close because both results depart from the tune in the same places by similar amounts. The direct comparison with the published data is the median bin ratio of Table 6 and the residual map of Fig. 8.

7 Higher-dimensional extensions: available energy and momentum transfer

Unbinned OmniFold adds an observable by adding an input to the step-1 and step-2 classifiers, without an explicit migration matrix or per-bin response tensor. It avoids the combinatorial binning penalty of matrix unfolding but still has a dimensional cost: classifier training, acceptance and support coverage, sample sparsity and uncertainty quantification become more demanding as observables are added. The primary methodological motivation is the absence of a practical analog of D'Agostini iterative Bayesian unfolding (IBU). We extend the measurement to $d^3\sigma/(dp_T\, dp_\parallel\, dE_{\text{avail}})$ with available energy as a third axis.

MINERvA has published binned triple-differential results (Fig. 15, Table 34), including recent comparisons between low- and medium-energy beams [44]. The latter measure p_T, $p_\parallel$, and available recoil energy for a quasielastic-like signal, for which the available energy is the sum of proton kinetic energies. This differs from the inclusive signal and species-dependent energy definition used here. What distinguishes this analysis is not the number of dimensions by itself but the unbinned, simultaneous CC-inclusive unfolding of three, four and five recorded high-level observables.

The comparisons with generator predictions in this section and in Sec. 8 are descriptive comparisons of central values and do not identify a nuclear or interaction mechanism (§7.7). In three and four observables they use central values without uncertainties: no 3D or 4D covariance is quoted, and no generator χ^2, p-value or significance (§§7.6 and 7.8).

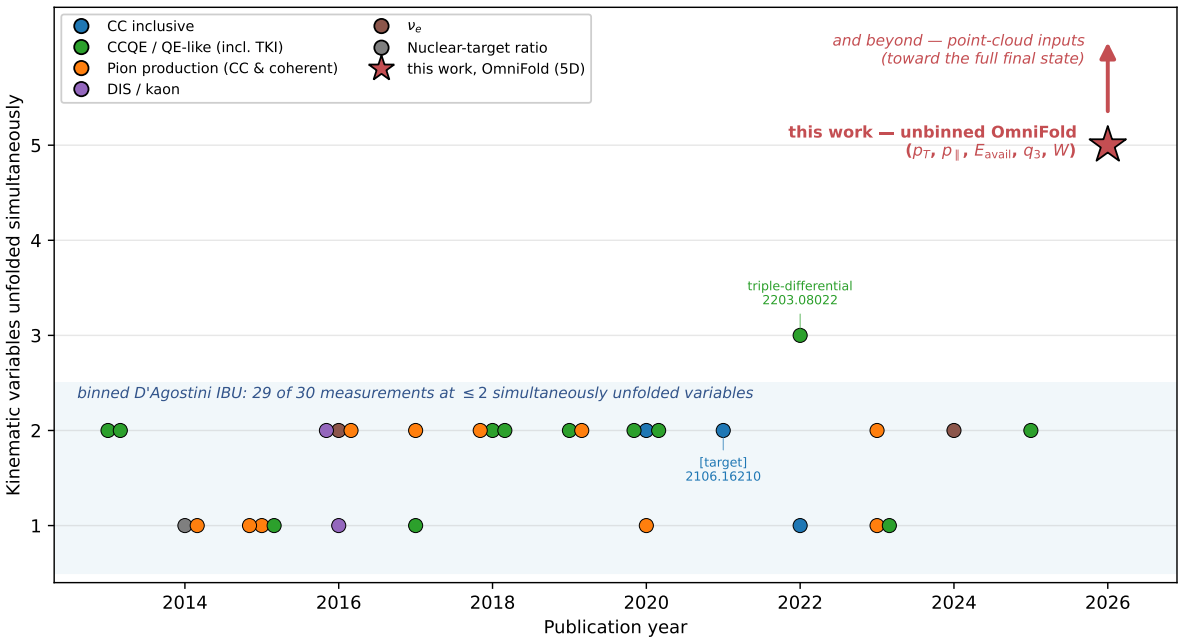


Figure 15: Number of kinematic variables unfolded *simultaneously* in surveyed MINERvA differential cross-section measurements, by year, colored by channel (a double-differential $d^2\sigma/dx\,dy$ counts as two). Prior results use explicit binned unfolding, including a triple-differential quasielastic-like measurement; this work extends an unbinned CC-inclusive unfolding of high-level event features to five recorded observables. Source: the MINERvA data-release page and the cited publications. The plotted inventory runs through 2025; newer measurements are listed in App. E. Classification uses each analysis's headline differential result; flux, reconstruction, generator-tuning and total-only entries are excluded.

7.1 Operational definition and provenance

Truth available energy is defined by its implementation, for which "low-recoil convention" is shorthand. In MeV, the truth function (`CVUniverse::GetEAvailableTrue()`) computes

$$E_{\text{avail}}^{\text{true}} = \sum_{p}(E - 938.27) + \sum_{\pi^{\pm}}(E - 135) + \sum_{\pi^0} E + \sum_{\gamma} E. \tag{1}$$

Protons and charged pions contribute the indicated kinetic-energy-like terms; neutral pions and photons contribute total energy. All other final-state species contribute zero, including neutrons, the primary muon, secondary $e^{\pm}$, neutrinos, kaons, antiprotons, strange baryons, other mesons and nuclear-remnant records. Reconstructed available energy is the MAT calorimetric quantity

$$E_{\text{avail}}^{\text{reco}} = 1.17\left[(E_{\text{blob}}^{\text{tracker}} - E_{\mu\text{-fuzz}}^{\text{tracker}}) + (E_{\text{blob}}^{\text{ECAL}} - E_{\mu\text{-fuzz}}^{\text{ECAL}})\right], \tag{2}$$

implemented in the reconstruction function (`CVUniverse::NewEavail()`), with no final clamp to non-negative values.

Both definitions are inherited from MAT-MINERvA rather than chosen for this analysis; the comparisons below record differences between conventions, not the origin of the inherited one. There is no unique published E_{avail} definition. Rodrigues *et al.* [5] give a closed list of five species: proton

and charged-pion kinetic energy plus π^0, electron and photon total energy. Equation (1) includes four of those five species and excludes secondary $e^{\pm}$. The later definition of Ascencio *et al.* [8] is open: it includes all other final-state particles except neutrons, subtracting a nucleon mass for strange baryons, and so includes species omitted here. The charged-pion term is kinetic energy in Rodrigues and in Henry/Su *et al.* Eq. 4 [45], and charged-pion total energy in the `minerva-ml` implementation (App. F). We therefore do not claim that our definition matches "the published definition."

The charged-pion term subtracts the inherited constant 135 MeV rather than the charged-pion mass of about 139.57 MeV, and the production keeps that constant. Relative to the charged-pion mass, it moves 0.65 % of the simulated signal weight into another of the seven true-E_{avail} bins of the fine grid and changes no such bin by more than 0.97 %. This effect has not been assessed against the projection of the five-dimensional covariance of the configuration-(a) central value. On the sixteen-cell result of Sec. 10, from configuration (e), to first order in the unfolding and without retraining, replacing the constant by the charged-pion mass moves the cross section by at most 0.19σ in any cell and by 0.42σ as one coherent shift, the norm of the shift under the reported covariance (0.21σ and 0.42σ without the run-period block).

The reconstructed coordinates share inputs. In units with $c = 1$, the implemented definitions (with reconstructed muon quantities) are

$$\begin{aligned} E_\nu^{\text{reco}} &= E_\mu + q_0^{\text{reco}}, \\ Q^2_{\text{reco}} &= \max\{0,\, 2E_\nu^{\text{reco}}(E_\mu - p_\mu \cos\theta_\mu) - m_\mu^2\}\,, \\ q_3^{\text{reco}} &= \sqrt{Q^2_{\text{reco}} + (q_0^{\text{reco}})^2}, \\ W^{\text{reco}} &= \sqrt{\max\{0, M^2 + 2M q_0^{\text{reco}} - Q^2_{\text{reco}}\}}, \end{aligned} \tag{3}$$

where $m_\mu^2 = E_\mu^2 - p_\mu^2$ and M is the average nucleon mass. The recoil-energy estimate q_0^{reco} differs from the tracker-plus-ECAL available energy $E_{\text{avail}}^{\text{reco}}$; truth E_{avail} also omits energy carried by species such as neutrons and is not the full energy transfer. Thus five recorded coordinates do not establish five independently constrained physical directions. Adding a coordinate, supplying a new detector input, and resolving a new truth direction are separate questions.

Unfolding and binning. Both definitions are evaluated for all twelve ME FHC playlists, and, for configuration (a), the 2D unfolding code is run with three classifier inputs, $(p_T, p_\parallel, E_{\text{avail}})$. Table 8 lists the configuration (§3.2) of each result. The cross section is extracted with the 3D bin volume $\Delta p_T\, \Delta p_\parallel\, \Delta E_{\text{avail}}$, and the flux Φ is still applied per p_T bin, with the same value for every $p_\parallel$ and E_{avail} bin. The E_{avail} binning $[0, 0.1, 0.2, 0.4, 0.8, 1.5, 3.0, 100]$ GeV ends in a wide final bin, the catch bin, so that the E_{avail} marginal contains the full CC-inclusive recoil tail. For configuration (a), the overall completeness of the unfolding input (§3.4) is $c = 1.0000$, as in 2D.

7.2 The available-energy spectrum

The unfolded $d\sigma/dE_{\text{avail}}$, the 1D projection of the 3D result (Fig. 16), falls monotonically from 2.18×10^{-38} cm^2/GeV/nucleon at the low-recoil peak to 6.7×10^{-41} in the catch bin, the expected CC-inclusive shape with most of the rate at $E_{\text{avail}} \lesssim 0.8$ GeV.

Table 8: Configuration of each result in this section (Table 2). (e) and (f) use refined negative-weight subtraction with a 400-tree, 31-leaf refinement; (a) uses purity-based subtraction.

Result	Configuration
3D and 4D central values, unless stated otherwise	(e), deterministic LightGBM (the learner of (c))
2D result compared in §7.3	(f), LightGBM at seed 42 (the learner of (a))
Figs. 17 and 23 and numbers read from them; 3D closure; classifier-family cross-check	(a), LightGBM

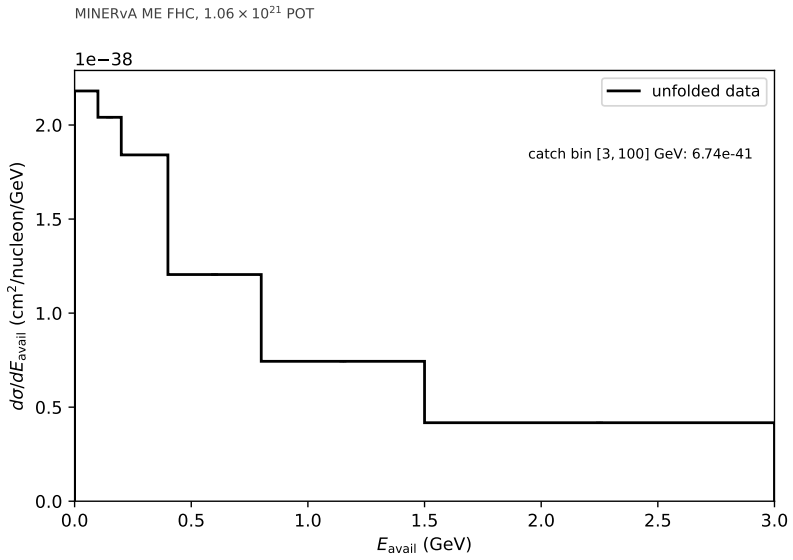


Figure 16: Unfolded $d\sigma/dE_{\text{avail}}$, the 1D projection of the 3D measurement. The wide $[3, 100]$ GeV catch bin is excluded from the axis; its value is annotated.

7.3 Validation against the 2D measurement

Marginalizing over E_{avail} should reproduce the 2D result of §6, our validation reference in the absence of a directly matched published CC-inclusive 3D reference for this signal and phase space. The 3D marginal (configuration (e)) and the 2D result ((f)) share the refined negative-weight subtraction but differ in learner (Table 8). The *normalization* agrees closely: the marginal's area-weighted total over the 205 reported cells agrees with the 2D result's to 0.141 %, and the median per-bin ratio is 1.0010. The *shape* agrees less closely. The per-bin ratio of the marginal to the 2D result scatters by $\sim 3.7\,\%$ (standard deviation over the 205 cells). With the published covariance, the indicative distance of the marginal from the published result is $\chi^2/\text{ndf} = 4.52$, above the 2D values: 2.27 for the quoted LightGBM central value and 3.66 for the exact-split cross-check. Figure 17 (configuration (a)) gives 4.98. Because the two results share data and their cross-covariance is unavailable, this number is not a calibrated goodness-of-fit statistic.

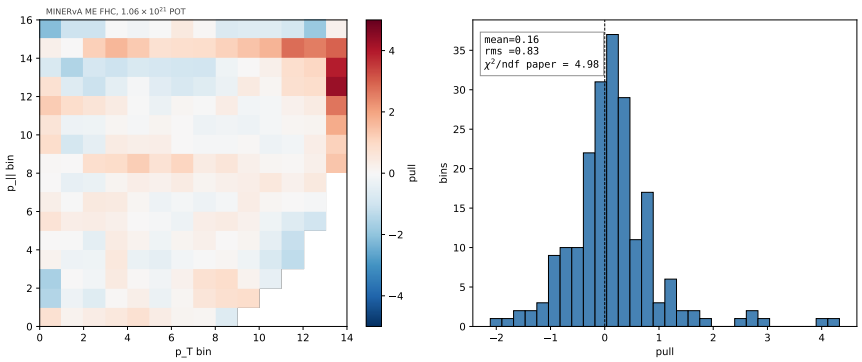


Figure 17: Standardized per-bin differences of the E_{avail}-marginal, scaled by published uncertainties, on the 14×16 grid and in distribution. Shared data and unavailable cross-covariance make these descriptive; they are not formal independent-result pulls. Configuration (a), with purity-based subtraction (§3.2).

7.4 Closure: the one injected E_{avail} deformation is recovered

A reweighting closure tests the 3D unfolding of configuration (a) only, on one injected deformation. The unfolding recovers a known +30 % Gaussian bump injected into pseudo-data in *truth* E_{avail} (center 0.3 GeV, width 0.15 GeV). The recovered step-2 mean weight is 1.048 against the injected 1.0485. Residuals against reweighted truth on the E_{avail}-marginal have median 0.9999 and standard deviation 0.0006 (0.06 %); the per-3D-bin residual standard deviation is 3.2 % from MC statistics in sparse high-recoil corners. The closure's marginal scatter is far below the 3.7 % data-versus-2D scatter. Because the closure uses configuration (a) and the data comparison sets (e) against (f), the ratio of the two scatters is not quoted. Total normalization matches to better than 1 %. In the metric of §5 (E_{avail} projection) the residual is 1.84 % against 18.1 % injected (ratio 0.102). No acceptance threshold set before the test is documented, so it has no pass or fail verdict.

The test covers one shape, this Gaussian, injected into the pseudo-data and unfolded *with the same detector response that the unfolding uses*. For this injection the configuration-(a) estimator neither removes the injected feature nor creates a spurious one. That result is shown for configuration (a) only and does not extend to (e); since the data-versus-2D scatter sets (e) against (f), it is not interpreted as data–simulation structure resolved by the available-energy axis. The test does *not* exclude estimator bias for other deformations (§5.1). The hidden-variable closure exceeds its threshold at the largest amplitude (§5, App. D).

7.5 Classifier-family cross-check (GBDT vs. NN)

Our gradient-boosted decision trees differ from the neural-network classifiers used in published OmniFold analyses [1, 18]. This cross-check uses the configuration-(a) GBDT unfold, with purity-based subtraction. The weight $w = p/(1-p)$ holds only if the GBDT output is a calibrated posterior. The step-1 reco classifier (data vs. MC reco) has a near-diagonal reliability curve, Brier ≈ 0.25 at AUC ≈ 0.537. Data and reco MC are nearly indistinguishable, so the reweight is a modest correction. A full OmniFold cross-check with a Keras multilayer perceptron (MLP), the same weighted 3D inputs and the same five-iteration loop gives a total ratio of 1.0078 to the GBDT result and median projection differences of 0.7 % to 1.4 % (Fig. 18). In the populated bulk the configuration-(a) result therefore agrees between the two tested classifier families.

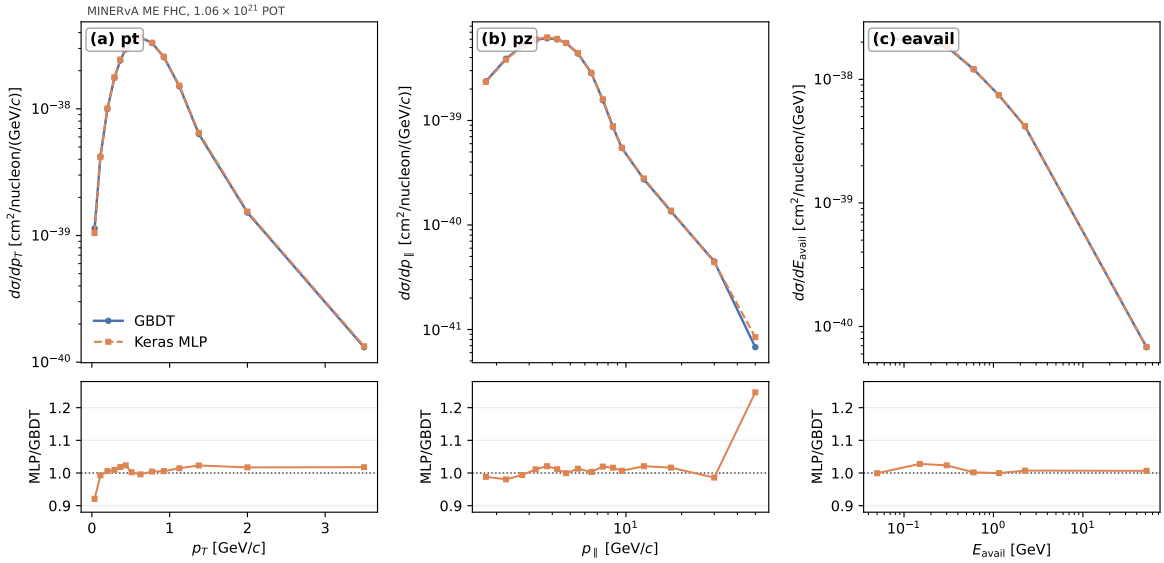


Figure 18: Five-iteration OmniFold cross-check with identical weighted 3D inputs and extraction, changing only the classifier family. Top: absolutely normalized GBDT (configuration (a)) and Keras-MLP projections; bottom: binwise ratio. The MLP/GBDT total-cross-section ratio is 1.0078; median absolute projection differences are 1.20 % in p_T, 1.36 % in $p_\parallel$, and 0.66 % in E_{avail}. Larger excursions occur only in sparsely populated outermost p_T and $p_\parallel$ bins.

7.6 Generator comparison in 3D

The published ancillary files provide only binned 2D model histograms. Quantitative 3D comparisons, or comparisons in arbitrary projections, require generating truth-level events and histogramming them as $d^3\sigma$ in the same fiducial phase space with the same truth E_{avail} definition. Because the unfolded result is at truth level, the comparisons need no detector simulation. The GENIE, NuWro and GiBUU samples were generated and processed for this analysis, and MINERvA Tune v1 was extracted from the analysis simulation's truth with its full tune weight; no spectra were digitized or externally supplied.

- **GENIE 2.12.10 central value**, generated with `gevgen` on a CH target with the MINERvA ME FHC flux, using the DefaultPlusValenciaMEC cross-section splines [4] but gevgen's default event-generator list, which excludes MEC, so the sample has no 2p2h events (§7.7). It carries no MINERvA reweights, and its events are reweighted to the analysis flux as described below.
- **MINERvA Tune v1**, the analysis tune (base GENIE + RPA, low-recoil 2p2h, and non-resonant-pion reweights), taken directly from the truth of the analysis simulation, which reproduces the normalization of the published ancillary curve to 0.01 %.
- **NuWro 21.09** [6], an independent generator on C with the same flux (after reweighting below) and phase space.
- **GiBUU 2019** [7], a transport-theory generator on C, built and run natively, without a container, with the same flux and phase space.

NEUT [43] is not included.

Flux repair and normalization. The GENIE (CV and +Valencia 2p2h) and NuWro samples were generated from a flux that had no PPFX weights, was capped at 50 GeV, and stored flux *densities* in variable-width bins that were sampled and normalized as *contents*, which under-weighted every wide high-energy bin and suppressed the high-$p_{\parallel}$ predictions and the integrated rates (Ref. [37], R1.3). The repair reweights each event exactly to the analysis PPFX central-value flux on 0–100 GeV, with correctly sampled 50–100 GeV supplements. The repaired GENIE and NuWro predictions are used in §§7–8, except in three studies, flagged where they appear, that keep the unrepaired samples: the FSI-dial studies, the departure ratio of §9.2 and the NuWro-shaped prior of §12. GiBUU reads its own flux and receives an energy-dependent per-event reweight to the same convention, which changes its integrated cross section by a factor 0.9966. Its sample stops at $E_\nu \approx 20$ GeV, so it is empty above $p_{\parallel} \approx 20$ GeV/c. The analysis simulation, MINERvA Tune v1, is not affected. Each prediction is normalized by its own flux-averaged total CC cross section per nucleon.

Figure 19 shows the comparison. Totals over the full measurement phase space follow GiBUU (2.22×10^{-38}) < NuWro (2.66×10^{-38}) < Tune v1 (2.71×10^{-38}) < GENIE CV (2.76×10^{-38}) < data (3.07×10^{-38} cm^2/nucleon). All four under-predict the rate by 10 % to 28 % of the data. Integrating the reported E_{avail} range with the catch bin dropped gives the same order. All four closely track the p_T and $p_{\parallel}$ projections but differ in E_{avail}: NuWro and Tune v1 *suppress* the lowest-E_{avail} (quasielastic) bin below the untuned GENIE CV through RPA and nuclear effects, while GiBUU does not. Data lie *above* all four across the quasielastic–Δ dip, $0.1 \leq E_{\text{avail}} < 0.4$ GeV (in the lowest bin GiBUU reaches the data). Of the displayed projections, only E_{avail} locates this excess; the 2D $(p_T, p_{\parallel})$ measurement does not resolve it.

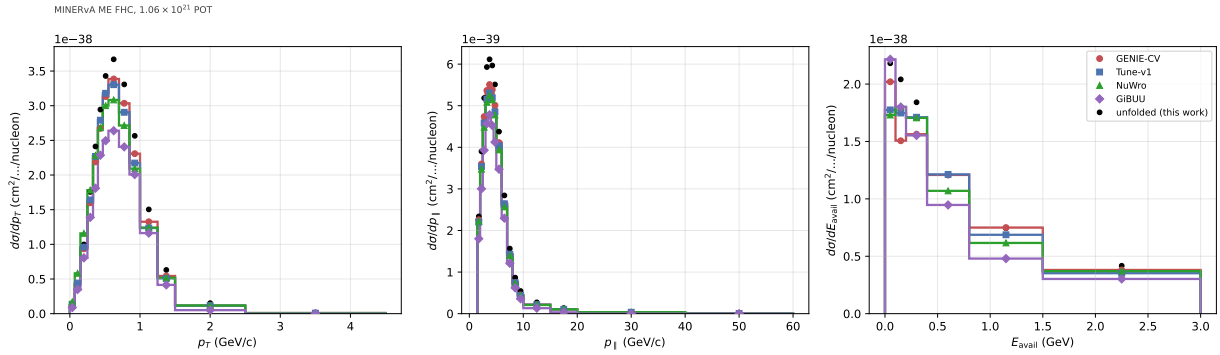


Figure 19: 1D projections of the unfolded 3D central cross section (black) versus truth-level predictions from GENIE 2.12 CV, MINERvA Tune v1, NuWro 21.09, and GiBUU 2019 (GENIE and NuWro: the flux-repaired predictions, §7.6). Models track the p_T and $p_{\parallel}$ shapes but differ on E_{avail}; all under-predict the quasielastic–Δ dip ($0.1 \leq E_{\text{avail}} < 0.4$ GeV). The wide E_{avail} catch bin is omitted from the axis. Data have no error bars because no 3D covariance is quoted (§7).

For the integrated E_{avail} rate without the catch bin, GENIE CV, Tune v1, NuWro and GiBUU central values lie respectively 7.2 %, 9.5 %, 12.8 % and 22.3 % below the data. The exact projection of the five-dimensional covariance of §9.6 would supply a 3D covariance; it has been computed but is not quoted (§7) or used for any result in this note.

7.7 The low-available-energy excess and the known low-recoil deficit

We compare the low-E_{avail} excess with candidate model changes, both by reweighting and by generating the relevant physics. MINERvA first isolated the rate enhancement between quasielastic

and Δ peaks, with multi-nucleon final states, in a low-three-momentum-transfer subsample [5]. The empirical tune was built to absorb it; our central values in muon kinematics and E_{avail} show the predictions falling short in the same region.

The tested pion-FSI variations do not account for it. Reweighting 2M central-value GENIE events with GENIE's `grwght1p` changes $d\sigma/dE_{\text{avail}}$ by less than a percent for both tested pion-FSI dials. `FrInel_pi` shifts the spectrum by $\leq 0.74\,\%$ (peak in 0.10–0.20 GeV) and `FrAbs_pi` by $\leq 0.82\,\%$ (peak in the lowest bin). Both are ratios within the unrepaired GENIE sample (§7.6), in which the flux defect cancels to first order. Neither was recomputed. The integrated rate moves by ± 0.02–$0.03\,\%$ at $\pm 1\sigma$. [1] A dial that moves the spectrum by less than 1 % cannot account for an excess of 8 % to 36 % per bin over the GENIE central value below $E_{\text{avail}} = 0.4$ GeV. This conclusion is limited to these dials and their range; other transport models or proton-FSI effects are not excluded. Recent MINERvA quasielastic-like measurements point toward overestimated proton and charged-pion FSI in their model comparisons [44]. Their different signal definition prevents directly identifying that effect with our inclusive discrepancy.

Comparison with a missing 2p2h component. The base GENIE central value uses the default event-generator list and contains *no* meson-exchange-current (2p2h/MEC) events. None of the 1.48×10^6 CC events of the generated sample is a MEC event, and the 50–100 GeV supplement is MEC-free as well. Before the flux reweight of §7.6, the sample's unweighted modes are QE 11 %, RES 22 %, DIS 66 % and COH 0.9 %. 2p2h must therefore be *added*; reweighting cannot supply it. The mode decomposition multiplies each bin of the central-value $d\sigma/dE_{\text{avail}}$ by the fraction of its events in each mode. Overlaid with data, it places the largest positive data−CV gaps in the quasielastic–Δ dip, $[0.10, 0.20)$ and $[0.20, 0.40)$. The CV is much closer to data in the QE-dominated $[0, 0.10)$ and $[0.40, 1.50)$ regions. At central value, 69 % of the positive deficit lies at $E_{\text{avail}} \leq 0.4$ GeV, where 2p2h events populate. Closing the integrated deficit requires 2p2h at $\approx 45\,\%$ of the QE rate (68 % to 71 % in the two dip bins). This is the standard magnitude of the empirical and Valencia MEC models; the FSI shifts are below one percent (Fig. 20).

GENIE with Valencia 2p2h enabled. We regenerated the GENIE central value with Valencia 2p2h enabled (`-event-generator-list Default+CCMEC`, 2M events). The Nieves–Simo–Vacas MEC channel [46, 47] is present in the splines but absent from the total-CC cross-section graph written by GENIE's `gspl2root`, so we normalize by the non-MEC CC count: QE+RES+DIS+COH is fixed to the known total, and MEC is added through $1/(1 - f_{\text{MEC}})$ with $f_{\text{MEC}} = 2.87\,\%$. 2p2h falls in the dip and brings all three low-E_{avail} bins toward data (Fig. 21). In central values, it closes 52 % of the data−CV gap at $E_{\text{avail}} \leq 0.4$ GeV and 63 % of the integrated deficit (CV $-7.2\,\% \to -2.7\,\%$). The prediction still falls short in the dip, especially $[0.10, 0.20)$, but its integral over the reported E_{avail} range is closer to data than Tune v1 ($-2.7\,\%$ against $-9.5\,\%$). Tune v1 adds MINERvA's empirical low-recoil 2p2h *enhancement* and RPA to the stock Valencia MEC model. The central values are consistent with the known need for that enhancement, and with the NuWro and GiBUU predictions, whose native 2p2h is not enhanced, also falling below the data in the dip; they do not establish that need.

2p2h leaves the high-E_{avail} $[1.50, 3.00)$ bin unchanged. There, data exceed the GENIE CV by about 10 % (data/CV 1.10) and exceed Tune v1 as well; the high-E_{avail} excess over Tune v1 is

[1] `FrInel_pi` is disabled in the standard MAT-MINERvA set of systematics. This is MINERvA's current practice (private communication from the collaboration, 2026), and the reason is a degeneracy (of overlapping pion-FSI dials, one must be dropped), not a defect of that dial.

localized on the (E_{avail}, W) plane in §8.

The generator's 2p2h event category is also not the same as a two-body current: recent NEUT studies, for example, include two-body currents that contribute to one-particle–one-hole final states and examine their interplay with axial form factors [48]. Visible-energy comparisons can, moreover, depend on both the primary interaction and the subsequent hadronic transport.

Ascencio *et al.* [8] report the same deficit in MINERvA's dedicated low-recoil inclusive $d^2\sigma/(dq_3\, dE_{\text{avail}})$ measurement for $q_3 < 1.2\,\text{GeV}$ at the same $\langle E_\nu \rangle \sim 6\,\text{GeV}$, finding GENIE and NuWro underpredicting their low-E_{avail} data without enhanced 2p2h. Our muon-kinematics extension integrates over all q_3 instead of restricting q_3, which limits the comparison to the *shape* and direction in E_{avail}, without absolute normalization. In those terms our central values are consistent with that established low-recoil deficit. A bin-identical comparison of central values is in §7.8.

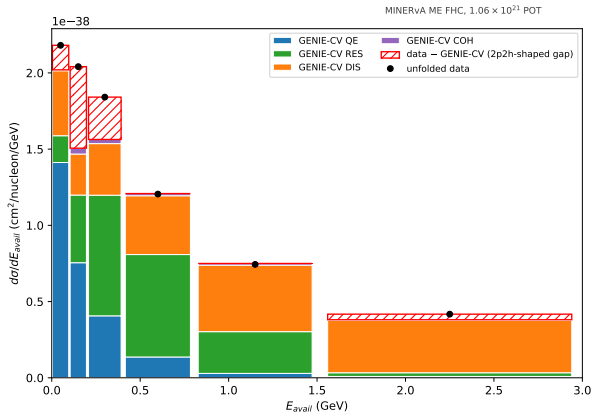


Figure 20: Central-value $d\sigma/dE_{\text{avail}}$ by interaction mode, overlaid with unfolded data. The excess lies in the quasielastic–Δ dip ($E_{\text{avail}} \leq 0.4\,\text{GeV}$), where a missing 2p2h component would populate. Data carry no error bars because no 3D covariance is quoted (§7).

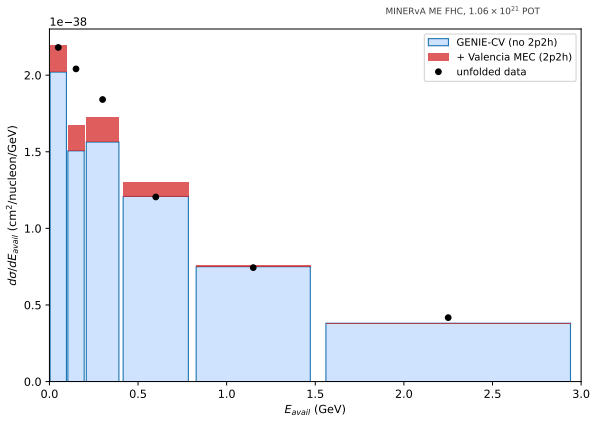


Figure 21: GENIE regenerated with Valencia 2p2h (red, stacked on no-2p2h CV; both flux-repaired) closes about half the low-E_{avail} data−CV gap in central values (52 %). The remainder occupies the region the MINERvA tune's empirical enhancement is built to absorb. Data carry no error bars because no 3D covariance is quoted (§7).

7.8 Fourth axis: three-momentum transfer q_3

The fourth axis adds one feature (§7), three-momentum transfer $q_3 = |\vec{q}|$, chosen because its (E_{avail}, q_3) plane separates quasielastic, 2p2h and resonance processes and permits bin-compatible comparison to MINERvA's dedicated low-recoil $d^2\sigma/dq_3\, dE_{\text{avail}}$ measurement (Ascencio *et al.* [8]). Truth q_3 comes from the generator (`CVUniverse::GetQ3True()`); reconstructed q_3 uses the muon Q^2 and the calorimetric recoil (`CVUniverse::RecoQ3()`). Both are evaluated for all twelve playlists, as for E_{avail}. Lateral systematic shifts change q_3 but leave E_{avail} unchanged, so a shifted q_3 is stored for each lateral universe instead of reusing the central value. The classifier inputs are $(p_T, p_\parallel, E_{\text{avail}}, q_3)$, with deliberately coarse q_3 binning $[0, 0.2, 0.4, 0.6, 0.8, 1.2, 2.0, 100]\,\text{GeV}$ (seven bins including a final catch bin), giving $14 \times 16 \times 7 \times 7 = 10\,976$ four-dimensional bins (4830 populated).

Marginalization check (4D→3D). In configuration (a), the q_3 marginalization reproduces the 3D E_{avail} spectrum and low-recoil excess (Fig. 23, right). The E_{avail}-marginal data/Tune v1 ratio is

1.21 in the lowest bin, dips to 1.01 across the quasielastic–Δ region, and rises to 1.16–1.23 in the high-E_{avail} DIS tail (integrated data/Tune v1 = 1.13). These ratios are to Tune v1, whereas §7.7 uses untuned GENIE. Recovering the 3D result as a by-product of the higher-dimensional unfolding is the 4D→3D analog of the 3D→2D marginalization check (§7.3).

The (E_{avail}, q_3) plane. The (E_{avail}, q_3) plane (Fig. 23, left) shows structure that the q_3 marginal, integrated over E_{avail}, does not resolve; the marginal's data/Tune v1 ratio is 1.03–1.19 (center). At fixed E_{avail} in the quasielastic–Δ dip, data/Tune v1 *rises* with q_3. The central value over-predicts low-q_3, moderate-E_{avail} cells (ratio down to 0.66); data excess concentrates at low E_{avail} (all q_3) and toward high q_3. The muon-only or fixed-q_3 projections considered here do not resolve this correlation. It bears on the bin-compatible comparison with Ascencio *et al.* The map's comparator is the MINERvA Tune v1 prior, distinct from the untuned GENIE sample of §7.7.

No 4D uncertainty. Neither a 4D uncertainty nor a ratio of the covariance from simultaneous variation of every reweightable band and the flux multisim (the unified throw of §4.5) to the block covariance is reported. A corrected 4D covariance would use the actual minus and plus variation endpoints; MAT's mean-centered covariance blocks, normalized by $1/N$; one fixed estimator seed; and mean-centered joint throws, with the mean shift reported separately. It would not subtract a scalar jitter floor. Because the single-pass shifted observables exist only for events selected at the central value, re-selected event loops (§4.7) would be needed to bound migration effects.

Bin-identical comparison to the published low-recoil measurement. Supplemental data from Ref. [8] (44-cell $d^2\sigma/(dE_{\text{avail}}\, dq_3)$ with its full covariance, distributed inside the public arXiv source package) permit a direct quantitative cross-check. Both measurements are merged onto their maximal common grid, the edges present in *both* binnings. Our $(p_T, p_\parallel)$ marginalization is restricted to $p_\parallel < 20\,\text{GeV/c}$ to mirror the Ascencio muon selection ($\theta_\mu < 20°$, $1.5 < p_\mu < 20\,\text{GeV}$). Together, these give two super-cells, $E_{\text{avail}} < 0.4\,\text{GeV}$ in $q_3 \in [0.4, 0.6)$ and $[0.6, 1.2)\,\text{GeV}$: the low-recoil 2p2h region. Our central result lies 9.2 % and 6.0 % above Ascencio. No pulls or full-covariance χ^2 are reported (§7; Fig. 22). Both share MINERvA flux/detector systematics, treated here as independent. The p_μ and $p_\parallel$ cuts differ by $\lesssim 6\,\%$ kinematic smearing at the 20 GeV boundary. The two analyses also define true E_{avail} differently (§7.1): in this analysis's simulation, Ascencio's broader definition moves 11.0 % of the population of our lowest true-E_{avail} bin ($E_{\text{avail}} < 0.1\,\text{GeV}$) out of it. That shift has the sign and location of the residual. The two common cells, however, span our three lowest bins ($E_{\text{avail}} < 0.4\,\text{GeV}$), and the aggregate shift over those bins has not been computed. The definition difference therefore remains an alternative explanation that is not excluded. At the 10 % scale of the residual, the comparison neither refutes nor validates the result. A comparison on all 44 cells has not been made; it requires re-unfolding on the Ascencio edges, which is straightforward for the unbinned method, *and* a systematic sweep on that binning.

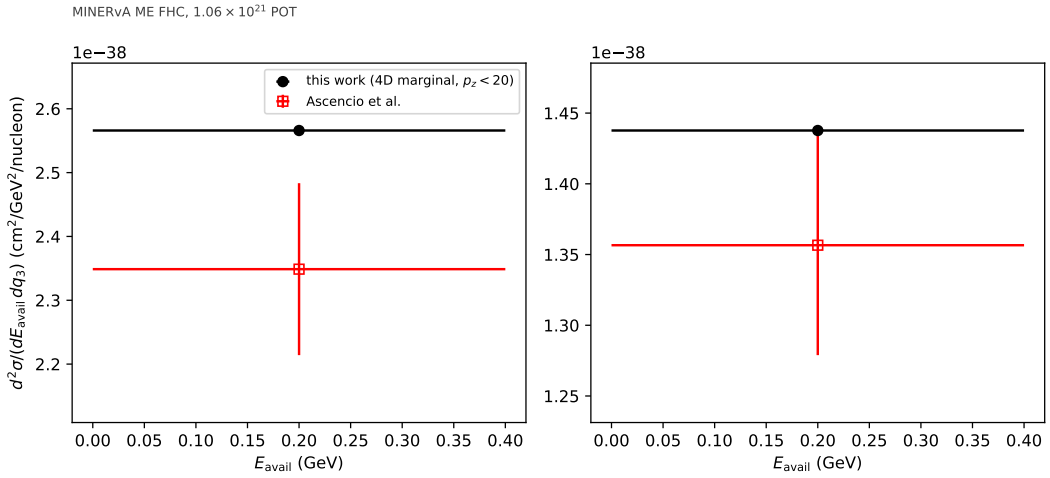


Figure 22: Bin-identical comparison to Ascencio *et al.* [8] on the maximal common (E_{avail}, q_3) grid. Our points have no uncertainty or χ^2 drawn because no 4D covariance is quoted (§7); the published points carry the published covariance.

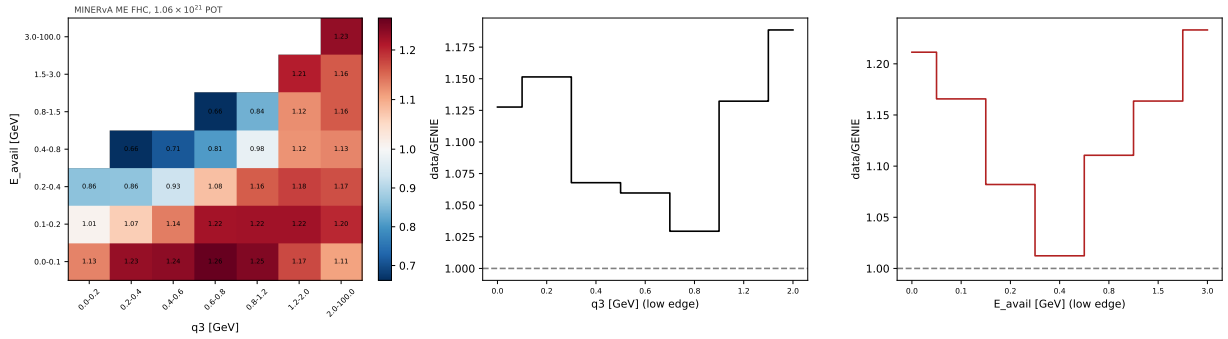


Figure 23: 4D data/Tune v1 ratio ("data/GENIE" in the figure) projected onto (E_{avail}, q_3) (left; blank cells are kinematically empty), with q_3 (center) and E_{avail} (right) marginals. The E_{avail} marginal reproduces the 3D low-recoil excess. In the displayed joint plane the data/Tune v1 ratio rises with q_3 across the quasielastic–Δ dip. Configuration (a), with purity-based subtraction (§3.2).

8 Fifth axis: W and the (E_{avail}, W) plane

For the central values, Valencia 2p2h closes about half of the low-recoil gap (§7.7) but leaves a separate excess over the MINERvA Tune v1 prior at high E_{avail}. There the predicted 2p2h contribution is zero, and the two tested pion final-state-interaction variations change the prediction by less than one percent, too little to explain the excess.

GiBUU is low throughout the E_{avail} tail. Its W-integrated five-dimensional data/generator ratio is 1.59 at $E_{\text{avail}} \in [0.8, 1.5)$ and 1.37 at $[1.5, 3.0)$, in configuration (a), the LightGBM unfolding with purity-based subtraction (§3.2); its sample stops at $E_\nu \approx 20\,\text{GeV}$ (§8.5). GENIE CV instead matches data at $[0.8, 1.5)$ and is low at $[1.5, 3.0)$, where its three-dimensional data/CV ratio is 1.10 (§7.7) in configuration (e), the deterministic learner with refined negative-weight subtraction.

Adding the hadronic invariant mass W as the fifth OmniFold axis maps where in W the high-

E_{avail} excess lies. The validation of §9.1 includes a model-dependence study; it does not establish a bias–variance tradeoff across estimator settings and provides no new total uncertainty for this projection.

8.1 W as a fifth axis

Truth W is the experimenter's invariant mass $W = \sqrt{M^2 + 2M(E_\nu - E_\mu) - Q^2}$, with struck-nucleon mass M. Reconstructed W combines the calorimetric energy transfer q_0 of the reconstructed-q_3 estimator and the muon-kinematic Q^2 with the average nucleon mass. The external generator predictions instead compute truth W with the neutron mass for every event and place events with $W^2 \leq 0$ in the lowest W bin, whereas the analysis uses the struck-nucleon mass and excludes such events from its five-observable truth sample. In the GENIE and NuWro samples this moves 0.14–0.15% of the events into another W bin. In GENIE CV and NuWro (without their 50–100 GeV samples) it changes the coarse-cell predictions by about 0.1 % at the median and at most 2.2 %. In any cell, that is at most about one Monte Carlo standard error of the prediction. The change was not evaluated at cell level for GENIE+MEC and cannot be evaluated for GiBUU.

Like q_3 (§7.8), and unlike E_{avail}, W changes under the lateral systematic shifts, so a shifted W is stored for each lateral universe. The event-loop output files also store truth hadronic diagnostics: the proton and pion multiplicities and the polar angle of the summed momentum of all final-state particles other than muons and neutrinos. The W edges $[0, 1.1, 1.4, 1.8, 2.2, 3.0, 100]$ GeV bracket the quasielastic peak, the Δ, the transition region and the DIS tail.

The configuration-(a) five-dimensional unfolding in $(p_T, p_\parallel, E_{\text{avail}}, q_3, W)$ agrees under marginalization with the lower-dimensional results: integrating over W reproduces the configuration-(a) 4D normalization to 0.11 % (5D/4D total = 1.0011) with median differences per axis of 0.3 % to 1.5 %, and $d\sigma/dW$ is finite and non-negative in all six bins.

8.2 Implemented definitions of the five observables

Table 9 gives the implemented truth and reconstructed definitions of the five unfolded observables, as used in the companion journal article. E_μ, p_μ, and θ_μ are the muon energy, momentum, and angle; q_0 is the calorimetric estimate of the energy transfer; M is the struck-nucleon mass, $\bar{M} = (1.5M_n + M_p)/2.5$. Exact Q^2 is $-(k - k')^2$ from the generator's four-vectors; truth W instead uses the massless-muon Q^2 with θ_μ from the beam axis, a difference whose effect on W is given below.

Reconstructed q_3 and W combine the muon kinematics with the calorimetric estimate q_0 of the energy transfer, which is separate from reconstructed E_{avail}. Since $W^2_{\text{reco}} = (\bar{M} + q_0)^2 - q^2_{3,\text{reco}}$ and $q_{3,\text{reco}}$ increases with q_0 at fixed muon kinematics, reconstructed W is an exact function of $(p_T, p_\parallel, q_3)$, apart from the clamp at $W^2 = 0$, which applies to 1.5 % of simulated events.

At truth level, q_3 is computed from the true neutrino and muon four-vectors, but θ_μ, and with it p_T and $p_\parallel$, is measured from a fixed beam axis tilted by 58.87 mrad from the detector axis rather than from the true neutrino direction. Truth W is therefore fixed by $(p_T, p_\parallel, q_3)$ only up to that event-by-event angle, of about 1 mrad, and the choice of M: the residual is within ±2 MeV for 68% of events and within −49 to +67 MeV for 99.9%; 0.25 % of events (weighted) change their coarse W bin, and 0.15 % have no real solution.

The five reconstructed inputs therefore share information and do not, by themselves, provide five independent constraints: the result is five simultaneous observables, not five independent measurements. The companion journal article distinguishes the five classifier inputs, the lower-dimensional support they populate, and the five reported observables. Only reconstructed W is

Table 9: The five observables as implemented.

	Truth definition	Reconstructed definition	Shared inputs and conventions
p_T, $p_\parallel$	$p_\mu \sin\theta_\mu$, $p_\mu \cos\theta_\mu$, θ_μ from the beam axis	the same, from the MINERvA–MINOS muon	muon only
E_{avail}	proton kinetic energy, $E_{\pi^\pm} - 135\,\text{MeV}$, π^0 and γ energy (§7.1)	1.17× calorimetric recoil energy in the tracker and electromagnetic calorimeter, muon deposits removed	its own calorimetric estimator; 135 MeV convention
q_3	$\sqrt{Q^2 + (E_\nu - E_\mu)^2}$, exact Q^2	$\sqrt{Q^2 + q_0^2}$, $Q^2 = 2E_\nu(E_\mu - p_\parallel) - m_\mu^2$, $E_\nu = E_\mu + q_0$	muon kinematics and q_0
W	$\sqrt{M^2 + 2M(E_\nu - E_\mu) - Q^2}$, $Q^2 = 4E_\nu E_\mu \sin^2(\theta_\mu/2)$	$\sqrt{\bar{M}^2 + 2\bar{M}q_0 - Q^2} = \sqrt{(\bar{M} + q_0)^2 - q_3^2}$	fixed by $(p_T, p_\parallel, q_3)$ when reconstructed; nearly so at truth

fixed by the other inputs; reconstructed E_{avail} comes from a separate calorimetric estimator and is not a function of them. The nearly degenerate correlation directions of the joint-test metric (§11.1) are consistent with this redundancy, but that origin has not been shown, and the information added by W has not been measured.

The analysis excludes events with negative truth W^2, a weight fraction of 4.9×10^{-5}, from its five-observable truth sample and normalization; §8.1 gives the external predictions' convention. The generators measure p_T and $p_\parallel$ about their own neutrino axis; their p_T differs from the analysis truth p_T by ± 4 MeV for 68% of events. Over the 109 joint-test cells, this convention changes the analysis simulation's cell contents by factors between 0.9965 and 1.0035 (median 1.000008), at the level of its Monte Carlo precision, and shifts GENIE's central prediction by between -0.47 and $+0.53$ of each cell's null standard deviation. For GENIE with MEC, NuWro and GiBUU its effect is not reliably determined (Ref. [37], R1.18).

8.3 Generator predictions and the target difference

Table 10 lists the generator predictions compared in this note and tested in §11. All are per nucleon of the stated target and are binned with the same fine-grid-to-cell map.

Table 10: Generator predictions.

Prediction	Target	E_ν coverage and flux	Test cells
Tune v1 (GENIE 2.12.6)	CH	analysis simulation	109
GENIE 2.12.10 CV	CH	0–50 GeV, reweighted; 50–100 GeV added	109
GENIE 2.12.10 + MEC	CH	as GENIE CV	109
NuWro 21.09	C	0.5–50 GeV, reweighted; 50–100 GeV added	109
GiBUU 2019	C	below 20 GeV, reweighted	72

Hydrogen has no quasielastic, 2p2h, or coherent ν_μ charged-current channel, so normalizing per nucleon does not remove the difference between the targets. In GENIE's own CH sample, the carbon-only cross section per nucleon exceeds that of CH by 8.3 % for $W < 1.1$ GeV and is 2.2 % below it for $1.1 \le W < 1.4$ GeV (5.6 % and -4.6 % with the overall normalization removed). The NuWro and GiBUU comparisons contain a target difference of this kind, which the tests of §11, as

run, do not correct. Section 11.9 re-evaluates them with the correction.

8.4 Distribution of the excess in the (E_{avail}, W) plane

Figure 24 shows the unfolded five-dimensional result projected onto (E_{avail}, W), with the MINERvA Tune v1 prediction subtracted. That prediction is the OmniFold prior, the simulation's truth weighted by its full tune weight; the untuned GENIE CV of §8.5 is a separate prediction. In cell-integrated terms, the positive excess lies mostly at high E_{avail} and high W. The two highest-E_{avail} bands carry 57 % of it, and the region $E_{\text{avail}} \geq 0.8\,\text{GeV}$ carries 67 %, with 83 % of that at $W \geq 1.8\,\text{GeV}$. The largest cell, the corner $E_{\text{avail}} > 3\,\text{GeV}$, $W > 3\,\text{GeV}$, holds 22 %. A secondary excess lies at $W < 1.1\,\text{GeV}$, with a small deficit at $W \simeq 1.4$–$1.8\,\text{GeV}$.

These statements hold under the nominal detector response, and the regularization bias of Sec. 9 is largest at high E_{avail} and high W. Estimator-seed shifts of central values are a small fraction of the projected uncertainties (§9.6). The seed sensitivity of the shares, which are ratios of central-value differences, has not been measured separately. The response-mismatch closure of Appendix D is designed to test whether an incorrect hadronic response can move strength into or out of the low-W component. It has not yet produced a result.

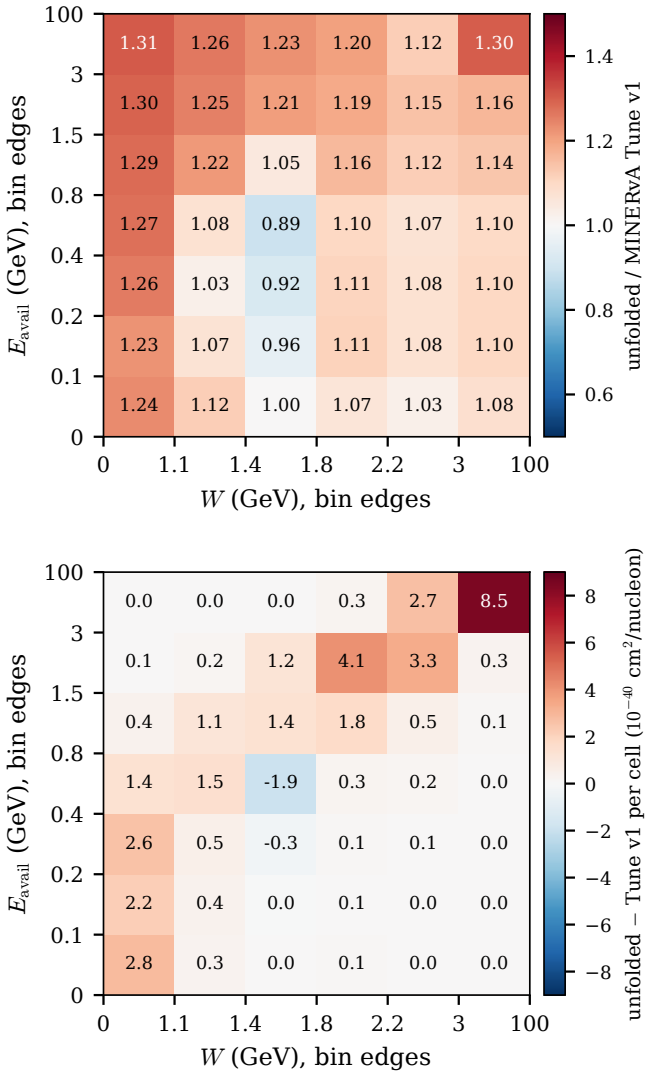


Figure 24: Unfolded data and MINERvA Tune v1 (GENIE 2.12.6 with analysis tune weights, the OmniFold prior) in (E_{avail}, W). Top gives dimensionless per-cell data/Tune on a 0.5–1.5 colour scale; bottom gives per-cell data − Tune in units of $10^{-40}\,\text{cm}^2$/nucleon. Equal-size display cells are labelled by physical bin edges on the 7×6 reporting grid, with E_{avail} edges $[0, 0.1, 0.2, 0.4, 0.8, 1.5, 3.0, 100]$ GeV and W edges $[0, 1.1, 1.4, 1.8, 2.2, 3.0, 100]$ GeV. The final interval on each axis, $[3, 100]$ GeV, is a wide catch bin. The density ratio is independent of cell area; the bottom panel instead subtracts *cell-integrated* cross sections, $(\text{d}^2\sigma/\text{d}E_{\text{avail}}\,\text{d}W) \times \Delta E_{\text{avail}}\,\Delta W$ in cm^2/nucleon, so the catch-bin cells, which are wide, carry correspondingly larger values. Together the panels locate the positive excess at high E_{avail} and high W, with a secondary component at $W < 1.1$ GeV and a small deficit at $W \simeq 1.4$–1.8 GeV. Quoted percentage shares refer to cell-integrated positive excess in the bottom panel.

8.5 Comparison with four external generators

All four external generators fall below the data in the high-E_{avail}, high-W corner. Figure 25 compares the (E_{avail}, W) predictions of GENIE CV, GENIE+Valencia 2p2h/MEC, NuWro and GiBUU. As in §7.6, each is normalized by its flux-averaged total CC cross section per nucleon, and those of GENIE

and NuWro are flux-corrected. The 3D and (E_{avail}, W) comparisons use the same five-dimensional histogram of each generator. GENIE CV totals $2.76 \times 10^{-38}\,\text{cm}^2$/nucleon in the measurement phase space in both; GENIE+MEC uses the normalization convention of the 3D comparison.

In the corner ($E_{\text{avail}} \geq 0.8\,\text{GeV}$, $W \geq 1.8\,\text{GeV}$), data/generator = 1.14, 1.14, 1.16 and 1.61, respectively. For GENIE and NuWro the corner ratios exceed their integrated ratios (1.11, 1.07 and 1.15) by 1 % to 7 %, compared with 16 % for GiBUU, whose integrated ratio is 1.39 ($\sigma = 2.22 \times 10^{-38}\,\text{cm}^2$/nucleon, against 3.07×10^{-38} for the data).

GiBUU's larger contrast comes from its sample's energy limit: it was generated only up to $E_\nu \approx 20\,\text{GeV}$ (§7.6). In GENIE and NuWro, neutrinos above this energy carry 14.3 % to 14.8 % of the corner cross section and 6.8 % to 7.4 % overall. Removing them raises these generators' corner-to-integrated contrast to 9 % to 16 %, bracketing GiBUU's. If GiBUU is instead supplied with the shares that GENIE and NuWro carry above this energy, data/GiBUU becomes about 1.37 in the corner and 1.29 integrated. With the GENIE and NuWro fluxes corrected and GiBUU supplied with the shares above this energy, each generator's central-value corner ratio lies close to its integrated ratio. GiBUU is lower than the others everywhere.

Valencia 2p2h raises the GENIE+MEC integrated prediction while leaving the corner essentially unchanged (1.142 → 1.139), since it populates low W. At $W \in [2.2, 3.0)\,\text{GeV}$, data/generator = 1.12, 1.11, 1.13 and 1.34, again close to the integrated ratios. At $1.4 \leq W < 1.8\,\text{GeV}$, instead, both GENIE predictions lie 9 % to 12 % above the data. These are central-value ratios. No compatibility metric based on the configuration-(a) five-axis covariance is quoted, because the estimator-seed sensitivity of its construction, measured with two companion covariances, exceeds the limit set before production (§9.6); §10.6 gives a sixteen-cell χ^2 comparison.

The localization in §8.4 is relative to MINERvA Tune v1; data exceed all four external generators both overall and in the corner.

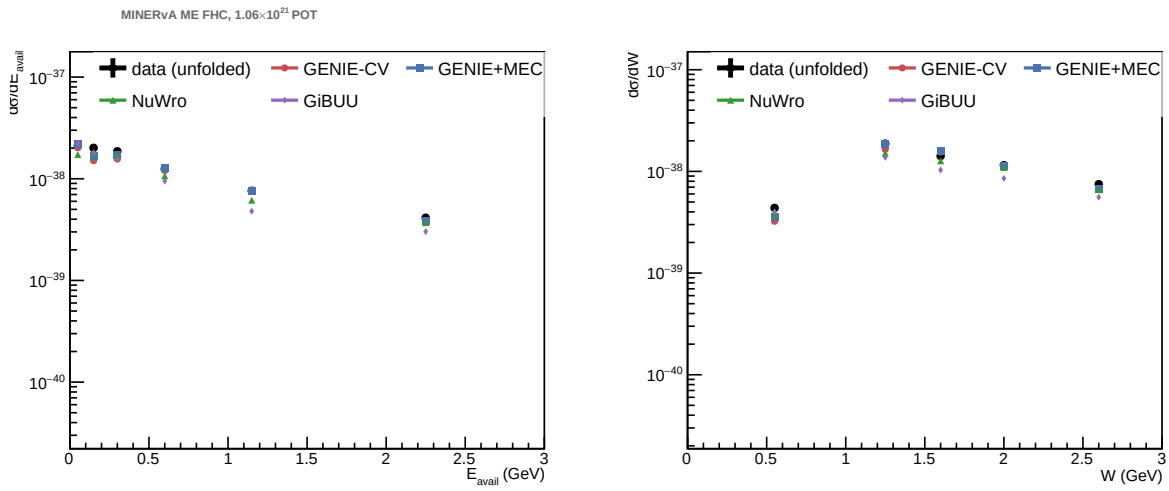


Figure 25: Projections of the unfolded (E_{avail}, W) cross section onto E_{avail} (left) and W (right), with the predictions of four generators (GENIE CV, GENIE+MEC, NuWro, GiBUU; GENIE and NuWro flux-corrected, §7.6). In the high-W, high-E_{avail} corner, which the projections do not resolve, every generator lies below the data (§8.5) — GENIE and NuWro by about their overall normalization shortfall, GiBUU by more; adding 2p2h leaves the corner essentially unchanged because 2p2h populates low W.

MINERvA shallow-inelastic neutrino and antineutrino measurements also report shape and

normalization disagreements with generators [49]. They motivate tests across the resonance–DIS transition, but their selections and reported observables differ from this inclusive projection, so they provide physics context, not an independent confirmation of the same localized excess. A later significance for the location identified in the central-value map must account for any selection of the tested region made with these data. The published low-recoil cross-check in (E_{avail}, q_3) (§7.8) gives central values 9.2 % and 6.0 % above the published measurement on the maximal common grid.

9 Five-dimensional estimator: validation and uncertainty

This section validates the five-dimensional estimator behind the central values of §§7–8. It covers what each test isolates (§9.1), background subtraction and generator-sized truth changes (§9.2), why the estimator does not follow such changes (§9.3), model dependence and the reporting definitions it rules out (§9.4), the coverage test (§9.5) and the status of the five-axis covariance of the configuration-(a) central value (§9.6). The sixteen-cell result of configuration (e) follows in §10, and the calibrated generator tests built on a variant of this estimator in §11. The unbinned negative-weight background method itself is derived in App. C.

9.1 Which source of error is being tested

Prior reweighting changes the frequencies of existing simulated truth events while retaining their detector-response simulation; a response mismatch instead changes how a given truth is reconstructed. Finite classifier capacity and iteration count approximate the inverse and can leave residuals in either case. Coverage asks whether the resulting intervals contain their specified target at the stated rate (§9.5). These mechanisms are separate: an incorrect generator prediction alone does not diagnose an interval failure. Each test below isolates one intervention.

All 2D closures in §5 unfold reweighted simulated signal without background, so they do not test background subtraction.

Table 11 lists the validation studies of the article and of this note (detector response: §5.1 and §11.2). Letters refer to Table 2.

9.2 Background subtraction and generator-sized truth changes

Background subtraction. End-to-end five-dimensional pseudo-experiments add background-simulation events and rebuild per-bin purity weights as in the measurement. At nominal truth they return the highest-W column of (E_{avail}, W) 3.5 % to 4.1 % low and the highest-E_{avail}, highest-W cell 1.7 % high; without background, the same experiments recover the truth to 0.22 %. Relative to these cells' total uncertainty in the configuration-(a) covariance, the bias is 0.36–1.07 standard deviations. The bias comes from the purity-based subtraction (§C), enters no quoted covariance, and has no established size on data. Refined unbinned negative-weight subtraction (App. C), applied with a deterministic variant of the five-dimensional LightGBM production estimator and a refinement of 100 trees and 8 leaves, brings the highest-W column to 0.11 % to 0.24 %, the corner cell to 0.24 % and every other (E_{avail}, W) cell to at most 0.39 % over sixty nominal-truth experiments, which are not paired with the purity-based ones (Fig. 37). Some joint cells remain up to 1.0 % high, up to 14 standard errors, against −0.2 % without background; the mechanism is not established. On data, the two treatments differ by a median of 0.36 % and at most 1.4 % over the reported functionals. These differences measure method sensitivity, not bias. Negative-weight subtraction is not used for the configuration-(a) five-dimensional central value; the sixteen-cell result uses it (§10).

Table 11: Validation studies of the article and of this note.

Study	Truth change	Backgrounds	Domain	Criterion
Two-dimensional reproduction (f), (a), (b)	none: data against the publication	data	205 published bins	none; descriptive
Three-, four-observable closure (a)	injected bump in truth E_{avail} (three observables) or q_3 (four)	none	one-dimensional marginals	descriptive: residual ratio 0.10 (E_{avail}; §7.4, §5.1)
Background closure (c), (d)	none (nominal truth)	included	42 (E_{avail}, W) and 109 joint cells	none; studies during estimator development
Generator-sized change (d)	GiBUU/GENIE E_{avail} shape ratio	included	42 (E_{avail}, W) cells	recovery requirement; not met
Shapes held out of development (e)	NuWro and GENIE + MEC shapes, not used while (e) was chosen	included	42 (E_{avail}, W) cells	compared with the configuration-(a) total σ
Coverage (c)	nominal; signal rate lowered by about 10% with a weak E_{avail} slope (weight 0.900–0.906 below 3 GeV)	included	153 cell and projection integrals	upper bound on coverage at least 0.66 (0.94)
Test calibration (e)	each null hypothesis	included	109 (GiBUU 72) joint cells	size set by the calibration (§11.3)

Generator-sized truth changes. The unfolding does not follow a GiBUU-to-GENIE E_{avail} shape reweighting of the truth (0.70–1.33 per E_{avail} bin, total preserved). The GENIE prediction behind the ratio uses a mis-sampled flux (§7.6), so the ratio is a valid known-truth deformation of generator size but not a measured GiBUU–GENIE difference. The (E_{avail}, W) residuals have a median of 9.7 %; they reach 74 % in the highest-W column and several percent in the largest cells. The 74 % residual is in the cell $0.8 \leq E_{\text{avail}} < 1.5$ GeV, $W \geq 3$ GeV, whose share of the cross section the deformation study did not record. A q_3 deformation preserving the (E_{avail}, W) truth shifts these cells by up to 37 %. Signal-only experiments with the same seeds reproduce both results. The residuals are therefore a property of the estimator (five iterations, the GBDT configuration of §3), not of the background treatment; their status in the quoted uncertainties is given in §9.3.

9.3 Why the estimator does not follow a generator-sized change

Both OmniFold steps behave as intended: the first matches the reconstructed pseudo-data within statistical noise, and the second reproduces the truth-level weight sums to 0.5 %. Yet the unfolded result folds back onto the data less well than the true model does. With noise-free pseudo-data the unfolding recovers more than 99 % of the change that the departure produces in the reconstructed distribution, whereas the median (E_{avail}, W) truth residual ranges from 5 % to 11 %, with 74 % in the worst cell. The unrecovered reconstructed-level difference is small relative to the departure but statistically significant at the analysis exposure.

With noise-free pseudo-data, thirty iterations reduce the E_{avail} residual by 26 %; the E_{avail} residual continues to fall slowly, while the q_3 residual grows with iterations. A larger classifier improves the reconstructed-level fit without reducing the E_{avail} truth residual, and removing the regression for events that fail reconstruction does not reduce that residual either. No single cause is established, and no estimator change within the scope of the study meets the recovery requirement, set before the test, for the GiBUU-to-GENIE departure. The E_{avail} and q_3 residuals, a regularization (unfolding-model) bias of the estimator, enter no uncertainty quoted for the configuration-(a) central

value. For the sixteen-cell result (§10), the model term of the reported covariance is the change in the unfolded data when the prior is reweighted to each generator (GENIE + MEC, NuWro, GiBUU; §10.3). This term is unscaled and is not a calibrated bound. The fake-data study of §10.5 measures the estimator's bias and compares it with the reported covariance. The pseudo-experiments of the paper's recovery figure, sixty for the development variant at the GiBUU-to-GENIE E_{avail} shape and twenty for each of three shapes not used in the joint-test estimator's selection, use seed sets that were not held out from the estimator's development.

Comparison with exact binned unfolding. A diagnostic in simulation compares the boosted-tree residual with that of an exact binned iterative unfolding. The binned unfolding unfolds the same simulated events, with their exact efficiency and the response binned in the 10,499 occupied five-dimensional reconstructed cells, starting from the nominal prior. The comparison uses noise-free signal-only pseudo-data, the 109 supported fine joint cells (J cells) and the fixed historical truth deformations D1, W1, W3 and the q_3 stress, pooled with W2 (all defined in §9.4). The boosted-tree comparator for W2 includes background; the outcome is unchanged without it. A residual is eligible when its diagnosis is stable under the truth resolution: of the 376 J-cell and deformation combinations with residuals above 2 %, 232 changed classification with the truth resolution or the response weighting and were excluded, leaving 144 (38 % of the 376). At five iterations the J-cell medians of the absolute residual agree: 9.5 %, 5.4 %, 5.6 % and 4.3 % for the boosted trees, against 8.3 %, 5.4 %, 4.9 % and 5.3 % for the exact binned unfolding, for D1, W1, W3 and the q_3 stress. Of the 144 eligible residuals, pooled over the deformations, 15 are dominated by the tree approximation. Only 2 of these 15 lie in identified quantities (as defined in the diagnostic record cited below), a share of 0.014 of the 144. Of the 144, 121 lie in quantities that the binned response constrains weakly at the analysis exposure. Their Cramér–Rao widths exceed 10 % at both truth resolutions tested (median 31 % at the coarser resolution, that of the J cells themselves), so the regularization trades a large variance for the bias. Longer noise-free exact iteration removes much of the residual for D1 (J median 3.3 % after 200 iterations and 0.8 % after 10^5) and part of it for W1 (3.3 % and 2.8 %). It does not do so for W3 and the q_3 stress (4.4 % and 6.6 % after 200 iterations, 8.9 % and 6.7 % after 10^5), whose within-cell binning bias grows with the iteration number. The long runs do not converge (only 9–34 of the 109 J quantities converge for each traced deformation), so these values are last iterates. The coarser muon projection retained only three cases and is uninformative. Within this comparison, the residual is consistent with the regularization of the inverse problem at this binning and prior, and the tree approximation is not supported as its main cause. This conclusion holds for a binned, signal-only and noise-free comparison at the simulation's detector response, on the development truths, and for these 144 residuals only. It does not address what unbinned or biased estimators could recover, the coverage failure of §9.5 or the validity of any interval, and it establishes no impossibility. The precision thresholds that the diagnostic proposes are not acceptance criteria for any result in this note. The exclusions, definitions and numerical verification are in the binned response diagnostic record (Ref. [37], R1.14).

Higher-capacity background refinement and data stability. The refinement classifier underfits, which causes the residual background bias (App. C). A candidate with raised refinement capacity, configuration (e) of the sixteen-cell result (§10), passes the nominal background-inclusive closure on 40 fresh pseudo-experiments (largest deviation 2.7 standard errors) and a pooled test of its statistical intervals (pull width 1.02). The pooled test does not establish coverage and width for every functional. The candidate fails the data-stability test, whose criterion was set before it was run: input perturbations within floating-point rounding move functionals by a median of 0.8 and

at most 2.6 statistical standard deviations. The baseline estimator's shifts, a median of 0.8 and at most 2.5, place this instability in the estimator family. The instability also explains the difference of at most 1.2 % on data between the driver and the development pipeline. The data bootstrap, which refits the full chain per replica, already includes this rounding-level variability: each replica carries the same spread (median 0.23 % per joint cell), about 37 % of the bootstrap variance and below a tenth of any total uncertainty. Central values are therefore reproducible only to within this spread. The paired perturbation study of §9.4 tests whether the bootstrap contains this variability, for this candidate only. A declared probe on simulated pseudo-data of two-fold cross-fitting and of two tree regularisations found no variant that met the data-stability requirement, so the estimator is unchanged.

Unfolding-model dependence. For historical shapes held out while this candidate was assessed (NuWro; GENIE with enhanced meson-exchange currents), its unfolding-model dependence is median 2 % to 6 % and at most 16 % to 31 %, within the configuration-(a) total uncertainty in 20 of the 42 (E_{avail}, W) cells. No uncertainty from this assessment is used for a quoted result.

9.4 Model dependence, variability and reporting resolution

The scalar five-dimensional estimator studied here uses production LightGBM.

These studies measure three different quantities: truth-dependent responses to iterations and capacity, small repeat variability at one setting, and appreciable prior sensitivity. Unfolding benchmarks also assess these quantities separately [50]. The studies do not establish a bias–variance tradeoff across estimator settings, and regularization bias can invalidate intervals based solely on estimator variability [51]; neither reference supplies the variance measurement that is missing for this estimator.

The noise-free study unfolds the full simulation's reweighted reconstructed events without background. Baseline classifiers and the missed-event regressor use 100 trees and 8 leaves; the capacity probe uses 400 trees and 31 leaves. Repeated experiments use estimator R, the raised-refinement-capacity candidate of §9.3: five iterations, baseline OmniFold estimators and a separate 400-tree, 31-leaf background-refinement classifier. With seed 42, its 40 nominal experiments and 20 at each of five departures include Poisson pseudo-data, changing MC splits and background refinement. These experiments are not an independent validation: their samples were used during development, and their assessment seeds, although separate, were later also used in the estimator's development.

Saved fine-grid outputs are integrated into the supported J, H2 and EW cells of the table below. Unsupported cells remain in the full-domain rate. The interval assessment keeps all 42 EW cells, including the three that the support rule excludes (definitions and normalization checks in App. B.2).

Label	Meaning in this section
J	109 supported fine joint cells, subdividing all five axes
H2	27 supported coarser joint cells, also subdividing all five axes, not nested within J
EW	39 supported (E_{avail}, W) cells (42 with the three unsupported cells)
RD1–RD3	reporting definitions at J, H2 and EW resolution ("Reporting definitions" below)
D1, W1–W3	fixed historical truth deformations (next paragraph)
R	the raised-refinement-capacity estimator of §9.3: baseline OmniFold classifiers with a 400-tree, 31-leaf background-refinement classifier

The truth labels denote fixed historical deformations, built before the generator flux was repaired: D1 is a GiBUU/GENIE E_{avail} ratio, W1 a NuWro/GENIE E_{avail} ratio, W2 a GENIE MEC/CV (E_{avail}, W) ratio, and W3 a NuWro/GENIE $(p_T, p_\parallel, E_{\text{avail}})$ ratio. Excluding D1 does not make the others corrected generator comparisons. The q_3 stress changes the conditional distribution at fixed (E_{avail}, W) with amplitude 0.3.

Recovery changes, but not uniformly. Figure 26 shows the traces over the recorded iteration ranges (later checkpoint entries are excluded, App. B.2); none is extrapolated. D1 and W1 improve slowly. W3's J-cell median absolute residual rises from 5.57 % at iteration 5 to 6.57 % at 200, and its maximum from 32.9 % to 44.8 %. The corresponding H2 medians are 3.42 % and 3.68 %. The persistence of the W3 residual is established over this finite scan only, not for unlimited iterations or for every possible classifier. The q_3 stress shows how much the functional matters: its EW median residual rises from 4.99 % at iteration 5 to 6.42 % at 30 even while some joint-cell summaries improve. The capacity probe changes recovery. At five iterations, the most the computing budget allowed at that capacity, the larger classifiers lower the largest residual proxy over the development truths (lower row of Fig. 26) by 3% on J and 8% on H2, short of the improvement of at least 20% required before the larger capacity would be used. The joint-test estimator keeps five iterations under a rule fixed before the comparison. The rule would have replaced five iterations by 10, 15 or 20 only if one of these lowered its measure of model dependence, the largest residual proxy over the development truth changes, on the J or the H2 cells. None did. Because the measure is a maximum over the truth changes, the slow gains for D1 and W1 do not lower the measure: on H2, over the noise-free traces of Fig. 26, it is 0.314 at five iterations and at least 0.384 at 10, 15 or 20. The sixteen-cell result is reported on the same H2 grid after a purity merge of its supported boxes, with total uncertainties (§10.1).

Reporting definitions for a five-dimensional measurement. The five-dimensional precision study assessed three reporting definitions at the development resolution: the 109 J cells (RD1), the 27 H2 cells (RD2) and the 39 EW cells (RD3). Its bounded unfolding-model component, the largest shift over five generator-sized truth departures at the smallest envelope the vertices allow, has median 12.0 %, 9.6 % and 13.4 % respectively (90th percentile 38.6, 22.5 and 39.2 %), with GiBUU the dominant vertex in most cells. The study set its total-width target, a median 68% half-width of at most 10 %, and its selection procedure before it was run. The study adds the model component linearly to the probabilistic half-width, cell by cell. The model component alone therefore exceeds the total-width target in RD1 and RD3, and in RD2 the target is exceeded once the flat 1.4 % normalization is added (median at least 11.0 %). The model-component target, a median of at most 5 %, and the recovery target, recovery of at least 75% of a generator-sized departure (a residual proxy of Fig. 26 of at most 0.25, against 0.31–0.63 measured), fail in every definition. The selection procedure then takes RD3 as the fallback, and RD3 fails the total-width target. None of the three reporting definitions can meet these precision targets with any estimator configuration the study's selection procedure permits within its computing budget. The limitation is the measured prior dependence, about 10–13% per coarse joint cell, which is comparable to the generator differences a user would need to resolve. It motivates the calibrated joint tests of §11, a separate inference that reports no measured cross section.

Variability and interval behavior at five iterations. Figure 27 compares ensemble residual means and repeat standard deviations at the same setting. At historical W3 truth, EW29 has mean residual 23.67 % and repeat SD 0.56 %; EW7 has −6.81 % and 0.30 %. Small spread does not imply

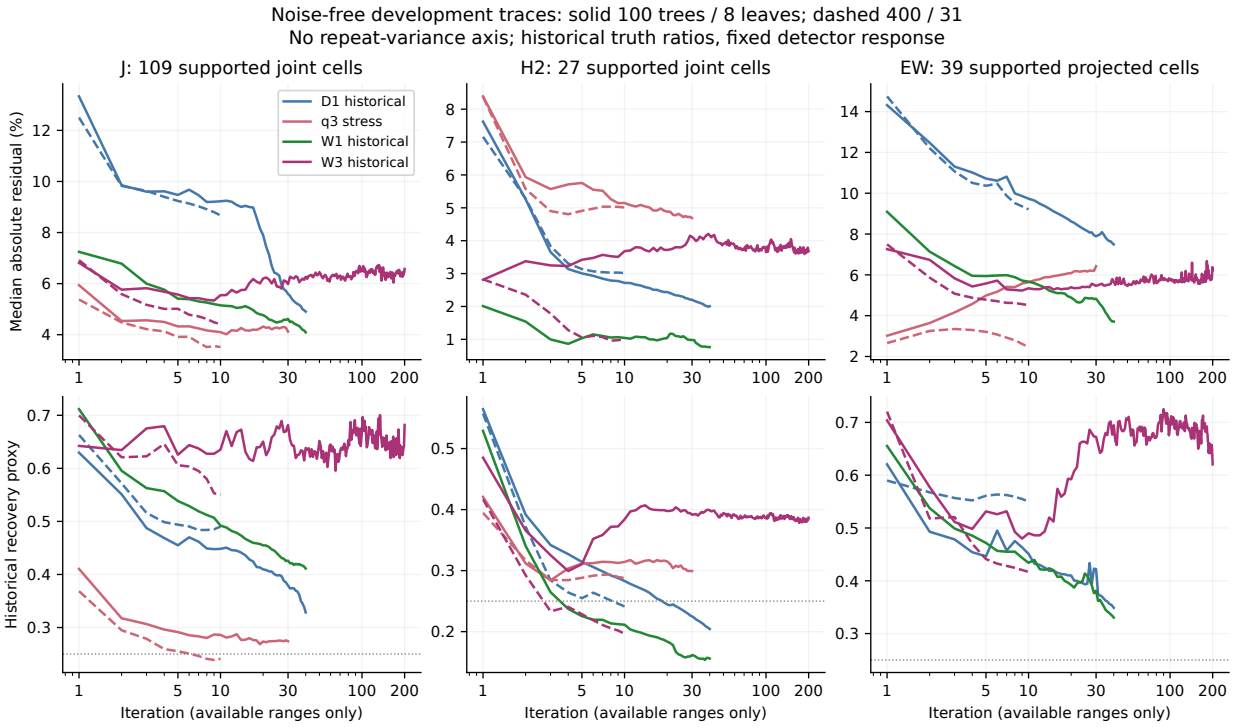


Figure 26: Noise-free five-dimensional recovery of historical truths at fixed simulated response. Solid curves use 100 trees/8 leaves; dashed use 400/31 for OmniFold estimators. D1 and W1 extend to 40 iterations, q_3 stress to 30, W3 to 200, and capacity probes to 10; some baseline traces end at their last saved checkpoint. The upper row gives median $|\hat{f}/f_{\text{true}} - 1|$ over supported cells. The lower row gives the study's residual proxy median$(|\hat{f}/f_{\text{true}} - 1|/|f_{\text{true}}/f_0 - 1|)$ for departures above 1%, with the required bound 0.25 dotted. Different denominators prevent interpreting one minus the proxy as an exact recovered fraction. The q_3 stress has no eligible EW departure. These traces measure no repeat variance.

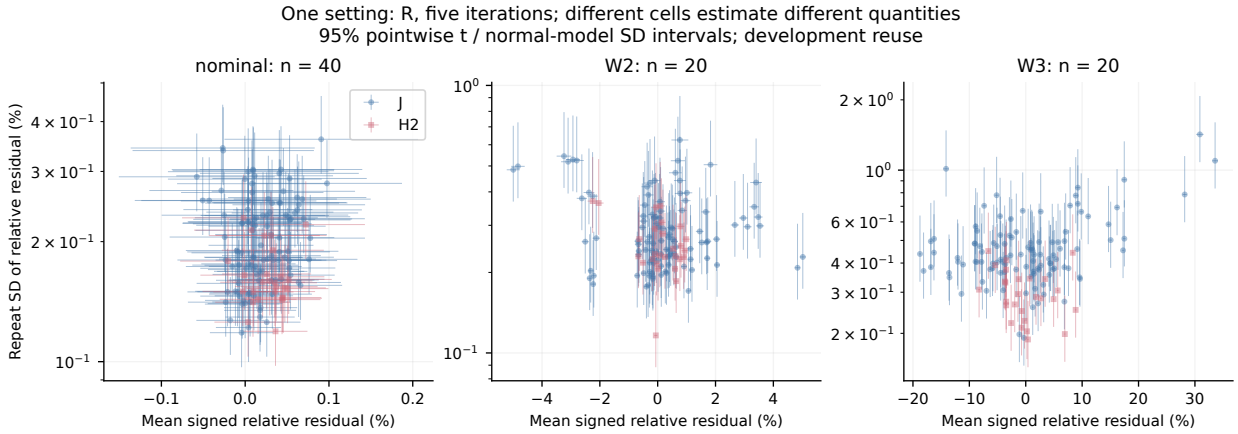


Figure 27: Estimator R at five iterations. Each supported J or H2 cell gives the mean and SD of $(\hat{f} - f_{\text{true}})/f_{\text{true}}$ in the same background-inclusive ensemble: 40 nominal experiments, 20 each for W2 and W3. Bars give pointwise 95% Student-t intervals for means and normal-model intervals for SDs, conditional on saved simulation. They supply neither simultaneous bounds nor total uncertainties. Truth panels and reporting cells form no frontier across estimator settings.

small bias. Coarser cells can have smaller residuals and spread, but they estimate different integrals, so their smaller residuals and spread are not an improvement in precision on an unchanged quantity. No matching repeat ensembles exist at more iterations or larger OmniFold capacity, so no tradeoff can be inferred.

The statistical intervals of the development assessment take their width from 100 bootstrap replicas of one nominal development experiment and do not recompute it for each assessment experiment (Fig. 28). At nominal truth, EW41 contains the truth in 15 of 40 68% intervals (37.5%, pointwise exact 95% interval [22.7%, 54.2%]), so a pooled pass cannot qualify this cell. A 160-experiment check with the width averaged over six bootstrap estimates changes the pull scales but does not validate the real-data procedure, which uses the data's own width. Finite replica noise explains much of the apparent variation among the six estimates.

A paired perturbation study supports the containment of rounding variability in the refit-per-replica bootstrap of R. Twenty edge-safe data jitters (uniform input perturbations within half a single-precision unit in the last place that keep the fine-grid membership fixed) give a J-cell median SD of 0.228 %, against a bootstrap SD of 0.368 %. Fifty bootstrap replicas with jittered partners give a median consistency ratio of 1.02 and an estimated numerical share of 0.37 of the bootstrap variance. In this design, adding the numerical variance to the bootstrap variance would count it twice. The stability failure and the study's scope are in §9.3, interval coverage in §9.5.

Prior sensitivity is not a measured real-data bias. Real-data prior shifts correlate with minus the noise-free residual at the same historical truth (Fig. 29). For W3 the correlation over J cells is 0.92, yet |shift| ≥ |residual| holds in only 71% of supported J cells (52% of H2 cells, 54% of EW cells). Maximum shifts over five historical vertices have medians 12.0%, 9.6% and 13.4% on J, H2 and EW; excluding D1 gives 11.2%, 4.3% and 9.9%. These envelopes are sensitivities: they provide no probabilities, covariance components or guaranteed bias bounds, and the truth of the data has not been shown to lie within their span.

Deciding the tradeoff would require sharing pseudo-data and MC splits across a small set of iteration and capacity settings and measuring bias, variance and mean squared error for common

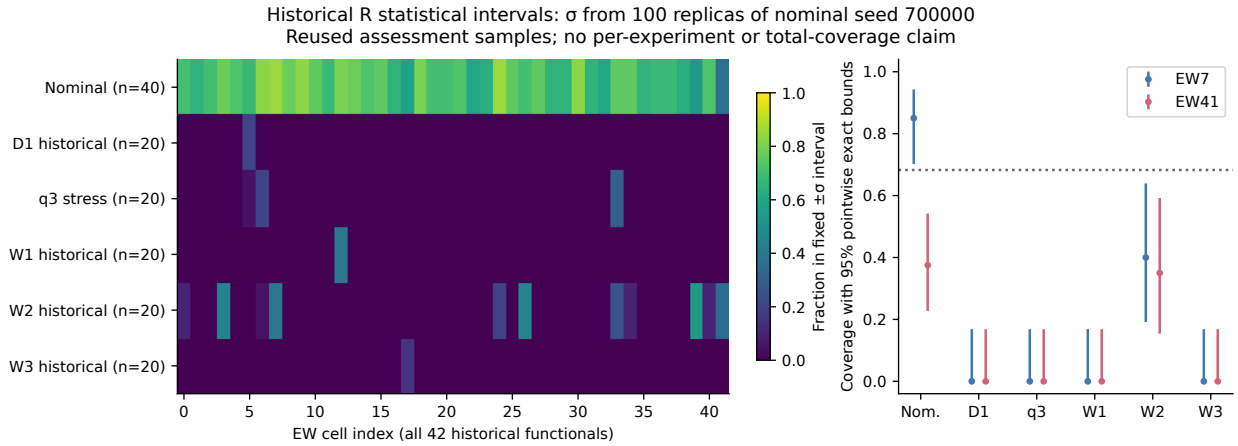


Figure 28: Development assessment of R: containment of the truth in $\hat{f} \pm \sigma$, with σ fixed from the nominal bootstrap, for all 42 EW functionals. Right shows EW7 ($E_{\text{avail}} \in [0.1, 0.2)$, $W \in [1.1, 1.4)$ GeV) and EW41 (both above 3 GeV), with pointwise 95% Clopper–Pearson bounds. The dotted line gives nominal Gaussian 68.27% coverage. Zero of 20 intervals contain the truth; the upper 95% bound is 16.8%, which does not imply zero population coverage. The truth ratios are the historical deformations. No total uncertainty or newly validated interval is shown.

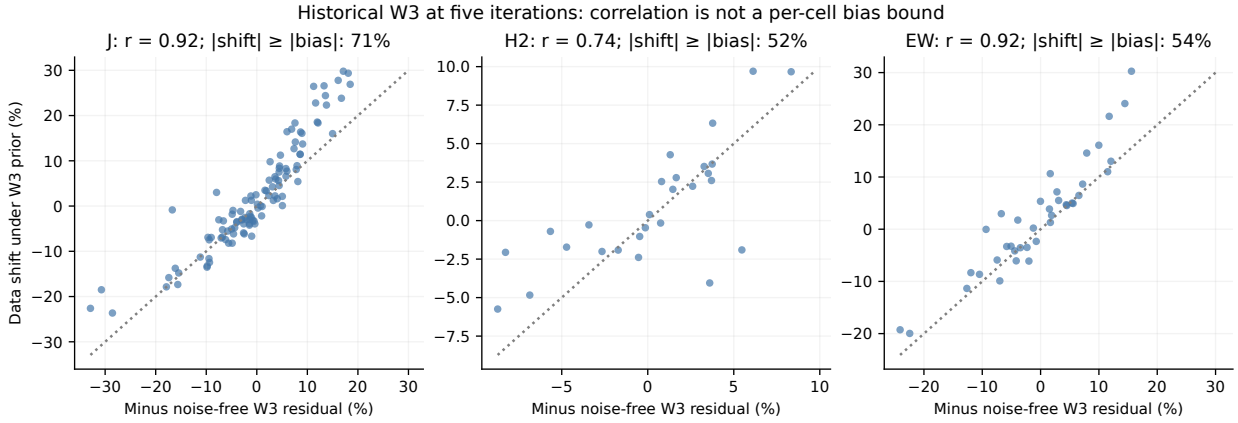


Figure 29: Historical W3 at five iterations. Horizontal gives minus the noise-free relative truth residual; vertical gives the relative shift in real-data R when its MC prior changes to W3, normalized to CV-prior results. Points are supported cells, not independent repeated experiments. The diagonal marks equality. Strong correlation supplies no per-cell bound; shifts do not measure unknown real-data bias.

functionals. That comparison would not validate intervals, which also requires an explicit treatment of physical nuisances and model dependence. No new ensemble enters these studies.

9.5 Five-dimensional coverage test

An end-to-end independent-truth test assessed statistical intervals of a deterministic five-dimensional estimator variant with nominal-truth bias correction (1,200 pseudo-experiments over three truths). It failed its futility criterion, set before the test was run, at the nominal truth and under a 10% reduction of the signal rate with a weak E_{avail} slope (below). No frequentist coverage is therefore claimed for any five-dimensional interval. Figure 36 shows the coverage of each reported integral under the rate reduction. The intervals divide each estimate by $1+\beta$, where β is the relative closure bias at nominal truth, and add the standard error of β times the corrected estimate in quadrature to the statistical standard deviation. A quantity fails when the upper confidence bound on its coverage lies below 0.66 for 68% intervals or below 0.94 for 95% intervals. The second truth (the E_{avail} tilt) multiplies the true signal by $1 + 0.2\,(E_{\text{avail}}/100\,\text{GeV} - \frac{1}{2})$ without renormalization, with the background unchanged; below $E_{\text{avail}} = 3\,\text{GeV}$ the weight is 0.900–0.906. Under that change, 51 of the 153 reported cell and projection integrals have 68% coverage confidently below nominal, with coverage as low as 0.060, and 42 of the 153 fail at 95%. At nominal truth two of the 153 fail at 95%, the lowest coverage being 0.850.

Why the intervals miscover under the rate reduction. The diagnosis below uses the stored outputs and the code. The rate change itself is recovered: in the one locally stored pseudo-experiment the total changes by $-9.58\,\%$ against $-9.56\,\%$ injected, and over the 400 the total covers at 0.74 (68%), so the miscoverage is not a normalization failure. Nor are the intervals too narrow: they use one nominal bootstrap σ at every truth, which for 10% fewer events is slightly too wide apart from the background not being redrawn (inferred spread about 1.15σ at $W \geq 3\,\text{GeV}$), and setting the spread equal to the half-width leaves the failing quantities' coverage at 0.06–0.22. The purity subtraction does not apply a nominal signal fraction to the reduced data: it removes the absolute nominal background expectation, which is unchanged, and in the stored pseudo-experiment the subtracted total agrees with the pseudo-signal to 60 of 3.2 million events. The miscoverage is a residual bias that the nominal-truth bias correction does not remove. The rate change adds a positive bias that grows with W, about $0.2\,\%$ for $1.4 \leq W < 2.2\,\text{GeV}$, $0.4\,\%$ for $2.2 \leq W < 3\,\text{GeV}$, and 0.7–$1.1\,\%$ for $W \geq 3\,\text{GeV}$, which is 1.3–2.6 standard deviations against intervals of 0.16–0.43%, whatever the sign of the nominal bias β; removing that bias restores coverage to 0.62–0.82 at 68% and 0.92–0.99 at 95%. These per-quantity biases and spreads are inferred, not measured: the stored output records only hit counts, so each quantity's mean shift and spread come from a Gaussian fit to its 68% and 95% coverage, checked against one recorded value (EW17: +1.04% fitted against +1.05%). The source of the W-graded term is not established: its size and ordering roughly match, under an approximate assignment of background fractions to W bins, the rise of the background fraction by $0.111\,f_{\text{bkg}}$, but the term may also arise from the estimator alone. A rerun of five of the same pseudo-experiments, without background and with the background scaled by 0.9 so that the background-to-signal ratio is restored, did not decide between the two: restoring the ratio raised the residual for $W \geq 3\,\text{GeV}$ rather than removing it, and without background the residual persists. Agreement in the marginal normalizations thus coexists with a substantial residual bias under the tested deformation, and whether that bias has the same origin as the regularization bias of §9.3 is not established.

9.6 The five-axis covariance of the configuration-(a) central value and its (E_{avail}, W) projection

Status. The five-axis covariance used in this note is the variant centered on the central value (`cv`); its (E_{avail}, W) projection has 42 cells. The file and the projection are identified by their SHA-256 checksums (Ref. [37], identities already fixed). The covariance does not meet the acceptance criteria of its specification, and its metadata mark it as not passing and as not to be used. The specification requires the null bound to be fixed before production, which was not done for this covariance, so it cannot meet that requirement. This note nevertheless uses this covariance and its projection. The use does not extend to any other covariance built to the same specification, including the two companion covariances of the measurements below.

Construction. The covariance projects the corrected five-axis statistical covariance as $MC_{5D}M^{\top}$, without rebuilding it in the destination bins, and uses corrected systematic endpoints. Re-selected event loops bound the selection migrations that the single-pass shifted observables omit (§4.7). On the 10 694 reported cells of $(p_T, p_{\parallel}, E_{\text{avail}}, q_3, W)$, the covariance is

$$C_Z \;=\; D\left(\textstyle\sum_V C_b\right) D \;+\; \textstyle\sum_R C_b \;+\; \sum_A L_b \;+\; C_{\text{stat}} \;+\; C_{\text{ML}}, \tag{4}$$

with the inflation $D = \operatorname{diag}(g)$,

$$g_i \;=\; \sqrt{\frac{\max\left(v^{\text{cv}}_{\text{uni},i},\, v_{\text{blk},i}\right)}{v_{\text{blk},i}}} \;\geq\; 1, \qquad g_i \equiv 1 \;\text{ where } v_{\text{blk},i} = 0. \tag{5}$$

Here $v_{\text{blk}} = \operatorname{diag}\left(\sum_V C_b\right)$ is the vertical block-sum diagonal; $v^{\text{cv}}_{\text{uni}}$ is the unified-throw covariance diagonal *centered on the central value*. Where the denominator is zero, $g = 1$ is set by definition, not computed. The band groups V (vertical), R (residual) and A (re-selected lateral) must be disjoint and exhaust the 45 bands. The same inflation of the vertical bands can be transferred onto a sweep block, $\mathbf{C}^{\text{syst}} \to (\mathbf{C}^{\text{syst}} - \mathbf{C}^{\text{vert}}) + \mathbf{G}\,\mathbf{C}^{\text{vert}}\mathbf{G}$ ($\mathbf{G} = \operatorname{diag} g$), only if the low-rank treatment, the mean shift and the selection-support bound above are documented (a condition set in July 2026). This transfer is positive-semidefinite by construction and never falls below the block baseline, whereas a direct rank-N swap is not positive-semidefinite.

Centering, explicitly. Per-band blocks C_b use the MAT-conformant mean-centered $\pm 1\sigma$ estimator of §A.2.5. Inflation in the CV-centered variant uses $v^{\text{cv}}_{\text{uni}}$. The unified-throw centerings differ by the outer product of measured joint mean shift **ms** (joint throw mean minus central value),

$$C^{\text{cv}}_{\text{uni}} = C^{\text{mean}}_{\text{uni}} + \mathbf{ms}\,\mathbf{ms}^{\top} \qquad \Longrightarrow \qquad v^{\text{cv}}_{\text{uni},i} - v^{\text{mean}}_{\text{uni},i} = \mathbf{ms}_i^2, \tag{6}$$

so CV centering retains the mean shift as variance and mean centering removes it. The covariance used here is the CV-centered variant, which retains the shift deliberately; the mean-centered variant (`mean`) has a $\sqrt{\operatorname{Tr} C}$ lower by 7.13 %. A file's variant is identified by its metadata, not by its path.

The projection to (E_{avail}, W). The quoted 42-cell covariance is $C_{EW} = M\,C_Z\,M^{\top}$, with M of shape $42 \times 10\,694$ over kept axes (E_{avail}, W). Each entry of M is the product of the dropped axes' bin widths, giving a differential density in $\text{cm}^2/\text{GeV}^2/\text{nucleon}$. Integrating a destination cell requires its own $\Delta E_{\text{avail}}\,\Delta W$. Rows follow the 7×6 (E_{avail}, W) grid in measured C order. No source cell is dropped, and the result is exactly symmetric. Support and pinned masks and row-index arrays accompany the covariance and are byte-identical in the two variants. The covariance is paired with

the central values of Fig. 24 by checksum: the central-value array of the projected source and that of the figure are byte-identical, with numerical agreement only as a cross-check.

Conditioning, and why no rank is quoted. Measured extremes $\lambda_{\min} = +4.359104 \times 10^{-92}$ and $\lambda_{\max} = 1.488215 \times 10^{-77}$ give $\lambda_{\min}/\lambda_{\max} = 2.9 \times 10^{-15}$ and condition number $\approx 3.4 \times 10^{14}$, at double precision's edge. No eigenvalue is negative, but the smallest lie at the floating-point noise floor of a 10 694-term weighted sum. Their sign carries no information: $n_{\text{negative}} = 0$ describes this computation and does not show that the source's negative directions were cured. Rank is a property of the cutoff, not of the matrix. The 42 eigenvalues decay smoothly across about fifteen orders of magnitude with no plateau. Retained counts are 26 at relative cutoff 10^{-6}, 36 at the code default 10^{-12}, 41 at `numpy.linalg.matrix_rank`'s default $n\epsilon = 9.3 \times 10^{-15}$, and 42 without a cutoff. No rank is therefore quoted for this covariance; in particular, 36 is the count at one cutoff, not a number of degrees of freedom. An inverse requires a declared regularization, and none is applied implicitly.

What the covariance supports. Four measurements limit what the covariance supports. Items 1, 2 and 4 use two companion covariances built at two estimator-seed settings; only item 3 describes this file.

1. Estimator-seed sensitivity exceeds its limit. Companion covariances at seed offsets $\{0, 1200\}$ give the correlation-sensitive measure $s_{\text{proj}} = 6.145\,\%$, against the movement limit of $5\,\%$, fixed on 2026-09-18 before production as a direct tolerance at the scale of the ensemble's own precision. A 160-throw ensemble determines a standard deviation only to a relative precision of about $1/\sqrt{2(N-1)} \approx 5.6\,\%$ for independent Gaussian throws, so a smaller seed-induced drift could not be resolved against the width it would change. The limit is a declared stability tolerance, not a derived confidence threshold for a maximum over correlated projections. The diagonal-only measures pass ($s_{\text{agg}} = 0.471\,\%$, $s_{\text{med}} = 0.485\,\%$); the failing one probes the off-diagonal structure used by projections.

2. Within the limits stated below, the sensitivity is not the statistic's own resampling noise, which would fall with N. With the throw ensemble fixed and only the seed changed, the sensitivity does not fall over the tested ensemble sizes: mean $6.02\,\%$ over four disjoint 40-throw subsets, $6.02\,\%$ over two 80-throw halves, and $6.145\,\%$ over all $N = 160$ throws. The same-seed resampling floor falls steeply from $20.91\,\%$ at $N = 40$ to $7.57\,\%$ at $N = 80$, with two-point exponent 1.467; no floor measurement exists at $N = 160$. Three limits rule out a statement about infinite ensembles. First, the 40- and 80-throw points are nested subsets of the same 160 throws, not independent ensembles. The flatness therefore holds within one ensemble, over a factor of four in N; the behaviour at larger N is unmeasured. Second, the quoted $\pm 0.39\,\%$ is subset scatter at a single seed pair (offsets $\{0, 1200\}$); the width of the distribution over seed pairs is unmeasured. Third, the statistic is a maximum over the declared offset set, so additional pairs can only raise it.

3. Five of the covariance's 45 bands fall outside the seed variation. The endpoint convention fixes the estimator seed for the muon-energy, muon-resolution and beam-angle blocks, so the offset cannot reach them. They carry $26.0\,\%$ of $\sqrt{\operatorname{Tr} C}$ and move by exactly zero; their seed sensitivity is unmeasured. Allowing them to vary could move the total in either direction, and neither is measured, because s_{proj} maximizes relative changes over functionals rather than summing nonnegative component contributions.

4. Seed shifts in central values are a small fraction of their uncertainty. Across the comparison's 43 projection functionals (42 destinations plus the all-ones total rate), shifts reach 0.761 %, versus relative uncertainties of order ten percent. They reach at most 6.02 % of quoted uncertainty, with median 1.06 %. The largest shift occurs in the destination cell driving the covariance result. Individual five-axis bins reach 49.8 % of their uncertainty; summing bins suppresses this movement in projections.

The estimator-seed magnitude criterion specified before production *has not been computed* for this covariance. It was computed only for the two companion covariances built for that purpose, and their outcomes do not transfer to this covariance (Ref. [37], R1.4).

10 The sixteen-cell five-observable measurement

This section gives a five-dimensional cross-section result with total uncertainties: the data result of configuration (e) (§3.2) on 16 cells, R00–R15, with the reported covariance $C_{\text{total}} + C_{\text{resp}} + C_{\text{PL,5D}}$. It is a separate product from the configuration-(a) five-dimensional central value of Secs. 8 and 9 and from the five-axis covariance used with that central value (§9.6). Statements in those sections about that central value and that covariance do not describe this result, and statements here do not describe them. The section gives the cells and the result (§10.1), the result's quadrant sums against the lower-dimensional results (§10.2), the covariance and what it is built from (§10.3), the fake-data tests of the estimator's bias against the covariance (§10.5) and the comparison with generator predictions (§10.6).

10.1 Cells and result

The cells are formed from the H2 grid of §9.4, which divides each axis once into 32 boxes, with edges $p_T = (0, 0.7, 4.5)$, $p_\parallel = (1.5, 4.5, 60)$, $E_{\text{avail}} = (0, 0.8, 100)$, $q_3 = (0, 2, 100)$ and $W = (0, 1.8, 100)$, in the units of App. B.2. Of these, 27 meet that appendix's support rule (minimum expected signal and truth weight). Each supported box with purity below $\tau = 0.1$ is merged into a neighbour, in 12 predefined steps. The merges absorb 1 unsupported box into cell R09; the other 4 are not reported. Each of the 16 resulting cells meets the support rule. Four of them, R02, R06, R10 and R15, have a reconstruction stability of 0.16–0.25 (the other twelve, 0.26–0.81): of the simulated signal events generated in such a cell and reconstructed, that fraction is reconstructed in the same cell. No stability criterion is applied in forming the cells. Table 12 gives each cell's box, its cell-integrated cross section f and its relative uncertainty.

10.2 Quadrant sums against the lower-dimensional results

The sixteen cells form four $(p_T, p_\parallel)$ quadrants, divided at $p_T = 0.7$ and $p_\parallel = 4.5$ GeV/c, of four cells each. In each quadrant the sum S_{16} of the sixteen-cell result is compared with the sum S_{2D} of the adopted two-dimensional result over the same region, through $d = S_{16} - S_{\text{2D}}$. On the sixteen-cell side the covariance is the reported one; on the two-dimensional side it is the four-block covariance of §4.5 in the first comparison below, and that covariance with its generator-shape term C_{model} in the second. The two results share the data, the flux and most systematic variations, and come from different estimators: configuration (e) for the sixteen cells and configuration (f) for the two-dimensional result (§3.2). The three-observable marginal of configuration (e), compared with the two-dimensional result over all 205 reported cells, agrees with it to 0.141 % in area-weighted total (§7.3); that is a different quantity from the quadrant differences below, of which QA's is +0.141 %.

Table 12: The sixteen reported cells: boxes (p_T, $p_\parallel$ and q_3 in GeV/c; E_{avail} and W in GeV), cell-integrated cross section f in units of $10^{-40}\,\text{cm}^2$/nucleon, and per-cell relative σ (%) under C_{total}, under C_{resp} alone, under $C_{\text{total}} + C_{\text{resp}}$ (the covariance without the run-period block), under $C_{\text{PL,5D}}$ alone and under the reported $C_{\text{total}} + C_{\text{resp}} + C_{\text{PL,5D}}$ (§§10.3 and 10.4).

cell	p_T	$p_\parallel$	E_{avail}	q_3	W	f	C_{total}	C_{resp}	$C_{\text{total}} + C_{\text{resp}}$	$C_{\text{PL,5D}}$	reported
R00	0–0.7	1.5–4.5	0–0.8	0–2	0–100	32.6	6.79	2.07	7.10	0.40	7.11
R01	0–0.7	1.5–4.5	0–100	2–100	1.8–100	36.7	6.69	2.35	7.10	2.71	7.59
R02	0–0.7	1.5–4.5	0.8–100	0–2	0–1.8	6.31	13.69	2.95	14.00	1.14	14.05
R03	0–0.7	1.5–4.5	0.8–100	0–2	1.8–100	4.30	12.50	1.85	12.64	1.50	12.73
R04	0–0.7	4.5–60	0–0.8	0–2	0–100	49.0	4.14	0.70	4.20	2.31	4.80
R05	0–0.7	4.5–60	0–100	2–100	1.8–100	9.98	6.49	3.71	7.47	3.16	8.12
R06	0–0.7	4.5–60	0.8–100	0–2	0–1.8	5.57	5.60	1.81	5.89	2.88	6.56
R07	0–0.7	4.5–60	0.8–100	0–2	1.8–100	3.21	7.82	1.55	7.98	2.79	8.45
R08	0.7–4.5	1.5–4.5	0–0.8	0–100	0–1.8	13.9	5.13	3.58	6.25	0.28	6.26
R09	0.7–4.5	1.5–4.5	0–100	0–100	1.8–100	39.6	5.84	2.26	6.26	2.77	6.85
R10	0.7–4.5	1.5–4.5	0.8–100	0–2	0–1.8	7.83	8.16	2.69	8.59	1.29	8.69
R11	0.7–4.5	1.5–4.5	0.8–100	2–100	0–1.8	5.61	13.39	2.85	13.69	2.36	13.89
R12	0.7–4.5	4.5–60	0–0.8	0–2	0–100	22.0	6.25	2.20	6.63	2.18	6.98
R13	0.7–4.5	4.5–60	0–100	2–100	0–1.8	9.04	10.87	2.08	11.07	2.65	11.38
R14	0.7–4.5	4.5–60	0–100	2–100	1.8–100	49.6	4.94	1.82	5.27	2.13	5.68
R15	0.7–4.5	4.5–60	0.8–100	0–2	0–100	11.3	7.77	1.64	7.94	2.37	8.29
median							6.74	2.14	7.29	2.34	7.86

Without the cross covariance. The first comparison bounds the unknown correlation ρ between the two sums: over $\rho \in [0, 1]$, σ_d lies between $|\sigma_{16} - \sigma_{2D}|$ and $(\sigma_{16}^2 + \sigma_{2D}^2)^{1/2}$. A quadrant agrees if $|d| \le 2|\sigma_{16} - \sigma_{2D}|$, which holds at every ρ; it agrees below a correlation if $|d|$ lies between that and $2(\sigma_{16}^2 + \sigma_{2D}^2)^{1/2}$, and ρ^* is the correlation at which $|d| = 2\sigma_d$; it differs if $|d|$ exceeds $2(\sigma_{16}^2 + \sigma_{2D}^2)^{1/2}$, beyond two standard deviations at every non-negative ρ. The rule was fixed before the comparison was made. Table 13 gives each quadrant. All four agree, QA at every ρ and the others below correlations of 0.9902 (QC) to 0.9997 (QB). Since σ_{16} and σ_{2D} are close in every quadrant, the band valid at every ρ is narrow, which is why three quadrants agree only below a correlation.

Table 13: The quadrant sums of the sixteen-cell result against the two-dimensional result, without their cross covariance: S_{16} and S_{2D} with their σ (in $10^{-39}\,\text{cm}^2$/nucleon), d/S_{2D}, $z = d/(\sigma_{16}^2 + \sigma_{2D}^2)^{1/2}$ (the value at $\rho = 0$), the reading and ρ^*, under the reported covariance and, printed only, without $C_{\text{PL,5D}}$ (against the two-dimensional covariance, which keeps its own run-period block, so not a like-for-like pair; it is printed only, and no outcome is read from it). A ρ^* above 1 means agreement at every ρ.

						reported		without $C_{\text{PL,5D}}$	
quadrant	cells	S_{16} (σ)	S_{2D} (σ)	d/S_{2D} (%)	z	reading	ρ^*	reading	ρ^*
QA	R00–R03	7.9912 (0.5464)	7.9799 (0.5319)	+0.141	+0.015	agree	1.0003	agree below a correlation	0.9999
QB	R04–R07	6.7751 (0.3276)	6.7899 (0.3280)	−0.217	−0.032	agree below a correlation	0.9997	agree	1.0115
QC	R08–R11	6.6907 (0.3565)	6.7911 (0.3473)	−1.478	−0.202	agree below a correlation	0.9902	agree below a correlation	0.9903
QD	R12–R15	9.1925 (0.5036)	9.1265 (0.5152)	+0.724	+0.092	agree below a correlation	0.9982	agree	1.0035

With the cross covariance. The second comparison evaluates the covariance between the two results from what they share, and tests the four differences jointly. The cross covariance sums paired terms: 34 of the 38 systematic bands, universe by universe, after their signal weights were checked to agree event by event between the two analyses; the 5 lateral shifts; the normalization; the run-period terms, paired period by period by construction; and the C_{model} shifts, paired generator by generator.

Two parts could not be paired. The statistical correlation is not measured: the paired bootstrap replicas give cross correlations that do not fit the statistical blocks of the two results, which were built from other replica sets, and the joint matrix they form is not positive semidefinite. The weight check failed in AGKYxF1pi, AhtBY, BhtBY, MinosEfficiency, on non-finite ratios only, between the two analyses' universe-to-central weight ratios. These five enter as $r\,S_{16}^{1/2}S_{2D}^{1/2}$, with S each side's own block, for $r \in [0, 1]$: a one-parameter family that keeps the joint covariance positive semidefinite, not every admissible correlation. The rule, fixed in advance, is that the results agree if the joint χ^2 of the four differences has $p \geq 0.05$ and every $|z| \leq 2.50$; it is read at every point of a 10 001-point grid in r, and 99 structural and block-by-block controls passed. Table 14 gives the result. The two results agree at every r and in every reading. In the primary reading $\chi^2 = 1.170$ ($p = 0.883$) at $r = 0$ and 1.240 ($p = 0.871$) at $r = 1$, and the largest $|z|$ is 0.985, in QC at $r = 1$. Each $|z|$ is largest at an end of the family, since its variance is linear in r, and χ^2 is convex in r, so it is largest, and p smallest, at an end; the value of χ^2 between the ends is not recorded. The implied correlation between the two quadrant sums is 0.77–0.92 at $r = 0$ and 0.96–0.99 at $r = 1$.

Table 14: The quadrant differences of the sixteen-cell result from the two-dimensional result, with the cross covariance between them, at the two ends of the family of bounded correlations ($r = 0$ and $r = 1$). Top: in the primary reading, the reported sixteen-cell covariance against the two-dimensional four-block covariance with its generator-shape term, each quadrant's $z = d/\sigma_d$ and the implied correlation between the two sums. Bottom: the joint χ^2 for four degrees of freedom, its p, and the largest $|z|$ of either end, in the primary reading, against the four-block covariance alone (so the C_{model} shifts are not paired), and with the paired run-period terms set to zero.

quadrant	z, $r = 0$	z, $r = 1$	correlation, $r = 0$	correlation, $r = 1$
QA	+0.051	+0.163	0.92	0.99
QB	−0.106	−0.312	0.91	0.99
QC	−0.415	−0.985	0.77	0.96
QD	+0.275	+0.694	0.89	0.98

reading	χ^2, $r = 0$ (p)	χ^2, $r = 1$ (p)	largest $\|z\|$
primary	1.170 (0.883)	1.240 (0.871)	0.985
four-block covariance alone	0.782 (0.941)	0.926 (0.921)	0.720
run-period pair set to zero	0.802 (0.938)	1.163 (0.884)	0.467

Summed over each of the four $(p_T, p_\parallel)$ quadrants divided at $p_T = 0.7$ and $p_\parallel = 4.5$ GeV/c, the sixteen-cell result lies 0.2–0.8 % below the three-observable central value of §7, the largest difference being 0.17 of the sixteen-cell uncertainty of the quadrant sum. This comparison is descriptive, not a test: the three-observable central value carries no uncertainty here, since no 3D covariance is adopted, so it reads as agreement or disagreement in neither direction. The two quadrants with $p_T < 0.7$ GeV/c include unreported boxes at their central value, a share of at most 1.1×10^{-7} of the sum.

10.3 The reported covariance

The reported covariance is $C_{\text{total}} + C_{\text{resp}} + C_{\text{PL,5D}}$, with $C_{\text{total}} = C_{\text{e}} + C_{\text{model}}$; the run-period block $C_{\text{PL,5D}}$ is described in §10.4. C_{e} sums the statistical bootstrap of configuration (e), 50 replicas, each of which refits the whole unfolding with its own rounding-level perturbation of the inputs; the PPFX flux universes, weighted by the neutrino–electron and inverse-muon-decay constraint; the interaction-model and detector bands, which reweight simulated events; the lateral bands, which

shift reconstructed kinematics (§4); the normalization; and the training-seed block. With full-sample binning and deterministic training, the unfolds at the other seeds are bitwise equal to the central one, so the measured training-seed block is zero in every cell. Table 15 gives each group's per-cell relative σ.

For C_{model}, the prior is reweighted in turn by the ratio of GENIE + MEC, NuWro or GiBUU to GENIE CV, and the data are unfolded again. Each generator's shift is the relative change of the data result, and C_{model} is the unscaled sum of their outer products. It is generator-shape model dependence, not a calibrated bound; its coverage at truths held out of its construction has not been validated. The GiBUU reweighting is applied only below $p_{\parallel} = 10$ GeV/c; above it the prior is unchanged. Between $p_{\parallel} = 6$ and 10 GeV/c, the GiBUU sample lacks the contribution from $E_\nu > 20$ GeV, which is at least 4.9–5.3% of the cross section there in the GENIE CV and NuWro samples (which stop at $E_\nu = 50$ GeV). The GiBUU shift contributes 0.890 of the trace of C_{model}, the NuWro shift 0.095 and the GENIE + MEC shift 0.014. In every cell, 77–100% of the prior weight lies next to a discontinuity larger than 0.1 in the reweighting ratio, created by the ratio's statistical gate or by a cut; the shifts are used unmodified.

The coverage of C_{model} is not validated because an earlier route to a calibrated model term with validated coverage did not pass its first two tests and was not pursued further, after which C_{model} was adopted unscaled (the J and H2 cells and estimator R are those of §9.4). The first, Stage 0, tested that earlier term on the J and H2 cells, not C_{model} on the sixteen cells. It asked whether a model term built from the data result's shifts under generator reweightings of the prior, with its scale calibrated on estimator R's bias at the truth deformations assigned to the other generator families, covers that bias, cell by cell, at a family held out of the calibration, requiring a covered fraction of at least 0.60 at every held-out family and of at least 0.68 pooled over the families. This first part failed on both sets of cells: on the H2 cells, whose grid the sixteen cells are formed from, GiBUU reached 0.259, although the pooled 0.697 met its bar; on the J cells, NuWro reached 0.570 and GiBUU 0.541, and the pooled 0.645 did not meet its bar. Its second part, on the term's size and on whether truth departures stay distinguishable from the nominal with the term in the covariance, passed on both. With the first part failed, Stage 0 stopped. The second test, a reconstruction-information test on the J cells, asked whether more reconstructed information, from finer reconstructed binning or added reconstructed variables, narrows a width floor and reduces the residual of a binned iterative unfolding at truth departures. None of its 11 configurations met its residual requirement, and the test ended incomplete, neither passing nor stopping.

For C_{resp}, the simulation's reconstructed recoil energy is scaled by one standard deviation separately for four deposit categories: protons, mesons, electromagnetic deposits and other deposits. C_{resp} sums the covariance contributions of the four resulting shifts of configuration (e)'s data result. It does not cover neutron or muon deposits, the effects of particle species, energy dependence and resolution beyond a scale, the null distributions of the joint tests, the configuration-(a) products or the two-, three- and four-dimensional results. The median per-cell relative σ of C_{resp} is 2.14%, and adding C_{resp} to C_{total} raises the per-cell σ by a factor of 1.011–1.220 (median 1.048). A cross-talk term, reported beside C_{resp} and not summed into it, has a median per-cell relative σ of 0.36%. Of the trace of $(C_{\text{total}} + C_{\text{resp}})/ff^T$, C_{total} contributes 0.9262 and the four deposit categories 0.0168, 0.0436, 0.0071 and 0.0062.

Table 16 gives the correlation matrix of the reported $C_{\text{total}} + C_{\text{resp}} + C_{\text{PL,5D}}$; with the central values and the relative uncertainties of Table 12, it gives the reported covariance to the printed precision.

Every χ^2 of this section inverts the full covariance of the cells compared, 16×16 or 8×8 for GiBUU's cells, by Cholesky decomposition and without truncation, so the inverted dimension p equals the number of cells and is fixed in advance; the shape-only χ^2 inverts the same matrix with

Table 15: Per-cell relative σ (%) of the covariance groups. C_{e} is the sum of the first six groups and the training-seed block (zero in every cell); $C_{\mathrm{total}} = C_{\mathrm{e}} + C_{\mathrm{model}}$. The four deposit categories of C_{resp} are in the last four columns.

cell	stat.	flux	interaction	detector	lateral	norm.	C_{e}	C_{model}	p	meson	EM	other
R00	0.20	3.93	0.98	1.56	5.01	1.40	6.77	0.43	1.10	1.69	0.39	0.28
R01	0.23	3.58	1.18	2.14	4.85	1.40	6.66	0.67	0.66	2.02	0.80	0.61
R02	0.33	3.75	3.56	1.96	5.08	1.40	7.65	11.36	0.44	2.27	1.39	1.18
R03	0.33	3.52	5.39	1.66	4.70	1.40	8.27	9.38	0.29	1.49	0.54	0.91
R04	0.14	3.47	0.58	1.31	0.57	1.40	4.05	0.88	0.52	0.46	0.09	0.04
R05	0.37	4.14	2.13	1.68	1.85	1.40	5.48	3.46	1.45	3.15	1.13	0.68
R06	0.33	3.48	2.25	1.21	0.73	1.40	4.61	3.19	1.67	0.42	0.39	0.42
R07	0.30	3.90	5.08	1.76	1.75	1.40	7.02	3.45	1.45	0.09	0.14	0.52
R08	0.22	3.74	1.52	2.17	0.77	1.40	4.86	1.63	2.51	2.46	0.56	0.46
R09	0.18	3.52	1.85	2.18	1.08	1.40	4.87	3.24	0.99	1.85	0.60	0.58
R10	0.25	3.42	5.53	2.42	0.43	1.40	7.10	4.03	0.04	2.31	0.95	0.98
R11	0.31	3.35	5.16	2.56	2.91	1.40	7.42	11.15	0.13	2.37	1.08	1.14
R12	0.18	3.48	1.95	1.68	4.25	1.40	6.23	0.51	1.66	1.41	0.33	0.12
R13	0.28	3.85	5.93	2.08	3.93	1.40	8.47	6.81	0.26	1.69	0.87	0.82
R14	0.16	3.73	1.22	2.14	1.34	1.40	4.87	0.82	0.69	1.54	0.52	0.45
R15	0.22	3.48	3.19	1.82	3.16	1.40	6.13	4.78	0.90	1.07	0.62	0.59
median	0.24	3.55	2.19	1.89	2.38	1.40	6.45	3.34	0.79	1.69	0.58	0.58

Table 16: Correlation matrix of the reported covariance $C_{\mathrm{total}} + C_{\mathrm{resp}} + C_{\mathrm{PL,5D}}$ on the sixteen cells (lower triangle; the diagonal is one).

	R00	R01	R02	R03	R04	R05	R06	R07	R08	R09	R10	R11	R12	R13	R14
R01	0.69														
R02	0.52	0.30													
R03	0.53	0.32	0.81												
R04	0.49	0.53	0.15	0.21											
R05	0.34	0.70	−0.03	−0.14	0.57										
R06	0.36	0.51	0.64	0.57	0.59	0.43									
R07	0.41	0.40	0.40	0.75	0.55	0.19	0.64								
R08	0.66	0.30	0.24	0.31	0.73	0.13	0.31	0.40							
R09	0.19	0.57	−0.24	−0.07	0.77	0.65	0.42	0.48	0.40						
R10	0.37	0.13	0.73	0.42	0.41	0.16	0.56	0.21	0.36	−0.06					
R11	0.06	0.07	−0.40	−0.43	0.39	0.46	−0.04	−0.15	0.16	0.26	0.14				
R12	0.02	0.03	−0.01	0.00	0.73	0.22	0.35	0.26	0.60	0.51	0.29	0.34			
R13	0.03	0.05	−0.26	−0.38	0.51	0.50	0.13	−0.09	0.21	0.31	0.34	0.92	0.46		
R14	0.24	0.52	0.18	0.21	0.74	0.65	0.65	0.48	0.38	0.72	0.26	0.11	0.64	0.25	
R15	0.08	0.05	0.61	0.43	0.52	0.11	0.71	0.41	0.32	0.20	0.83	−0.04	0.54	0.23	0.52

$100\, f f^T$ added and has one degree of freedom fewer. The blocks inverted are of unlike kinds. Two are sample covariances of an ensemble: the statistical block, from 50 bootstrap replicas with the unbiased $1/(N-1)$ normalization, and the flux block, from 100 PPFX universes with constraint weights, normalized by the sum of the weights (the weighted analogue of $1/N$; effective number of universes 19). The 2p2h band has three universes. The other bands and the lateral shifts are two-point shifts, the normalization is one outer product, C_{model} sums three chosen generator shifts, C_{resp} four $\pm 1\sigma$ shifts, and $C_{\mathrm{PL,5D}}$ is built from three exposure-weighted periods; the training-seed block is zero. Each comparison with a prediction adds V_g, its diagonal sum of squared event weights, and each fake-data comparison the variance of the mean over its pseudo-experiments. No finite-ensemble correction is applied to any block. The standard correction is derived for a single unweighted sample covariance, which none of these sums is; the flux bootstrap below measures the scatter of the χ^2 from estimating the flux block, and is not a correction.

10.4 The run-period block

On the sixteen cells, the unfolded results of the three run-period groups (playlists 1A–1F, 1G–1M and 1N–1P), each over its own simulated truth (the double ratio), differ from one another by more than the scatter of their bootstrap replicas. The offsets of the twelve playlists differ between the three period groups by more than expected from their playlist-to-playlist scatter. The cause is not known, and these data cannot establish which period, if any, is right. As for the two-dimensional result (§4.5, Table 3), the block $C_{\mathrm{PL,5D}}$ is built from these double ratios with no reference period: each group's offset from the exposure-weighted mean of the three, applied to the central value, with a reliability-weight factor for the weighted spread. It has rank 2 and is not debiased for the replica noise it contains. With three groups its own size is poorly determined: the 68 % interval on the ratio of its true to its estimated σ is 0.73–2.56, and in cells R00, R08 it is dominated by that noise.

Each of the twelve playlists was also unfolded on its own, with its own simulation and bootstrap replicas, in a study whose tests, bounds and readings were fixed before any playlist result was read. A playlist's offset is its double ratio on the sixteen cells relative to the exposure-weighted mean of the twelve. The statistic F of a division of the playlists into groups is the share of the offsets' exposure-weighted squared size, measured with the inverse of the relative $C_{\mathrm{total}} + C_{\mathrm{resp}}$, that the groups' exposure-weighted mean offsets carry. For the three run-period groups $F = 0.70$, the largest of all 9,240 divisions of the twelve playlists into groups of six, three and three, so the permutation p-value is $1/9{,}240 = 1.08 \times 10^{-4}$, the smallest the test can give. The permutation is approximate: the playlists differ in exposure and hence in noise, so their labels are not exchangeable, and the test's validity rests on a simulated null. In each of 2,000 simulated datasets the playlists' offsets are drawn independently from one covariance, estimated from the twelve playlists and scaled by 0, 0.5, 1 or 2, with each playlist's own bootstrap and simulation noise added; the share of datasets with $p < 0.05$ is 0.0395–0.0565 over the four scales, within the bound of 0.0646. For an alternative chronological division into groups of five, two, two and three playlists (1A–1E, 1F and 1G, 1L and 1M, and 1N–1P), the same statistic gives $p = 5.17 \times 10^{-3}$ over its 83,160 divisions; this comparison is descriptive.

In the form of the two-dimensional test (§4.5), the five-observable unfolds of the run-period groups give the double ratio of 1G–1M to 1A–1F a $p_{\parallel}$ step of (6.28 ± 0.60) % across 8 GeV/c, 10.5 times its bootstrap scatter. Each playlist's own step, against 1A–1F (leaving the playlist out when it is one of them), was classed as consistent with none within two standard deviations (FREE), larger than half of the 1G–1M group's step, recombined from its three playlists, by two standard deviations (CARRIES), or neither (PARTIAL). The classes are CARRIES for 1L and 1M; PARTIAL for 1G, 1N and 1P; and FREE for 1A–1F and 1O. The study's two patterns are that all three of 1G, 1L

and 1M carry the step, or that some carry it and the others are free, each with 1A–1F free. Neither holds, so no statement of where the step lies is drawn from these classes. None of these readings identifies a cause.

The block was chosen after the period differences had been seen. Before it was added, an audit examined its overlap with the existing blocks, especially the flux and detector-response ones; its noise treatment, with bootstrap replicas of each period; why its cell pattern differs from the projection of the two-dimensional period offsets, although their sizes agree; and the stability of its rank-two construction. The audit's stopping condition was fixed before any of its numbers were read, and it did not fire. The addition therefore followed the data. Passing the audit establishes neither the block's size nor the absence of overlap: descriptively, one detector-response pair, a pion-interaction band with the meson response shift, reproduces one period's offset pattern within noise and part of the other two. Built from the observed differences, the block contains them by construction, and its size and coverage have not been independently validated. The per-cell uncertainties, the fake-data tests and the generator comparison are given here with and without it.

Added to $C_{\text{total}} + C_{\text{resp}}$, the block raises the median per-cell relative σ from 7.29 % to 7.86 %, by about 7.8 %. The median of the cells' own increases is 4.8 %, and the largest is 14.1 %, in R04. The block's median per-cell relative σ is 2.34 %, at most 3.16 %, and its share of the reported variance has a median of 9.0 % over the cells and a maximum of 23.2 %. With the block, each prediction's total χ^2 is lower by 3.6–29.3 % of its value without it, and each shape-only χ^2 by 7.4–40.4 % (Table 19). We infer from these falls that the predictions' residuals have components along the directions the rank-2 block adds; the comparison does not test this interpretation. Under the reported covariance one fake-data truth fewer is listed (§10.5), with the bias unchanged.

10.5 Fake-data tests of the estimator's bias

The fake-data tests compare the estimator's bias at truths other than the simulation with the reported covariance. Pseudo-data are generated at 12 truths and unfolded with configuration (e). The mean unfolded result is compared with each truth on the 16 cells, through χ^2 with 16 degrees of freedom and the per-cell pull z. The truths are the simulation's prior; the five hybrid truths of the generator tests (Sec. 11), each a prediction on the fine grid; the historical shapes W1, W2 and W3 of §9.4; a GiBUU-to-GENIE E_{avail} shape; a noise-free closure at the NuWro shape; and a synthetic tilt of the q_3 distribution at fixed (E_{avail}, W). A truth from a C_{model} generator is compared with the covariance without that generator's shift. A truth is listed as failing when $p < 0.01$ or when any cell has $|z| > 2$; these criteria were fixed before any of the tests was graded against them. These are not held-out truths, and the detector response is not probed. Table 17 gives each truth under the reported covariance, without $C_{\text{PL,5D}}$, and under C_{total} alone. The bias of each truth is the same in all three; only the covariance differs.

Under the reported covariance, 2 truths meet the listing criterion ($p < 0.01$, or a cell with $|z| > 2$): the GiBUU hybrid truth, by p and by its cells R02, R06; and the q_3 tilt, by p alone ($p = 9.1 \times 10^{-6}$). Without $C_{\text{PL,5D}}$, 3 truths meet it: the GiBUU E_{avail} shape is also listed, by its cell R06 ($z = 2.01$, against 1.76 with the block), and the GiBUU hybrid truth by its cells R02, R06, R07. The fake-data bias therefore exceeds the stated thresholds at the GiBUU hybrid truth and the q_3 tilt under the reported covariance, and also at the GiBUU E_{avail} shape without $C_{\text{PL,5D}}$; the bias is the same in both, so the difference is the covariance's, not a smaller bias or a validated uncertainty. At the other 10 truths the bias meets both criteria under the reported covariance.

The truths form three groups, evaluated separately, not pooled: the 8 truths from the C_{model} generators; the other truths, the GENIE CV hybrid and the q_3 tilt; and the prior truths, the simulation prior and the Tune v1 hybrid. Under the reported covariance, the C_{model}-generator truths

Table 17: Fake-data tests on the sixteen cells: χ^2 (16 degrees of freedom), p and the cells with $|z| > 2$ of the bias of the mean against the reported $C_{\text{total}} + C_{\text{resp}} + C_{\text{PL,5D}}$, against $C_{\text{total}} + C_{\text{resp}}$ (without the run-period block), and against C_{total} alone, each with the truth's own generator shift removed where the second column says so (−). n is the number of pseudo-experiments.

truth	covariance	n	reported			without $C_{\text{PL,5D}}$			C_{total}		
			χ^2	p	cells	χ^2	p	cells	χ^2	p	cells
simulation prior (MINERvA Tune v1)	full	40	0.00	1.00	—	0.00	1.00	—	0.00	1.00	—
MINERvA Tune v1, hybrid	full	16	0.00	1.00	—	0.01	1.00	—	0.01	1.00	—
GENIE CV, hybrid	full	16	15.58	0.48	—	16.14	0.44	—	17.37	0.36	—
GENIE + MEC, hybrid	−GENIE + MEC	16	13.65	0.62	—	14.54	0.56	—	16.18	0.44	—
GENIE + MEC, (E_{avail}, W) shape (W2)	−GENIE + MEC	20	0.50	1.00	—	0.57	1.00	—	0.77	1.00	—
NuWro, hybrid	−NuWro	16	23.86	0.093	—	24.92	0.071	—	29.96	0.018	—
NuWro, E_{avail} shape (W1)	−NuWro	20	4.82	1.00	—	5.01	1.00	—	5.69	0.99	—
NuWro, $(p_T, p_\parallel, E_{\text{avail}})$ shape (W3)	−NuWro	20	27.41	0.037	—	30.83	0.014	—	36.19	0.0027	—
NuWro, closure (noise-free)	−NuWro	1	10.66	0.83	—	11.60	0.77	—	13.08	0.67	—
GiBUU, E_{avail} shape	−GiBUU	20	24.57	0.078	—	25.46	0.062	R06	35.75	0.0031	R06
GiBUU, hybrid	−GiBUU	16	171.01	5.3×10^{-28}	R02, R06	175.13	8.0×10^{-29}	R02, R06, R07	246.75	$< 10^{-30}$	R02, R06, R07
synthetic q_3 tilt	full	20	52.48	9.1×10^{-6}	—	53.42	6.4×10^{-6}	—	59.83	5.6×10^{-7}	—

have 2 of 128 cells with $|z| > 2$ (Gaussian reference 5.82) and 1 of 8 truths with $p < 0.01$ (reference 0.08); the other truths 0 of 32 and 1 of 2; and the prior truths 0 of 32 and 0 of 2. The Gaussian references are context only, because the bias is deterministic.

One control check failed: the offset check, which compares the offset corrections of two hybrid truths, gives $\chi^2 = 245.74$ for 16 degrees of freedom. The other control checks (numbers 0, 1, 2, 3, 4, 5, 6, 8) passed. As fixed in advance, the six power alternatives built on that correction are therefore descriptive and not judged against the failure criteria; Table 18 gives them, with two descriptive variants of each tested truth.

Table 18: Descriptive fake-data numbers, none of them a test result. In-sample: against the reported $C_{\text{total}} + C_{\text{resp}} + C_{\text{PL,5D}}$ including the truth's own generator shift, which is not evidence (— where the test already uses the full matrix). Model-only: against the statistical block and C_{model} without the truth's own generator shift, omitting the interaction, flux, detector, lateral and normalization blocks, so a lower bound on C_{model}'s reach. Power alternatives: the NuWro $(p_T, p_\parallel, E_{\text{avail}})$ shape (P1), a NuWro q_3 shape at fixed (E_{avail}, W) (P2) and the GENIE + MEC (E_{avail}, W) shape (P3), each on the Tune v1 truth (r) and on the GENIE CV truth (g).

truth	in-sample χ^2, p		model-only χ^2	power alternative	n	χ^2	p
simulation prior (MINERvA Tune v1)	—	—	0.39	P1r	193	38.93	0.0011
MINERvA Tune v1, hybrid	—	—	0.20	P2r	195	33.89	0.0056
GENIE CV, hybrid	—	—	1794.19	P3r	195	2.09	1.00
GENIE + MEC, hybrid	10.89	0.82	1006.75	P1g	199	75.53	1.1×10^{-9}
GENIE + MEC, (E_{avail}, W) shape (W2)	0.33	1.00	117.49	P2g	193	36.45	0.0025
NuWro, hybrid	23.85	0.093	3028.41	P3g	172	21.60	0.16
NuWro, E_{avail} shape (W1)	4.82	1.00	1647.66				
NuWro, $(p_T, p_\parallel, E_{\text{avail}})$ shape (W3)	26.43	0.048	5552.98				
NuWro, closure (noise-free)	10.56	0.84	2890.73				
GiBUU, E_{avail} shape	22.10	0.14	7990.52				
GiBUU, hybrid	138.83	1.2×10^{-21}	24824.55				
synthetic q_3 tilt	—	—	11642.59				

Two truths are held out of the analysis: the GENIE v3 tunes G18_10a_02_11b and AR23_20i_00_000, generated for this test. Each tune enters as its normalized shape ratio to GENIE CV on the fine grid, constructed as the reweighting ratios of the C_{model} shifts are. Where the tune's sample fails the ratio's statistical gate, the ratio is pooled over a coarser cell or set to one. The ratio is applied to the simulated truth, under the simulation's detector response. At each tune, 20 Poisson pseudo-experiments on half the simulation and a noise-free closure are tested against

the full reported covariance, as truths with no shift in C_{model}, in a fourth group, not pooled with the others. The bias meets both criteria in all four records, the pseudo-experiment mean and the closure at each tune. The mean of the pseudo-experiments gives $\chi^2 = 9.26$ ($p = 0.90$; largest $|z|$ 0.61, in R11) at G18_10a_02_11b and 14.06 ($p = 0.59$; 0.61, in R07) at AR23_20i_00_000; the closures give 9.14 ($p = 0.91$; 0.60, in R11) and 13.43 ($p = 0.64$; 0.57, in R07). In every cell of the four records, the bias is below 0.62 of the cell's reported σ; in cell R11 at G18_10a_02_11b, the bias lies below the reported uncertainty only with C_{model} included. This group has 0 of 64 cells with $|z| > 2$ (Gaussian reference 2.91) and 0 of 4 records with $p < 0.01$ (reference 0.04). The four records are the pseudo-experiment and closure results for two truths, so the Gaussian references, which count them as independent, are context only.

The two tunes use the same quasielastic cross-section model, GENIE's `NievesQELCCPXSec`, AR23_20i_00_000 with a z-expansion form factor, and the MEC model of G18_10a_02_11b is of the Valencia family used in the GENIE + MEC shift. Both are GENIE tunes, so they are partial evidence and not independent tests. Descriptively, without the GENIE + MEC shift, the pseudo-experiments give $\chi^2 = 11.55$ ($p = 0.77$) and 18.79 ($p = 0.28$), and the closures 11.07 ($p = 0.81$) and 17.22 ($p = 0.37$). The model-only χ^2, defined as in Table 18, are 817–922 in the four records, with cell R00 beyond 2σ at AR23_20i_00_000 in both its records. Rerunning the tests with the two tunes added reproduces every earlier fake-data result exactly. The test is a bias test, not a coverage test: its Poisson pseudo-experiments do not draw the systematic parameters, so they cannot test the interval widths, and the detector response is not probed. A coverage test at these two tunes, declared in advance for the covariance without the run-period block, with pseudo-experiments that also draw the flux, interaction and detector uncertainties (the lateral and normalization ones as surrogates, and C_{model} and C_{resp} not at all), was stopped before any coverage pseudo-experiment ran. At one of its gates, the ratio of each cell's uncertainty reproduced through the pseudo-experiments to that of the drawn covariance lay outside 0.9–1.1 in 11 of the 16 cells, ranging from 0.70 to 2.30. Replacing the pseudo-experiments' values by the covariance's for any three or fewer of the 38 drawn bands does not bring every cell inside that window. For the interaction uncertainties, a later investigation at the nominal truth found that the gate compared unlike objects. Its pseudo-experiments reproduce in every cell the spread of the response alone, with the simulated truth held fixed, whereas the full interaction universes from which the covariance is built give a different spread at the same truth. Separately, for the interaction uncertainties, in all but one cell the covariance's construction gives a different spread at the nominal truth than at the data, where the covariance is made. Pseudo-data that take each full interaction universe as the truth, unfolded with the nominal simulation, give a spread whose ratio to that of the covariance's construction at the nominal truth lies within the gate's window, 0.9–1.1, in all 16 cells: under this declared swap the universes' second-order departures at $\pm 1\sigma$ are below the window, as expected, since closure fixes the ratio at one to first order. This concerns the test; it neither validates nor undermines the reported interaction block, and neither finding shows which spread the reported intervals should carry. In cells R00 and R05 the interaction uncertainties carry only 0.57 and 0.36 of the squared ratio's departure from one, so this account does not explain the gate's departure there, and this account says nothing about the parts carried by the flux and detector uncertainties. The coverage of the result's intervals has not been validated. A staged coverage test at the two tunes, with the covariance rebuilt by the same procedure at each tune, ended without a coverage result: its first stage stopped under a fail-closed rule declared in advance.

10.6 Comparison with generator predictions

Table 19 compares the result with five generator predictions. Each gives the χ^2 against the reported covariance plus V_g, $C_{\text{total}} + C_{\text{resp}} + C_{\text{PL,5D}} + V_g$, where V_g is the prediction's finite-sample (diagonal) covariance, and the shape-only χ^2, with the normalization free, and both again without $C_{\text{PL,5D}}$ (§10.4). No diagonal-only χ^2 is formed. GiBUU's sample has no events above $E_\nu = 20$ GeV, so it is compared on the 8 cells below $p_\parallel = 4.5$ GeV/c only, and no value is formed on the other 8. In the GENIE CV and NuWro samples, at least 11.71% and 12.59%, respectively, of the cross section above $p_\parallel = 4.5$ GeV/c comes from $E_\nu > 20$ GeV. Between $p_\parallel = 1.5$ and 4.5 GeV/c, they are 0.83% and 0.77%. These GENIE CV and NuWro fractions, not GiBUU values, are what a sample stopping at $E_\nu = 20$ GeV would miss. All five predictions are in the analysis definitions of W and of the muon angle, and NuWro and GiBUU, generated on carbon, are corrected to CH by adding each generator's own free-proton prediction with weight 1/13 per nucleon (§10.7). MINERvA Tune v1 is the simulation's prior, not an independent prediction. Because C_{model} holds the GENIE + MEC, NuWro and GiBUU shifts, those comparisons each use a covariance built in part from the compared generator, and the GENIE CV comparison does so through the reference of the ratios.

Table 19: χ^2 against the reported covariance plus V_g, $C_{\text{total}} + C_{\text{resp}} + C_{\text{PL,5D}} + V_g$, where V_g is the prediction's finite-sample (diagonal) covariance, with the shape-only χ^2 beside it, and the same two without the run-period block, against $C_{\text{total}} + C_{\text{resp}} + V_g$. All five predictions are in the analysis definitions, with NuWro and GiBUU corrected to CH; the paper prints these values. The ndf of each block is given in its heading.

	reported covariance		without $C_{\text{PL,5D}}$	
prediction	χ^2	shape-only χ^2	χ^2	shape-only χ^2
All 16 cells: ndf 16, shape-only 15				
GENIE 2.12.10 CV	107.18	102.11	111.22	110.32
GENIE 2.12.10 + Valencia MEC	229.09	229.00	277.94	263.74
NuWro 21.09 (CH)	127.69	107.52	180.62	180.52
MINERvA Tune v1	262.06	261.88	329.72	311.19
8-cell subset, $p_\parallel < 4.5$ GeV/c: ndf 8, shape-only 7				
GiBUU 2019 (CH)	47.78	22.19	61.41	24.66

Table 20 gives each cell's pull and $(g - f)/f$.

The fake-data bias exceeds the stated thresholds at the GiBUU hybrid truth under the reported covariance, and also at the GiBUU E_{avail} shape without $C_{\text{PL,5D}}$ (§10.5), so the GiBUU comparison carries that limitation.

The sensitivity of the reported χ^2 to the correlations is probed by shrinking one group's correlations toward the diagonal by a fraction α of the way, with its variances kept. These are descriptive sensitivities, not alternative results. Shrinking only the flux correlations by 10 % lowers the χ^2 of the 16-cell predictions by 23–38 %. The same change to the detector or lateral correlations lowers it by at most 18 %. With the correlations of the interaction, flux, detector and lateral groups removed, every 16-cell prediction keeps $\chi^2 \geq 41$ for 16 degrees of freedom. Table 21 gives each group's sensitivity, GiBUU's eight-cell values included. The flux correlations are themselves estimated from the 100 PPFX flux universes (§10.3). Rebuilding the flux block from 2000 bootstrap replicates of those universes, with its variances kept, gives 16–84 % percentile bands of the total χ^2: GENIE 2.12.10 CV 104–110 around 107, GENIE 2.12.10 + Valencia MEC 222–235 around 229, NuWro 21.09 (CH) 126–130 around 128, MINERvA Tune v1 253–272 around 262, and GiBUU 2019 (CH), on its 8 cells, 47–48 around 48. In each, the bootstrap median lies above the reported value, by 0.26, 0.32, 0.20,

Table 20: Per-cell pull $(f-g)/\sigma$ against $C_{\text{total}}+C_{\text{resp}}+C_{\text{PL,5D}}+V_g$ and $(g-f)/f$ (%) for each prediction of Table 19; GiBUU on its eight cells only.

cell	GENIE 2.12.10 CV		GENIE 2.12.10 + Valencia MEC		NuWro 21.09 (CH)		GiBUU 2019 (CH)		MINERvA Tune v1	
	pull	$(g-f)/f$	pull	$(g-f)/f$	pull	$(g-f)/f$	pull	$(g-f)/f$	pull	$(g-f)/f$
R00	1.49	-10.6	0.12	-0.8	2.18	-15.5	2.66	-19.2	0.85	-6.0
R01	1.48	-11.2	1.45	-11.0	1.17	-8.9	3.96	-30.1	1.72	-13.1
R02	-0.88	12.4	-1.11	15.6	0.83	-11.7	3.70	-51.9	-0.27	3.7
R03	0.03	-0.4	0.10	-1.2	-0.44	5.6	3.52	-44.8	1.20	-15.2
R04	1.75	-8.4	-0.53	2.6	2.41	-11.6	—	—	0.52	-2.5
R05	1.73	-14.2	2.11	-17.3	-0.16	1.3	—	—	2.27	-18.4
R06	-2.28	15.1	-2.87	19.1	1.45	-9.6	—	—	-1.52	10.0
R07	-0.23	2.0	0.17	-1.4	-1.10	9.3	—	—	1.62	-13.7
R08	2.77	-17.4	2.22	-13.9	4.36	-27.3	3.49	-23.0	3.24	-20.3
R09	0.94	-6.4	0.90	-6.2	2.60	-17.9	2.70	-18.5	1.70	-11.6
R10	0.84	-7.3	0.73	-6.3	2.27	-19.7	4.45	-39.0	1.92	-16.7
R11	0.26	-3.7	0.26	-3.6	0.90	-12.5	1.31	-18.3	1.29	-17.9
R12	2.31	-16.1	1.76	-12.3	3.34	-23.4	—	—	2.16	-15.1
R13	0.37	-4.2	0.26	-3.0	1.09	-12.4	—	—	1.55	-17.6
R14	2.97	-17.1	2.71	-15.5	4.08	-23.3	—	—	3.41	-19.4
R15	0.53	-4.4	0.37	-3.1	2.15	-17.8	—	—	1.65	-13.7

1.00 and 0.07 in the same order. These bands measure the estimation scatter of the reported χ^2 given these universes, not how much it would change if the flux correlations were wrong, which the shrinking above probes; they are not intervals for the χ^2 under the true correlations.

The χ^2 can be decomposed along the eigenmodes of each prediction's covariance $C_{\text{total}}+C_{\text{resp}}+C_{\text{PL,5D}}+V_g$ (Table 22). The decomposition is a breakdown, not an alternative result; each mode is an eigenvector of that covariance, and its dominant block is the group with the largest share of the mode's variance. Figure 30 shows, for each sixteen-cell prediction, the three modes that carry most of its χ^2 and the prediction's distance along each.

Cell R04 ($p_T<0.7$, $p_\parallel\geq 4.5$, $E_{\text{avail}}<0.8$, $q_3<2$) has the smallest relative σ of the sixteen, 4.80 %, of which flux is the largest group at 3.47 %. Under the flux block alone its correlation with the 7 cells that load with it in the top modes is 0.972–0.998; under the reported covariance it is 0.41–0.77.

10.7 The predictions' target and definitions

The predictions of Table 19 differ from the generators' own products in two ways. First, NuWro and GiBUU, generated on carbon, are corrected to CH per fine cell as $x_{\text{CH}}=\frac{12}{13}x_{\text{C}}+\frac{1}{13}x_{\text{H}}$, with x_{H} the generator's own free-proton product, as in §11.9. Second, all four use the analysis definitions of W and of the muon angle: W is computed with the struck nucleon's mass, or an average nucleon mass where the generator records no single struck nucleon, and the muon angle is measured from the beam axis, each event's neutrino direction being drawn from the analysis simulation's distribution in its E_ν bin. The prediction is the mean over five such draws. With the generators' own definitions restored, every sample reproduces its committed product bit for bit. The change of definitions moves no cell of any of the four by more than 0.063 of the cell's reported σ, and the standard deviation over the five draws is at most $0.052\,\sigma$. MINERvA Tune v1 is already in the analysis definitions and is unchanged.

Table 23 gives the carbon values and the sensitivities. On carbon, in the analysis definitions, NuWro gives 129.98 (shape-only 112.93) and GiBUU 45.49 (24.94). The correction lowers NuWro's total χ^2 and raises GiBUU's by 2.3, while GiBUU's shape-only χ^2 falls. On the joint cells of §11.9 both corrections move the prediction toward the data; in the sixteen-cell comparison GiBUU's total

Table 21: χ^2 against $C_{\text{total}} + C_{\text{resp}} + C_{\text{PL,5D}} + V_g$ with one group's correlations shrunk by α toward the diagonal (α of the declared grid; the first column is the reported χ^2). Degrees of freedom as in Table 19.

group	prediction	$\alpha = 0$	$\alpha = 0.1$	$\alpha = 0.25$	$\alpha = 0.5$	$\alpha = 1$
interaction	GENIE 2.12.10 CV	107.18	93.42	84.01	75.61	68.96
	GENIE 2.12.10 + Valencia MEC	229.09	209.08	190.39	169.49	149.11
	NuWro 21.09 (CH)	127.69	113.76	104.56	95.69	89.53
	GiBUU 2019 (CH)	47.78	47.10	46.39	45.55	44.70
	MINERvA Tune v1	262.06	240.57	217.78	190.31	157.90
flux	GENIE 2.12.10 CV	107.18	82.27	67.10	54.97	45.86
	GENIE 2.12.10 + Valencia MEC	229.09	151.92	109.98	80.36	56.70
	NuWro 21.09 (CH)	127.69	90.88	72.77	60.86	57.17
	GiBUU 2019 (CH)	47.78	45.68	45.17	47.36	59.63
	MINERvA Tune v1	262.06	161.75	114.12	83.50	64.25
detector	GENIE 2.12.10 CV	107.18	96.12	86.81	78.39	70.27
	GENIE 2.12.10 + Valencia MEC	229.09	205.48	181.78	156.96	129.91
	NuWro 21.09 (CH)	127.69	116.06	106.45	97.29	88.46
	GiBUU 2019 (CH)	47.78	47.44	47.25	47.53	49.41
	MINERvA Tune v1	262.06	232.46	204.63	176.12	146.19
lateral	GENIE 2.12.10 CV	107.18	88.07	77.31	69.88	66.18
	GENIE 2.12.10 + Valencia MEC	229.09	194.16	171.44	152.95	138.34
	NuWro 21.09 (CH)	127.69	117.68	110.10	102.88	110.21
	GiBUU 2019 (CH)	47.78	43.64	40.65	38.35	36.82
	MINERvA Tune v1	262.06	223.84	200.25	179.78	165.30
the four together	GENIE 2.12.10 CV	107.18	69.00	55.05	45.66	41.91
	GENIE 2.12.10 + Valencia MEC	229.09	125.93	86.12	61.63	45.40
	NuWro 21.09 (CH)	127.69	82.48	65.58	57.35	65.06
	GiBUU 2019 (CH)	47.78	43.23	43.21	47.99	75.67
	MINERvA Tune v1	262.06	136.69	91.79	66.60	57.56

Table 22: Eigenmode breakdown of the reported χ^2: the fewest modes, in χ^2 order, carrying 50/90/99 % of it, and the top mode's share of the χ^2, its relative σ and its dominant block.

prediction	modes for 50/90/99 %			top mode (% of χ^2)	σ (%)	block
GENIE 2.12.10 CV	2	6	11	31.8	3.24	interaction
GENIE 2.12.10 + Valencia MEC	2	5	9	44.9	1.18	interaction
NuWro 21.09 (CH)	3	7	12	24.7	2.19	interaction
GiBUU 2019 (CH)	2	5	7	37.5	14.02	flux
MINERvA Tune v1	2	5	10	30.1	1.13	detector

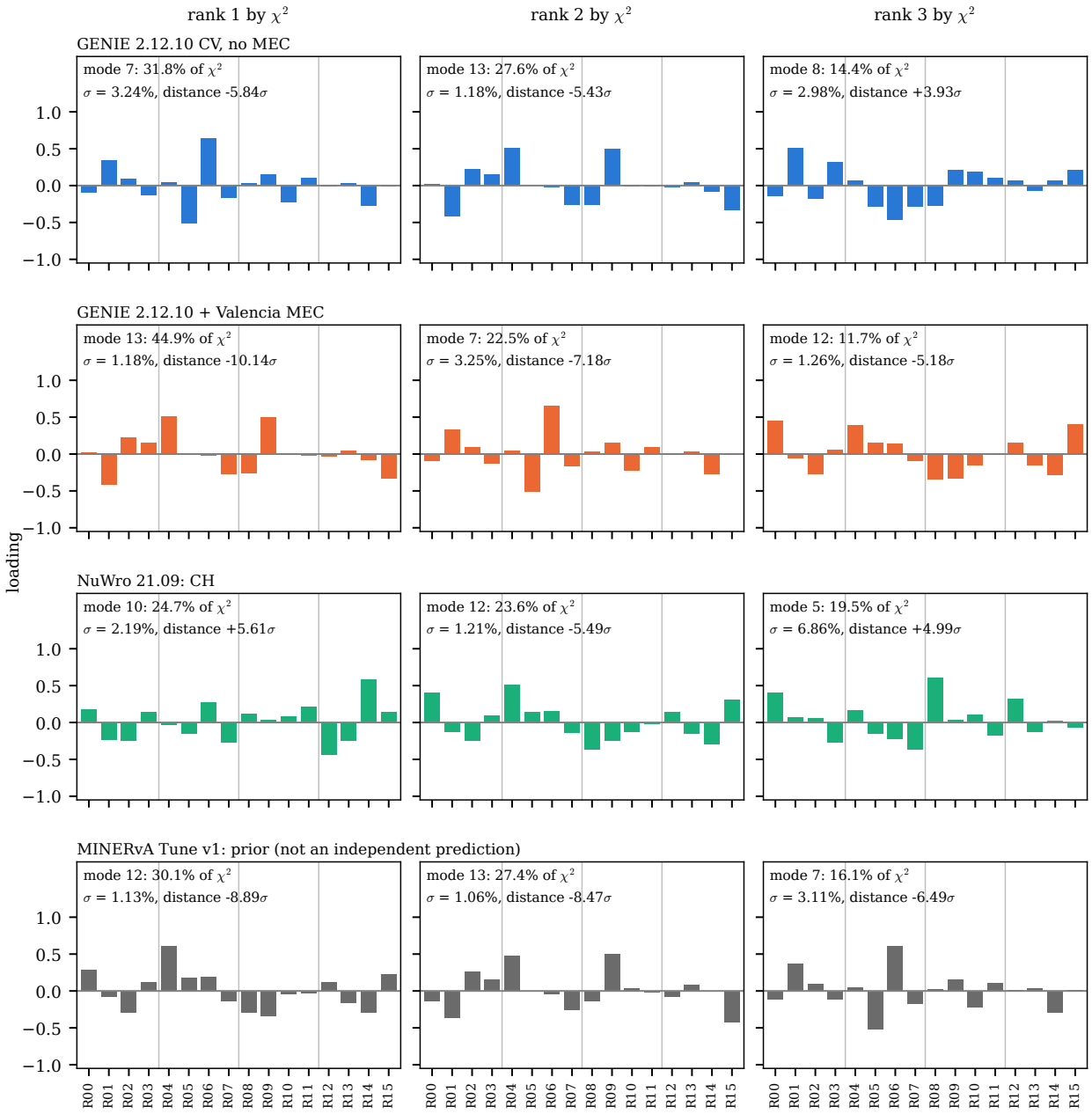


Figure 30: The three eigenmodes that carry the largest shares of each sixteen-cell prediction's χ^2 (columns, by rank), for GENIE CV, GENIE + MEC, NuWro and MINERvA Tune v1 (rows), from the reported comparison on $C_{\text{total}} + C_{\text{resp}} + C_{\text{PL,5D}} + V_g$ relative to f. The bars are each mode's unit loadings on R00–R15, signed so that the largest loading is positive. Each panel gives the mode's share of the χ^2, its relative σ and the prediction's distance along it, the projection over σ. The residual is $(f - g)/f$, so a positive distance means that the data lie above the prediction along the mode as signed. GiBUU, compared on 8 cells, has no modes here. The predictions are in the analysis definitions.

χ^2, on its eight cells, rises, because the CH prediction is paired with C_{model}'s NuWro and GiBUU shape ratios from carbon (below). With the reported C_{model}, that rise also depends on the hydrogen sample: with GENIE CV's hydrogen in place of GiBUU's own, in the generators' own definitions, GiBUU's total is 44.19, below its carbon value of 45.25. GENIE's hydrogen sample is thin, so that spread is largely its own Monte Carlo noise, and the GENIE-hydrogen rows are sensitivities, never the comparison. GiBUU's CH comparison carries the label of §11.9: the sampling-closure check failed on its hydrogen sample, a bias check of that check's premise, added post hoc, passed at $\leq 0.1\,\sigma_c$, and deep-inelastic events, 46 % of its hydrogen sample, do not conserve energy.

A native CH generation of NuWro, with only the target changed in the production card, was compared cell by cell with the combination under a rule fixed in advance: $\sum z^2 < 32.0$ over the sixteen cells and no $|z| > 3.5$, with z from the two samples' Monte Carlo variances. It agrees in the generators' definitions ($\sum z^2 = 17.9$, largest $|z| = 2.2$) and in the analysis definitions (16.9 and 2.2). Its χ^2 is 134.02 against the combination's 128.41; the comparison uses the combination. A diagnostic that was not a declared test: in one chunk of the generation's probe, NuWro's total cross section per nucleon is 2.1 % above $(12\sigma_{\text{C}} + \sigma_{\text{H}})/13$ computed from one chunk each of the production carbon sample and the free-proton sample, mostly in deep-inelastic events.

The C_{model} components of the reported covariance are built from the generators' samples in their own definitions, with NuWro and GiBUU on carbon, not from the predictions of Table 19. Table 24 measures the effect of that target. There, NuWro's and GiBUU's shape ratios in C_{model} are taken from their CH combinations in the generators' own definitions (the "CH (generator)" rows of Table 23), with everything else fixed; the reported covariance keeps them on carbon. Under the reported covariance the total χ^2 changes by -0.92 (GENIE CV), -5.56 (GENIE + MEC), $+0.17$ (NuWro) and -2.68 (GiBUU). Over both matrices and both kinds, the largest change is -6.07, in GENIE + MEC's shape-only χ^2 without $C_{\text{PL,5D}}$. Only the NuWro and GiBUU components change, but every prediction's χ^2 moves, because the components enter one matrix. With the CH shape ratios, GiBUU's CH total is 45.10, below its carbon value of 45.49 with the reported covariance, so the rise described above comes from pairing the CH prediction with the carbon shape ratios. For the scale of C_{model} itself, whose size is not calibrated, the table also gives the χ^2 with C_{model} removed and doubled. Every χ^2 falls as C_{model} grows. Removing C_{model} raises GiBUU's total most, to 166.83 from 47.78. With the CH shape ratios, the reported per-cell σ falls in 15 of the 16 cells and rises in R12, by 0.3 %. It falls most in R02, where σ/f goes from 0.1405 to 0.1273 (-9.4 %; -9.5 % without $C_{\text{PL,5D}}$). C_{model}'s own σ/f changes most in R02, from 0.1136 to 0.0968, and most in relative terms in R12, where it is small: from 0.0051 to 0.0075 ($+47.8$ %). This measures C_{model}'s dependence on the target only: its definitions stay the generators' own, its coverage is still not validated, and GiBUU's CH shape ratio carries the hydrogen label of §11.9. The changes are those of these data; their statistical uncertainty, from the data entering the unfolds that build C_{model}, is not measured. Table 25 gives the comparison in the generators' own definitions with NuWro and GiBUU on carbon.

Table 23: The carbon values and the sensitivities of the CH correction. Rows marked (generator) are in the generators' own definitions; the others in the analysis definitions. GENIE CV's hydrogen is a sensitivity only. The native NuWro rows are a check of the combination, which the comparison uses.

	reported covariance		without $C_{\mathrm{PL,5D}}$	
prediction	χ^2	shape-only χ^2	χ^2	shape-only χ^2
NuWro 21.09, 16 cells				
carbon	129.98	112.93	187.42	187.39
CH, Table 19	127.69	107.52	180.62	180.52
carbon (generator)	130.68	113.73	188.03	188.00
CH (generator)	128.41	108.40	181.21	181.13
CH, GENIE CV's hydrogen (generator)	120.47	101.20	169.10	168.98
native CH (generator)	134.02	111.98	193.37	193.29
native CH	134.71	112.01	194.46	194.35
GiBUU 2019, 8 cells with $p_{\parallel} < 4.5$ GeV/c				
carbon	45.49	24.94	58.65	27.86
CH, Table 19	47.78	22.19	61.41	24.66
carbon (generator)	45.25	24.75	58.46	27.69
CH (generator)	47.55	22.01	61.22	24.49
CH, GENIE CV's hydrogen (generator)	44.19	21.95	56.68	24.37

Table 24: The χ^2 of the four predictions of Table 19 with C_{model} removed ($\times 0$), as reported ($\times 1$, with NuWro's and GiBUU's shape ratios from carbon), doubled ($\times 2$), and with NuWro's and GiBUU's shape ratios from CH (CH), everything else fixed; Δ_{CH} is the CH column minus the reported one, from unrounded values. Each χ^2 includes V_g, as in Table 19; GiBUU is compared on its 8 cells.

prediction	χ^2	$\times 0$	$\times 1$ (reported)	$\times 2$	CH	Δ_{CH}
reported covariance						
GENIE 2.12.10 CV	total	121.76	107.18	104.98	106.27	−0.92
	shape-only	118.10	102.11	99.69	101.58	−0.54
GENIE 2.12.10 + Valencia MEC	total	261.54	229.09	223.88	223.53	−5.56
	shape-only	261.03	229.00	223.82	223.33	−5.67
NuWro 21.09 (CH)	total	140.25	127.69	126.87	127.86	+0.17
	shape-only	120.84	107.52	106.77	108.07	+0.54
GiBUU 2019 (CH)	total	166.83	47.78	45.50	45.10	−2.68
	shape-only	139.76	22.19	19.00	18.61	−3.58
without $C_{\mathrm{PL,5D}}$						
GENIE 2.12.10 CV	total	126.90	111.22	108.86	110.40	−0.82
	shape-only	126.64	110.32	107.82	109.68	−0.64
GENIE 2.12.10 + Valencia MEC	total	316.63	277.94	271.63	273.15	−4.79
	shape-only	297.63	263.74	258.24	257.67	−6.07
NuWro 21.09 (CH)	total	192.42	180.62	179.93	183.20	+2.58
	shape-only	192.38	180.52	179.84	183.18	+2.66
GiBUU 2019 (CH)	total	175.39	61.41	58.95	57.73	−3.68
	shape-only	143.87	24.66	21.20	20.61	−4.05

Table 25: The comparison of Table 19 as first made, with the predictions in their generator definitions and NuWro and GiBUU on carbon, with no correction to CH: against the reported covariance plus V_g and without the run-period block, as in Table 19, and in the last column against $C_{\text{total}} + V_g$, without the response and run-period blocks.

	reported covariance		without $C_{\text{PL,5D}}$		C_{total}
prediction	χ^2	shape-only χ^2	χ^2	shape-only χ^2	χ^2
All 16 cells: ndf 16, shape-only 15					
GENIE 2.12.10 CV	105.42	100.57	109.35	108.51	160.62
GENIE 2.12.10 + Valencia MEC	223.55	223.46	272.12	258.07	318.26
NuWro 21.09	130.68	113.73	188.03	188.00	298.69
MINERvA Tune v1	262.06	261.88	329.72	311.19	360.79
8-cell subset, $p_\parallel < 4.5$ GeV/c: ndf 8, shape-only 7					
GiBUU 2019	45.25	24.75	58.46	27.69	68.44

11 Calibrated joint tests of generator predictions

This section tests five generator predictions jointly in the five observables $(p_T, p_\parallel, E_{\text{avail}}, q_3, W)$, each in total and in shape. Each test evaluates a hybrid null hypothesis about one prediction (§11.1) against a null distribution obtained by simulating the full analysis chain. The tests are hypothesis tests, not a measured cross section, and they add no reportable uncertainty. They neither use nor validate the five-axis covariance of the five-dimensional central value unfolded with purity-based background subtraction (configuration (a), §9.6). The section gives the design, the six conditions of the declared model (§11.4), the decisions (§11.5; lost pseudo-experiments: §11.6) and the later sensitivity studies. Reference [52] gives the specification and the decisions, and each later study's own reference is cited where the study is described.

Two properties must be distinguished. The coverage of unfolded intervals is the rate at which they contain the true cross section. The calibration of a test is the rate at which it rejects a true hypothesis. Calibration can be obtained without validated intervals by simulating the whole analysis chain at the hypothesis, so that the estimator's bias enters the simulated null distribution. A calibration obtained in this way is valid to the extent that the simulated chain reproduces the analysis of the data. Unfolded results can be compared with any prediction at truth level without simulating each generator through the detector. For hypothesis tests the advantage is smaller. A test at reconstruction level would avoid the regularization bias, but it would still need the same response model. Each external prediction would have to be folded through the analysis simulation's response, which is the hybrid construction of §11.1, with the same composition below the fine grid. The tests of this section instead calibrate the unfolded statistic in simulation, so that the bias enters the null distribution rather than the decision. We have not compared their power with that of a reconstruction-level test.

11.1 Hypotheses, grid and statistics

For each of five predictions G (the MINERvA Tune v1, GENIE 2.12.10 without and with MEC, NuWro 21.09 and GiBUU 2019; §8.5), the null $H_0(G)$ is that the true cross section equals G on every cell of a fine five-dimensional grid. Below that grid, the null truth has the MINERvA Tune v1 simulation's event composition, varied in each pseudo-experiment by the drawn interaction-model universes. Every fine-cell integral is held fixed in that variation. The background is the Tune v1 simulation's, with its uncertainty bands. This *hybrid null* is the hypothesis the calibration simulates. For the Tune v1, which is the analysis simulation itself, no external prediction is combined with the Tune v1 composition. Three requirements are distinct. The hybrid null is defined exactly on the finite grid, and no convergence is needed to define it. Whether it represents the physical generator prediction depends on condition (i) (§11.4). Whether the calibration reproduces the analysis of the data is calibration fidelity (conditions (ii)–(iv)). Each prediction is tested in total (rate and shape together) and in shape alone: 10 tests. The fine grid is the analysis binning of §3.5, $14 \times 16 \times 7 \times 7 \times 6$ bins in $(p_T, p_\parallel, E_{\text{avail}}, q_3, W)$; the test cells below are its integrals.

The tests are evaluated on a coarse 3^5 grid in $(p_T, p_\parallel, E_{\text{avail}}, q_3, W)$ with edges $p_T \in [0, 0.55, 0.85, 4.5]\,\text{GeV/c}$, $p_\parallel \in [1.5, 3.5, 6, 60]\,\text{GeV/c}$, $E_{\text{avail}} \in [0, 0.4, 1.5, 100]\,\text{GeV}$, $q_3 \in [0, 1.2, 2, 100]\,\text{GeV}$ and $W \in [0, 1.4, 2.2, 100]\,\text{GeV}$. Each inner edge is the fine-grid edge nearest a tertile of the weighted simulated truth distribution of the signal sample in that coordinate, so the cells keep physical bin widths. That distribution is taken from the development simulation. The tests are restricted to the 109 supported cells (J). For GiBUU, whose sample lacks high-energy neutrinos, the domain is the 72 cells with $p_\parallel < 6\,\text{GeV/c}$. The grid specifies a finite hybrid hypothesis and a reporting statistic. It does not turn OmniFold's event-level training into binned unfolding.

Two statistics are formed from the unfolded cell values F and the prediction μ_G. The total (rate-and-shape) statistic is $T_{\text{tot}} = r^\top \Sigma^{-1} r$ with $r = F - \mu_G$. The shape statistic T_{sh} has the same form on the normalized cell fractions, with the normalization Jacobian applied to Σ and one cell dropped. The metric $\Sigma = V + \operatorname{diag} \operatorname{Var}(\mu_G)$ adds the prediction's own finite-sample variance to V. Here V is a Ledoit–Wolf-shrunk covariance [53] of 194 pilot pseudo-experiments at the Tune v1 null. The metric defines how hypotheses are compared, and so it affects power. The null calibration below, not the metric, sets each test's size.

The prediction variance is diagonal because each simulated event of a prediction falls in one cell. The pilot's seeds are disjoint from those of the calibration, but it reuses the same simulated events and omitted the MINOS-efficiency band. The diagonal of V matches the calibration's null variances (median ratio 1.00). The shrinkage, of intensity 0.015, floors about 80 of the 109 nearly degenerate correlation directions. At the Tune v1 null the statistic therefore has a whitened trace of about 47 rather than 109. Since V was fixed before the calibration, this affects power only (§11.5). The power at other shrinkage intensities has not been measured.

11.2 Calibration within the implemented simulation model

The null distribution of each statistic is obtained from pseudo-experiments that run the full analysis chain at the null truth, using the joint-test estimator (§9.3). That estimator is deterministic LightGBM OmniFold with five iterations and the negative-weight background treatment, whose refinement classifier has 400 trees and 31 leaves. Each pseudo-experiment

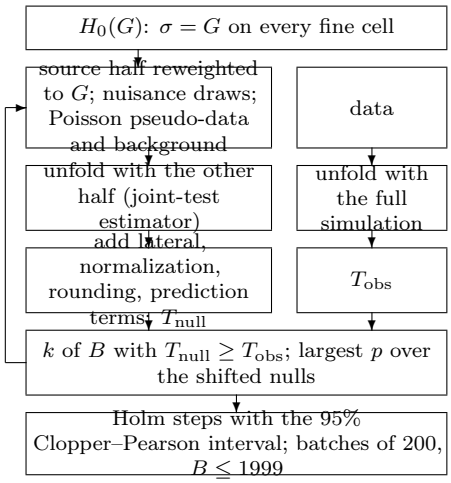
- draws Poisson pseudo-data from one half of the simulation together with an independent background source, refits the background-refinement classifier, and unfolds with the estimator trained on the other half;
- draws one of 100 flux universes and every one of the 34 interaction-model bands at one of its universes;
- draws the weight-based detector bands (MINOS muon-matching efficiency and the GEANT4 neutron, pion and proton interaction reweights) and the five muon and beam-angle lateral bands through a linear surrogate;
- draws a 1.4 % normalization, the data's rounding noise (taken from 20 real-data re-unfolds with rounding-level input changes), and the prediction's finite-sample residual.

The data are unfolded once, with the full simulation. The Monte Carlo p value of a test is $(k+1)/(B+1)$, where k of B null pseudo-experiments reach the observed statistic. This p value is never zero. The p value on which each decision rests, the *claim* p value, is the largest over the declared *claim variants*, which shift the null. The variants move it by an upper bound on the simulation-size difference between the pseudo-experiments and the data unfold (condition (iv) below), and by $\pm 2\,\delta_{\text{fine}}$ (condition (i)). The external nulls are strongly non-central, with non-centralities λ of 318–6344 in these units. They are non-central because the calibration reproduces the estimator's pull toward its prior (§9.3). Each observed statistic is compared with that displaced distribution (Fig. 31).

Generative model of one pseudo-experiment. Each calibration pseudo-experiment splits the simulation at random into two halves (Table 26) and refits the background refinement. Its truth equals G on every fine cell that the source half populates. Cells it leaves empty lower a coarse cell by at most 0.2 %, or about 4 % of the null standard deviation (the per-experiment spread is below 1.5 % of it). On the signal, each interaction-model band is renormalized within each fine truth cell,

so it moves only the composition within the cell, and with it the efficiency and the migrations. The flux universe and the band universes are drawn independently, and their weight ratios multiply. The muon-energy, muon-resolution, and beam-angle lateral bands enter as $\sum_b z_b \Delta_b$, with Δ_b from endpoint unfoldings of the data. The detector weight bands include the GEANT4 hadron-interaction reweights. Because the Δ_b come from the observed data, the null distributions are conditional on these data-derived inputs, which would differ in a repeated experiment. That conditioning has not been validated.

Table 26: Generative model of the calibration. Left: the flow of one test, repeated over B pseudo-experiments. Right: what each element of a pseudo-experiment does, and the corresponding element of the data analysis. Nuisance draws act on the pseudo-data source only.



Element	Pseudo-experiment	Data analysis
Truth	source half reweighted per fine cell to G	unknown
Unfolding simulation	other half at Tune v1 weights (the prior)	full simulation at Tune v1 weights
Background	Poisson source and template from disjoint halves; Tune v1 with its bands	full background template
Flux	one of 100 universes on the source (changes the rate, not σ)	central flux
Interaction model	34 bands, one universe each; signal renormalized per fine truth cell; two bands clipped	Tune v1
Detector	four weight bands at $\pm 1\sigma$; lateral bands by linear surrogate	nominal
Added terms	1.4 % normalization; rounding noise; prediction residual	none
Statistic	T_{tot}, T_{sh} with fixed Σ	the same

11.3 Multiplicity, stopping rule and design

The ten tests are carried out with the Holm procedure [54] at familywise level $\alpha = 0.05$. A determinacy requirement also applies: a step rejects only when the 95% Clopper–Pearson interval [55] of k/B lies entirely below its Holm threshold. Calibration proceeded in batches of 200 pseudo-experiments per null, to at most 1999, under a sequential rule that stopped a null once both of its decisions were determined. A null stopped when a 99.5% Clopper–Pearson interval of k/B lay entirely on one side of each of its thresholds and met a fixed precision requirement. The decision rule uses the

upper end of the two-sided 95% interval, a one-sided 97.5% upper bound. Two assumptions are made: (A1) each true null's calibrated tail probability is uniformly distributed or conservative, and (A2) the pseudo-experiments' exceedances are independent given the data. Under A1 and A2, the probability that this upper end falls below the threshold, at any number of pseudo-experiments up to 1999, is below the threshold. This holds whatever rule determines when sampling stops, including the budget extensions. The decision rule therefore controls the familywise error rate of the ten decisions at 5% under these assumptions. That rate is the probability of falsely rejecting at least one true hypothesis among them. The error control applies to the decisions, not separately to the point estimates $(k+1)/(B+1)$. Under these assumptions, the bound FWER $\leq 0.8902\,\alpha < 0.9\,\alpha$ is derived in §B.5. A2 requires the lost pseudo-experiments to have been lost independently of their outcomes, which has not been shown. When rerun, however, none of them reaches the observed statistic (§11.6). The calibrated size is averaged over the nuisance priors: at a fixed true value of the nuisance parameters it can exceed α.

The joint-test estimator (§11.2) was chosen before the design was fixed. The hypotheses, statistics, variants, multiplicity rule, stopping rule and every input were fixed before any production pseudo-experiment or observed statistic, with two data-informed elements. First, the design's reviewers had seen descriptive per-cell residuals of the data before it was fixed. Second, a bounded extension of the production budget was decided during production, while the calibration's monitoring output showed observed claim p values, so that the sequential rule, rather than the budget, could set the stop. No statistic, variant, threshold or stopping rule was changed, and the familywise bound above holds whatever rule determined when sampling stopped. An independent recomputation from the stored products, with separately written code, reproduces every p value and decision of the production evaluation. It verifies the calculation, not the adequacy of the calibration. The calibration is valid for the simulated null to the extent that the pseudo-experiment chain reproduces the analysis applied to the data. No rejection rate has been measured for null pseudo-data processed like the data, which would require an independent simulation sample.

Three studies followed the decisions: the scan of the shift multiplier (§11.5), the rerun of the lost pseudo-experiments (§11.6), and the matched coarse-projection comparison (§11.7). They are for reporting only, outside the decision rule. Each was specified before it ran; the projection comparison was specified after the five-dimensional outcomes were known.

11.4 Conditions

Each decision holds under six conditions of the declared model. They are restated here in the terms of this note, numbered (i)–(vi) as in the test specification, and fall into three kinds.

Numerical convergence. Condition (i) concerns the null truth below the fine grid. There it has the Tune v1 composition of the hypothesis definition (§11.1), varied by the drawn interaction-model universes, with every fine-cell integral held fixed at that of G, including the finite-sample fluctuations of G's simulation. The fine-grid convergence check has not shown convergence. In the L_2 norm of the per-cell relative changes, the last refinement, from a twofold-merged grid to the fine grid, is 0.50–0.86 of the whole change from the coarse test grid to the fine grid. In that norm the successive refinement changes are therefore not shown to shrink, and the residual below the fine grid is unmeasured and potentially material. The design assumes that each twofold refinement of the grid multiplies the refinement change, measured in a proxy of the test metric, by a factor of 0.45–0.76. This factor comes from the review of the design, and its derivation is not given. If the change kept shrinking geometrically at that rate, the residual would be about one to three times the last refinement change, δ_{fine}. Here δ_{fine} is the change of the noise-free unfolded result when the null truth is refined from the

twofold-merged grid to the fine grid. That estimate sets the claim variants at $\pm 2\,\delta_{\text{fine}}$, the central extrapolated estimate, and the robustness variants, for reporting only, at $\pm 3\,\delta_{\text{fine}}$. The extrapolation assumes geometric convergence. The measured L_2 ratios also allow slower rates and larger residuals. Measured with V fixed before any observed statistic, the claim variants move the external nulls by 0.55–2.0 null standard deviations and the robustness variants by 1.07–3.6. For the hybrid null as simulated, the residual plays no role. The variants carry the rejections over to G itself only if replacing the Tune v1's shape below the fine grid by G's moves the noise-free unfolded result by no more than $\pm 2\,\delta_{\text{fine}}$ along δ_{fine}. Condition (vi) concerns the predictions' own finite simulated samples, which enter the metric as $\operatorname{diag}\operatorname{Var}(\mu_G)$ and the calibration as a drawn residual term.

Calibration validity. The calibration's validity, and what the independent recomputation verifies, are stated in §11.3. Two known departures are bounded. Condition (iv) concerns the simulation size: the pseudo-experiments unfold with half of the simulation and the data with all of it. The resulting shift D, estimated from 16 pairs, is at noise level per cell, with a median $|D|/f$ of 0.02–0.075%. Along the null's bias direction it is not always consistent with zero (proxy a/se of +4.7 for the Tune v1 and −3.4 for NuWro), but it is at most 0.068 null standard deviations. That bound was measured with V fixed before any observed statistic, and it supersedes the specification's estimate of about 0.05. The shift enters as the bias-aligned upper-bound variant. That shift was measured for the mean only, in a process without noise or nuisance draws. A separate check reran the first 200 calibration pseudo-experiments of the NuWro, GENIE 2.12.10, and GENIE 2.12.10 + MEC nulls with a quarter of the simulation. Between half and quarter simulation, it resolved no change of the null distributions' location, width, or tails: the medians agree within 0.074 null standard deviations, and the widths lie within 0.975–1.010 of the half-simulation values. The only change it resolved is a mean paired shift of 0.006–0.034 null standard deviations. Under an assumed extrapolation from half to full simulation that is conservative by design, all three shape decisions, and so all ten, remain rejected. The check concerns the simulation size only and is not a test of the data process. Every pseudo-experiment resamples the same simulated events, so the null distributions are conditional on that simulation. The decisions closest to their boundary are the shape tests of NuWro and GENIE 2.12.10. With the Holm order held fixed, their observed statistics exceed the value at which the decision would change by 1.11 and 1.30 null standard deviations, and by 1.01 and 0.67 with the robustness variants at $\pm 3\,\delta_{\text{fine}}$. The other eight tests lie 2.7 to 203 null standard deviations from that value, and 2.0 to 203 with the robustness variants. Crossing that value would make a test undetermined, its Clopper–Pearson interval straddling the threshold, rather than not rejected. The data unfold's rounding-level nondeterminism, measured over the 20 re-unfolds, leaves every observed p value unchanged except NuWro's shape test (2/1752 to 3/1752).

Physical-model adequacy. Condition (ii) concerns the detector model. It is the simulation's per event, varied only by the drawn detector bands (MINOS muon-matching efficiency and the GEANT4 neutron, pion and proton reweights, each at $\pm 1\sigma$) and by the linear surrogate for the five lateral bands. The surrogate has been checked only for the symmetry of its endpoints. The analysis's systematic set (App. A) has no recoil-energy-scale band, so the completeness of the hadronic response is not established (§11.8). Condition (iii) concerns the interaction-model prior, the analysis's set of 34 interaction-model reweighting bands, with every band drawn in every pseudo-experiment. Negative weights in two of them, LowQ2_1 and HighQ2_1, are clipped per universe. Measured on the pilot, the effect of clipping them rather than keeping them is at most 0.35% of a band's shift in whitened norm. Condition (v) concerns the neutrino-energy range. The GiBUU sample contains no events with $E_\nu > 20\,\text{GeV}$ (§8.5), and its truth outside the test domain ($p_\parallel \geq 6\,\text{GeV/c}$,

with truncated E_ν) enters through the migrations into the domain cells. No prediction contains events with $E_\nu > 100\,\mathrm{GeV}$; only the data's flux integral stops there, and the omission is negligible. The use of the Tune v1 composition below the fine grid in (i) is also a modelling choice of this kind.

11.5 Decisions

Table 27 lists the ten decisions. Every test is rejected at familywise $\alpha = 0.05$; nine have $k = 0$, and NuWro's shape test has $k = 1$. Each rejection also survives the robustness variants at $\pm 3\,\delta_{\mathrm{fine}}$, the upper end of the extrapolation in condition (i). Figure 31 shows each null's calibrated distribution with the observed statistic. The Tune v1 null is centred at small values, whereas the external nulls are displaced far upward by the estimator's pull toward its prior.

G	test	B	k	claim p	Holm thr.	95% CP interval of k/B
NuWro 21.09	total	1751	0	1/1752	0.005	$[0, 0.00210]$
GENIE 2.12.10	total	1366	0	1/1367	0.00556	$[0, 0.00270]$
GENIE 2.12.10	shape	1366	0	1/1367	0.00625	$[0, 0.00270]$
MINERvA Tune v1	total	1365	0	1/1366	0.00714	$[0, 0.00270]$
MINERvA Tune v1	shape	1365	0	1/1366	0.00833	$[0, 0.00270]$
GiBUU 2019	total	1351	0	1/1352	0.01	$[0, 0.00273]$
GiBUU 2019	shape	1351	0	1/1352	0.0125	$[0, 0.00273]$
GENIE 2.12.10 + MEC	total	1343	0	1/1344	0.0167	$[0, 0.00274]$
GENIE 2.12.10 + MEC	shape	1343	0	1/1344	0.025	$[0, 0.00274]$
NuWro 21.09	shape	1751	1	2/1752	0.05	$[0.0000145, 0.00318]$

Table 27: Joint five-dimensional generator tests, in Holm order; every test is rejected. B is the number of completed calibration pseudo-experiments at the stop set by the sequential rule (1400 submitted per null, NuWro 1800; the rest were lost to time limits and later recovered for reporting only, §11.6), and k the number at least as extreme as the data; claim $p = (k + 1)/(B + 1)$ is never zero.

Sensitivity to the sub-fine-grid residual. The multiplier κ scales the last measured refinement change δ_{fine}: $\kappa = 1$ means one such change, not one standard deviation or a confidence level. A scan of κ along the direction δ_{fine} of the claim variants [56] (for reporting only) found, in the calibration ensembles used for the decisions, all ten decisions to remain determinate rejections at every sampled multiplier up to 5.25. Determinate rejection is first lost between the sampled multipliers 5.2547 and 5.2563, and between 5.1688 and 5.1703 with the recovered draws included. It is lost by the GENIE 2.12.10 + MEC shape test, whose interval then contains its Holm threshold, and, through the Holm step-down, by the GENIE 2.12.10 shape test. Neither is then a determinate failure to reject. The scan probes one prescribed direction. It establishes neither convergence nor a bound on the actual below-grid residual (condition (i), §11.4).

Power. Because every test rejects, power serves here only as context: the test specification uses power only to interpret a failure to reject. At the Tune v1 null, two of the three declared alternatives are detected in shape with power 1.0 at the 0.005 level, and the third, a MEC-like (E_{avail}, W) shape, with 0.231. At the GENIE 2.12.10 null the shape powers at 0.005 are 0.739, 0.02 and 0.00. The binomial 95% intervals are $[0.98, 1]$ for the first two Tune v1 alternatives, $[0.17, 0.30]$ for the third, and $[0.67, 0.80]$, $[0.01, 0.05]$ and $[0, 0.02]$ at the GENIE 2.12.10 null. These values use the power sets as evaluated for the decisions. Each of those sets had lost some of its 200 alternative

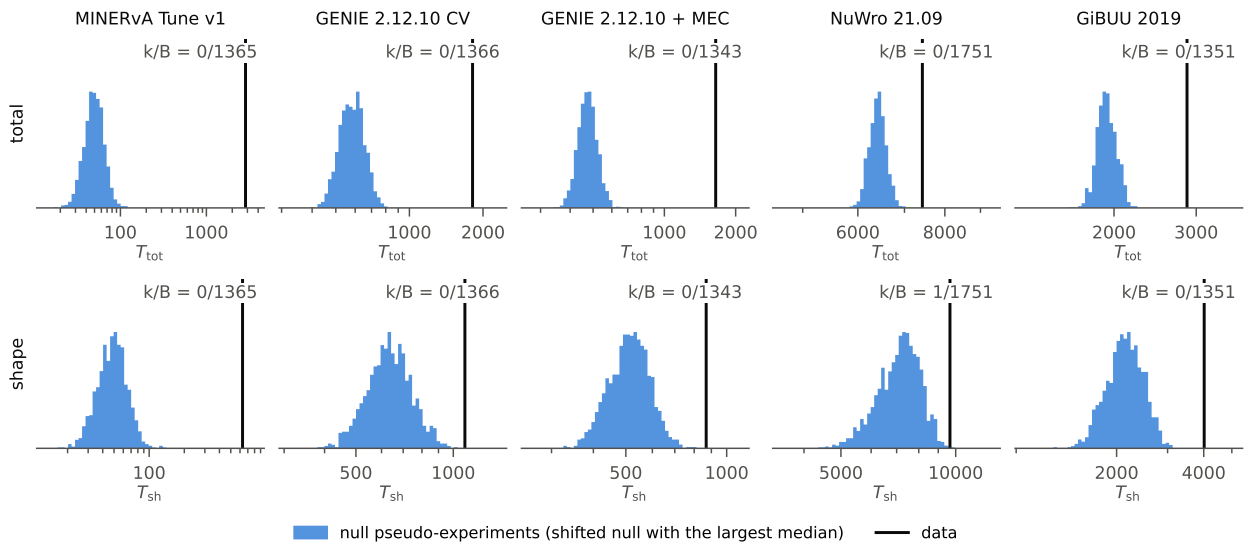


Figure 31: Calibrated null distributions of the joint statistics for each prediction (columns), total statistic (top) and shape statistic (bottom), shown for the declared claim variant whose null distribution has the largest median among those used in the decisions; the vertical line is the observed statistic and k/B is as in Table 27. The horizontal axis is logarithmic. Each shifted null adds the same fixed vector s to every null draw before the statistic is formed, with $s \in \{0, \frac{1}{2}D, D, +2\,\delta_{\text{fine}}, -2\,\delta_{\text{fine}}\}$; the p value used for the decision is the largest over these shifts and need not come from the one shown. Plotted from the released sufficient inputs (App. G), which reproduce every recorded p value and decision.

pseudo-experiments to time limits (172–199 retained). Evaluated against the null ensembles used for the decisions, the complete power sets, recovered as in §11.6, agree with them within 0.01. When the GENIE 2.12.10 null is also recomputed on all draws, the shape power at 0.005 against the first alternative at that null is 0.72.

11.6 Lost pseudo-experiments and their recovery

Computing time limits interrupted or prevented 224 of the submitted calibration pseudo-experiments (111 interrupted, 113 never started) and 53 power pseudo-experiments. The decisions in Table 27 were evaluated on the completed draws. A sensitivity analysis computed before the lost outcomes were known showed that every decision could change under a realizable assignment of them. Had every interrupted draw exceeded the observed statistic, no test would have been rejected (p 0.013–0.019). Among completed draws, runtime is not associated with the statistic ($|\rho| \leq 0.06$), which supports but does not certify the decisions.

The lost seeds were rerun under a procedure that was fixed, and reviewed independently, before the rerun [57]. A determinism check fixed in advance (16 of 16 completed seeds rerun bitwise identically) shows that each recovered draw is the lost draw, so every submitted draw has been observed. No recovered draw reaches the observed statistic in any test, claim variant or robustness variant, so no assignment of the lost draws changes any decision. The closest margin is 4.6 units of T below the observed value of 868.6 (GENIE 2.12.10 + MEC shape, $\pm 3\,\delta_{\text{fine}}$ variant). The next is 49.8 (GENIE 2.12.10 shape, same variant). With complete batches the predeclared rule would have stopped the Tune v1, GENIE 2.12.10, GENIE 2.12.10 + MEC and GiBUU nulls one batch earlier ($B = 1200$), with the same decisions.

The recovered draws were evaluated in two ways: with the variant shifts of the primary evaluation held fixed (the *fixed-shift* evaluation), and with the shifts recomputed on all draws (the *recomputed-shift* evaluation). The recovery's checks are narrower in scope than the main recomputation. Only the fixed-shift evaluation was cross-checked independently. The recomputed-shift evaluation was reproduced only by a standalone rerun of the same code, a consistency check rather than an independent recomputation. The earlier stops rest on the calibration run's own re-application of the sequential rule and have no independent check. The decisions of Table 27 remain those of the evaluation before recovery, at the B at which its rule stopped. The observed outcomes of the lost draws supersede the finding of the sensitivity analysis described above.

11.7 Matched coarse projections

The same calibrated tests are restricted to matched 3×3 coarse projections of the same cells, onto $(p_T, p_\parallel)$ and onto (E_{avail}, W). The comparison uses the same calibration ensembles, with the recovered draws included, and the same variants and claim rule. Its p values were reproduced independently [58]. The projected tests do not enter the Holm family and change no decision of Table 27. The comparison and its criterion were specified after the five-dimensional outcomes were known but before the comparison was run. Under the criterion, a test would count as evidence of additional joint discrimination only if both of its projected claim p values were at least 0.05 in both the fixed-shift and the recomputed-shift evaluations. No test meets it, and the values are identical in the two evaluations. Every test has $p < 0.05$, without a multiplicity correction, in at least one of the two projections. This study therefore provides no evidence of additional discrimination from the joint five-observable information. Table 28 gives every projected claim p value. In the muon-only $(p_T, p_\parallel)$ projection, GENIE 2.12.10 is not rejected ($p = 0.48$ in total, 0.63 in shape; a post hoc observation), whereas its (E_{avail}, W) projection is. Hadronic information therefore participates in that rejection.

This does not establish added power of the joint test over all matched projections, or accept GENIE as true. The powers of the joint and projected tests have not been compared.

G	test	$(p_T, p_\parallel)$: claim p (k/B)	(E_{avail}, W): claim p (k/B)
MINERvA Tune v1	total	0.00071 (0/1400)	0.00071 (0/1400)
MINERvA Tune v1	shape	0.00071 (0/1400)	0.00071 (0/1400)
GENIE 2.12.10	total	0.483 (676/1400)	0.00071 (0/1400)
GENIE 2.12.10	shape	0.629 (880/1400)	0.00071 (0/1400)
GENIE 2.12.10 + MEC	total	0.0014 (1/1400)	0.00071 (0/1400)
GENIE 2.12.10 + MEC	shape	0.0014 (1/1400)	0.0021 (2/1400)
NuWro 21.09	total	0.00056 (0/1800)	0.00056 (0/1800)
NuWro 21.09	shape	0.00056 (0/1800)	0.0033 (5/1800)
GiBUU 2019	total	0.00071 (0/1400)	0.00071 (0/1400)
GiBUU 2019	shape	0.00071 (0/1400)	0.00071 (0/1400)

Table 28: The joint tests restricted to matched 3×3 coarse projections of the same cells (for GiBUU, 6 and 8 cells of its domain), with all submitted calibration draws, the same variants and the same claim rule. Every test has $p < 0.05$ in at least one projection. The GENIE 2.12.10 values in the $(p_T, p_\parallel)$ column are a post hoc observation.

11.8 Hadronic-response condition and recoil-energy sensitivity

MINERvA's recoil-energy analyses assign a hadronic-energy response uncertainty derived from test-beam calorimetry [59], separate from the GEANT hadron-interaction reweights [3, 8]. In the low-recoil measurement the resulting hadronic-energy uncertainty rises to 10 % at its highest three-momentum transfers, $0.9 < q_3 < 1.2\,\text{GeV}$ [8]. The declared nuisance model here includes the reweights but no calorimetric response band. Such a band would act on all five nulls together, through the shared detector simulation and unfolding. A response mismodelling could move each observed statistic in either direction, and a band drawn in the calibration would also broaden and could shift each null distribution. The net effect on each p value is unmeasured. The published comparison in muon kinematics alone, whose observables do not use the reconstructed recoil energy, also disfavours the same generator families [3]. It provides context for a disagreement involving muon observables, but the hypotheses tested here use a different estimator and calibration. The GENIE 2.12.10 projection result of §11.7 shows that the information rejecting that prediction involves the hadronic axes, on which a response band would act.

In an exploratory sensitivity study of the response [60], the data are re-unfolded with the simulation's reconstructed recoil energy scaled by ±4 %, and the ten tests are re-evaluated against the null distributions used for the decisions. This scale is the level of data–simulation agreement reported for test-beam protons, pions and electrons of 0.35–2.0 GeV/c [59]. Neutrons and higher-momentum hadrons, which matter at high W, lie outside that sample. The ±4% scale is not a validated uncertainty prescription. A re-unfold with unscaled recoil reproduces the decisions of Table 27, with every claim p value inside the range spanned by the data's rounding variations. At both −4% and +4% every one of the ten rejections holds. The observed statistics move by up to about 11 % from their recorded values. Nine tests keep $k = 0$. NuWro's shape test, the closest to its null, changes from $k = 1$ to $k = 3$ of 1751 at +4% ($p = 0.0023$, against a Holm threshold of 0.05). The evaluation was reproduced independently. Table 29 gives the observed statistics.

The study measures the sensitivity of the observed statistics and decisions to one coherent scale of reconstructed available energy and recoil energy, applied after the energy calibration at a single

	T_{tot}				T_{sh}			
G	recorded	−4%	+4%	null median	recorded	−4%	+4%	null median
MINERvA Tune v1	2852	2856	2540	49	631	647	600	49
GENIE 2.12.10	1813	1796	1721	528	1086	1104	1040	582
GENIE 2.12.10 + MEC	1647	1652	1595	406	869	893	839	450
NuWro 21.09	7427	7380	7361	6324	9660	9690	9540	7110
GiBUU 2019	2867	3129	2665	1791	4013	4154	3901	1975

Table 29: Observed joint statistics with the simulation's reconstructed recoil energy scaled by ±4%, beside the recorded values and the median of each unshifted null distribution, which is not regenerated. Every test keeps $k = 0$ at both signs except NuWro's shape test ($k = 1$ of 1751 at −4%, 3 at +4%); all ten remain rejected. The values are descriptive.

magnitude. It leaves untested (a) the response of the null distributions, which were not regenerated with the variation although they are simulated through the same detector model, so the size of this omission is unmeasured for these strongly non-central nulls; (b) particle-species, energy-dependent or resolution effects, and a response difference applied before calibration; and (c) larger variations. The rejections are therefore conditional on the simulation's hadronic response, and the response model is not shown to be complete.

11.9 Hydrocarbon target correction for NuWro and GiBUU

NuWro 21.09 and GiBUU 2019 were generated on carbon, while the data and the other predictions are per nucleon of CH. This study corrects the two predictions to CH and re-evaluates their tests against the null distributions used for the decisions. These target-corrected evaluations are descriptive; the decisions of Table 27 are unchanged.

Construction. Each prediction is rebuilt per fine cell as $x_{\mathrm{CH}} = \frac{12}{13}\, x_{\mathrm{C}} + \frac{1}{13}\, x_{\mathrm{H}}$. Here x_{H} is the generator's own free-proton product, run through the unchanged conversion, flux-repair and binning chain. Each prediction's Monte Carlo variance is combined with the squares of the same weights. The hydrogen fraction is $f_{\mathrm{H}} = 5.08\%$ for NuWro over the grid and 4.77% for GiBUU over its 72 cells. Over the reported cells, the ratio $r_c = \mu_{\mathrm{CH}}/\mu_{\mathrm{C}}$ has minimum, median, and maximum 0.923, 0.963, and 1.039 for NuWro (109 cells) and 0.923, 0.963, and 1.036 for GiBUU (72 cells). Per cell, f_{H} reaches 0.112 and 0.109. On these cells both corrections move the prediction toward the data; in the sixteen-cell comparison, GiBUU's total χ^2 on its eight cells rises with the correction, because the corrected prediction is paired with C_{model}'s NuWro and GiBUU shape ratios from carbon (§10.7).

Re-evaluated tests. With the metric used for the decisions, NuWro's total statistic falls by 3.83 null standard deviations and its shape statistic by 0.37. At its original Holm position (threshold 0.005) the total test would then lie 1.11 null standard deviations below its boundary, and so be undetermined. It remains rejected only because its larger p value moves it to the last Holm position (threshold 0.05). In the $\kappa = 3$ family, which is for reporting only, it is not rejected ($p = 0.075$). NuWro's shape test stays rejected, 0.74 null standard deviations above its decision boundary. GiBUU's statistics fall by 4.67 (total) and 1.33 (shape) null standard deviations. Both tests stay rejected at their original Holm positions, including at $\kappa = 3$. Their margins shrink from 7.86 to 3.18 and from 3.58 to 2.24 null standard deviations. With each CH prediction's own variance in the metric, no decision changes for either generator.

Generation checks. For GiBUU, the sampling-closure check failed on the hydrogen sample. A bias check of that check's premise was added post hoc. Its criterion, a change of at most $0.1\,\sigma_c$ in every cell, was declared before the check was computed. The check met the criterion: it moved the CH prediction by at most $5.1 \times 10^{-4}\,\sigma_c$. Here σ_c is the standard deviation of the cell's result over NuWro's null distribution used for the decisions ($B = 1751$). The GiBUU re-evaluation is reported under the label "B3 (sampling closure) failed on hydrogen; a post-hoc bias check of its premise passed at $\leq 0.1\,\sigma_c$". It has a further limitation: GiBUU's deep-inelastic events on hydrogen, 46% of its hydrogen sample, do not conserve energy. They lose a median of 45 MeV, with a median transverse-momentum imbalance of 31 MeV, while its other hydrogen events balance exactly. The cause has not been identified. The available energy of those events may be low, and the non-conservation may also affect GiBUU's carbon prediction, which cannot be checked in this way. In NuWro's hydrogen generation, four chunks aborted and were rerun on alternate seeds. The fraction of events this excludes is estimated as 1.8×10^{-5} in the main sample and 3.8×10^{-5} in the 50–100 GeV supplement, or 1.9×10^{-5} and 6.2×10^{-5} including the first attempt. These estimates rest on four to six events. The supplement's 95% upper bound of 1.8×10^{-4} exceeds the 10^{-4} level set for it, so the supplement is flagged rather than assumed negligible.

11.10 What the tests establish

The comparisons recover the generator disagreements already established by the published two-dimensional measurement on the same data [3]. Their methodological contribution is calibration within the implemented simulation model, which puts the estimator's bias toward its prior and the declared nuisances into each null distribution. The rejections are global assessments of the five specified predictions under the declared model. Assigning a discrepancy to a kinematic region or nuclear mechanism would require separate tests. The rejections apply to the generator versions listed in §8.5, and they retain all six conditions of §11.4. Extending the inference would require a response band drawn through the calibration and a demonstrated bound on the residual below the fine grid.

12 Extended fiducial phase-space study

This section gives a preliminary scalar central value for an extended fiducial domain, with a three-prior dependence envelope, a hidden-variable closure and a 200-pseudo-experiment containment diagnostic for the extension. The signal remains ν_μ CC in the tracker interaction fiducial volume. The truth-muon kinematic cuts are removed, but the reconstructed MINOS-based selection is unchanged, so the domain is an extended *fiducial* one, not unrestricted neutrino phase space. The result consists only of central values and these validation tests, without a quoted covariance.

12.1 Acceptance and the extended domain

The published cuts ($\theta_\mu < 20°$, $1.5 < p_\parallel < 60\,\mathrm{GeV/c}$, $p_T < 4.5\,\mathrm{GeV/c}$) exclude a third of the signal. Of the weighted true ν_μ CC rate in the tracker fiducial volume, 66.4 % lies inside the cuts, 22.3 % at $p_\parallel < 1.5\,\mathrm{GeV/c}$, 11.3 % at $\theta_\mu > 20°$, and the high-$p_T/p_\parallel$ tails are negligible (Fig. 32). OmniFold's treatment of events that fail reconstruction already corrects for acceptance, so removing the truth-level kinematic cuts is a change of configuration only: the truth denominator grows from 32.8 million to 49 million events. In regions without acceptance the unfolding extrapolates through the step-1 regressor for missed events; the validation tests below quantify the resulting prior dependence.

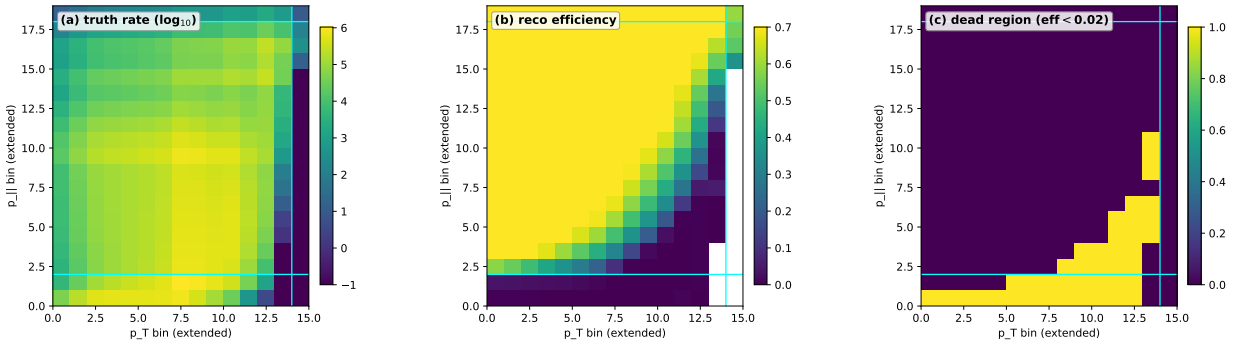


Figure 32: Extended-fiducial acceptance for all twelve data-taking periods: (a) weighted truth rate ($\log_{10}$), (b) reconstruction efficiency, and (c) the low-acceptance region, $\epsilon < 2\,\%$, on the extended $(p_T, p_\parallel)$ grid (bin number; the overflow bins are too wide for a linear axis). Cyan lines mark the published boundary. Efficiency is ~ 0.65 and flat through the published region, degrades smoothly across the $\theta > 20°$ wedge, and falls to zero in the $p_\parallel < 1.5\,\mathrm{GeV/c}$ strip and the high-p_T corner. The low-acceptance region carries 27.7 % of the total truth rate.

12.2 Validation tests

Four tests characterize the extension (Fig. 33):

1. **Published-region check.** The control unfolds the standard 2D input on the published binning, with settings identical to the extended-domain unfolding's. In the published sub-block, the extended-domain unfolding reproduces it with integral ratio 0.9994, median cell ratio 1.0013 and median absolute relative deviation from the control 0.57 % over all 185 reported cells. The control's total, $3.073 \times 10^{-38}\,\mathrm{cm}^2$/nucleon, agrees at the printed precision with the purity-subtracted 2D totals (exact-split and LightGBM). The quoted 2D result, which uses refined negative-weight subtraction, is $3.069 \times 10^{-38}\,\mathrm{cm}^2$/nucleon, 0.14 % lower, so the control does not reproduce it.

2. **Closure.** Extended-grid self-closure using simulated reconstruction as pseudo-data returns recovered/truth = 1.0000 in every cell, including the extension, with no event-accounting or normalization bias from the enlarged denominator. In *hidden-variable* closure, a Gaussian truth bump is injected in true E_{avail}, unseen by the $(p_T, p_\parallel)$ unfolding. The per-cell absolute relative deviation of the recovered cross section from the bump-reweighted $(p_T, p_\parallel)$ truth has a median of 0.17 % (90th percentile 0.65 %) in the published cells and a median of 0.77 % (maximum 4.05 %) in the extension, both well inside the three-prior envelope.

3. **Three-prior envelope.** The total cross section varies by ±1.5 % across three priors (MINERvA Tune v1, bare GENIE and NuWro-shaped). The NuWro-shaped prior reweights the truth $(p_T, p_\parallel)$ shape to two million raw NuWro events without phase-space cuts. The sample's mis-sampled flux (§7.6) suppresses its high-$p_\parallel$ shape, and this prior was not rebuilt after the flux repair. The per-cell half-spread of the three results, divided by their mean, has a median of 2.9 % in the published region and 7.9 % (90th percentile 62 %) in the extension, concentrated in zero-acceptance cells (Fig. 34).

4. **Containment diagnostic.** With a fixed estimator seed and one disjoint 100/100 split into calibration and evaluation sets, widths learned from the calibration set contain the fixed truth in 68.67 % of evaluation (bin, pseudo-experiment) pairs, compared with the 68.27 % Gaussian

nominal. Bin correlations and estimated widths preclude a global binomial error or a calibrated test, and the closure pseudo-experiments fluctuate statistically only, so a full uncertainty band is not tested. The statistical-band, pull and regional fractions are in App. B.3; none is used to certify a covariance.

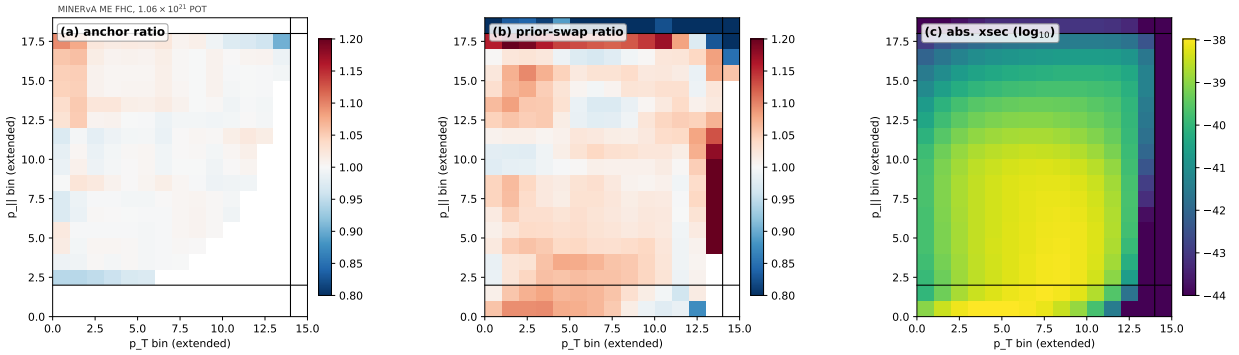


Figure 33: Extended-fiducial validation tests (all data-taking periods): (a) ratio to the control in the published sub-block (median 1.0013), (b) MINERvA Tune v1/bare-GENIE prior-swap ratio, and (c) extended-fiducial cross section ($\log_{10}$). Panels (a) and (b) are *ratios* on a diverging scale centered at unity; panel (c) gives the *absolute* magnitude on a sequential scale. Black lines mark the published boundary; axes are bin number on the extended grid.

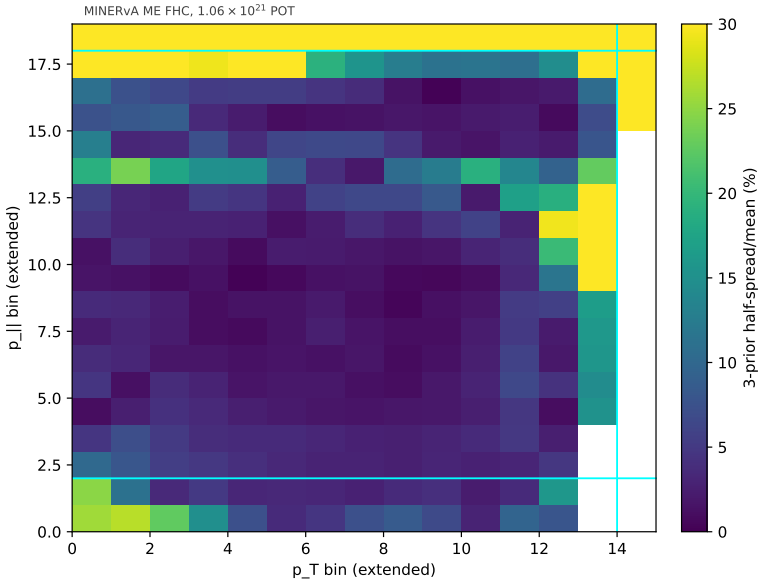


Figure 34: Three-prior dependence envelope (per-cell half-spread/mean, in percent). Prior dependence is at the few-percent level wherever the detector has acceptance and reaches tens of percent only in low-acceptance cells, the quantitative basis for the two-tier reporting.

12.3 Central value and two-tier reporting

The preliminary extended-fiducial CC-inclusive central value on the tracker fiducial volume is

$$\sigma_{\text{ext}} = 4.502 \times 10^{-38}\,\text{cm}^2/\text{nucleon} \quad \text{(central value)},$$

compared with $3.073 \times 10^{-38}\,\mathrm{cm}^2$/nucleon for the control unfolding in the restricted phase space. The extended-fiducial central value was produced with the binned purity correction, not with refined negative-weight subtraction, the treatment designated for the extended-fiducial analysis (App. C). We adopt **two-tier reporting**. *Tier 1* comprises the acceptance-supported cells ($\epsilon \gtrsim 2\%$): the published region and the measurable part of the extension, which carries $\sim 6\,\%$ of the rate, chiefly in the $\theta > 20^\circ$ wedge and the $p_\parallel \in [0.75, 1.5]\,\mathrm{GeV/c}$ edge. *Tier 2*, prior-extrapolated, comprises the low-acceptance cells ($\sim 28\,\%$ of the rate) and is flagged separately with an explicit additional three-prior envelope band. Both tiers enter the sum; only tier 1 is interpreted as an acceptance-supported measurement. No uncertainty magnitude is assigned beyond the three-prior envelope, because no extended-domain covariance has been constructed with re-selected lateral bands under the procedure of §4.5.

13 Summary and outlook

Summary. Beyond the results stated with their scopes in the executive summary (p. 1), the body supports two further conclusions.

- The refined negative-weight subtraction, which the sixteen-cell result uses (§10), largely removes the bias that purity-based background subtraction leaves in the highest-W cells (§9.2). In a two-dimensional closure test with backgrounds, the purity and refined estimates differ by more than one statistical standard deviation in 14 of the 205 cells. In those cells the purity estimate lies up to $5.9\,\sigma_{\mathrm{stat}}$ from the truth, and the refined one within $0.44\,\sigma_{\mathrm{stat}}$. On this basis, the two-dimensional result uses the refined estimator (App. C).
- The joint tests' design was fixed before any production pseudo-experiment or observed statistic, with two data-informed elements (§11.3). The tests recover established discrepancies and show no evidence of discrimination beyond matched coarse projections (§11).

Principal limitations. The executive summary lists the principal limitations (p. 2), each with the section that states it. The quoted two-dimensional statistical block has had no coverage test (§4.2, §5.2).

Outlook. The scalar extended-fiducial study (§12) yields a preliminary central value $\sigma_{\mathrm{ext}} = 4.502 \times 10^{-38}\,\mathrm{cm}^2$/nucleon, against $3.073 \times 10^{-38}\,\mathrm{cm}^2$/nucleon for its control unfolding in the restricted phase space. New acceptance-supported cells contribute about 6 % of the rate; separately labeled prior-dominated extrapolation contributes about 28 %. Further refinements are a fifth external generator, NEUT, and a finer 44-cell comparison with the Ascencio *et al.* measurement (the maximal-common-grid comparison is in §7.8). A flux and muon-energy-scale posterior derived with MAT-MINERvA, independently of the published analysis, would determine the correlation ρ between the leading flux mode and the muon-energy-scale band (§4) entirely within this analysis.

Reusing the unbinned result. The public ATLAS charged-particle phase-space release [61] exemplifies unbinned distribution. The emerging `omnifold_publication` proposal [62] emphasizes weight-family metadata, normalization, row alignment, and checksums. A MINERvA release should likewise specify event identity, estimator and uncertainty-family definitions, units, support, normalization, and a reproducible projection example. App. G specifies this release for our analysis, including which of its objects already have fixed identities and which are still to be produced. A local release candidate exists, with checksums and a verifier that recomputes and checks its results;

no release has been tagged or deposited. Rebuilding this note's PDFs assembles documents and reproduces no scientific result (App. G). New projections require their own applicability checks; nominal weights alone cannot certify arbitrary downstream inference (App. B.4). The release design and its public example have no bearing on the choice of the five-axis covariance of the configuration-(a) central value (§9.6).

Acknowledgments

[funding statement pending]

This analysis was carried out on the Perlmutter system at the National Energy Research Scientific Computing Center (NERSC), a U.S. Department of Energy Office of Science User Facility, under project m3246. It builds on the MINERvA Analysis Toolkit (MAT) and the MINERvA-101 cross-section framework, and we thank their authors. We are grateful to the MINERvA Collaboration for the public release of the medium-energy CC-inclusive data product on which the reproduction rests.

We thank Callum Wilkinson for connecting us with this dataset and for many helpful discussions. We thank Kevin McFarland for suggesting a detector-response-mismatch closure targeting correlations among the reconstructed hadronic observables.

The analysis code, statistical procedures, computing campaigns, independent recomputations and reviews, figures, and manuscript drafts were produced with substantial assistance from AI language-model tools, used through the Claude Code and OpenAI Codex command-line agents: Anthropic Claude (Opus 4.7, 4.8, 5, and 5.5; Fable 5 and 5.1) and OpenAI GPT models (GPT-5.5, GPT-5.6 Sol and Luna, GPT-6 Astra, and GPT-6.1), including those used for independent review. The authors directed this work through written specifications fixed before each study. Results were accepted only after verification against committed evidence: predeclared criteria, independent recomputation by separately written code (for example, the joint-test evaluation and the lost-seed recovery), and independent read-only reviews. AI tools were also used for methodological discussion. The figures were drawn by plotting code from the analysis outputs; no image was generated or edited by a generative-AI image tool. AI tools are not authors. The authors take responsibility for the content.

Guide to the appendices

The appendices fall into three groups. The first gives methods and context for the body: statistical methods and the 2D comparison (App. A) and those for the multidimensional results and inference (App. B); unbinned background subtraction (App. C); the protocol for a detector-response-mismatch closure, which rescales the reconstructed available energy of the pseudo-data while OmniFold keeps its nominal response (App. D); and the survey of MINERvA measurement dimensionality (App. E). The second, on future work, covers point-cloud (PET) representations (App. F). The third describes the result package and what it reproduces (App. G) and the code layout (App. H); it is reference material and is not needed to follow the physics results. The provenance of the records the note cites, the fixed identities and audit record of the five-axis covariance of the configuration-(a) central value, the test-suite discipline, and superseded values and withdrawn results are kept in the analysis repository's records document, Ref. [37]; each citation names the entry's number or heading.

Labels used in the repository. The body describes each object in words: the learner backends (§3.2), the background-subtraction modes (§3.1, App. C), the two centerings of the five-axis covariance (§9.6), the re-selected universes (§4.7) and the two universe sweeps (§4.4). Where a repository label is

needed to find the object, it is given once in parentheses at first use. The learner backends are labelled `lgbm`, `exact` and `hist`: LightGBM (the quoted central values and their uncertainty ensembles; also the joint-test estimator), the scikit-learn exact-split estimator (the exact-split cross-check) and the scikit-learn histogram estimator (§3.2). Products described as flux-corrected are those repaired for an audit finding that the N-dimensional code divided every flux universe by the central-value flux (App. C). The calls behind Eq. (10), §A.2.3 and §A.1 (default cutoff) are `np.cov(Xrep, rowvar=False, ddof=1)`, `np.random.default_rng().poisson(1.0)` and `np.linalg.pinv`; the earlier MAT check without the normalization band (§A.2.6) used the option `--add-norm 0`.

A Statistical methods and the 2D comparison

This appendix gives the statistical methods of this note's measurements, centered on the 2D comparison: covariance inversion; the training-seed, bootstrap and universe estimators, with the Poisson limit and the $\pm1\sigma$ pair covariance; their MAT-conformant construction and combination; the χ^2, pull and shape-only conventions; the rank and conditioning of the covariances; the anatomy of the tension in the 2D comparison; the closure procedures; and a summary of the 2D comparison numbers. It covers analysis-specific conventions, not general statistics. The χ^2 convention used throughout is Eq. (17) in §A.3.2. The N-D declarations, scalar-5D reductions, extended-fiducial fractions and unfolded-event inference are in App. B; the 2D fixed-truth coverage test is in §5.2 and the same-ensemble pull diagnostic in §5.3.

A.1 Numerical conventions: inverting a covariance (Cholesky, ridge jitter, pseudo-inverse)

Equation (17) requires inverting $\mathbf{C}$.

Cholesky and ridge jitter. $\mathbf{C}$ is positive-definite (PD) iff its Cholesky factorization $\mathbf{C} = \mathbf{L}\mathbf{L}^\top$ exists with lower-triangular $\mathbf{L}$; then Eq. (17) is well-posed. Numerical noise that leaves a PD matrix barely non-PD is repaired with an isotropic ridge $\varepsilon\,\mathbf{I}$, set to

$$\varepsilon \;=\; 10^{-12}\,\frac{\mathrm{Tr}\mathbf{C}}{n}, \tag{7}$$

twelve orders below the mean diagonal, far below any real signal in $\mathbf{C}$ and safely above floating-point round-off.

Rank deficiency at $N \le n$. A sample covariance from $N \le n$ replicas in n bins has rank at most $N-1$. The 300-replica bootstrap ($N = 300 > 205$) is full-rank. The rank of the systematic block is limited at any universe count (§A.4).

Pseudo-inverse. For the χ^2 against the paper covariance, the comparison replaces $\mathbf{C}^{-1}$ with the Moore–Penrose pseudo-inverse computed via SVD, $\mathbf{C} = \mathbf{U}\Sigma\mathbf{V}^\top$ with $\Sigma = \mathrm{diag}(\sigma_1, \ldots, \sigma_n)$:

$$\mathbf{C}^{+} \;=\; \mathbf{V}\,\Sigma^{+}\,\mathbf{U}^\top, \qquad \Sigma^{+}_{ii} \;=\; \begin{cases} 1/\sigma_i & \sigma_i > \tau, \\ 0 & \text{otherwise.} \end{cases} \tag{8}$$

The cutoff is $\tau = 10^{-15}\,\sigma_{\max}$, not the $n\,\sigma_{\max}\varepsilon_{\mathrm{machine}}$ convention. Modes below the cutoff receive inverse weight 0 rather than ∞, keeping the quadratic form stable for zero or near-zero eigenvalues.

A.2 Variance-source procedures

Table 30 summarizes the four variance sources, their MAT reference and combination. Definitions and otherwise unreported values follow; §4 describes the measurement's block construction, and Table 7 gives the 2D budget.

Source	What it estimates	Procedure	Ensemble	Value / where reported
Training-seed variation (ML)	Variance from random choices inside the GBDT optimizer at matched inputs (the same 20 bootstrap replicas at each setting; the "ML noise floor")	Full 5-iteration ME FHC LightGBM unfolds (the backend of the quoted central value and of the bootstrap and universe ensembles) on the same 20 bootstrap replicas at each of five settings of the GBDT seed triplet (step-1 classifier, step-2 classifier, miss regressor); sample covariance of the five per-setting means	5×20	median 0.048%; Table 32
Poisson bootstrap (stat)	Data- and MC-statistical variance, propagated coherently through the full pipeline	Per-event Poisson(1) factors from independent data (seeded from k), signal-MC (seeded from $k + 10^7$) and background-MC (seeded from $k + 2 \cdot 10^7$) sub-RNGs; one MC factor on both reco and truth copies of each event; 5-iteration re-unfold with the GBDT seed pinned at the central value's (42)	$N = 300$	median 0.676%; Table 32
Universes (syst)	Flux, detector and interaction-model uncertainty, by re-unfolding under reweighted MC	The event loop writes truth and reco weights per (band B, universe u); each (B, u) is re-unfolded with a fixed seed against a matched CV job (same event-loop output file, backend, seed, iteration count and flux, so ML noise cancels); MAT per-band covariance, Eq. (11), plus a rank-1 normalization	Quoted sweep: 187 universes, 44 bands; band sum 43 bands + norm	median 5.935%; Table 31
Run period	Inconsistency between the run periods' results	Separate seed-42 unfolds of three run periods; spread of each period's unfolded-to-simulated ratio about the exposure-weighted mean, applied to the central value (§4.5)	3 periods	median 2.087%; Table 32
MAT reference	How MINERvA builds its published covariance	Read from the MAT source (`MnvVertErrorBand::CalcCovMx`, `MnvH1D::GetTotalErrorMatrix`); our universe block compared with it element by element, Flux with MAT's universe weights	the 43 bands of the universe block	largest difference 1.3×10^{-16} of the largest element
Block sum	Total uncertainty covariance	Element-wise sum of the four blocks, Eq. (16)	—	median 6.414%; Table 32

Table 30: Variance-source procedures for the 2D measurement (205 reported bins).

A.2.1 Training-seed variation

The training-seed block is built from five settings of the GBDT seed triplet, each unfolded on the same 20 bootstrap replicas (at the central value's seed, 42, the first 20 of the 300 of §A.2.2). With $\bar{\mathbf{X}}^{(s)}$ the mean of the 20 replica 205-vectors at setting s and $\bar{\bar{\mathbf{X}}}$ the mean over the five settings,

$$\mathbf{C}^{\mathrm{ML}} = \frac{1}{4} \sum_{s} \left(\bar{\mathbf{X}}^{(s)} - \bar{\bar{\mathbf{X}}}\right)\left(\bar{\mathbf{X}}^{(s)} - \bar{\bar{\mathbf{X}}}\right)^{\top}, \tag{9}$$

a sample covariance of rank 4 that includes the bootstrap × seed interaction at its 1/20 share. The relative spread of its total cross section is $\boxed{0.0010\%}$, the per-bin median relative spread $\boxed{0.048\%}$, and $\sqrt{\mathrm{Tr}\mathbf{C}^{\mathrm{ML}}} = 1.351 \times 10^{-41}$.

A.2.2 Poisson bootstrap

A coherent bootstrap carries each data factor through step-1 classifier training and final cross-section construction. Each MC factor is shared by the reco-tree and truth-tree copies, preserving migration consistency. The three sub-RNGs (data, signal MC, background MC; Table 30) make the sources uncorrelated by construction, and pinning the GBDT seed isolates the bootstrap from the ML noise of §A.2.1. A Poisson(1) factor can be zero, so some events are dropped from each replica, mimicking a resample with replacement; Poisson(1) is the large-N limit of the multinomial counts (§A.2.3). Each replica's 14×16 histogram becomes a 205-vector $\mathbf{X}^{(k)}$ of reported bins, row-major over $(p_T, p_\parallel)$, with unbiased sample covariance

$$\left(\mathbf{C}^{\mathrm{boot}}\right)_{ij} = \frac{1}{N-1}\sum_{k=1}^{N}\left(X_i^{(k)} - \bar{X}_i\right)\left(X_j^{(k)} - \bar{X}_j\right), \qquad \bar{X}_i = \frac{1}{N}\sum_k X_i^{(k)}, \tag{10}$$

checked with the Cholesky factorization and ridge jitter of §A.1. The quoted $N = 300$ block varies only the bootstrap seed and carries no training-seed variation.

For these replicas (205 bins, ME FHC LightGBM, seed 42) the relative standard deviation of the total cross section is $\boxed{0.061\%}$, the per-bin median relative spread $\boxed{0.676\%}$, and the per-bin p_{84} 1.602%. With 300 replicas the per-bin σ has a relative standard error of about 4.1% by the Gaussian formula $1/\sqrt{2(N-1)}$; $N = 300$ is chosen for full rank.

A.2.3 Multinomial → independent Poisson bootstrap limit

Drawing N events with replacement from an N-event sample gives counts $(n_1, \ldots, n_N) \sim \mathrm{Multinomial}(N, \mathbf{p})$ with uniform $p_i = 1/N$ and $\sum_i n_i = N$. Each marginal $n_i \sim \mathrm{Binomial}(N, 1/N)$ has mean 1, variance $(N-1)/N \to 1$, and converges to Poisson(1) as N grows (many trials, small success probability, fixed expected count). The $\sum n_i = N$ constraint induces $\mathcal{O}(1/N)$ correlations that vanish at large N. The multinomial is an independent Poisson process conditioned on N counts; removing that constraint gives independent Poisson(1) variables (Poissonization). One Poisson(1) factor is drawn per event. For $N \gtrsim 10^6$, our regime, the relative total-count error is $\mathcal{O}(1/\sqrt{N})$, far smaller than the per-bin Poisson variance.

A.2.4 Universes

The universe sweep used by the quoted budget (the July sweep) has 187 universes across 44 bands, dominated by the 100-universe PPFX flux band (hadron-production sampling). The quoted band sum drops one band (Rvp1pi) and holds 43 bands plus the normalization. Almost all other bands are $\pm 1\sigma$ pairs ($N_u = 2$). They include the MINOS and lateral MINERvA muon energy scales, the MINOS muon-match efficiency, the GENIE parameters `MaRES`, `MvRES` and `MaCCQE`, and the lateral GEANT FSI responses for neutrons, pions and protons. Where selection migration matters, the five kinematic bands (the two beam-angle components, the muon resolution and the two muon energy scales) require re-selected event loops, run with one universe active at a time (weight-only bands: §4.7). The quoted universe block takes these five bands from re-unfolds of re-selected universes.

MINERvA's MAT framework (`PlotUtils/MnvVertErrorBand::CalcCovMx`, §A.2.6) uses the mean-centered, biased sample covariance of the 205-vectors $\mathbf{X}^{(B,u)}$, referenced to the universe mean with normalization $1/N_u$,

$$\boxed{\mathbf{C}^B \;=\; \frac{1}{N_u}\sum_{u=1}^{N_u}\big(\mathbf{X}^{(B,u)} - \langle\mathbf{X}\rangle_B\big)\big(\mathbf{X}^{(B,u)} - \langle\mathbf{X}\rangle_B\big)^\top, \qquad \langle\mathbf{X}\rangle_B = \frac{1}{N_u}\sum_u \mathbf{X}^{(B,u)}.} \tag{11}$$

uniformly for all $N_u \geq 2$, with no special $\pm1\sigma$ pair formula. At $N_u = 2$ it reduces to rank-1 $\frac{1}{4}(\mathbf{X}_+ - \mathbf{X}_-)(\mathbf{X}_+ - \mathbf{X}_-)^\top$. Mean-centering removes the symmetric component, correctly giving near-zero variance for a one-sided or asymmetric pair where naive CV-centering would give rank 2 (§A.2.5). The 1.4% target-nucleon normalization uncertainty (Aliaga NIM, arXiv:1305.5199) lies outside `MnvVertErrorBand` and is added as a fully-correlated rank-1 outer product. Bands are assumed independent, with different physics sources and RNGs,

$$\mathbf{C}^{\text{norm}} \;=\; (\sigma_N\,\mathbf{X}^{\text{CV}})(\sigma_N\,\mathbf{X}^{\text{CV}})^\top, \qquad \sigma_N = 0.014, \tag{12}$$

$$\mathbf{C}^{\text{syst}} \;=\; \sum_B \mathbf{C}^B \;+\; \mathbf{C}^{\text{norm}}. \tag{13}$$

Independence is a modeling assumption with one known correlated exception, the Bashyal flux ↔ Muon_Energy_MINOS joint block (§A.4).

Equation (13) omits a larger correction. The band sum is linear, but OmniFold responds nonlinearly to input perturbations, so joint re-unfolding produces a cross term beyond any pairwise correlation. A unified throw measures it through N re-unfolds, simultaneously perturbing every reweightable band and the flux multisim per event ($w_{\text{CV}}\prod_B \rho_B^{g_B}$, $g_B \sim \mathcal{N}(0,1)$), then re-extracting the cross section. The corrected construction is in §4.5; its use in the five-axis covariance of the configuration-(a) central value is in §9.6 (Eq. (5)).

Band	Per-bin median rel. σ
MINOS muon energy scale (Muon_Energy_MINOS)	2.358%
MINERvA muon energy scale (Muon_Energy_MINERvA)	1.435%
MINOS muon-match efficiency (MinosEfficiency)	1.505%
Flux (PPFX, $N_u = 100$, constraint-weighted)	3.712%
MaRES	0.627%
MvRES	0.362%
MaCCQE	0.336%
Systematic total (43 bands + norm)	$\boxed{5.935\%}$ (median) $\sqrt{\text{Tr}\mathbf{C}} = 2.689 \times 10^{-39}$

Table 31: Top contributing bands of the quoted universe block (the quoted universe sweep, with the lateral bands from re-unfolds of re-selected universes and Rvp1pi dropped): per-bin median relative σ_i/X_i^{CV} over the 205 reported bins. Flux carries MAT's ν–e and inverse-muon-decay constraint weights, and Muon_Energy_MINOS is divided by its own flux.

Grouped by category as in Fig. 4, which follows Figs. 6–7 of Ref. [3], the medians are: Muon reconstruction 3.437%, Models 1.401%, Flux 3.712%, Hadronic response 0.743%, Statistical 0.676%, Normalization 1.400%, Training seed 0.048%, Run period 2.087%. (MINOS+MINERvA) and the MINOS-efficiency band dominate; these are the bands that the Ruterbories paper identifies as flux-correlated (§A.4).

A.2.5 The $\pm1\sigma$ pair covariance: mean-centered (MAT-conformant)

For two universes $u = 0, 1$ at $\pm1\sigma$, with vectors $\mathbf{X}_0$, $\mathbf{X}_1$ and mean $\langle\mathbf{X}\rangle = \frac{1}{2}(\mathbf{X}_0 + \mathbf{X}_1)$, Eq. (11) at $N_u = 2$ gives

$$\mathbf{C}^B_{\text{mean}} \;=\; \tfrac{1}{2}\sum_{u=0,1}(\mathbf{X}_u - \langle\mathbf{X}\rangle)(\mathbf{X}_u - \langle\mathbf{X}\rangle)^\top \;=\; \tfrac{1}{4}(\mathbf{X}_0 - \mathbf{X}_1)(\mathbf{X}_0 - \mathbf{X}_1)^\top, \tag{14}$$

a rank-1 outer product of the half-difference. A common offset from X_{CV} contributes nothing. Naive CV-centering instead uses $\boldsymbol{\Delta}_u = \mathbf{X}_u - \mathbf{X}_{\text{CV}}$, $\boldsymbol{\Delta}_0 = \mathbf{m} + \mathbf{a}$, $\boldsymbol{\Delta}_1 = \mathbf{m} - \mathbf{a}$,

$$\mathbf{C}^B_{\text{CV-centered}} \;=\; \tfrac{1}{2}\big(\boldsymbol{\Delta}_0\boldsymbol{\Delta}_0^\top + \boldsymbol{\Delta}_1\boldsymbol{\Delta}_1^\top\big) \;=\; \mathbf{m}\mathbf{m}^\top + \mathbf{a}\mathbf{a}^\top. \tag{15}$$

The estimators agree only for CV-symmetric pairs ($\mathbf{m} = 0$), both reducing to $\mathbf{a}\mathbf{a}^\top$. For asymmetric or one-sided pairs, CV-centering adds the rank-1 $\mathbf{m}\mathbf{m}^\top$ common-offset direction. Every per-band block C_b here is mean-centered to match MAT (§A.2.6); this matters for BeamAngle, MuonResolution and the one-sided NC band.

Scope of mean-centering. Mean-centering applies to the $\pm1\sigma$ band estimator only. The five-axis covariance of the configuration-(a) central value deliberately retains the CV-centered term: the unified-throw covariance diagonal that enters its inflation is centered on the central value (§9.6, Eqs. (5) and (6)).

A.2.6 How MINERvA builds its published covariance

MAT's C++ framework defines the covariance construction that the published paper leaves unspecified. It is the mean-centered $1/N$ per-band sample covariance and element-wise band sum used here.

Ruterbories 2021, Section V. The paper evaluates three uncertainty categories (flux, detector response and the neutrino-interaction model) by "re-extracting the cross section using modified simulations", i.e. the universe method, and gives no covariance formula. It reports a flux uncertainty below 4% across most of phase space as a result of the ν-e-scattering constraint (Valencia, arXiv:1906.00111); a 1.4% target-nucleon normalization; an in-situ test-beam hadron response with per-particle GEANT inelastic variations; and a 1% muon energy scale constrained by the low-recoil fit of Bashyal (JINST 16 P08068). It states that *"the fit causes the flux and muon energy scale uncertainties to be correlated and that correlation is propagated to the final result"*; this is the only correlation between bands that the paper specifies. The only covariance in its equations is V in the model-test χ^2 (Eq. 1, "the measurement covariance matrix"), with 205 degrees of freedom (Table I); the paper also gives a log-normal χ^2 variant, motivated by the highly correlated flux normalization (Peelle's Pertinent Puzzle). The paper itself gives no per-band covariance construction, no $\pm1\sigma$ pair formula and no band-independence or block-sum statement. MAT's code supplies the per-band construction and the independent band sum, but no separate pair formula (below). Nor does it state the rank, condition number and regularization of V, or any bootstrap or coverage validation (it uses D'Agostini iterative unfolding).

MAT per band and in total. `MnvVertErrorBand::CalcCovMx` implements Eq. (11), referenced to the universe mean $\langle x\rangle_u$ rather than X_{CV}, normalized by biased $1/N$ rather than Bessel's $1/(N-1)$ (a 1% offset for PPFX, $N = 100$), with no special case for $N = 2$. The conventional CV-centered pair

formula ("MINERvA-101 $\pm1\sigma$ pair"), $\frac{1}{2}(\mathbf{\Delta}_+\mathbf{\Delta}_+^\top + \mathbf{\Delta}_-\mathbf{\Delta}_-^\top)$, agrees with it only for CV-symmetric shifts. The quoted universe block, filled into a `PlotUtils::MnvH1D` with MAT's universe weights on Flux, agrees with `MnvH1D::GetTotalErrorMatrix` to 1.3×10^{-16} of its largest element; the quoted Flux band carries MAT's ν–e and inverse-muon-decay constraint weights. An earlier check compared `MnvH1D::GetTotalErrorMatrix` with our band sum without the normalization band, over 44 bands / 187 universes, and gave a maximum difference of 5.5×10^{-17} of the largest element (machine ε) and $\sqrt{\mathrm{Tr}\mathbf{C}}$ matching to seven digits. That check could not detect that the Flux band omitted those constraint weights, because it filled the `MnvH1D` without MAT's per-universe weights (`SetUnivWgts`), so both sides were unweighted; it also ran on a sweep from before the flux fix, not on a quoted product.

`MnvH1D::GetTotalErrorMatrix` sums bands and the statistical matrix element-wise, assuming band independence. Cross-band correlations exist only if encoded in a combined band, such as the Bashyal joint fit handled outside this loop. MAT applies no ridge, shrinkage, SVD truncation or rank-deficiency handling to the published total covariance. The statistical block makes V full-rank; its systematic-only rank is 120, below the 138 of our universe block (eigenvalues above $10^{-12}\lambda_{\max}$; §A.4).

Upstream references that set each band's $\pm1\sigma$ amplitude.

- Flux / PPFX (100 universes), Aliaga, PRD 94 092005 (2016), arXiv:1607.00704;
- ν-e flux constraint ($< 4\%$), Valencia, PRD 100 092001 (2019), arXiv:1906.00111;
- muon-E scale $\leftrightarrow$ flux fit, Bashyal, JINST 16 (2021) P08068;
- test-beam / GEANT hadron response, Aliaga, NIM A 789 28 (2015), arXiv:1501.06431;
- GENIE reweighting parameters, Andreopoulos, NIM A 614 87 (2010), and the GENIE manual arXiv:1510.05494;
- 1.4% normalization (detector mass), Aliaga, NIM A 743 130 (2014), arXiv:1305.5199.

A.2.7 Block-sum combination

The bootstrap varies events through distinct data, signal-MC and background-MC sub-RNGs; universes vary model weights at a pinned GBDT seed; the training-seed block varies the GBDT seed over five settings, each on the same 20 bootstrap replicas (at seed 42, replicas of the statistical block); the run-period block compares separate unfolds of three run periods. The bootstrap and the universes use the central value's LightGBM seed (42), and the training-seed settings include it. The block sum is

$$\mathbf{C}^{\mathrm{tot}} = \mathbf{C}^{\mathrm{ML}} + \mathbf{C}^{\mathrm{stat}} + \mathbf{C}^{\mathrm{syst}} + \mathbf{C}^{\mathrm{PL}}. \tag{16}$$

The comparison adds OmniFold covariances to the paper's `TotalCovariance` by this rule. $\mathbf{C}^{\mathrm{ML}}$ is the smallest block (Table 32); it is included in the quoted total but numerically negligible.

The combined $\sqrt{\mathrm{Tr}\mathbf{C}}$ is the quadrature sum of the component square-root traces. The per-bin median 6.414% is not the quadrature sum of the component medians; both combined entries are evaluated on the summed covariance. The relative entries divide by the quoted central value, the LightGBM seed-42 central value of the universe sweep; dividing the combined covariance by the exact-split cross-check instead gives a combined median of 6.462 %. The universe, statistical and training-seed blocks are LightGBM ensembles, the first two at the central value's training seed and the third over five settings that include it (§3.2). The universe and statistical blocks together give 5.982%, again evaluated on their sum rather than combined from the printed medians.

Tables 31 and 32 give the blocks of the quoted 2D covariance (§4).

Component	$\sqrt{\mathrm{Tr}\mathbf{C}}$	Per-bin median rel. σ
ML noise (LightGBM training-seed settings, $n = 5$)	1.351×10^{-41}	0.048%
Statistical (Poisson bootstrap, $N = 300$)	1.886×10^{-40}	0.676%
Systematic (universe sweep + norm)	2.689×10^{-39}	5.935%
Run period (three periods)	8.314×10^{-40}	2.087%
Combined (block sum)	2.821×10^{-39}	6.414%
Paper `TotalCov` (reference)	2.676×10^{-39}	6.85%

Table 32: Component traces and per-bin median relative spreads of the 2D uncertainty envelope (205 reported bins): the full-precision form of Table 7.

A.3 Comparing to the paper

These conventions and tables support the comparison and interpretation limits in §6.2.

A.3.1 The 205-bin selection

The paper reports $\mathrm{d}^2\sigma/(\mathrm{d}p_T\,\mathrm{d}p_\parallel)$ on 14 p_T bins × 16 $p_\parallel$ bins = 224 cells (edges in `minerva_paper_anc/bin_mapping.txt`). Of these, 205 have $x^{\mathrm{paper}} > 0$ in `data_result_ptpl_2D_minerva_inclusive_6GeV.txt`. The paper suppresses the remaining 19 through an efficiency cut; every χ^2 and pull excludes them on both sides.

A secondary strict-interior mask, used only as a sensitivity check, keeps cells with $p_T^{\mathrm{high}}/p_\parallel^{\mathrm{low}} \leq \tan 20°$, removing the 20 reported cells that straddle the $\theta_\mu < 20°$ kinematic cut. The 185-bin check tests whether the main tension comes from the boundary. The main metric throughout uses the 205-bin set.

A.3.2 Covariance-aware χ^2/ndf

With $\boldsymbol{\Delta} \equiv \mathbf{x}^{\mathrm{ours}} - \mathbf{x}^{\mathrm{paper}}$ on the 205-bin set and $\mathbf{C}$ the chosen covariance,

$$\chi^2 \;=\; \boldsymbol{\Delta}^\top \mathbf{C}^+ \boldsymbol{\Delta}, \qquad \mathrm{ndf} \;=\; n_{\mathrm{reported}} = 205, \qquad \chi^2/\mathrm{ndf} \;=\; \chi^2/205. \tag{17}$$

The comparison applies the pseudo-inverse of §A.1 to the 205×205 block and computes the quadratic form. Options add OmniFold covariances via Eq. (16) or subtract the paper's stat block to avoid counting it alongside the bootstrap (§A.4).

Dimension-conditional ndf. Setting ndf = n_{reported} requires nearly that many well-determined directions. The choice rests on the 205-bin 2D truncation scan rather than being assumed: the scan rises smoothly with the retained rank r, 0.39 (r=50) → 1.04 (100) → 1.76 (180) → 2.27 (205) (exact-split cross-check: 0.69 → 2.35 → 3.30 → 3.66), with no plateau or late jump, indicating an effective rank comparable to the dimension. This evidence is specific to 2D and does not transfer to N-D covariances (§B.1).

Worked example: comparison to paper (quoted numbers).

Metric (205 bins)	Paper	OmniFold (combined)
Width-weighted total σ, $\sqrt{\mathbf{w}^\top \mathbf{C}\, \mathbf{w}}/(\mathbf{w}^\top \mathbf{x})$	4.61%	4.76%
Per-bin median relative σ	6.85%	6.414%
χ^2/ndf	2.27 (paper-only)	$\boxed{0.929}$ (paper+ours)
Log-normal χ^2/ndf	—	0.945

The log-normal χ^2 uses $\delta_i = \ln x_i^{\text{ours}} - \ln x_i^{\text{paper}}$ and $\mathbf{C}_{ij}^{\ln} = \left(\texttt{TotalCov} + \mathbf{C}^{\text{tot}}\right)_{ij}/(x_i^{\text{paper}} x_j^{\text{paper}})$; both covariances are scaled by the paper's values. This is the definition used for the quoted value: of five definitions recomputed, only this one reproduces 0.945. The others give 1.197 (our covariance scaled by our own values), 0.958 (both scaled by ours), 0.963 (scaled by the geometric mean) and 1.015 ($\ln(1 + \mathbf{C}/(x_i x_j))$ in place of the ratio). Adding the OmniFold error budget lowers the summed-covariance distance from 2.27 to 0.929 per bin. It is a descriptive distance, not a validation result, and three limits apply to every use of it.

(i) Missing cross-covariance prevents calibration. For estimates x and y from the same data,

$$\text{Cov}(x - y) \;=\; C_x + C_y - C_{xy} - C_{yx}, \tag{18}$$

and the published IBU-era and OmniFold results share MINERvA data and flux, detector and GENIE systematics. Without C_{xy}, the summed $C_x + C_y$ double-counts shared systematics and omits their correlation. Its net error has unknown size and sign. No value of the statistic, including 1.5 or 1, can be converted to a goodness-of-fit probability or a p-value.

(ii) Near-one realizations are not evidence of correct uncertainties. Under calibrated assumptions, χ^2/ndf has expectation 1 and spread of order $\sqrt{2/\text{ndf}}$, about 0.10 for ndf = 205. One realization near 1, such as the quoted 0.929, is not evidence of correct uncertainties. Nor is the exact-split cross-check's 1.47, computed with an earlier covariance built before the flux-band repair, evidence against them on its own (Ref. [37], superseded values).

(iii) It does not address coverage (§4.6, §5.2). The covariance and the central value share the LightGBM estimator and, in the universe and statistical blocks, its training seed (§3.2). Agreement claims use per-bin standardized differences and ratios (§A.3.3); this scalar is only a distance and ratio to the paper. The log-normal value 0.945 depends on the definition given with the worked example.

External statistical floor of the *quoted* combined χ^2. The quoted 0.929 inverts the *sum* of the paper's covariance and ours, not our blocks alone. That sum contains the paper's `StatOnlyCov`, which has rank 205 and is near-diagonal (§A.4), so it places a variance floor under *every* eigendirection of the combined matrix. It is the larger of the two statistical contributions. Our statistical block is 0.407 of it in *square-root* trace, and the square of that factor in *trace* (operands in §A.4). Each ratio states which power of the trace it uses, because the two ratios describe the same comparison and differ by more than a factor of two. An external, published block, rather than any ensemble of ours, therefore sets the small-eigenvalue directions on which a finite-N debiasing factor would act. This bounds the quoted number's exposure to declaration (v) of §B.1.

Trace share alone cannot justify this protection: trace weights eigenvalues by λ, precision by $1/\lambda$, so even a negligible-trace block can dominate an inverse. Without the external floor, our finite-N blocks fill $\mathbf{C}^{\text{syst}}$'s null space (rank 138 → 205, eigenvalues above $10^{-12}\lambda_{\max}$) and, inverted directly, yield χ^2/ndf = 22.29 (§A.4). Protection depends on the source of the smallest retained eigenvalues.

Edge-bin sensitivity ($p_\parallel$-min cut). The comparison recomputes χ^2/ndf on shrinking subsets using only the paper covariance, with no OmniFold addition, for the exact-split cross-check (not

recomputed for the LightGBM central value),

$p_\parallel \geq$ (GeV/c)	N_{bins}	χ^2	χ^2/ndf	median ratio
1.5	205	750.49	3.661	1.0064
2.0	196	685.36	3.497	1.0059
2.5	186	545.87	2.935	1.0059
3.0	175	474.84	2.713	1.0059
3.5	163	452.95	2.779	1.0073
4.0	151	428.72	2.839	1.0068

The residual is concentrated in the low-$p_\parallel$, near-edge cells; the high-$p_\parallel$ tails carry sub-2% shape disagreement. This is a robustness check, not a separate method.

A.3.3 Pull distribution

For each of the 205 reported bins,

$$\text{pull}_i \;=\; \frac{x_i^{\text{ours}} - x_i^{\text{paper}}}{\sqrt{C_{ii}^{\text{tot}}}}, \qquad \mathbb{E}[\text{pull}_i] = 0, \quad \text{Var}[\text{pull}_i] = 1 \text{ under the null}, \tag{19}$$

summarized by mean, RMS, a 14×16 heatmap and histogram. The combined covariance (paper TotalCov + the quoted 2D covariance of §4) gives mean $\boxed{0.034}$, centered on zero with no global offset, and RMS $\boxed{0.388}$. For the exact-split cross-check, on the earlier covariance built before the flux-band repair, they are 0.051 and 0.405. RMS < 1 is descriptive and does not establish conservatism: the summed covariance is wrong by an unknown amount (Eq. (18)).

The paper covariance alone gives mean 0.060 (similar, no bias) and RMS 0.535, slightly wider before our uncertainties are added.

A.3.4 Shape-only χ^2 with Jacobian propagation

This tests whether the *shape* of the $(p_T, p_\parallel)$ cross section matches the paper independently of absolute normalization, which depends on the flux integral, target nucleon count, and POT (each can drift by $\mathcal{O}(\%)$); the shape is a pure unfolding diagnostic. For bin widths $w_j = (\Delta p_T)_j\,(\Delta p_\parallel)_j$ and total $T(\mathbf{x}) = \sum_{j\in\text{mask}} w_j x_j$, the unit-area shape is $s_i(\mathbf{x}) = x_i/T(\mathbf{x})$, satisfying $\sum_i w_i s_i = 1$, with

$$\frac{\partial s_i}{\partial x_j} \;=\; \frac{1}{T}\big(\delta_{ij} - s_i\, w_j\big), \qquad \mathbf{J} \;=\; \frac{1}{T}\big(\mathbf{I} - \mathbf{s}\,\mathbf{w}^\top\big), \qquad \mathbf{C}^{\text{shape}} \;=\; \mathbf{J}\,\mathbf{C}^{\text{paper}}\,\mathbf{J}^\top. \tag{20}$$

The unit-area constraint removes one degree of freedom: $\mathbf{w}^\top\mathbf{J} = \mathbf{0}^\top$. The "all-bins-up" direction is $\mathbf{J}$'s kernel, leaving $\mathbf{C}^{\text{shape}}$ one zero eigenvalue and rank $n-1$. Thus $\chi^2_{\text{shape}} = (\mathbf{s}^{\text{ours}} - \mathbf{s}^{\text{paper}})^\top (\mathbf{C}^{\text{shape}})^{+} (\mathbf{s}^{\text{ours}} - \mathbf{s}^{\text{paper}})$ drops that direction, with ndf $= n_{\text{bins}} - 1 = 204$ for 205 bins. For the exact-split cross-check (not recomputed for the LightGBM central value), with the paper's `TotalCovariance` (Jacobian-propagated) the shape χ^2/ndf on 205 bins is $\boxed{3.596}$: essentially the absolute-normalization 3.661 *minus* the contribution of overall scale. The total-cross-section ratio $\sigma_{\text{tot}}^{\text{ours}}/\sigma_{\text{tot}}^{\text{paper}} = 1.0111$ is a 1% scale difference. The remaining residual comes from $\sim 1\%$ shape disagreement (median bin ratio 1.0064; 94% of bins within 10% of the paper).

A.4 Ill-conditioning and rank deficiency

The obstruction. A data-to-model χ^2/ndf with this analysis's covariance alone, χ^2_{ours}, requires inverting $\mathbf{C}^{\text{ours}} = \mathbf{C}^{\text{syst}} + \mathbf{C}^{\text{boot}} + \mathbf{C}^{\text{ML}} + \mathbf{C}^{\text{PL}}$ instead of the paper's V. Its universe block is rank-deficient,

$$\text{rank}(\mathbf{C}^{\text{syst}}_{\text{sample}}) = 138/205, \qquad \text{cond}(\mathbf{C}^{\text{syst}}) = 9.78 \times 10^{11}, \tag{21}$$

counting eigenvalues of $(\mathbf{C} + \mathbf{C}^{\top})/2$ above $10^{-12}\lambda_{\max}$, with cond the ratio of $\lambda_{\max}$ to the smallest retained eigenvalue (124 and 9.45×10^9 above $10^{-10}\lambda_{\max}$). Every rank and condition number of our blocks quoted below uses this definition. Band count, not universe count, limits the rank: Eq. (11) gives each $\pm 1\sigma$ pair rank 1 and the 100-universe PPFX band rank ≤ 99. PPFX is effectively rank-1 (§4.1). The 43 bands cannot span 205 bins. The deficit does not come from missing bands. The paper's systematic-only covariance (`TotalCov` − `StatOnlyCov`) has rank 120, below our 138. We use the same `GetStandardSystematics` and propagate at least as many independent systematic directions. The statistical block (`StatOnlyCov`, rank 205, near-diagonal) alone makes the paper's V full-rank (rank = 205, with 204 eigenvalues above $10^{-12}\lambda_{\max}$; cond = 1.5×10^{12}). Our 300-replica bootstrap similarly gives rank($\mathbf{C}^{\text{syst}} + \mathbf{C}^{\text{boot}}$) = 205 (eigenvalues above $10^{-12}\lambda_{\max}$; cond = 7.91×10^{11} by the definition above).

The real obstruction is statistical, not rank. Even at full rank (205) a defensible single χ^2_{ours} is out of reach, for a *statistical-precision* reason. The bootstrap block is smaller in *global scale*: its square-root trace is 0.407 of the paper's ($\sqrt{\text{Tr}\mathbf{C}^{\text{boot}}} = 1.886 \times 10^{-40}$ vs $\sqrt{\text{Tr}\,\texttt{StatOnlyCov}} = 4.639 \times 10^{-40}$). The paper's full-rank, near-diagonal `StatOnlyCov` supplies a variance floor in every reported eigendirection; because the covariance structures differ, this should *not* be read as a uniform per-mode effect. The direct inverse of $\mathbf{C}^{\text{syst}} + \mathbf{C}^{\text{boot}}$ gives χ^2/ndf = 22.29 (21.77 for all four blocks) for the LightGBM central value, against the paper covariance's 2.27 with its larger stat floor, so a usable χ^2_{ours} depends on the *statistical* block, not on more systematic bands. The gap is *not* a missing MC-statistical term: our bootstrap already draws independent Poisson(1) multipliers on data, signal MC and background MC per replica (§A.2.2; Practical Guide 2507.09582 §5.1). The remaining candidates are a real difference in statistical efficiency between the unbinned reweighting and the published binned D'Agostini unfold, a difference in what the paper's `StatOnlyCov` encompasses, or a regularization effect in our own estimator that suppresses dispersion without improving information. A split bootstrap of the earlier rebuilt bootstrap (Ref. [37], superseded values: retired material) separated the first two from an MC-statistics omission but not from the third, and it has not been repeated for the quoted statistical block.

Data/MC split bootstrap. The data/MC split bootstrap was run on the earlier rebuilt bootstrap. For the quoted statistical block, $\sqrt{\text{Tr}\mathbf{C}^{\text{boot}}}/\sqrt{\text{Tr}\,\texttt{StatOnly}} = 0.407$ and the per-bin median ratio of the two σ is 0.425. The fixed-truth test (§5.2, §4.2) does not decide between the explanations above. The measured statement is the ratio; "higher statistical efficiency" is one of at least three explanations of it and is not established here.

The inverse of $\mathbf{C}^{\text{ours}}$ remains regularizer-dependent, so pulls and the scan lead the reporting. χ^2_{ours} uses the truncated-spectral pseudo-inverse (retaining $\lambda > 10^{-10}\lambda_{\max}$), the convention of the earlier 3D generator scan over all 1431 reported bins.

Regularization scans (earlier covariance; all cutoff-dependent). For the exact-split cross-check and an earlier $\mathbf{C}^{\text{ours}}$ over the 205 reported bins, neither recomputed for the quoted central value and covariance, no regularizer gives a stable single number:

Method	Setting	Result
Eigenvalue truncation	$\lambda/\lambda_{\max} > \{10^{-3}, 10^{-4}, 10^{-5}\}$	rank $\{19, 49, 82\}$; $\chi^2/205 = \{0.13, 1.12, 3.08\}$
Ridge $\mathbf{C} + \varepsilon\lambda_{\max}\mathbf{I}$	$\varepsilon = \{10^{-3}, 10^{-4}, 10^{-5}\}$	$\chi^2/205 = \{0.35, 1.38, 4.30\}$
Diagonal-target shrinkage	$\lambda = 0.05$	rank $178 \to 204$, cond 1.3×10^{11}, $\chi^2/205 = 3.29$

Each method spans roughly two decades of $\chi^2/205$ across a two-decade scan of its regularization parameter, with no defensible single cutoff. The useful conclusion is diagnostic, not a number: the naive inverse is dominated by tiny, poorly-determined modes, and in the retained high-variance modes the central-value difference is not large. We therefore report direct per-bin pulls (§A.3.3) and this scan rather than a single unqualified χ^2_{ours}.

Bootstrap/paper-stat double-count. The combined distance (§A.3.2) adds $\mathbf{C}^{\text{boot}}$ to `TotalCov`, double-counting the paper's included statistical block. Subtracting that block uses `TotalCov` − `StatOnlyCov` as baseline before our additions, changing χ^2/ndf $0.929 \to 6.72$. It over-corrects: our $\sqrt{\text{Tr}\mathbf{C}^{\text{boot}}} = 1.886 \times 10^{-40}$ is 0.407 of the paper's $\sqrt{\text{Tr}\,\texttt{StatOnlyCov}} = 4.639 \times 10^{-40}$, leaving no comparable replacement. The double-counted 0.929 is kept as the combined distance, a well-defined measure of what OmniFold's uncertainty estimate adds on top of the published V; the subtraction result is a secondary diagnostic.

Remaining refinements. The known band-independence exception, the Bashyal flux ↔ Muon_Energy_MINOS joint block, has been rederived ($\rho = 0.900$). It preserves systematic rank because its directions already lie in the universe covariance's span (§4.1). The quoted 2D covariance therefore remains the block sum, with the cross term as a diagnostic. A fitted $(\Phi_{\text{norm}}, \Phi_{\text{shape}}, \text{MuonE}_{\text{scale}})$ posterior from MAT-MINERvA could supply a paper-independent ρ. The collaboration uses the same truncated-spectral pseudo-inverse as χ^2_{ours} (private communication), so the convention is collaboration practice rather than one introduced by this analysis. No published reference for it has been identified.

What is reported. The combined $\chi^2/\text{ndf} = 0.929$ is the descriptive distance of §A.3.2, not a quoted result or validation statistic, with the limits stated there (Eq. (18)). The per-band sum is MAT-conformant, with the Flux band weighted as MAT weights it (the earlier machine-precision check of §A.2.6 covers the unweighted bands) and the 1.4% normalization as a rank-1 band. The truncated-spectral χ^2_{ours} uses $\mathbf{C}^{\text{ours}}$, which for the quoted covariance has rank 197 above $10^{-10}\lambda_{\max}$; the regularization scan shown above is of an earlier $\mathbf{C}^{\text{ours}}$. The 2D agreement claims are the total-cross-section ratio, per-bin ratio distribution and published-σ standardized differences; no covariance-based goodness-of-fit is claimed.

A.5 Anatomy of the paper-covariance tension

This derivation supports §6.4. Its numbers are for the exact-split cross-check, the quoted 2D central value when the derivation was made (§3.2). For the quoted LightGBM central value, §6.4 gives the same decomposition from the same code: a smaller distance (2.27), an 8× rather than 10× off-diagonal inflation, 9.6 % rather than 3 % in the ten smallest-λ modes, and its leading poorly determined modes in the highest-$p_{\|}$ bins, while its moderate-λ modes remain at low-to-moderate p_T.

The paradox. The central values agree well ($\sigma_{\rm tot}^{\rm ours}/\sigma_{\rm tot}^{\rm paper} = 1.011$, median bin ratio 1.006), and the per-bin pull RMS against the paper covariance is only 0.598 for the exact-split cross-check (§A.3.3 gives 0.535 for the quoted value). A diagonal-only χ^2, ignoring all off-diagonal correlations, is therefore

$$\chi^2_{\rm diag}/{\rm ndf} = \frac{1}{n}\sum_i \frac{\Delta_i^2}{C_{ii}} = {\rm RMS(pull)}^2 + {\rm mean(pull)}^2 \approx 0.366, \tag{22}$$

versus full-covariance $\chi^2/{\rm ndf} = 3.661$. The 10× inflation comes entirely from published `TotalCov`'s off-diagonal and small-eigenvalue structure.

Eigenmode decomposition. Diagonalizing the reported-block paper covariance, $\mathbf{C} = \sum_k \lambda_k \mathbf{u}_k \mathbf{u}_k^\top$ ($\lambda_1 \geq \cdots \geq \lambda_n$), and projecting $c_k = \mathbf{u}_k^\top \mathbf{\Delta}$,

$$\chi^2 = \mathbf{\Delta}^\top \mathbf{C}^+ \mathbf{\Delta} = \sum_k \frac{c_k^2}{\lambda_k}, \qquad \text{“eigen-pull”}_k \equiv \frac{c_k}{\sqrt{\lambda_k}}, \tag{23}$$

reproduces the pseudo-inverse $\chi^2 = 750.49$ to 3×10^{-9}. The tension is neither a few pathological null directions nor the absolute smallest eigenvalues. The largest contributions lie at $\lambda_k/\lambda_{\rm max} \sim 10^{-4}$–$10^{-5}$, in the lower-middle spectrum, $\sim$ 7–8 decades above the condition floor ($\lambda/\lambda_{\rm max} \sim 10^{-12}$). Their standardized-component magnitudes 5–6 are descriptive coordinates in the published basis, without calibrated significance for independent results. The ten *smallest-*λ modes carry only 3% of the χ^2 and about 90 modes are needed to reach 90%: the χ^2 is broadly distributed, not localized to the ill-conditioned tail. A scan of the pseudo-inverse cutoff excludes a numerical artifact: the 73 best-determined directions ($\lambda_k/\lambda_{\rm max} > 10^{-4}$) give $\chi^2 = 291$ ($\chi^2/205 = 1.42$), and 139 directions ($> 10^{-6}$) give $\chi^2 = 572$ ($\chi^2/205 = 2.79$), each divided by the number of reported bins, not by the number of directions kept. The rank-truncation scan of §A.3.2 rises smoothly with no late jump, so the tension does not depend on the last few least-determined modes.

Shape contribution. Splitting $\mathbf{\Delta}$ along $\mathbf{x}^{\rm paper}$ and its orthogonal remainder gives $\chi^2_{\rm norm} = 0.10$ for scale, $\chi^2_{\rm shape} = 750.1$ for shape, and cross term 0.28; the split is not $\mathbf{C}^+$-orthogonal. The +1.1% normalization offset is statistically irrelevant to the tension; it is a correlated shape difference, consistent with Jacobian shape-only $\chi^2/{\rm ndf} = 3.60$ (§A.3.4).

Location in the $(p_T, p_\parallel)$ plane. Mapping dominant eigenvectors and per-bin contributions $\Delta_i(\mathbf{C}^+\mathbf{\Delta})_i$ to the 14×16 grid (numbers for the exact-split cross-check; Figs. 11 and 12 show the LightGBM central value) locates tension on the low-p_T peak ridge ($p_T \lesssim 0.4$ GeV/c, p_T-bins 2/7/10 carry 16/11/12%) and low-$p_\parallel$ near-edge cells ($p_\parallel$-bins 0/1/8). Carrier modes oscillate in sign across the peak. They form neither vertical p_T-columns characteristic of a flux $1/\Phi(p_T)$ normalization error nor the $\theta_\mu < 20°$ acceptance boundary. Flux and Muon_Energy, the largest bands and only named cross-band correlation (Bashyal, §A.4), dominate there. Tight bin-to-bin correlations leave little freedom to absorb a coherent $\sim$ 1–2% shape difference, so a small OmniFold-vs-IBU shape difference becomes a large χ^2.[2]

[2]The pipeline uses one global ${\rm POT_{data}/POT_{MC}}$ ratio; per-playlist ratios span 0.1707–0.2371. Sensitivity at 1–2% motivated a direct mixture check: $R_{\rm glob}\sum_p {\rm MC}_p$ versus $\sum_p R_p\,{\rm MC}_p$ over twelve playlists changes reco shape by at most 0.07% in p_T, 0.14% in $p_\parallel$, and 0.03% on the low-p_T ridge. The residual +0.12% is normalization. Nearly shape-identical playlists make even large mixture reweighting almost pure normalization, leaving the statement above unaffected at the 14–30× level.

A methodological component (estimator and iteration). Changing the GBDT backend on the same data and normalization ($\Sigma\sigma$ within 0.1%) shifts the χ^2 against the paper covariance by ~ 1 unit. The integrated agreement does not mean the bins agree: the LightGBM seed-1 and exact-split central values differ by a median of 0.97 % per bin, 1.3 statistical standard deviations (seed 42: 0.98 %, 1.32; §3.2). Iteration-count effects are negligible,

OmniFold configuration	χ^2/ndf, paper covariance	pull RMS
exact-split, 5 iter (cross-check; earlier central value)	3.66	0.598
HistGBT, 5 iter (10-seed mean)	2.70 ± 0.04	0.56
LightGBM, 5 iter, seed 42, refined subtraction (quoted)	2.270	0.535
LightGBM, 5 iter, seed 42	2.654	0.564
LightGBM, 5 iter, seed 1	2.65	0.563
LightGBM, 10 iter, seed 1	2.66	0.565

All rows except the quoted one use the purity-based subtraction. (i) **Iteration count is negligible**: LightGBM 5 $\rightarrow$ 10 iters moves the χ^2 by +0.01 (2.645 $\rightarrow$ 2.657), so OmniFold is well-converged at 5 iters and the tension is *not* an under-iteration artifact. (ii) **The GBDT estimator carries a $\sim$ 1-unit band**: the exact-split cross-check (3.66) is the outlier, while both histogram-binned backends agree at 2.65–2.70 (HistGBT is a 10-seed mean, range 2.63–2.74; the $\pm$0.04 ML-seed jitter is $\sim 25\times$ smaller than the 0.96 exact$\rightarrow$hist gap, so that gap is attributable to the estimator, not to seed noise). The exact-gradient trees fit finer shape structure (less implicit smoothing) and so deviate more from the paper's IBU-regularized result. This is direct evidence that ~ 1 χ^2-unit of the exact-split cross-check's tension is due to OmniFold's implicit regularization rather than to a physical data–MC disagreement.

Conclusion. For the exact-split cross-check, the tension is a broadly distributed, correlated shape difference concentrated on the low-p_T peak dominated by flux/Muon_Energy bands; for the LightGBM central value the distance is smaller and a larger share lies in poorly determined directions (§6.4). Normalization and small-λ inversion artifacts do not drive it. A $\sim$ 1-χ^2-unit GBDT estimator/iteration regularization band adds a methodological component. Both point to the same structure: the Bashyal flux $\leftrightarrow$ Muon_Energy joint block (§A.4) is the covariance structure that governs the region carrying the tension.

A.6 Closure tests: detecting bias

The variance methods of §A.2 quantify the scatter of the unfolded result. Closure tests measure its bias, a consistent offset from a known injected truth, which the variance methods cannot show. Each of three tests injects a known MC deformation, unfolds, and compares the median per-bin |residual| of the recovered answer with a threshold. The thresholds lie comfortably above the 0.166% ML floor of the earlier ten-seed scan, which was in force when the tests were graded (Ref. [37], superseded values: retired material). All use playlist-1A LightGBM 5-iteration unfolds. Thresholds and results are in §5; procedures and otherwise unreported values follow.

A.6.1 Truth-reweight closure

For known $f(p_T, p_\parallel)$, both reco and truth MC event weights are multiplied by f. Reweighted pseudo-data are unfolded with the unchanged CV response and compared with the reweighted truth marginal. Two shapes are tested: `gauss_pt`, $f = 1 + A\exp\big[-((p_T - p_T^0)/\sigma)^2\big]$ with $A = 0.2$,

$p_T^0 = 0.4$ GeV/c, $\sigma = 0.1$ GeV/c; and `tilt_pz`, $f = 1 + \alpha(p_\parallel - p_\parallel^{\text{ref}})$ with $\alpha = 0.1$, $p_\parallel^{\text{ref}} = 5$ GeV/c. Both residuals are at least 30× below the threshold: OmniFold recovers the injected truth shape.

A.6.2 Hidden-variable closure

OmniFold sees reco and truth $(p_T, p_\parallel)$ but excludes resolution variable $\Delta p_T \equiv p_T^{\text{reco}} - p_T^{\text{truth}}$. A bias confined to this hidden axis tests leakage into the measured marginal. Pseudo-data receive

$$w^{\text{reco}} \;\to\; w^{\text{reco}} \cdot \Big(1 + A \exp\big[-((\Delta p_T - c)/\sigma_{\Delta p_T})^2\big]\Big), \tag{24}$$

with $c = +0.1$ GeV/c, $\sigma_{\Delta p_T} = 0.05$ GeV/c, and amplitude scanned over $A \in \{0.05, 0.10, 0.30\}$. Truth weights are *unchanged*, so the reference is the unaltered CV truth marginal. For median per-bin signed residual $\bar{r}(A)$, a linear fit gives $\bar{r}(A) \approx \kappa_{\text{leak}} A$, with $\kappa_{\text{leak}} = \partial\bar{r}/\partial A$, the fraction of hidden Δp_T bias entering the visible answer. The 3.0% acceptance bound is looser than the truth-reweight bound because the weak Δp_T–p_T correlation prevents complete removal of the leakage.

On the 205 reported bins the per-amplitude median residuals, 0.57%/1.14%/3.43% for $A = 0.05/0.10/0.30$, are linear in A, as expected for small perturbations, with fitted slope $\kappa_{\text{leak}} = \boxed{0.113}$: about 11% of a hidden Δp_T bias leaks into the measured $(p_T, p_\parallel)$ marginal. Against the 3.0% acceptance bound, two of the three arms pass and the largest does not: $A = 0.30$ gives 3.43%, which *exceeds* 3.0% (§5). The scan establishes the linearity and the coefficient: the acceptance bound is satisfied for hidden-Δp_T deformations up to roughly $A \simeq 0.26$ and is violated above it. No non-closure uncertainty is assigned from this scan either way (§5). A Δp_T-shaped mismodeling at the 30% level would leak more than the analysis declared tolerable; this is an open sensitivity, not a passing test.

A.6.3 Alternative-model closure

The OmniFold response is trained on the CV weights and the pseudo-data are built from an alternative-universe reweighting: `MaCCQE:0` ($+1\sigma$), which probes an interaction-model shape change, or `Flux:50` (PPFX index 50), which probes flux reweighting — two distinct kinds of model dependence. The reference is the in-acceptance-extrapolated alternative truth,

$$\tilde{X}_i^{\text{ref}} \;=\; X_i^{\text{CV,truth}} \cdot \frac{X_i^{\text{alt,in-acc}}}{X_i^{\text{CV,in-acc}}}, \tag{25}$$

scaling the CV truth marginal by the alternative-to-CV in-acceptance count ratio to isolate model-shape response from acceptance/efficiency effects. In-acceptance biases are −1.68% for `MaCCQE:0`, removed by extrapolation, and −7.32% for `Flux:50`. Both residuals are below the 2.0% bound by at least a factor of 5, showing robustness to model variations of order $\pm 1\sigma$ GENIE and full PPFX flux universes.

A.7 Summary of the 2D comparison numbers

Across 205 reported bins, combined OmniFold + paper uncertainties give $\chi^2/\text{ndf} = 0.929$ and pull mean/RMS 0.034/0.388. This distance is descriptive, without calibrated goodness-of-fit (§A.3.2). Same-ensemble toy pulls are approximately Gaussian; the 0.048% training-seed block is more than an order of magnitude below the systematic block.

Sources of the 2D comparison numbers

Number	Section	Procedure
$\chi^2/\mathrm{ndf} = 0.929$	§A.3.2	Eq. (17) with $\mathbf{C}^{\mathrm{paper}} + \mathbf{C}^{\mathrm{syst}} + \mathbf{C}^{\mathrm{stat}} + \mathbf{C}^{\mathrm{ML}} + \mathbf{C}^{\mathrm{PL}}$
$\mathbf{C}^{\mathrm{syst}}$	§A.2.4	Universe sweep (187 universes; 43 bands in the band sum), M
$\mathbf{C}^{\mathrm{stat}}$	§A.2.2	Poisson bootstrap, 300 replicas, seed 42, three streams, Eq. (1
$\mathbf{C}^{\mathrm{ML}}$	§A.2.1	LightGBM training-seed settings, 5×20 replicas
$\mathbf{C}^{\mathrm{PL}}$	§4.5	Run-period block, three run periods
$\mathbf{C}^{\mathrm{paper}}$	§A.2.6	Paper's published `TotalCovariance`
Pull RMS = 0.388	§A.3.3	Per-bin standardized residual on combined $\mathbf{C}$
Combined median rel. = 6.414%	§A.2.7	$\sqrt{\mathrm{diag}\mathbf{C}^{\mathrm{tot}}}/\mathbf{X}^{\mathrm{CV}}$, $\mathbf{X}^{\mathrm{CV}}$ the LightGBM seed-42 central value
Fixed-truth $C_2 = 0.912$ (nominal 0.954)	§5.2	Earlier statistical band vs. fixed truth, $N_{\mathrm{toy}} = 100$: fails
Same-ensemble $\lvert r\rvert \leq 1$: 68.71%	§5.3	Pull diagnostic, $N_{\mathrm{toy}} = 200$

Standing assumptions

1. Pulls and χ^2 approximate bins as Gaussian: weighted sums of $\mathcal{O}(10^4)$ MC events justify the central-limit approximation.
2. **Independence of error sources.** Eq. (16) requires no cross-correlations between ML, statistical, systematic and run-period blocks. The bootstrap pins the GBDT seed (so ML noise cancels at first order); universes pin the seed too; the training-seed block averages the same 20 bootstrap replicas at each seed setting and includes the bootstrap × seed interaction at its 1/20 share. Among the first three, cross-terms are higher-order. The run-period block is built from separate unfolds of the same data, and its noise share is not subtracted (§4.5).
3. Same-ensemble 68.71% (§5.3) checks Gaussianity; it does not certify $\mathbf{C}^{\mathrm{stat}}$ at the half-percent level. The fixed-truth test (§5.2) found the diagonal of the first-built $\mathbf{C}^{\mathrm{stat}}$ undercovering in its tails ($C_2 = 0.912$, unevenly across bins); the band was rebuilt in response, and the rebuilt band has not been independently re-tested. The statistical block quoted here is a different ensemble and has not been tested either.
4. **Band independence in $\mathbf{C}^{\mathrm{syst}}$.** Different physics sources are taken as uncorrelated; the one known exception, the Bashyal flux ↔ muon-E joint block, is treated in §A.4.
5. Closure passes apply to tested perturbations; unphysical deformations far outside training have no guarantee of passing.

B Statistical methods for the multidimensional results and inference

This appendix gives the conventions for covariances and comparisons beyond the 2D result: rank, ndf and finite-ensemble declarations for N-D covariances, the scalar five-dimensional diagnostic reductions and interval definitions, the extended-fiducial band and pull fractions, and inference from unfolded events and selected projections. The 2D methods and their worked examples are in App. A.

B.1 Rank, ndf and finite-ensemble declarations for N-D covariances

The 2D justification in §A.3.2 does not transfer to N-D. An earlier 5D covariance from the standard production chain has 10 694 reported bins, effective positive rank **263** and infinite numerical condition number. It is distinct from C_Z, the five-axis covariance of the configuration-(a) central value, and was built before C_Z was chosen for use. Its 4825-bin 5D→4D projection retains rank 263 exactly.

The covariance is a sum of about 45 rank-one outer products, one for each two-endpoint MAT systematic band, and of the statistical and machine-learning (ML) sample blocks; such a sum cannot span 10 694 directions, so the rank deficiency follows from the construction. Quoting $\chi^2/n_{\text{reported}}$ would understate tension by roughly 40× in 5D or 18× in 4D.

For an N-D covariance, ndf is the retained rank, not n_{reported}. Every N-D χ^2 must state, where it is quoted, (i) the inverse actually used, including the pseudo-inverse `rcond` or the truncation rank; (ii) the retained rank, as ndf; (iii) the rank-truncation scan of that covariance, which is expected to lack the smoothness that supports the 2D choice in §A.3.2; (iv) the identity of the covariance, since the earlier 5D covariance, its 4D projection and the published 2D block differ in rank; and (v) for each sample block in the sum, the ensemble size N, the biased $1/N$ or unbiased $1/(N-1)$ normalization, the effective inverted dimension p after truncation, and the finite-ensemble treatment, or that none is applied. The measurements, the derivation and the requirement that N-D difference statistics use a rank cut fixed in advance are in Ref. [37] (R1.15).

Finite-ensemble bias of the inverse. Item (v) records the ensemble size of each block; without it, an uncorrected finite-ensemble bias could enter unnoticed. The inverse of a noisy sample covariance is biased *upward*. For an unbiased $1/(N-1)$ estimator over N independent Gaussian realizations, the standard debiasing factor is $(N-p-2)/(N-1)$. For this analysis's production convention, universe-mean centered with biased $1/N$ normalization, it is $(N-p-2)/N$. The superseded unified-throw 5D covariance has $N=160$, truncated $p=100$, giving $58/160 \simeq 0.363$: uncorrected precision is too large by $\simeq 2.76$. For fixed residuals this inflates χ^2 and overstates tension; confidence regions from the same over-tight covariance become over-optimistic.

No generic correction is prescribed. A sum of deterministic rank-one band terms and statistical and ML sample blocks, such as the earlier 5D covariance from the standard production chain, has no single debiasing factor. The standard factor also assumes independent Gaussian realizations and a data-independent truncation dimension, neither established for a data-dependent cut. The appropriate treatment depends on the covariance and inverse actually used, and is to be determined from the quantities recorded under item (v).

B.2 Scalar-5D diagnostic reductions and interval definitions

The synthesis in §9.4 reuses saved outputs; it involves no new unfolding, training or pseudo-experiments. Fine-grid cross-section densities are converted to integrals by multiplying by all five bin widths before aggregation. The saved trace functionals on the EW cells are differential in (E_{avail}, W) and are multiplied by the cell areas; those on the J and H2 cells are already integrals. Direct tensor-slice integration checks the index-map sums and full-domain rate. Reporting boundaries coincide with fine edges. The J boundaries are $p_T = (0, 0.55, 0.85, 4.5)$, $p_\parallel = (1.5, 3.5, 6, 60)$, $E_{\text{avail}} = (0, 0.4, 1.5, 100)$, $q_3 = (0, 1.2, 2, 100)$ and $W = (0, 1.4, 2.2, 100)$; H2 uses interior boundaries $(0.7, 4.5, 0.8, 2, 1.8)$, respectively, with the same outer boundaries. Energy units are GeV and momentum units GeV/c. A cell is supported when both its expected reconstructed signal and selected truth weight at data exposure are at least 1000. This gives 109 J, 27 H2 and 39 EW cells; the unsupported EW11, EW36 and EW37 are omitted from the supported-cell summaries only.

For experiment i and cell f,

$$r_{if} = \frac{\hat{f}_i - f_{\text{true},i}}{f_{\text{true},i}}, \qquad \bar{r}_f = \frac{1}{n}\sum_i r_{if}, \qquad s_f^2 = \frac{1}{n-1}\sum_i (r_{if} - \bar{r}_f)^2. \tag{26}$$

The truth of the stored pseudo-data source varies with the MC split, so s_f measures relative-residual spread, not unconditional fixed-truth cross-section variance. The saved table also reports

bias-retaining $\sqrt{n^{-1}\sum_i r_{if}^2}$. Pointwise 95% mean intervals use $t_{0.975,n-1}s_f/\sqrt{n}$; SD intervals use $s_f\sqrt{(n-1)/\chi^2_{0.975,n-1}}$ and $s_f\sqrt{(n-1)/\chi^2_{0.025,n-1}}$. These normal-model approximations condition on finite simulation and make no simultaneous claim across cells. A deterministic noise-free trace from one simulation receives no error band; its cell-to-cell variation is not sampling uncertainty.

The development assessment includes an interval diagnostic. It uses σ_f, the sample standard deviation of 100 refitted bootstrap replicas of the nominal development pseudo-experiment (experiment 700000). Assessment results are scored against $[\hat{f}_i - \sigma_f, \hat{f}_i + \sigma_f]$ and $[\hat{f}_i - 1.96\sigma_f, \hat{f}_i + 1.96\sigma_f]$, without bias correction or model envelope. The table gives integer hits, both empirical fractions, exact pointwise 95% binomial intervals and fixed half-width relative to mean truth. It conditions on estimated bootstrap widths without propagating finite-replica uncertainty. Coverage fractions are not pooled across cells to certify them. The examples shown are a populated low-energy cell (EW7) and the corner cell EW41, already known to fail; they are not freshly selected validation targets.

The rounding study keeps the fine-grid membership fixed under uniform input perturbations within half a float32 unit in the last place. With 20 base jitters, 50 unperturbed bootstrap replicas f_b and their jittered partners f'_b, the reported quantities are

$$s_{\text{base}} = \text{SD}(f_j), \quad s_{\text{boot}} = \text{SD}(f_b), \quad s_{\text{pair}} = \text{SD}\Big((f'_b - f_b)/\sqrt{2}\Big)\,. \tag{27}$$

The consistency ratio is $s_{\text{pair}}/s_{\text{base}}$; the estimated numerical share is $s^2_{\text{pair}}/s^2_{\text{boot}}$. The paired-model $\sqrt{2}$ factor introduces no physical noise source. Containment, as defined before the study ran, requires a median consistency ratio in $[0.7, 1.4]$ and $s^2_{\text{boot}} \geq s^2_{\text{pair}}$ in at least 95% of supported cells. All supported cells meet the inequality. This finite-sample diagnostic supports the construction without proving coverage or independence from every physical uncertainty.

For prior vertex v, data sensitivity is $d_{vf} = \hat{f}_{\text{data},v}/\hat{f}_{\text{data},0} - 1$, with envelope $h_f = \max_v |d_{vf}|$. Its denominator differs from the simulated residual's. Neither h_f nor an outer product of closure residuals is given a probability model. The reductions keep the per-cell values, the identity of each input and the exclusions, and distinguish complete outputs from checkpoints that carry no metadata. From a checkpoint, only the iterations that reproduce the originally recorded values are used; later entries are excluded. The analyst cross-checked the reductions numerically.

B.3 Extended-fiducial statistical-band and pull fractions

For the containment diagnostic in §12, the independent scalar statistical-covariance band contains 77.6 % of closure bin–toy cells and exceeds the closure-toy spread. The same-ensemble pull fraction is 68.9 %, with mean absolute pull $\langle|r|\rangle \approx 0.79$; it remains a diagnostic of the pulls and their Gaussianity. The split-sample fractions in the published and extension regions are 68.5 % and 70.0 %, respectively; both are descriptive, and neither is used to certify a covariance.

B.4 Inference from unfolded events and selected projections

Unfolding correlates weighted events. Treating them as independent in a downstream likelihood can underestimate parameter uncertainties [63]; nominal weights do not specify uncertainty. Coherent unfolding replicas could propagate it through the fit. Resampling unfolded rows independently is not equivalent to resampling inputs and repeating the learned estimator. Classifier and permutation goodness-of-fit tests also require null calibration for their fitting and data reuse. These are requirements on such inference, with no new validation result here.

The uncertainty target matters: separate bin or projection coverage does not imply simultaneous coverage. A fixed linear projection $y = Px$ propagates covariance as $C_y = PC_xP^\top$, with integrated

bin widths in P for densities. The identity neither calibrates that covariance nor accounts for regions chosen after inspecting data.

Recent work constructs confidence intervals for selected functionals in constrained inverse problems [64] and joint confidence regions for multiple functionals [65], which could provide independent benchmarks for specified integrals or contrasts. Their guarantees depend on the forward model, noise law, and parameter constraints and do not transfer automatically to a learned response with finite simulation and detector nuisances. A rank-deficient forward operator is also distinct from a rank-deficient sample covariance; neither truncating the latter nor validating a few projections certifies the full distribution. Such a benchmark would supplement, not replace, the scalar five-dimensional uncertainty construction that a measurement requires.

B.5 Familywise error of the sequential joint-test decisions

This derivation supports the familywise statement of §11.3.

Probability model. Fix one test, and condition on the data and on the simulated sample. The calibration pseudo-experiments, taken in their fixed seed order, give exceedance indicators $X_b = \mathbf{1}[T_b \geq t_{\mathrm{obs}}]$. Assumption A2 is that, given the data, these are independent Bernoulli draws with a common probability p, the calibrated tail probability of the observed statistic; this requires the lost pseudo-experiments to be lost independently of their outcomes. Assumption A1 is that, for a true null, p is super-uniform over repetitions of the experiment, $P(p \leq u) \leq u$, with the nuisance parameters drawn from their priors; the size statement is therefore averaged over the nuisance priors. Let k_B be the number of exceedances among the first B draws and $U(k, B)$ the upper end of the two-sided 95% Clopper–Pearson interval, a one-sided 97.5% upper bound.

Crossing event. The rule rejects at threshold c only if $U(k_B, B) < c$ at the number $B \leq 1999$ at which sampling stopped. Whatever rule chose that stopping point, including stops decided with results visible, irregular looks, lost draws, and the budget extensions, the rejection lies in the event $E_c = \bigcup_{B \leq 1999} \{U(k_B, B) < c\}$ that the bound crosses below c at some B. Hence $P(\text{reject at } c) \leq R(c) \equiv E_p[g_c(p)]$, with $g_c(p) = P(E_c \mid p)$.

Uniform p is the worst case. Coupling the Bernoulli draws in p shows that $g_c(p)$ does not increase with p. For any super-uniform p its expectation is therefore at most its value for uniform p, $R(c) \leq \int_0^1 g_c(u)\, du$.

The decision rule, operationally. The tests are ordered by increasing $(k+1)/(B+1)$, with ties broken in the fixed listing order (total before shape for each null). At step i the threshold is $\alpha/(m - i + 1)$: the test is rejected if $U < \alpha/(m-i+1)$, it is not rejected if the lower end of its interval exceeds the threshold, and it is undetermined otherwise; a test that is not rejected or undetermined ends the step-down, and the tests after it take the same status. Equality $U = c$ does not reject.

Holm reduction without independence. Let m_0 of the $m = 10$ hypotheses be true. The first true null to be rejected is reached after at most $m - m_0$ rejections of false nulls, so its threshold is at most α/m_0. A familywise error therefore requires some true null to cross below α/m_0, and by the union bound $\mathrm{FWER} \leq m_0\, R(\alpha/m_0) \leq \max_{j \leq m} j\, R(\alpha/j)$. The argument uses neither independence between tests nor the order of the Holm steps, so the shared ensemble of a null's total and shape tests, their unequal stopping counts, and the ranking by $(k+1)/(B+1)$ with its ties do not affect it. The decision p value is the largest over shifted nulls that include the unshifted one, evaluated at the same B, so its exceedance count is at least that of the unshifted test; since U increases with k, the bound for the unshifted test covers the decision rule.

Certified constant. For each $c = \alpha/j$, $j = 1, \ldots, 10$, let $k^*(B)$ be the largest k with $U(k, B) < c$. For fixed p, a dynamic program propagates the distribution of k_B from $B = 1$ to 1999, moving

the probability of every state with $k_B \le k^*(B)$ into an absorbing "crossed" state at each B, and pooling the counts above $\max_B k^*(B)$, which can never cross because counts never decrease; the absorbed probability is $g_c(p)$ exactly. This is evaluated on a grid of 6000 points in $[0, 6c]$. Because g_c is nonincreasing, a left Riemann sum over-estimates its integral on each grid interval, and with the tail term $(1-6c)\,g_c(6c)$ it is a rigorous upper bound on $\int_0^1 g_c$, so the integration error is included in the bound. The largest ratio $R(c)/c$, at $j=1$, lies in $[0.8892, 0.8902]$; it decreases to $[0.5138, 0.5148]$ at $j=10$. Hence, for $\alpha = 0.05$, the value used here, FWER $\le 0.8902\,\alpha < 0.9\,\alpha$ under any stopping at or before 1999 pseudo-experiments. The bound rests on A1, the calibration's fidelity, which §11.4 discusses, and on A2.

C Unbinned background subtraction by negative-weight injection

Binned purity subtraction (§3.1) is the only step that uses analysis binning before final histogramming. Each data event in reconstructed bin b receives the purity weight $w_{\mathrm{pur}}(b) = \max(0,\, N_{\mathrm{data}}(b) - N_{\mathrm{bkg}}(b))/N_{\mathrm{data}}(b)$. Injecting background simulation on the measured side with negative weight gives a fully unbinned correction with the same bin-integrated estimand. Its agreement with the binned correction is quantified on the 2D measurement below. It is the continuous version of the purity weight recommended for nontrivial backgrounds in Ref. [18], and the natural treatment for the larger backgrounds of the extended-fiducial analysis.

Scope of the numbers. The numbers in this appendix are not quoted uncertainties, and the analytic derivation, the estimand identity and the displacement argument do not depend on any. Which background treatment each quoted result uses is stated in §3.1, and by configuration in §3.2 (Table 2). Negative-weight injection is studied here for its equivalence to binned purity and as the extended-fiducial treatment; this appendix establishes no supported production path and changes no default. The two covariance ratios are agreement diagnostics between two realizations of one subtraction. Two provenance limits apply to the production numbers: the driver revision that produced them is corroborated but not pinned, and no output records the job that produced them (Ref. [37], R1.7).

C.1 Estimator

OmniFold step 1 trains a classifier to separate simulated reconstruction (class 0, weighted density S) from the measured sample (class 1, weighted density ρ_1), then reweights simulation by the learned density ratio $r(x) = p(x)/(1-p(x))$. Its population optimum is $r^\star(x) = \rho_1(x)/S(x)$. Background treatment is entirely a choice of ρ_1:

- binned purity assigns every data event in reconstructed analysis bin b the same scalar. For $x \in b$,
$$\rho_1^{\mathrm{pur}}(x) \;=\; D(x)\, w_{\mathrm{pur}}(b) \;=\; D(x)\, \max\Big(0,\; 1 - \frac{B_b}{D_b}\Big), \tag{28}$$
where D is the data density, B the POT-scaled background density, and D_b, B_b their bin masses;
- negative-weight injection appends reconstructed background-simulation events to the measured side with weight $-w_{\mathrm{bkg}} \times$ (POT scale) and keeps data weights at $+1$, giving
$$\rho_1^{\mathrm{neg}}(x) \;=\; D(x) - B(x) \tag{29}$$
pointwise, without analysis binning.

Equations (28) and (29) differ as functions of x. A bin-constant rescaling reproduces $D(x) - B(x)$ inside bin b only when B/D is constant there. Otherwise, $D(x)\,(1 - B_b/D_b)$ and $D(x) - B(x)$ share a bin integral but differ within the bin. The constructions agree *exactly* on bin masses under the piecewise-constant sieve of §C.2; in general they differ by within-bin shape terms. Both target the *bin-integrated* estimand $r^\star(b) = (D_b - B_b)/S_b$; they do not share the continuous estimand $r^\star(x) = (D(x) - B(x))/S(x)$. §C.4 and §C.5 measure the effect of that difference. The injected background and data occupy the same reconstructed fiducial window, giving identical support.

The floor. Where $D - B < 0$, binned purity floors the correction at zero through $\max(0, \cdot)$. The unbinned classifier's bounded output $p \in (0, 1)$ forces $r = p/(1-p) \geq 0$. Non-negativity is structural, while the value at the floor follows from the bounded output and ordinary regularization rather than from a minimizer of the fit (§C.2). Locally negative class-1 mass makes the fit ill-posed and calls for a positive-target prescription. We apply the "Stay Positive" refinement of Ref. [66], then explicitly zero-clip the learned refinement factor before step-1 training. Agreement between plain injection and clipped refinement depends on the classifier.

With the histogram-based gradient-boosted classifier used in *this comparison*, they agree to the reported median per-bin ratio 1.000 at this measurement's sub-percent genuine-background scale. This histogram backend is neither the LightGBM estimator of the quoted 2D central value and its ensembles nor the exact-split cross-check (§3). The equivalence is therefore established on a third backend, and its transfer to the production configuration is assumed rather than demonstrated. Raw signed weights destabilize an exact-split boosted-tree classifier, inflating a full low-p_T cross-section row by up to a factor $\sim 5 \times 10^4$ where local background is large. Clipped Stay-Positive refinement removes this pathology and recovers binned-purity agreement to a median 0.999 (141/148 bins within 5 %). This clipped refinement is therefore what makes the unbinned subtraction robust across estimators, and it matters most for the larger, more structured backgrounds of the extended-fiducial analysis.

C.2 Binned limit: exact reduction to the purity correction

A step-1 classifier restricted to functions piecewise-constant on reconstructed analysis bins $\{b\}$ gives the population-level binned limit of the OmniFold/Richardson–Lucy correspondence in Refs. [1, 2]. Its output is a single p per bin. With common support and unnormalized class weights, step 1 weighted cross-entropy separates into per-bin objectives

$$L_b(p) = -W_1 \ln p - W_0 \ln(1 - p), \tag{30}$$

where $W_0 = S(b) > 0$ is the integral of simulated class-0 density S in bin b, and W_1 is the signed class-1 weight. Injecting background simulation at negative weight while keeping data at $+1$ gives $W_1 = D(b) - B(b)$. Here $D(b)$ and $B(b)$ are the bin masses of D and B, equivalently $N_{\text{data}}(b)$ and $N_{\text{bkg}}(b)$ in §C.1.

For $D(b) > B(b)$, objective (30) is strictly convex on $(0, 1)$ with unique interior minimizer $p = W_1/(W_0 + W_1)$, giving

$$r(b) = \frac{p}{1-p} = \frac{W_1}{W_0} = \frac{D(b) - B(b)}{S(b)}. \tag{31}$$

Purity gives exactly this reweight under the same sieve, since its class-1 weight $W_1 = D(b)\, w_{\text{pur}}(b) = \max(0,\, D(b) - B(b))$ equals $D(b) - B(b)$ on this domain. At $D(b) = B(b)$, the infimum lies at $p \to 0^+$ and $r(b) \to 0$, again giving purity when the boundary is admitted. The treatments therefore

Table 33: Binned purity versus unbinned negative-weight background subtraction on the 2D ME FHC measurement (common estimator and seed). The per-bin figures are the negative-weight/purity ratio over the populated cross-section bins.

	binned purity	negative-weight
total σ (cm^2/nucleon)	$3.0727e-38$	$3.0687e-38$
total ratio (neg/pur)	0.9987 ($-0.13\,\%$)	
median per-bin ratio	1.000	
per-bin RMS deviation	$1.4\,\%$	
largest single-bin deviation	$-12.6\,\%$ (low p_T, high background)	

agree bin by bin at the population optimum under the common sieve. Combined with the binned OmniFold/Richardson–Lucy correspondence, they give the traditional subtract-background-then-unfold update. Finite, regularized classifiers approximate this identity; their residual differences are tested below.

The reduction fails where $D(b) < B(b)$. With $W_1 < 0$, $L_b(p) = |W_1| \ln p - W_0 \ln(1-p)$ tends to $-\infty$ as $p \to 0^+$ and is unbounded below, with no minimizer. A minimizing sequence gives $r(b) \to 0$. The $\max(0, \cdot)$ floor is a prescription, not a minimizer's value. A finite regularized classifier's behavior depends on the estimator. Stay-Positive remains well behaved but can retain negative refined weights where the target density is genuinely negative. The zero clip used here is therefore an additional analysis prescription chosen to match the physical purity floor. Ref. [66] documents the same reweighting-loss divergence for negative targets; signed binary cross-entropy has an analogous pathology for sWeights [67].

The binned $D - B$ update, negative-weight classification of the measured side, and binned OmniFold/Richardson–Lucy correspondence already appear together in Ref. [2]. Equation (31) states their immediate binwise corollary and its precise validity domain for this analysis. It is a consistency check, not a separate method.

C.3 Closed-form toy

A one-dimensional toy uses signal simulation $\mathcal{N}(0,1)$, equally normalized data signal $\mathcal{N}(0.3, 1.1)$, and background $\mathcal{N}(2, 0.7)$ contributing $16.7\,\%$ of pseudo-data. The known target $(D-B)/S = \phi(x; 0.3, 1.1)/\phi(x; 0, 1)$ is smooth and positive. A signed-weight gradient-boosted classifier recovers it with median $6.8\,\%$ relative error, matching binned purity's $6.7\,\%$ and reproducing it bin-for-bin on the same problem. Omitting subtraction leaves $10.9\,\%$ residual-background bias. Results are stable across classifier seeds ($\pm 0.3\,\%$), with no numerical instability from negative weights. Moving background into the sparse signal tail and deliberately over-subtracting makes the learned reweight floor cleanly at zero where $B > D$, as anticipated.

C.4 Comparison on the 2D measurement

With identical unfolding settings, the histogram backend and a common classifier seed, the ME FHC comparison isolates background treatment (Table 33). Total cross sections agree to $-0.13\,\%$, an order of magnitude inside the $\sim 3\,\%$ purity correction. The per-bin ratio has median 1.000 and $1.4\,\%$ RMS scatter across the plane (Fig. 35). Only one low-p_T, high-background edge bin differs by more than a few percent ($-12.6\,\%$); background concentrates there and subtraction is largest. Unbinned subtraction reproduces the binned correction wherever it is well determined. Departures occur only where the binned $\max(0, \cdot)$ and structural floor of §C.1 govern the result.

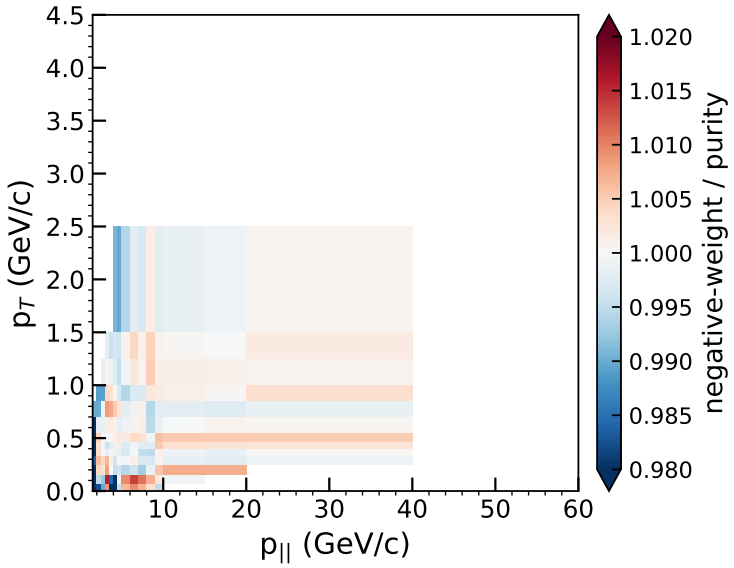


Figure 35: Negative-weight/binned-purity cross-section ratio in populated bins of the 2D ME FHC measurement, with common estimator and seed. The median is 1.000 (near-white). Only the low-p_T, high-background edge bin departs by more than a few percent. Background is largest there, and the $\max(0, \cdot)$/structural floor of §C.1 governs the result.

C.5 Propagation through systematics and statistics

Replacing binned subtraction requires reproducing its *uncertainty* budget as well as the central value. The full 2D ME FHC covariance was rebuilt end to end with negative-weight injection and compared to binned purity at identical settings, at full statistics with LightGBM, the estimator of the quoted 2D central value. The 187-universe systematic ensemble varied genuine background consistently within each universe instead of freezing it at nominal; a statistical bootstrap used matched seeds. Agreement is at the level predicted by the $(D - B)/S$ identity. The two covariances differ by one to two percent. The 187-universe systematic $\sqrt{\mathrm{Tr}}$ ratio (negative-weight/purity) is 0.986, from $2.9828 \times 10^{-39}\,\mathrm{cm}^2$/nucleon and $3.0242 \times 10^{-39}\,\mathrm{cm}^2$/nucleon, and the matched-seed statistical ratio is 0.982, from $1.7260 \times 10^{-40}\,\mathrm{cm}^2$/nucleon and $1.7576 \times 10^{-40}\,\mathrm{cm}^2$/nucleon. Residuals of 1.4 % and 1.8 % are consistent with "to within 2 %" in §3. They are bootstrap and classifier-seed sampling noise, with no method bias. Negative-weight injection reproduces the central cross section *per bin* (§C.4) and the covariance *in aggregate*. The covariance comparison gives two trace ratios; per-bin, correlation and eigenmode equivalence are not established.

The 2D central-value, closed-form recovery and full covariance comparisons, together with Stay-Positive robustness on larger extended-fiducial backgrounds, support negative-weight injection as an unbinned alternative at the 2D scale. The 2D result and its covariance, the 3D and 4D central values and the sixteen-cell five-dimensional result (§10) use its refined form (§3.1); binned purity remains the treatment of the configuration-(a) five-dimensional central value and its five-axis covariance and of the extended-fiducial value. Negative-weight injection is the *designated* extended-fiducial baseline (§12), where binning-independence and the continuous floor are essential.

C.6 Which background treatment each production chain uses

Agreement between the two subtractions does not make the choice between them arbitrary, and the production chains choose differently.

The standard N-dimensional chain uses purity. The entire standard five-dimensional chain, that of the configuration-(a) central value and its five-axis covariance, uses binned purity (§3.1): the central unfold, 169 vertical-band unfolds, 18 detector unfolds and 10 endpoint unfolds of the re-selected lateral bands.

The existing endpoint unfolds, produced in July 2026, used purity, as established by elimination from their job scripts and logs (Ref. [37], R1.17). Current production fixes the background treatment in a configuration object that rejects every other mode. The standard job script passes the driver's background-mode option (`--bkg-mode`) explicitly from that configuration, and the value is written into the record of every endpoint unfold as it is produced, so the background treatment of each new component can be read from the component itself; this changes how the treatment is recorded, not the treatment itself.

The standard chain uses purity by choice, on two grounds (Ref. [37], R1.8). First, a covariance that mixes background treatments is inferior to one that uses either treatment consistently: a lateral-band block built with negative-weight subtraction would be inconsistent with central and vertical-band unfolds built with purity subtraction. Second, switching would have required rerunning every unfold above and would have invalidated the flux-corrected covariance then just chosen for use; this consequence, rather than the computing cost, decided the choice. The background treatment has been revisited in two dimensions: the closure test below is its result, and the 2D result and its covariance use the refined estimator. For the configuration-(a) five-dimensional central value, which still uses purity, the revisit is open; the sixteen-cell result (§10) uses the refined estimator.

A two-dimensional closure test with backgrounds. Twenty pseudo-experiments at nominal truth (the truth model equals the prior), with background included and drawn as in the five-dimensional pseudo-experiments, were unfolded in two dimensions with the purity subtraction and with refined negative-weight subtraction (a 400-tree, 31-leaf refinement), both on the same pseudo-data. Differences are expressed in units of the statistical and total standard deviations, σ_{stat} and σ_{total}, of the purity-based 2D covariance on its 205 reported cells (the quoted covariance when the test ran). The mean difference between the two treatments exceeds $1\,\sigma_{\text{stat}}$ in 14 of the 205 cells, at most $5.45\,\sigma_{\text{stat}}$, in the cell $0.70 \le p_T < 0.85\,\text{GeV/c}$, $1.5 \le p_\parallel < 2.0\,\text{GeV/c}$. These cells are the whole lowest-$p_\parallel$ row (1.5–2.0 GeV/c) and five cells along the high-p_T, low-$p_\parallel$ boundary. In them, the purity estimate's mean residual against the truth is $+0.78$ to $+5.89\,\sigma_{\text{stat}}$, above $1\,\sigma_{\text{total}}$ in one cell (1.22); the refined estimate's mean residual is within $\pm 0.44\,\sigma_{\text{stat}}$ in every cell. The integrated difference between the treatments is $-0.198\,\%$. In the median cell, the pseudo-experiments' own spread is $0.46\,\sigma_{\text{stat}}$ (purity) and $0.58\,\sigma_{\text{stat}}$ (refined). The test is a closure with 20 pseudo-experiments, not a coverage test. On the data, the refined treatment moves the 2D result by at most $0.497\,\sigma_{\text{total}}$ per cell (median 0.060), by more than $1\,\sigma_{\text{stat}}$ in 43 of the 205 cells (at most $4.26\,\sigma_{\text{stat}}$), and by a median of 0.44 % per cell (84th percentile 1.13 %, at most 10.1 %); its integral moves by $-0.139\,\%$. The 2D result quoted in this note uses the refined treatment, configuration (f), with its covariance and its comparison with the published result rebuilt for it (§6).

What the quoted extended-fiducial number used. The $\sigma_{\text{ext}} = 4.502 \times 10^{-38}\,\text{cm}^2/\text{nucleon}$ in §12 was not produced with negative-weight injection. It comes from the extended-fiducial ME FHC battery of 2026-06-10. Its ten endpoint unfolds used the purity correction and are recorded as purity

controls, not publication inputs (Ref. [37], R1.9). Its central value, $4.502 \times 10^{-38}\,\mathrm{cm}^2$/nucleon, is recorded with its passed published-region check against the control. The extended-fiducial central value was therefore produced with the binned purity correction, not with refined negative-weight subtraction, the intended extended-fiducial baseline.

The extended-fiducial measurement and its control are designated different background treatments. For publication, the extended-fiducial analysis designates negative-weight injection with Stay-Positive refinement as its background treatment. The code fixes it, and a check stops the run if any endpoint's recorded mode differs or is missing; the number quoted in §12 was not produced this way. The standard chain uses purity throughout. Each side of the measurement and matched-control pair therefore has an internally consistent background treatment; uniformity across the two sides is not claimed.

C.7 Displacement argument and scope of the agreement

Displacement across universes. The displacement argument rests on a *measured* input; it assumes no identity of continuous estimands. The constructions share bin masses in the binned limit (§C.2) but differ by within-bin shape terms (§C.1). Switching background treatment produces a measured net per-universe displacement of order 0.1 %, including residual and shape effects, at this measurement's sub-percent genuine-background scale. Systematic covariance measures spread across universes about their own mean. Similar small shifts of each universe leave that spread nearly unchanged, explaining the 187-universe ratio 0.986, far from the purity correction's ∼3 %. This argument connects a handful-of-universes comparison to the 187-universe covariance; the full 187-universe rebuild in §C.5 tests it at 2D and passes.

Scope. The measured agreement concerns the 2D systematic and statistical covariances of §C.5 at the stated configurations. It is not a theorem that the two background treatments give the same covariance, because the underlying estimands are not pointwise identical, and covariance equivalence for the continuous estimator or in five dimensions is not established. The five-dimensional evidence is limited to the following two-universe spot check.

Limits of the five-dimensional evidence. There is no full five-dimensional 187-universe comparison of the two background treatments at the publication configuration (5 iterations, LightGBM). The spot check uses one iteration and the histogram backend on the stored five-dimensional universe file: under both treatments `2p2h:0` gives 3.014×10^{-38} with per-bin median 1.0005, and `MaCCQE:0` differs by −0.03 % with median 0.9990. Together with the displacement argument and the full-statistics 2D result, this supports quoting the standard chain with purity subtraction, with its background treatment stated. It does not show that the background treatment is irrelevant in five dimensions, which would require the full comparison. Raw negative weights destabilize the exact-split backend (§C.1), but the standard chain uses LightGBM, so neither treatment is exposed to that instability here.

C.8 Five-dimensional pseudo-experiments with both background treatments

The five-dimensional validation (§§9.1–9.3) reports end-to-end tests of both background treatments. They cover the purity bias in the highest-W column at nominal truth, its reduction with refined negative-weight subtraction, the data difference, and the estimator with higher refinement capacity. That estimator is configuration (e), which gives the 3D and 4D central values and the sixteen-cell

result (§10) but not the configuration-(a) 5D central value; the quoted 2D central value uses the same refinement capacity, in configuration (f).

Purity bias. Purity bias is 8–12 statistical standard deviations; signal-only experiments close. Subtraction is exact per reconstructed bin, but the widest bin ($W \geq 3\,\mathrm{GeV}$, background 13 % of signal) contains background shape resolved by the unbinned classifier and missed by bin-uniform purity. The within-bin shape term of §C.1 is therefore measurable, and it is direct evidence for revisiting the background treatment of the configuration-(a) five-dimensional central value (§C.6). The difference between five-dimensional purity and refined negative-weight subtraction on data is the least computationally expensive estimate of its size on data.

Refined negative-weight subtraction. Refinement used the unfold's deterministic GBDT configuration, independent injection and subtraction halves of background simulation, and bootstrap replicas that individually resampled every pseudo-event and refit refinement. It preserves the signed total to 3×10^{-6} and clips 0.01 % to 0.05 % of events. Corner bias falls from 1.66 % to 0.24 %, but residuals in some joint cells fail the nominal closure criterion. The criterion allows none of the 153 reported quantities a mean residual beyond 3.6 standard errors of the mean ($|t| > 3.6$, the two-sided 5% level, Bonferroni-corrected over the 153), or beyond half its bootstrap standard deviation with $|t| > 3$. The data difference's sign pattern matches simulated purity bias at an order of magnitude smaller size; the difference is method sensitivity, not a measured bias. With either treatment, the estimator does not follow generator-sized E_{avail} shape changes. No coverage test was run. Reducing one bias does not itself qualify an uncertainty, so this result does not close the revisit of the background treatment of the configuration-(a) five-dimensional central value (§C.6).

Refinement capacity. Stay-Positive classifier underfitting causes the residual nominal bias of the refined subtraction. At the unfold's GBDT setting (100 trees, 8 leaves), the positive-weight probability is underfit, leaving up to 4 % reco-level subtraction error per (E_{avail}, W) reco cell. With 400 trees and 31 leaves, error falls to 1.7 %, paired truth-level bias falls by more than 80 % in every affected cell, and nominal background-inclusive closure passes on 20 development and 40 fresh pseudo-experiments. Splitting background simulation into independent injection and subtraction halves is not the cause.

Background treatment and choice of covariance. The choice of background treatment is independent of the choice of covariance, and this appendix selects no five-dimensional covariance for use. The lateral-band component from re-selected event loops is built, with five bands and ten endpoint unfolds whose distinct checksums are listed in the record of the five-axis covariance of the configuration-(a) central value. The status of that five-axis covariance and the scope of its use are stated in §9.6. The choice of background treatment settles one of the conditions for using a component; every other condition still applies.

D Detector-response-mismatch closure test

The truth-shape and hidden-variable closures in §5 test recovery of deliberately modified event populations. They do not directly test truth-distribution bias from a mismatch between the pseudo-data detector response and OmniFold's nominal response. Kevin McFarland suggested this distinction and proposed stressing the hadronic record, especially for quasielastic events, to inspect low-W bias (private communication). This appendix gives a validation protocol for such a test, not a result or

an uncertainty component. Its first stage, the reconstruction-only test below, has been implemented but not yet run: no output of it has been produced and no pass threshold has been set. Closures of the same kind, from the shifted response samples of C_{resp}, have been run (below).

Reconstruction-only test. The test uses the ordinary five-dimensional closure sample. After the pseudo-data arrays are copied, and before background mixing or classifier training, it changes only the reconstructed pseudo-data record

$$E_{\text{avail}}^{\text{pseudo}} \longrightarrow (1+\epsilon) E_{\text{avail}}^{\text{pseudo}}. \tag{32}$$

The nominal reconstructed simulation, all truth coordinates, event weights, and event membership stay fixed. The pseudo-data therefore have a deliberately incompatible conditional response; OmniFold still uses the nominal response. Both signs of ϵ are required, at an amplitude fixed before the results are examined. A documented hadronic-response variation or an explicit quantitative assumption should justify that amplitude; the overall factor in the reconstructed-E_{avail} definition, Eq. (2), is not by itself a sufficient basis.

Outputs include the full five-dimensional unfolded/reference residual, one-dimensional E_{avail} and W projections, and an (E_{avail}, W) ratio map. The primary observable is the signed bias versus W, with $W < 1.1\,\text{GeV}$ reported separately from the high-W tail. The global integral is a control rather than the decision statistic. The implementation runs only as a pseudo-data closure and only on the five observables, and cannot be combined with truth-reweight injection or bootstrapping. It labels every output as a diagnostic that must not be quoted.

Closures from the shifted response samples. Noise-free, signal-only closures at the nominal truth take pseudo-data from simulation whose reconstructed recoil response is shifted, either for each deposit category of C_{resp} (§10.3) at its $\pm 1\sigma$ or as a $\pm 4\,\%$ scale of the whole recoil energy, and unfold them with configuration (e) on the unshifted response. Each closure's bias, less that of the unshifted closure, is compared on the sixteen cells with the per-cell σ of $C_{\text{total}} + C_{\text{resp}}$. For the deposit categories the median $|\Delta|/\sigma$ is 0.07–0.20 and the largest 0.40; for the whole-recoil scale the median is at most 0.50 and the largest 0.97. The closures are descriptive and carry no noise estimate: no pass threshold is set, because the closure settings were deferred. They do not establish the bias under the true detector response, validate C_{resp} or its category sizes, or describe response variants other than those tested.

Physically correlated follow-up. The reconstruction-only test is by design a computationally inexpensive adversarial stress test; it is not a detector model. A more faithful follow-up would select quasielastic events, shift the leading-nucleon energy up and down, and recompute reconstructed E_{avail}, q_3, W, and any affected selection from the altered low-level record. The current per-event outputs hold only scalar observables, not that low-level record, so the follow-up requires dedicated diagnostic output from the event processing. It is deferred unless the reconstruction-only test shows material bias or the collaboration asks for the stronger response model.

Interpretation. The test leaves the quoted central value and the current covariance construction unchanged. A small observed bias would support stability of the (E_{avail}, W) interpretation only within the tested response envelope. A large bias would require a decision on the response model or an uncertainty component, and would not be absorbed silently into a non-closure error.

E The MINERvA measurement-dimensionality survey

The survey behind Fig. 15 lists MINERvA differential cross-section measurements through the listed 2025 entries, with references and dimensionality: the number of kinematic variables unfolded simultaneously in each analysis's main differential result. The survey is nonexhaustive through 2026; recent additions follow.

Table 34: MINERvA differential cross-section survey used for the landscape figure, with the dimensionality (Dim.) defined in the text. Channels: CC-inclusive (incl.), CCQE/QE-like incl. transverse kinematic imbalance (QE), pion production CC and coherent (π), deep-inelastic / kaon (DIS), electron-neutrino (ν_e), nuclear-target ratio (ratio).

Year	Channel; observables	Dim.	Ref.
2013	QE; ν_μ CCQE $p_T, p_\parallel$	2	[68]
2013	QE; $\bar{\nu}_\mu$ CCQE $p_T, p_\parallel$	2	[69]
2014	ratio; CC on C/Fe/Pb to CH	1	[70]
2014	π; coherent $\pi^\pm$ (total & diff.)	1	[71]
2015	π; CC $1\pi^+$ (T_π, θ_π)	1	[72]
2015	QE; μ+p final states	1	[73]
2015	π; $\bar{\nu}_\mu$ single π^0	1	[74]
2016	ν_e; CCQE-like on CH	2	[75]
2016	π; $\nu_\mu/\bar{\nu}_\mu$ pion prod.	2	[76]
2016	DIS; partonic nuclear effects	2	[77]
2016	DIS; CC K^+ production	1	[78]
2017	π; CC single π^0	2	[79]
2017	QE; nuclear-dependence (multi-target)	1	[80]
2018	QE; $\bar{\nu}_\mu$ QE-like $p_T, p_\parallel$	2	[81]
2018	QE; μ-p mesonless TKI	2	[82]
2018	π; coherent $\pi^\pm$ on C	2	[83]
2019	QE; quasielastic-like ν_μ	2	[84]
2019	π; $\bar{\nu}_\mu$ single π^-	2	[85]
2020	incl.; CC-incl $\langle E_\nu \rangle \sim 3.5$ GeV $p_T, p_\parallel$	2	[86]
2020	QE; nuclear binding & TKI	2	[87]
2020	QE; high-statistics QE-like	2	[88]
2020	π; CC π^0 nuclear effects	1	[89]
2021	incl.; CC-incl $\langle E_\nu \rangle \sim 6$ GeV $p_T, p_\parallel$ **(target)**	2	[3]
2022	QE; QE-like $p_T, p_\parallel, \Sigma T_p$ (triple-diff.)	3	[90]
2022	incl.; CC-incl low-q_3 (E_{avail}, q_3)	2	[8]
2023	π; CC $1\pi^+$ (multi-target)	2	[91]
2023	π; coherent π^+ (multi-target)	1	[92]
2023	QE; $\bar{\nu}_\mu$ multi-neutron low-E_{avail}	1	[93]
2024	ν_e; low-Q^2 (E_{avail}, p_T)	2	[45]
2025	QE; A-dependence TKI (multi-target)	2	[94]
2026	**this work**; unbinned $p_T, p_\parallel$	2	—
2026	**this work**; unbinned $p_T, p_\parallel, E_{\text{avail}}$	**3**	—
2026	**this work**; unbinned $p_T, p_\parallel, E_{\text{avail}}, q_3$	**4**	—
2026	**this work**; unbinned $p_T, p_\parallel, E_{\text{avail}}, q_3, W$	**5**	—

Additional recent measurements. The low-/medium-energy comparison of quasielastic-like cross sections unfolds three variables simultaneously, p_T, $p_\parallel$, and available recoil energy [44]; it has the dimensionality of our three-dimensional result, with a different signal from our inclusive one. The shallow-inelastic measurement reports one-dimensional neutrino and antineutrino cross sections [49]. Other measurements report single-differential charged-pion production down to zero pion kinetic energy [95] and inclusive antineutrino cross sections versus p_T on several targets [96]. These channel-specific additions are not plotted in Fig. 15.

F Future work: point-cloud representations

We are developing unfolding with point-cloud transformer (PET) representations of the full event, which could use low-level detector and particle information that the scalar observables of this note summarize. That work, in progress, is documented separately. No central value, uncertainty or comparison from it is used or claimed in this note.

G Result package and reproduction limits

This appendix describes RC7, the existing local replay package, and specifies the broader event-level and covariance release, with the identities that are fixed and what each part does and does not reproduce. Rebuilding this note's PDFs assembles documents; it does not reproduce scientific results.

The code paths and file names in this appendix and in App. H refer to the snapshot of the code repository, `github.com/josephbaileyy/MINERvA-OmniFold`, identified at deposit: **[snapshot identifier pending deposit]**.

Release status. RC7, a local release candidate with a README, checksums and a replay verifier [97], has not been deposited or released under a publication tag. Run on the packaged files in an empty directory, RC7's own verifier reproduces the recorded observed statistics of the joint tests; the p value and count of every test and variant; the decision of every test; the robustness results and power figures; and every number quoted from a figure that it checks. The package, about 16 MB of sufficient inputs, reproduces the joint-test p values, decisions, robustness labels and power results; the same results with the recovered pseudo-experiments added to the calibration and the variant shifts recomputed on all draws (§11.6); the matched coarse-projection comparison (§11.7); and the fine-grid L_2 ratios of condition (i) (§11.4). From packaged arrays it also regenerates figures of the earlier article version for which it was assembled: the (E_{avail}, W) excess map (Fig. 24 of this note), the null distributions of the joint tests (Fig. 31), and one-dimensional generator projections, which neither this note nor the current article prints. RC7's two-dimensional curve is the exact-split one, which was the two-dimensional central value when the package was assembled. The current article's two-dimensional validation figure and its numbers were instead drawn separately, from the committed values and covariance of the refined LightGBM result, with RC7's plotting code extended to draw this analysis's band.

Contents and use. The package holds:

- the sufficient inputs of the joint tests, for the evaluation behind the decisions and for the evaluation with the recovered pseudo-experiments added;
- the arrays behind the figures of the earlier article version;
- the noise-free differences behind the L_2 ratios;
- an inventory of the analyzed AnaTuples;

- the replay and figure code, and the expected outputs.

The package's verifier (`publication/release/verify_rc.py` in the repository) runs from the top directory of the unpacked package, in a fresh Python environment with the package's pinned requirements (numpy, scipy and matplotlib). It performs every replay and figure check and prints `VERIFY: PASS` if all agree. In the two replays of the joint tests, floating-point values must agree to a relative 10^{-12}, and integers and decisions must agree exactly. Each number that the earlier article version quoted from its Figs. 1–3 must agree with its printed value to within half a unit of its last printed digit, and the L_2 ratios must reproduce their printed range. For the three figures themselves, the check is that their files are produced. The package also carries the 20 rounding jitters of the data, from which the stability of the observed p values (§11.4) can be recomputed; no packaged script recomputes it.

G.1 Scope of reproduction and required inputs

The public ancillary files of Ref. [3] supply the published inputs for the comparisons to its central values and covariance, the standardized differences, the tension anatomy, and the MINERvA Tune v1 model comparison, but not for this analysis's unfolded 2D uncertainty construction.

RC7 supports replay of the higher-dimensional inference from sufficient inputs. It does not support extraction from the archived pseudo-experiment products (about 2.3 GB), event-level unfolding, the quoted closure and bias measurements, or the measurements behind the covariance's status (§9.6). Nor does it reproduce the 2D uncertainty and coverage figures, the low-recoil comparisons, the recoil-response sensitivity, or the proxy convergence rate of joint-test condition (i). The multiplier scan of the joint tests is recomputable from RC7's sufficient inputs with the scan code of Ref. [56], but it is not part of RC7's own verification. An earlier version of the package, RC1, was replayed from an empty directory by a separate AI agent session (Opus 5.5) that had no access to the repository and used only the package and its README; its verifier passed. Event processing requires the earlier production of MINERvA open-data AnaTuples, event-loop outputs, and the generator samples used here.

All 2,374 inventoried input files (2,307 AnaTuples and 67 flux, reweight and parameter files) are identified by SHA256 checksum in Ref. [98]; RC7's own inventory lists their names, sizes and modification times, without checksums. As of that record (2026-10-10), verified copies of all 1,885 data, flux and parameter files are held on NERSC project storage; the record identifies the 489 simulation AnaTuples by checksum but holds no preservation copy of them. Neither the checksums nor the preservation copies are a public deposit, and they do not make the full event-level analysis reproducible from the small candidate package.

G.2 Release schema

The broader result package must contain every item below; RC7 does not contain all of them. A ✓ marks identities or values already fixed (bin edges: §G.3; checksums: Ref. [37], identities already fixed); the rest are packaging. A fixed identity specifies which object is meant, not whether the object is scientifically qualified.

1. **Reporting bin edges and units, per axis ✓**: p_T and $p_\parallel$ in GeV/c, E_{avail}, q_3 and W in GeV, with each wide terminal catch bin identified.
2. **The flattened-cell ordering ✓**: the dense $14 \times 16 \times 7 \times 7 \times 6$ grid in C order, with the reported subset supplied as an explicit index array, never inferred from a positivity cut.

3. **Central values**: the five-axis array and every projection quoted, each with its own checksum, on the density scale in cm^2/nucleon divided by the product of its kept axes' units.
4. **Covariances ✓**: the five-axis covariance of the configuration-(a) central value and its (E_{avail}, W) projection, with the algebra of Eqs. (4)–(5), the declared centering variant, and the band partition.
5. **Masks ✓**: the support mask, the pinned mask (bins where the inflation denominator vanished and g was declared 1), and the unreported-cell mask.
6. **Projection matrices ✓**: M itself, with its content checksum, its shape, and the explicit statement that its entries are the dropped axes' bin-width products, which is needed to specify the destination units.
7. **Generator predictions**: the four truth-level (E_{avail}, W) predictions (GENIE-CV, GENIE+Valencia 2p2h, NuWro, GiBUU) and the MINERvA Tune v1 ancillary model, as arrays on the release binning rather than figure curves, each with generator version, configuration, target and flux. GENIE and NuWro require the flux-repaired predictions of 2026-09-27 (per-event reweight to the analysis PPFX flux plus the 50–100 GeV supplements); GiBUU requires its reweight to the same flux convention (§7.6).
8. **Provenance**: for each object, the producing code revision, the job, and the checksum of the bytes read back from the closed file. Of these, only the following are fixed: the assembling revision and job of C_Z, the job of C_{EW}, and the read-back checksum of the row index. No producing revision or job is recorded for the other objects, and the assembled package carries none.
9. **One minimal worked example**: §G.4.
10. **Limiting measurements**: the four measurements of §9.6 that limit what the five-axis covariance of the configuration-(a) central value supports, stored with it in the released package, not only in this note.

G.3 Bin edges

The bin edges are the one thing a reader cannot recover from a checksum:

- p_T, GeV/c: 0, 0.07, 0.15, 0.25, 0.33, 0.40, 0.47, 0.55, 0.70, 0.85, 1.00, 1.25, 1.50, 2.50, 4.50 (14 bins).
- $p_\parallel$, GeV/c: 1.5, 2, 2.5, 3, 3.5, 4, 4.5, 5, 6, 7, 8, 9, 10, 15, 20, 40, 60 (16 bins).
- E_{avail}, GeV: 0, 0.1, 0.2, 0.4, 0.8, 1.5, 3.0, 100 (7 bins).
- q_3, GeV: 0, 0.2, 0.4, 0.6, 0.8, 1.2, 2.0, 100 (7 bins).
- W, GeV: 0, 1.1, 1.4, 1.8, 2.2, 3.0, 100 (6 bins).

The last bin on E_{avail}, q_3 and W is a catch bin to 100 GeV, and the $p_T \times p_\parallel$ grid carries the 19 unreported high-p_T, low-$p_\parallel$ acceptance cells of §3.5.

G.4 One minimal projection example

The released package is to support the following computation, which checks the package's ordering, mask and weights from end to end.

1. The five-axis central values $\mathbf{x}$ have 65 856 entries, and the reported-cell index array has 10 694 entries.

2. M has shape $42 \times 10\,694$, and C_Z has shape $10\,694 \times 10\,694$. The five-axis covariance of the configuration-(a) central value is the file `z-cv.npz`, with `sha256` 3d7465f6… and $\sqrt{\mathrm{Tr}C} = 5.6742 \times 10^{-38}$, so a loaded C_Z can be checked against both.

3. The projection $\mathbf{y} = M\mathbf{x}_{\mathrm{reported}}$, reshaped to 7×6 **in C order**, must reproduce the (E_{avail}, W) central-value map of Fig. 24. The F-order control establishes the ordering: C order agrees with the figure to a relative 1.7×10^{-16}; F order disagrees at relative 1.0.

4. The projected covariance $C_{EW} = M\,C_Z\,M^\top$ is checked by $\sqrt{\mathrm{Tr}C_{EW}} = 4.4552 \times 10^{-39}$ and exact symmetry ($\max|C - C^\top| = 0$). Its numerical rank depends on the cutoff, because the spectrum decays smoothly with no plateau: the retained count is 26 at 10^{-6}, 36 at the projector's hardcoded 10^{-12}, 41 at `numpy.linalg.matrix_rank`'s default, and 42 with no cutoff. The rank is therefore not an integrity check: with NumPy's default cutoff, an intact matrix fails a test for 36. The check is instead that $|\lambda_{\min}|/\lambda_{\max}$ is of order 10^{-15}, the edge of double precision; the matrix is ill-conditioned, without a definite rank. Do not check the sign of $\lambda_{\min}$, which sits at the noise floor and carries no information; a second host reproduces $\lambda_{\min}$ to only five significant digits.

5. The per-cell relative uncertainty is $\sqrt{(C_{EW})_{rr}}/y_r$. Because M carries the dropped axes' widths, $\mathbf{y}$ and C_{EW} are on the *density* scale. Converting cell r to an integrated cross section requires multiplying by that cell's own $\Delta E_{\mathrm{avail}}\,\Delta W$, and the catch bins make that factor large.

6. C_{EW} should *not* be inverted without a declared regularization: its spectrum runs continuously down to the double-precision noise floor, so an inverse is set by the regularizer rather than by the data. The estimator-seed sensitivity, which exceeds its limit (the first measurement of §9.6), is why no significance is computed from it.

Undeclared acceptance field. The record of the projection carries `acceptance_question: UNDECLARED`: when the projection was built, it had not been decided whether an acceptance correction is needed at the destination binning. The package must carry this field with its value until that is decided, so that no default in the data format makes that scientific choice.

H Codebase layout and terminology

The analysis chain has four components; the note refers to them by the names in italics below.

- *Event loop* (C++): `runEventLoopOmniFold`, built on the MINERvA Analysis Toolkit (MAT) and the MINERvA-101 framework. It reads production AnaTuples, applies selection and systematic-universe machinery, and writes per-event ROOT trees.
- *Event-loop outputs* (*omnifiles* in the code): per-event records for data, for simulated signal with paired true and reconstructed quantities, for simulated background, and for the truth-level efficiency denominator, together with the exposure in protons on target (POT). The `_universes_full` variants add per-universe weight and shifted-kinematics branches.
- *Driver* (Python): per-analysis orchestration, by a two-dimensional driver and its N-dimensional generalization. The driver builds training arrays from the event-loop outputs, subtracts the background by the method an option selects, the purity weight by default (§3.1), or negative-weight injection with or without the Stay-Positive refinement (App. C), runs the engine, and extracts the cross section.

- *Engine* (Python, `unbinned_unfolding/python/omnifold.py`): the OmniFold iteration, implemented independently of the learner backend: LightGBM for the central value and the uncertainty ensembles, scikit-learn's exact-split gradient boosting (`exact`) for the cross-check, its histogram gradient boosting (`hist`) as an alternate, and a multilayer perceptron (MLP) for a further cross-check.

Point-cloud (PET) code is under `nd-unfolding/pet/` (App. F).

The quoted training-seed block is assembled by the aggregation step of the 2D production, not by the scripts that run the training-seed variation.

Code identifiers cited in the note.

- Event loop (§2.4): `MINERvA101/MINERvA-101-Cross-Section/runEventLoopOmniFold.cpp`; its four trees are `mc_truth_denom` (efficiency denominator), `mc_signal_reco`, `mc_background` and `data`; the flux histogram is `pTmu_reweightedflux_integrated`.
- Cross-section extraction (§3.4): `extract_cross_section_2d` in the 2D driver, `2d-unfolding/unfold_2d_omnifold_unbinned.py`, and its N-dimensional generalization `nd-unfolding/xsec_nd.py`.
- Figures of §6: `2d-unfolding/plot_2d_paper_comparison.py` builds the simulated-truth curve of Fig. 9 from `hOFTruthDenom2D`; `2d-unfolding/compare_to_models.py` draws Fig. 14.

I Supplementary figure: coverage of a purity-subtracted variant

The coverage test of §9.5 measures, for each reported integral, the fraction of bias-corrected statistical intervals that contain the true value. Figure 36 shows it integral by integral at the truth with a reduced signal rate.

J Supplementary figure: closure with backgrounds

The closure of the five-dimensional pseudo-experiments with backgrounds, under purity-based and refined negative-weight subtraction (§9.2), is shown cell by cell in Fig. 37.

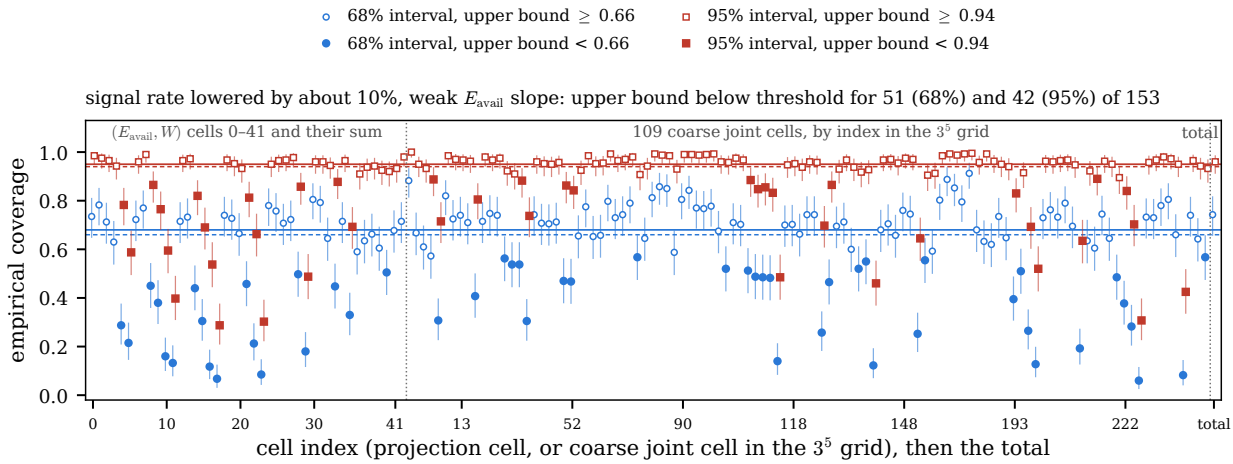


Figure 36: Empirical coverage of statistical intervals for the 153 reported cell and projection integrals, corrected for nominal-truth bias (§9.5). The study uses a deterministic LightGBM estimator with purity-based subtraction and 400 pseudo-experiments at a truth with an approximately 10 % lower signal rate and a weak E_{avail} slope. At nominal truth (not drawn) none lies below the 0.66 threshold, and two lie below the 0.94 threshold. Circles: 68% intervals; squares: 95% intervals. Bars are one-sided Clopper–Pearson lower and upper bounds, each at level 0.05/459, a Bonferroni correction over the 153 integrals and the three truths of the study. The third truth, a change of the q_3 dependence at fixed (E_{avail}, W), is omitted because its reweighting does not implement the intended change; the test's verdict does not depend on it. Solid lines mark the nominal coverage and dashed lines the acceptance thresholds 0.66 and 0.94; filled markers have an upper bound below the threshold. The integrals are ordered as the (E_{avail}, W) projection (cells 0–41 and their sum), the 109 coarse joint cells (labelled by their index in the 3^5 grid), and the total.

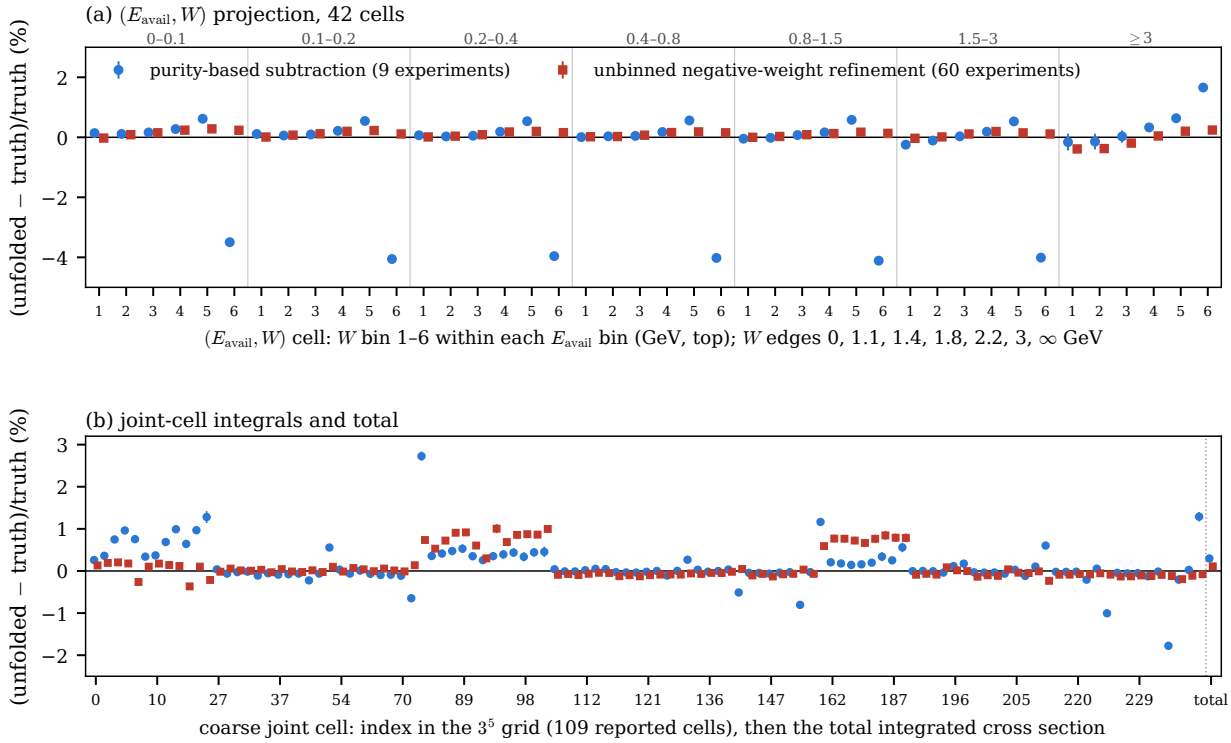


Figure 37: Fractional closure residual, (unfolded − truth)/truth, at nominal truth, for pseudo-experiments that include backgrounds. Both from a deterministic LightGBM estimator. Circles: purity-based background subtraction (9 pseudo-experiments). Squares: unbinned negative-weight subtraction with a refinement of 100 trees and 8 leaves (60 pseudo-experiments). The pseudo-experiments are not paired. (a) The 42 (E_{avail}, W) projection cells, grouped by E_{avail} bin. (b) The 109 coarse joint cells, labelled by their index in the 3^5 grid, then the total integrated cross section. Error bars are the standard error of the mean.

References

[1] A. Andreassen et al. "OmniFold: A Method to Simultaneously Unfold All Observables". In: *Phys. Rev. Lett.* 124.18 (2020), p. 182001. DOI: 10.1103/PhysRevLett.124.182001. arXiv: 1911.09107 [hep-ph].

[2] A. Andreassen et al. "Scaffolding Simulations with Deep Learning for High-dimensional Deconvolution". In: *ICLR simDL workshop* (2021). arXiv: 2105.04448 [stat.ML].

[3] D. Ruterbories et al. "Measurement of inclusive charged-current ν_μ cross sections as a function of muon kinematics at a mean neutrino energy of 6 GeV on hydrocarbon". In: *Phys. Rev. D* 104.9 (2021), p. 092007. DOI: 10.1103/PhysRevD.104.092007. arXiv: 2106.16210 [hep-ex].

[4] C. Andreopoulos et al. "The GENIE Neutrino Monte Carlo Generator". In: *Nucl. Instrum. Meth. A* 614 (2010), p. 87. DOI: 10.1016/j.nima.2009.12.009. arXiv: 0905.2517 [hep-ph].

[5] P. A. Rodrigues et al. "Identification of Nuclear Effects in Neutrino-Carbon Interactions at Low Three-Momentum Transfer". In: *Phys. Rev. Lett.* 116.7 (2016), p. 071802. DOI: 10.1103/PhysRevLett.116.071802. arXiv: 1511.05944 [hep-ex].

[6] T. Golan, J. T. Sobczyk, and J. Żmuda. "NuWro: the Wroclaw Monte Carlo Generator of Neutrino Interactions". In: *Nucl. Phys. B Proc. Suppl.* 229-232 (2012), p. 499. DOI: 10.1016/j.nuclphysbps.2012.09.136.

[7] O. Buss et al. "Transport-theoretical Description of Nuclear Reactions". In: *Phys. Rept.* 512 (2012), p. 1. DOI: 10.1016/j.physrep.2011.12.001. arXiv: 1106.1344 [hep-ph].

[8] M. V. Ascencio et al. "Measurement of inclusive charged-current ν_μ scattering on hydrocarbon at $<E_\nu> \sim 6$ GeV with low three-momentum transfer". In: *Phys. Rev. D* 106 (2022), p. 032001. DOI: 10.1103/PhysRevD.106.032001. arXiv: 2110.13372 [hep-ex].

[9] L. Aliaga et al. "Design, Calibration, and Performance of the MINERvA Detector". In: *Nucl. Instrum. Meth. A* 743 (2014), p. 130. DOI: 10.1016/j.nima.2013.12.103. arXiv: 1305.5199 [physics.ins-det].

[10] L. Aliaga et al. "Neutrino flux predictions for the NuMI beam". In: *Phys. Rev. D* 94.9 (2016), p. 092005. DOI: 10.1103/PhysRevD.94.092005. arXiv: 1607.00704 [hep-ex].

[11] K. Abe et al. "Measurements of neutrino oscillation parameters from the T2K experiment using 3.6×10^{21} protons on target". In: (2023). arXiv: 2303.03222 [hep-ex].

[12] M. A. Acero et al. "Improved measurement of neutrino oscillation parameters by the NOvA experiment". In: *Phys. Rev. D* 106.3 (2022), p. 032004. DOI: 10.1103/PhysRevD.106.032004. arXiv: 2108.08219 [hep-ex].

[13] K. Abe et al. "Hyper-Kamiokande Design Report". In: (2018). arXiv: 1805 . 04163 [physics.ins-det].

[14] B. Abi et al. "Deep Underground Neutrino Experiment (DUNE), Far Detector Technical Design Report, Volume II: DUNE Physics". In: (2020). arXiv: 2002.03005 [hep-ex].

[15] G. D'Agostini. "A Multidimensional unfolding method based on Bayes' theorem". In: *Nucl. Instrum. Meth. A* 362 (1995), p. 487. DOI: 10.1016/0168-9002(95)00274-X.

[16] M. Arratia et al. "Publishing unbinned differential cross section results". In: *JINST* 17.01 (2022), P01024. arXiv: 2109.13243 [hep-ex].

[17] N. Huetsch et al. "The Landscape of Unfolding with Machine Learning". In: *SciPost Phys.* 18 (2025), p. 070. arXiv: 2404.18807 [hep-ph].

[18] F. Canelli et al. "A Practical Guide to Unbinned Unfolding". In: *Eur. Phys. J. C* 86 (2026), p. 106. DOI: 10.1140/epjc/s10052-025-15265-9. arXiv: 2507.09582 [hep-ph].

[19] J. Chan and B. Nachman. "Unbinned Profiled Unfolding". In: *Phys. Rev. D* 108 (2023), p. 016002. DOI: 10.1103/PhysRevD.108.016002. arXiv: 2302.05390 [hep-ph].

[20] K. Datta, D. Kar, and D. Roy. "Unfolding with Generative Adversarial Networks". In: (2018). arXiv: 1806.00433 [physics.data-an].

[21] M. Bellagente et al. "How to GAN away Detector Effects". In: *SciPost Phys.* 8 (2020), p. 070. DOI: 10.21468/SciPostPhys.8.4.070. arXiv: 1912.00477 [hep-ph].

[22] M. Bellagente et al. "Invertible Networks or Partons to Detector and Back Again". In: *SciPost Phys.* 9 (2020), p. 074. DOI: 10.21468/SciPostPhys.9.5.074. arXiv: 2006.06685 [hep-ph].

[23] M. Backes et al. "An unfolding method based on conditional Invertible Neural Networks (cINN) using iterative training". In: (2022). arXiv: 2212.08674 [hep-ph].

[24] T. Adye. "Unfolding algorithms and tests using RooUnfold". In: *PHYSTAT 2011 Workshop on Statistical Issues*. 2011, p. 313. arXiv: 1105.1160 [physics.data-an].

[25] A. Höcker and V. Kartvelishvili. "SVD approach to data unfolding". In: *Nucl. Instrum. Meth. A* 372 (1996), p. 469. DOI: 10.1016/0168-9002(95)01478-0. arXiv: hep-ph/9509307.

[26] S. Schmitt. "TUnfold: an algorithm for correcting migration effects in high energy physics". In: *JINST* 7 (2012), T10003. DOI: 10.1088/1748-0221/7/10/T10003. arXiv: 1205.6201 [physics.data-an].

[27] H1 Collaboration. "Measurement of Lepton-Jet Correlation in Deep-Inelastic Scattering with the H1 Detector Using Machine Learning for Unfolding". In: *Phys. Rev. Lett.* 128.13 (2022), p. 132002. DOI: 10.1103/PhysRevLett.128.132002. arXiv: 2108.12376 [hep-ex].

[28] H1 Collaboration. "Unbinned Deep Learning Jet Substructure Measurement in High Q^2 ep collisions at HERA". In: *Phys. Lett. B* (2024). arXiv: 2303.13620 [hep-ex].

[29] R. Aaij et al. "Multidifferential study of identified charged hadron distributions in Z-tagged jets in proton-proton collisions at $\sqrt{s} = 13$ TeV". In: *Phys. Rev. D* 108 (2023), p. L031103. arXiv: 2208.11691 [hep-ex].

[30] G. Aad et al. "A simultaneous unbinned differential cross section measurement of twenty-four Z+jets kinematic observables with the ATLAS detector". In: *Phys. Rev. Lett.* 133 (2024), p. 261803. arXiv: 2405.20041 [hep-ex].

[31] A. Falcão and A. Takacs. "High-Dimensional Unfolding in Large Backgrounds". In: *JHEP* 07 (2026), p. 012. DOI: 10.1007/JHEP07(2026)012. arXiv: 2507.06291 [hep-ph].

[32] A. Butter et al. "Analysis-ready Generative Unfolding". In: (2025). arXiv: 2509.02708 [hep-ph].

[33] R. G. Huang et al. "Machine learning assisted unfolding for neutrino cross-section measurements with the OmniFold technique". In: *Phys. Rev. D* 112 (2025), p. 012008. DOI: 10.1103/sp1f-n9k2. arXiv: 2504.06857 [hep-ex].

[34] S. Agostinelli et al. "GEANT4 — a simulation toolkit". In: *Nucl. Instrum. Meth. A* 506 (2003), pp. 250–303.

[35] L. Grinsztajn, E. Oyallon, and G. Varoquaux. "Why do tree-based models still outperform deep learning on typical tabular data?" In: *Advances in Neural Information Processing Systems 35 (NeurIPS 2022), Datasets and Benchmarks Track*. 2022. arXiv: 2207.08815 [cs.LG].

[36] R. Milton et al. "Tools for Unbinned Unfolding". In: (2025). arXiv: 2503.09720 [hep-ph].

[37] J. Bailey and B. Nachman. *Records of the analysis note: provenance, fixed identities and audit record, test-suite discipline, and superseded values and withdrawn results*. Internal record of the MINERvA-OmniFold repository, not posted; built from `main_records.tex` by `build_all.sh`, https://github.com/josephbaileyy/MINERvA-OmniFold/tree/3c3a6f742d3aca583924903bd45f4cdea7b0b642/docs/analysis-note. 2026.

[38] H. Zhu et al. "Machine Learning-based Unfolding for Cross Section Measurements in the Presence of Nuisance Parameters". In: (2025). arXiv: 2512.07074 [stat.AP].

[39] A. Butter et al. "Simulation-Prior Independent Neural Unfolding Procedure". In: (2025). arXiv: 2507.15084 [hep-ph].

[40] A. Ore and T. Plehn. "Unfolding without Iterations, Adversaries, or Surrogates". In: (2026). arXiv: 2602.24282 [hep-ph].

[41] A. Bashyal et al. "Use of neutrino scattering events with low hadronic recoil to inform neutrino flux and detector energy scale". In: *JINST* 16.08 (2021), P08068. DOI: 10.1088/1748-0221/16/08/P08068. arXiv: 2104.05769 [hep-ex].

[42] *Two-dimensional uncertainty budget: defects in the earlier version and feasibility of a calibration test*. MINERvA-OmniFold repository record, https://github.com/josephbaileyy/MINERvA-OmniFold/blob/3892e769/docs/orchestration/state/next-preparation-20261009/two-d-followup/REPORT.md. 2026.

[43] Y. Hayato. "A neutrino interaction simulation program library NEUT". In: *Acta Phys. Polon. B* 40 (2009), p. 2477.

[44] D. Ruterbories et al. *Comparisons of triple-differential cross sections for quasielastic-like ν_μ-hydrocarbon interactions using $<E_\nu> \sim 3$ GeV versus ~ 6 GeV beams in MINERvA*. 2026. arXiv: 2606.00745 [hep-ex].

[45] S. Henry, H. Su, et al. "Measurement of Electron Neutrino and Antineutrino Cross Sections at Low Momentum Transfer". In: *Phys. Rev. D* 109.9 (2024), p. 092008. DOI: 10.1103/PhysRevD.109.092008. arXiv: 2312.16631 [hep-ex].

[46] J. Nieves, I. Ruiz Simo, and M. J. Vicente Vacas. "Inclusive Charged-Current Neutrino-Nucleus Reactions". In: *Phys. Rev. C* 83 (2011), p. 045501. DOI: 10.1103/PhysRevC.83.045501. arXiv: 1102.2777 [hep-ph].

[47] R. Gran et al. "Neutrino-nucleus quasi-elastic and 2p2h interactions up to 10 GeV". In: *Phys. Rev. D* 88.11 (2013), p. 113007. arXiv: 1307.8105 [hep-ph].

[48] T. Franco-Munoz et al. "Two-body current and axial form factor effects in charged-current quasielastic neutrino-nucleus scattering within the NEUT event generator". In: (2026). arXiv: 2605.00756 [nucl-th].

[49] A. Lozano et al. "Measurement of charged-current ν_μ and $\bar{\nu}_\mu$ cross sections on hydrocarbon in a shallow inelastic scattering region". In: *Phys. Rev. D* 114 (2026), p. 072001. DOI: 10.1103/vq38-kmxq. arXiv: 2503.20043 [hep-ex].

[50] L. Brenner et al. "Comparison of unfolding methods using RooFitUnfold". In: (2020). arXiv: 1910.14654 [physics.data-an].

[51] M. Kuusela and V. M. Panaretos. "Statistical unfolding of elementary particle spectra: Empirical Bayes estimation and bias-corrected uncertainty quantification". In: *Annals of Applied Statistics* 9.3 (2015), pp. 1671–1705. DOI: 10.1214/15-AOAS857. arXiv: 1505.04768.

[52] *Five-dimensional joint generator tests: specification and result*. MINERvA-OmniFold repository record, https://github.com/josephbaileyy/MINERvA-OmniFold/blob/5f4ec13a/docs/orchestration/RECORD-20261005-s5p-joint-5d-inference-result.md. 2026.

[53] O. Ledoit and M. Wolf. "A well-conditioned estimator for large-dimensional covariance matrices". In: *J. Multivariate Anal.* 88 (2004), pp. 365–411. DOI: 10.1016/S0047-259X(03)00096-4.

[54] S. Holm. "A simple sequentially rejective multiple test procedure". In: *Scand. J. Stat.* 6 (1979), pp. 65–70.

[55] C. J. Clopper and E. S. Pearson. "The use of confidence or fiducial limits illustrated in the case of the binomial". In: *Biometrika* 26 (1934), pp. 404–413. DOI: 10.1093/biomet/26.4.404.

[56] *Sensitivity of the joint-test decisions to the assumed residual below the fine grid*. MINERvA-OmniFold repository record, https://github.com/josephbaileyy/MINERvA-OmniFold/blob/fbf682f9/publication/kappa/RECORD-20261008-kappa-breakdown.md. 2026.

[57] *Five-dimensional joint generator tests: rerun of the pseudo-experiments lost to time limits*. MINERvA-OmniFold repository record, https://github.com/josephbaileyy/MINERvA-OmniFold/blob/5f4ec13a/docs/orchestration/RECORD-20261006-s5p-lost-seed-recovery-resolution.md. 2026.

[58] *Joint generator tests restricted to coarse two-dimensional projections*. MINERvA-OmniFold repository record, https://github.com/josephbaileyy/MINERvA-OmniFold/blob/5f4ec13a/docs/publication/w1/W1-RECORD-20261006.md. 2026.

[59] L. Aliaga et al. "MINERvA neutrino detector response measured with test beam data". In: *Nucl. Instrum. Meth. A* 789 (2015), pp. 28–42. DOI: 10.1016/j.nima.2015.04.003. arXiv: 1501.06431 [physics.ins-det].

[60] *Recoil-response sensitivity of the joint-test rejections*. MINERvA-OmniFold repository record, https://github.com/josephbaileyy/MINERvA-OmniFold/blob/2926e39e/publication/w2/W2B-REPORT-20261006.md. 2026.

[61] ATLAS Collaboration. *Unbinned measurement of Z+jets production, unfolded to full charged-particle phase space using the OmniFold algorithm*. CERN Open Data Portal. 2026. DOI: 10.7483/OPENDATA.ATLAS.EHNB.AXOS.

[62] A. Bajpai. *Omnifold—a Python package for publishing, and comparing unbinned, unfolded measurements*. PyHEP.dev 2026, 7–9 September 2026. 2026.

[63] K. Desai, O. Long, and B. Nachman. "Unbinned Inference with Correlated Events". In: *Eur. Phys. J. C* 85 (2025), p. 1089. DOI: 10.1140/epjc/s10052-025-14835-1. arXiv: 2504.14072 [physics.data-an].

[64] M. Stanley et al. "Confidence intervals for functionals in constrained inverse problems via data-adaptive sampling-based calibration". In: (2025). arXiv: 2502.02674 [stat.ME].

[65] P. Batlle et al. "Simultaneous Frequentist Calibration of Confidence Regions for Multiple Functionals in Constrained Inverse Problems". In: (2025). arXiv: 2510.11708.

[66] B. Nachman and J. Noll. "Stay Positive: Neural Refinement of Sample Weights". In: (2025). arXiv: 2505.03724 [physics.data-an].

[67] D. I. Glazier and R. Tyson. "Converting sWeights to Probabilities with Density Ratios". In: (2024). arXiv: 2409.08183 [physics.data-an].

[68] "Measurement of the ν_μ charged-current quasielastic cross section on hydrocarbon". In: (2013). arXiv: 1305.2243 [hep-ex].

[69] "Measurement of the $\bar{\nu}_\mu$ charged-current quasielastic cross section on hydrocarbon". In: (2013). arXiv: 1305.2234 [hep-ex].

[70] "Ratios of ν_μ charged-current cross sections on C, Fe, and Pb to CH". In: (2014). arXiv: 1403.2103 [hep-ex].

[71] "Coherent $\pi^\pm$ production in $\nu_\mu/\bar{\nu}_\mu$ interactions on carbon (1.5–20 GeV)". In: (2014). arXiv: 1409.3835 [hep-ex].

[72] B. Eberly et al. "Charged Pion Production in ν_μ Interactions on Hydrocarbon at a mean neutrino energy of 4.0 GeV". In: *Phys. Rev. D* 92.9 (2015), p. 092008. DOI: 10.1103/PhysRevD.92.092008. arXiv: 1406.6415 [hep-ex].

[73] "Muon-plus-proton final states in ν_μ interactions on hydrocarbon". In: (2015). arXiv: 1409.4497 [hep-ex].

[74] "Single neutral-pion production by $\bar{\nu}_\mu$ on hydrocarbon". In: (2015). arXiv: 1503.02107 [hep-ex].

[75] "Measurement of electron-neutrino quasielastic-like scattering on hydrocarbon". In: (2016). arXiv: 1509.05729 [hep-ex].

[76] "Cross sections for ν_μ and $\bar{\nu}_\mu$ induced pion production on hydrocarbon". In: (2016). arXiv: 1606.07127 [hep-ex].

[77] "Measurement of partonic nuclear effects in deep-inelastic ν_μ scattering". In: (2016). arXiv: 1601.06313 [hep-ex].

[78] "Charged-current ν_μ K^+ production on hydrocarbon". In: (2016). arXiv: 1604.03920 [hep-ex].

[79] "Measurement of ν_μ charged-current single π^0 production on hydrocarbon". In: (2017). arXiv: 1708.03723 [hep-ex].

[80] "Direct measurement of the nuclear dependence of charged-current quasielastic-like ν_μ interactions". In: (2017). arXiv: 1705.03791 [hep-ex].

[81] C. E. Patrick et al. "Measurement of the muon antineutrino double-differential cross section for quasielastic-like scattering on hydrocarbon at $E_\nu \sim 3.5$ GeV". In: *Phys. Rev. D* 97.5 (2018), p. 052002. DOI: 10.1103/PhysRevD.97.052002. arXiv: 1801.01197 [hep-ex].

[82] X.-G. Lu et al. "Measurement of final-state correlations in neutrino muon-proton mesonless production on hydrocarbon at a mean neutrino energy of 3 GeV". In: *Phys. Rev. Lett.* 121.2 (2018), p. 022504. DOI: 10.1103/PhysRevLett.121.022504. arXiv: 1805.05486 [hep-ex].

[83] "Coherent $\pi^\pm$ production in $\nu_\mu/\bar{\nu}_\mu$ charged-current interactions on carbon". In: (2018). arXiv: 1711.01178 [hep-ex].

[84] "Measurement of quasielastic-like ν_μ scattering on hydrocarbon". In: (2019). arXiv: 1811.02774 [hep-ex].

[85] "Measurement of antineutrino single π^- production on hydrocarbon". In: (2019). arXiv: 1906.08300 [hep-ex].

[86] MINERvA Collaboration. "Double-differential inclusive charged-current ν_μ cross sections on hydrocarbon in MINERvA at a mean neutrino energy of 3.5 GeV". In: *Phys. Rev. D* 101.11 (2020), p. 112007. DOI: 10.1103/PhysRevD.101.112007. arXiv: 2002.12496 [hep-ex].

[87] T. Cai et al. "Nucleon binding energy and transverse momentum imbalance in neutrino-nucleus reactions". In: *Phys. Rev. D* 101.9 (2020), p. 092001. DOI: 10.1103/PhysRevD.101.092001. arXiv: 1910.08658 [hep-ex].

[88] "High-statistics measurement of ν_μ charged-current quasielastic-like cross sections". In: (2020). arXiv: 1912.09890 [hep-ex].

[89] "ν_μ charged-current neutral-pion production and nuclear effects on hydrocarbon". In: (2020). arXiv: 2002.05812 [hep-ex].

[90] D. Ruterbories et al. "Simultaneous measurement of proton and lepton kinematics in quasielastic-like ν_μ-hydrocarbon interactions from 2 to 20 GeV". In: *Phys. Rev. Lett.* 129 (2022), p. 021803. DOI: 10.1103/PhysRevLett.129.021803. arXiv: 2203.08022 [hep-ex].

[91] MINERvA Collaboration. "Measurement of ν_μ charged-current single π^+ production on hydrocarbon". In: (2023). arXiv: 2209.07852 [hep-ex].

[92] "Measurement of coherent π^+ production in ν_μ charged-current interactions on nuclear targets". In: (2023). arXiv: 2210.01285 [hep-ex].

[93] MINERvA Collaboration. "Measurement of the Multi-Neutron $\bar{\nu}_\mu$ Charged Current Differential Cross Section at Low Available Energy on Hydrocarbon". In: (2023). arXiv: 2310.17014 [hep-ex].

[94] J. Kleykamp et al. "Measurement of the A dependence of the ν_μ charged-current quasielastic-like cross section as a function of muon and proton kinematics at a mean neutrino energy of 6 GeV". In: *Phys. Rev. D* (2025). arXiv: 2503.15047 [hep-ex].

[95] E. Granados et al. "High Statistics Measurements of ν_μ Charged-Current Single π^+ Production with Zero Pion Kinetic Energy Threshold in MINERvA". In: (2026). arXiv: 2605.24224 [hep-ex].

[96] A. Klustová et al. "Measurement of Inclusive Charged-Current $\bar{\nu}_\mu$ Scattering on C, CH, Fe, and Pb at $<E_{\bar{\nu}}> \sim 6$ GeV with MINERvA". In: (2026). arXiv: 2604.07091 [hep-ex].

[97] *Candidate release package for the article: replay of the joint tests and regeneration of the figures.* MINERvA-OmniFold repository record, https://github.com/josephbaileyy/MINERvA-OmniFold/blob/3892e769/docs/publication/release/RC7-README.md. 2026.

[98] *SHA256 checksums of the AnaTuple input files and preservation of the data files.* MINERvA-OmniFold repository record, https://github.com/josephbaileyy/MINERvA-OmniFold/blob/3892e769/docs/publication/corrections-20261008/RECORD-20261010-ost61-hashed-and-data-addition.md. 2026.